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QUESTIONS

VEENEETA DAYAL

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INTERVENTION

who

functional-wh

extraction

where

What

when

WH-SING / WH-PLURAL

why

maximality

ISLANDS

bias

D-linking

SELECTION

OXFORD SURVEYS IN SEMANTICS AND PRAGMATICS

Questions

OXFORD SURVEYS IN SEMANTICS AND PRAGMATICS

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For little Sameer

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General preface

Oxford Surveys in Semantics and Pragmatics aims to convey to the reader the life and spirit of the study of meaning in natural language. Its volumes provide distillations of the central empirical questions driving research in contemporary semantics and pragmatics, and distinguish the most important lines of inquiry into these questions. Each volume offers the reader an overview of the topic at hand, a critical survey of the major approaches to it, and an assessment of what consensus (if any) exists. By putting empirical puzzles and theoretical debates into a comprehensible perspective, each author seeks to provide orientation and direction to the topic, thereby providing the context for a deeper understanding of both the complexity of the phenomena and the crucial features of the semantic and pragmatic theories designed to explain them. The books in the series offer researchers in linguistics and related areas—including syntax, cognitive science, computer science, and philosophy—both a valuable resource for instruction and reference and a state-of-the-art perspective on contemporary semantic and pragmatic theory from the experts shaping the field.

This volume addresses the semantics and the pragmatics of questions, an absolutely fundamental topic in natural language meaning. What do questions denote? How precisely do the parts of an interrogative expression combine to form these denotations? How do questions differ across constructions in a given language, as well as across languages? This is a vast topic with a rich and varied literature, requiring a guide with wide-ranging experience and superb judgment. Dayal is exactly this kind of guide. She begins with a core standard fragment covering basic questions, and then explores other topics and other approaches as refinements and variants of the core model, including yes/no versus wh questions; single wh questions versus multiple wh questions; *mention-some* versus *mention-all* interpretations; pair-list and functional answers; focus; weak islands; and many other topics. This elegant approach does justice to the subtlety and the complexity of the subject at the same time that it provides a satisfying and solid foundation for researchers at any level who are interested in exploring this central and indispensable part of language.

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- making me aware of literature I did not know about;
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- helping me probe deeper into the facts;
- discussing issues that were sometimes unclear to me;
- arguing and/or playing devil's advocate for positions different from mine;
- making me feel that they were interested in what I had to say on this topic;
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More specific acknowledgments surface in the course of the discussions in this book.

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Last, but not least, thanks to my family and my friends outside academia for putting up with talk of “the book” for so long. For the most part they don't quite get what linguistics is all about, except perhaps for my daughter Karuna—but then she was exposed to it during the critical period. For the others, it's all a bit of mystery why it consumes me so. I don't believe this book is going to change that.

When I published my first book, my family informed me that they could not even understand its title: *Locality in Wh Quantification*. And who could blame them for that? Well, this time they will understand the title, but my father has already expressed some bewilderment, wondering how one could write a whole book on *Questions*. I thank my fellow researchers for making it possible to do so.

Introducing questions and answers

This chapter identifies the domain of inquiry undertaken in this survey, highlighting syntactic, semantic, and pragmatic aspects of questions. It gives a soft introduction to the semantics of questions, articulating the relationship between questions and answers. It also addresses the role of syntactic scope for *wh* expressions in semantic accounts and clarifies the extent to which this study engages with syntactic theory.

1.1 What is a question?

Questions are a quintessential interface phenomenon. They can be studied from the perspective of syntax, semantics, prosody, or pragmatics. Even the basic definition of a question involves a decision between these aspects of grammar. Consider:

- (1) a. What is your name?
b. Is the Pope Catholic?
c. Who are you to tell me what to do?

Syntactically, the examples in (1) would readily be classified as questions. They instantiate interrogative rather than declarative structures, marked by subject-auxiliary inversion and/or *wh* fronting. This structure typically goes along with a rising intonation, distinguishing it from the falling contour of standard declarative sentences.

The examples in (1) may not be classified as questions quite as readily, however, if we apply pragmatic criteria to them. The speech act of questioning involves a request for information. A felicitous use of a question requires that the speaker not be privy to the information and that he/she must believe that the addressee might be. Example (1a), under normal circumstances, is likely to meet the felicity requirements for the speech act of questioning, but not (1b). Anyone who understands the concept of pope-hood could not be ignorant about the content of the question.¹ Example (1c)

¹ Question (1a) addressed to a newborn would not be a question from the pragmatic perspective. Conversely, if someone ignorant about Catholicism asks (1b) it would qualify pragmatically as a question.

also does not appear to be a request for information. Rather, it seems to make a strong (negative) assertion. Natural discourse is full of interrogatives that are acts of questioning as well as interrogatives that do not fit that mold.

In the other direction, there are non-interrogative structures that may be considered questions from the speech act perspective:

- (2) a. It's raining?
- b. I wonder what time it is.

Example (2a) does not have inversion but it is marked by rising intonation, signaling that it is not an assertion. Its formal status as an interrogative could be challenged but it clearly calls for a response confirming/denying that it is raining. And (2b), a normal declarative sentence, could quite naturally lead to a response stating the time.

There is more one could say about such examples, but the general point should be clear. There is a non-trivial decision to be made even in delineating the empirical domain of this study. I take as my starting point what I believe is a fundamental assumption in the semantic literature. The interrogative is the canonical form for seeking information and the primary job of semantics is to explicate the relation between the interrogative form and the speech act of questioning. This does not mean that interrogatives that do not fulfill the felicity conditions for questioning, or non-interrogatives that do, are not legitimate objects of semantic inquiry. I take questions to include these cases but the tie between interrogatives and information-seeking provides a reasonable starting point.

Elaborating further on the distinction between form and use, although our primary examples will come from English, we note that there is considerable cross-linguistic variation in the forms used for requesting information. Consider (3)–(4), illustrating two question formation strategies attested in natural language:

- (3) a. tumhaaraa naam kyaa hai *Hindi*
 your name what is
 “What is your name?”
- b. ravi baahar hai?
 Ravi outside is
 “Is Ravi outside?”
- (4) a. Mary-ga nani-o kat-ta ka *Japanese*
 Mary what bought Q
 “What did Mary buy?”
- b. Mary-ga Aspects-o kat-ta ka
 Mary Aspects bought Q
 “Did Mary buy Aspects?”

Unlike English, Hindi *wh* expressions can occur in their normal argument position. Hindi is an SOV language so *kyaa* “what” appears in the preverbal

object position in (3a). Example (3b) is a yes-no/polar question which does not differ syntactically from its declarative counterpart. It is worth keeping in mind, however, that even when there are no discernible syntactic cues, intonation is used to distinguish interrogatives from declaratives.² Japanese manifests a different version of the in-situ strategy. The interrogative-declarative distinction is flagged by a particle on the clause, rather than on a noun phrase. That is, there is no direct correlate of English or Hindi *wh* expressions in Japanese questions. If *ka* were to occur next to *nani* in (4a), for example, it would be interpreted as “something” rather than “what.”³ There are many languages that follow in-situ strategies of the Hindi or the Japanese kind, as well as variations thereof.

There are also cross-linguistic differences in form within the class of wh fronting languages, a difference that emerges in multiple wh/constituent questions. While languages like English allow only one wh expression to occur clause initially, languages like Bulgarian require all of them to front:⁴

- (5) koj kakvo e kupila *Bulgarian*
 who what has bought
 ‘Who has bought what?’

The following schema indicates the range of possible variation in question formation strategies in natural language:

- [illegible]

This study does not explore these distinctions in any depth since its primary focus is on the semantics and pragmatics of questions.⁵ The issues discussed here should be applicable across all language types, once adjustments in the mapping from form to meaning are made to accommodate syntactic differences.

Turning now to the speech act involved, the felicity conditions for questioning can be spelt out as below:⁶

² Hindi *wh* expressions may not always appear in-situ since the language has free word order, though a preverbal position is considered the default (see Dayal 1996 and Kidwai 2000, among others). Also, the *wh* expression *kya* may optionally occur in polar questions.

³ See, for example, Nishigauchi (1990) on Japanese.

⁴ See Rudin (1988), among others, on multiple fronting.

⁵ Facts from languages other than English will be discussed if they bear on issues under discussion.

⁶ Adapted from *The Language Files*, 10th edition, Ohio State University. These conditions may not always apply, depending on the type of question: *rhetorical*, *quiz*, *echo*, etc. We briefly explore such questions in Chapter 9.

- (7) SPEECH ACT OF QUESTIONING—Speaker questions Hearer about proposition *p* iff
- i. S does not know the truth about *p*.
 - ii. S wants to know the truth about *p*.
 - iii. S believes H knows the truth about *p*.

Applying these criteria, we see how (1a) would qualify as a bona fide question if uttered by a speaker to an addressee she has just met. It is expected that the addressee will satisfy this request by stating her name correctly. It is also implicit in this view that a question must allow for more than one possible answer, representing the fact that the speaker is unaware of the true state of affairs.⁷

These felicity conditions also give us some insight into questions that appear to be special in some way. Consider echo or incredulity questions, both of which are triggered by statements in previous discourse. Such questions often do not involve fronting but they are marked by stress on the *wh* word, here indicated by small caps, marking them as intonationally distinct. Subscripting indicates an inaudible string:

- (8) a. John ate the _{meat}.
 b. John ate WHAT? (I couldn't hear you.)
- (9) a. John ate the meat.
 b. John ate WHAT? (I can't believe you.)

In the first case, the speaker does not hear some part of the previous utterance. The *wh* expression is a request for repetition, rather than a standard request for information about what John ate. In the second case, the speaker hears the utterance correctly but does not believe it. The question is not a request for information but signals that the information given is at odds with prior knowledge: perhaps John is known to have strong feelings against the slaughter of animals. Such interrogatives do not meet the conditions for the primary speech act of questioning. One might say that a secondary indirect speech act is ascribed to them in order to make sense of their discourse contribution.

Similarly, rhetorical questions arise because the conditions for the speech act of questioning are not met:

- (10) a. Is the Pope Catholic?
 b. Who enjoys doing such chores?

Polar questions such as (10a) cannot be answered negatively since that would go against world knowledge. A likely context for (10b) is one where the reference is to unpleasant chores that are assumed to be unappealing. Given that speaker and

⁷ Consider a contrived context where only people with the name *Kim Smith* are invited. Since this information would be known to the speaker, the question *what is your name?* would be infelicitous.

hearer are both aware of this, the only way for such questions to be meaningful conversational contributions is if the very fact of the determinacy of their answers was the message. They are ways of stating that the response to an issue under discussion should be self-evident or of noting that a previous assertion states the obvious.

Although there is considerable cross-linguistic variation in the form of interrogatives, the conditions in (7) are likely to be universal and will define the speech act of questioning in every language. Whenever these conditions are not met, interrogative forms will take on a specialized character. Even though these specialized interrogatives do not seek information, they fall within the realm of semantic investigation into questions. The conversational contribution of questions that are not requests for information are still calculated based on the semantics associated with canonical questions.

This might be the place for two clarifications. If you are a reader who has thought about this topic, you are probably dissatisfied with the imprecision surrounding almost every statement I have made so far. I expect this dissatisfaction will continue through the rest of this chapter. I could find no other way to bring the uninitiated into the topic than to unabashedly simplify and hedge. Hopefully, subsequent chapters where I discuss the insights of researchers will prove more satisfactory. The second clarification is terminological. The term *question* is going to be used as a cover term for the syntactic, semantic, and pragmatic object of this study. The reason for this is simply that we cannot always use *interrogative* when talking about the syntactic object. And, as we have already seen, there are requests for information that have declarative syntax. Similarly, we have seen that the *speech act of questioning* does not apply uniformly to every type of question. Finally, to use *question denotation* for the semantic object sometimes seems unnecessarily wordy. I opt for what sounds natural, allowing the context to make it clear what is intended by a particular use of the term *question*. The terms *interrogative*, *question denotation*, and *speech act of questioning* will refer unambiguously to syntactic, semantic, and pragmatic objects, respectively.

1.2 What do questions mean?

This is an introduction to a basic semantics for questions. It includes a step by step illustration of how question meanings are built up as well as some of the core intuitions that motivate this semantics. It presupposes some knowledge of truth conditional semantics, such as can be found in Chierchia and McConnell-Ginet (2000) or Heim and Kratzer (1998). Since determining the meaning of questions relies on intuitions about acceptable answers, it also includes a discussion of possible answers.

1.2.1 The syntax-semantics map

Let us start with an ordinary interrogative and assume that it is a simple request for information. We adopt the theory of questions proposed by Hamblin (1973)

and its development in Karttunen (1977), where questions are treated as denoting sets of propositions.⁸ Under this view, the question in (11a) denotes a set of possible answers. If there are two books in the domain, say *Emma* and *Persuasion*, the denotation of the question, (11b), would include the two propositions in (11c).^{9,10} An answer picks out the true propositions in the set. Thus, if Bill bought *Persuasion* but not *Emma*, (11d) would be an appropriate response. It would be truthful and it would provide the information requested. Presupposing a plurality of individuals in the domain of quantification guarantees that a question can represent a state of ignorance about which proposition(s) in the set happen to be true, an ignorance that the answer dispels:

- (11) a. Which book did Bill buy?
 b. $\lambda p \exists x [\text{book}(x) \wedge p = \wedge \text{bought}(b, x)]$
 c. $\{\wedge \text{Bill bought } E, \wedge \text{Bill bought } P\}$
 d. (Bill bought) *Persuasion*.

Let us work through a derivation, starting with the corresponding declarative. The structure on which interpretation is defined is what von Stechow (1996) calls *transparent Logical Form*, a level of syntax where all scope bearing expressions are at the position in which they are interpreted (see also Heim 1989; Bittner 1994, 1998).¹¹ In this particular case, I assume that quantified expressions have moved covertly at Logical Form (LF) via the rule of *Quantifier Raising* (QR) and adjoined at the clausal level, leaving behind a co-indexed trace in object position:

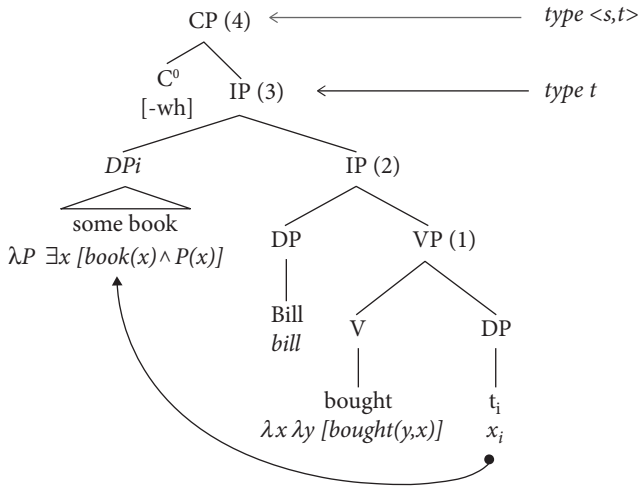
⁸ This section is meant for those without prior knowledge of the topic. In this exposition, we follow Hamblin in treating questions as denoting the set of all possible answers, not just the true ones, and Karttunen in taking *wh* expressions to be a type of existential quantifier. Differences between the two theories and reasons for opting for this particular blend are presented in Chapter 2.

⁹ $\wedge \text{Bill bought Emma}$ expresses the same proposition as $\lambda w \text{ bought}_w(\text{bill}, \text{emma})$, or $\lambda w \text{ bought}(w)(\text{bill}, \text{emma})$ i.e. the set of worlds in which Bill bought *Emma*. See Dowty et al. (1981) and Chierchia and McConnell-Ginet (2000) for discussion of the $\wedge$ operator, and Gallin (1975) for the version with explicit reference to worlds. I use this instead of adopting the practice of prefacing propositions with *that* since the latter seems to invoke embedded questions, for some readers at least. I will go back and forth between these two ways of expressing propositions, depending on what works better for expository purposes. If nothing else, it will serve to remind young readers of the system favored by Montague, without whom who knows what kind of semantics we would be doing now!

¹⁰ I use underlining to indicate the proposition(s) in the set that happen to be true in the world of evaluation.

¹¹ The LF structures in May (1985), for example, are not transparent since they allow two quantifiers $[Q_1 [Q_2 \dots]]$ to take scope over each other if they are sufficiently close structurally.

(12)



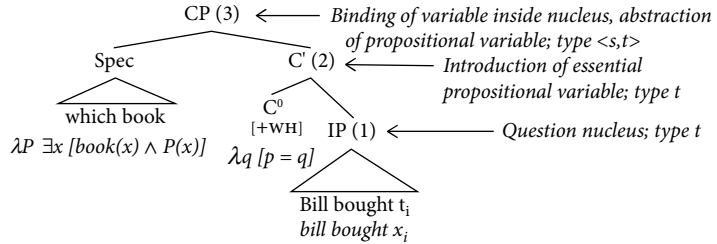
- $\llbracket 1 \rrbracket = \llbracket V \rrbracket (\llbracket DP \rrbracket) \Rightarrow \lambda x \lambda y [\text{bought}(y,x)] (x_i) \Rightarrow \lambda y [\text{bought}(y,x_i)]$ functional application
 $\llbracket 2 \rrbracket = \llbracket VP \rrbracket (\llbracket DP \rrbracket) \Rightarrow \lambda y [\text{bought}(y,x_i)] (\text{bill}) \Rightarrow [\text{bought}(\text{bill},x_i)]$ functional application
 $\llbracket 3 \rrbracket = \llbracket DP \rrbracket (\llbracket IP \rrbracket) \Rightarrow \lambda P \exists x [\text{book}(x) \wedge P(x)] (\lambda x_i [\text{bought}(\text{bill},x_i)])$ predicate abstraction
 $\Rightarrow \exists x [\text{book}(x) \wedge \lambda x_i [\text{bought}(\text{bill},x_i)] (x)]$ lambda conversion
 $\Rightarrow \exists x [\text{book}(x) \wedge \text{bought}(\text{bill},x)]$ lambda conversion
 $\llbracket 4 \rrbracket = \wedge \exists x [\text{book}(x) \wedge \text{bought}(\text{bill},x)]$ intensional abstraction

The trace in the internal argument position of the two-place verb *buy* is interpreted as an indexed variable of type e . The object and subject terms saturate the arguments of the verb and we get a sentence-level meaning. So far, we have simply done functional application. The raised indefinite DP is interpreted as a generalized quantifier (*type* $\langle\langle e,t \rangle, t \rangle$), denoting the set of those properties P , such that some x that is a book has that property. If the books are *Emma* and *Persuasion*, it would include properties common to both, such as *written by Jane Austen*, as well as those that only one of them has, such as *has a character named Wentworth* or *is named after a character in it*. This quantifier cannot, as it stands, semantically compose with its sister IP which is of type t . We take co-indexation between the quantifier and the variable inside IP to trigger lambda abstraction, turning a sentence-level meaning into a property (*type* $\langle e, t \rangle$). The two can now combine via functional application.¹² The final step of the derivation involves turning the sentence into a proposition, indicated by the addition of Montague's *up-operator* (see footnote 9) at the CP level. The sentence is true if the property of *being bought by Bill* is in the set denoted by *some book*.

¹² I will not represent these structures with a lambda appended, as is done in Heim and Kratzer: [*some book* _{i} [λ_i [*Bill bought t_i*]]]). This is to keep the exposition simple as nothing hinges on it. See discussion of Montague's quantificational rules in Dowty et al. (1981) and the *Predicate Abstraction Rule* in Heim and Kratzer (1998: 186).

With this perspective on quantification, when we return to interrogatives, the CP level becomes crucial. Though we treat the *wh* expression as an indefinite that moves overtly out of the core IP, it is taken to adjoin to CP, above C' , not to IP. This partitions the structure into three semantically significant parts:

(13)



$\llbracket 2 \rrbracket = \lambda q [p = q] (^{\wedge}\text{bought}(\text{Bill}, x_i))$	Intensional Functional Application
$\llbracket 3 \rrbracket = \lambda P \exists x [\text{book}(x) \wedge P(x)] (\lambda x_i [p = ^{\wedge}\text{bought}(\text{Bill}, x_i)])$	Functional Application
$\Rightarrow \exists x [\text{book}(x) \wedge \lambda x_i [p = ^{\wedge}\text{bought}(\text{Bill}, x_i)](x)]$	λ -conversion
$\Rightarrow \exists x [\text{book}(x) \wedge p = ^{\wedge}\text{bought}(\text{Bill}, x)]$	λ -conversion
$\Rightarrow \lambda p \exists x [\text{book}(x) \wedge p = ^{\wedge}\text{bought}(\text{Bill}, x)]$	Abstraction over p

The first significant part is IP, which provides the question nucleus. Its interpretation is exactly as in (12). It contains, in the case of a constituent question, the essential free variable contributed by the trace of the fronted expression. As before, we assume that the type of the trace will be compatible with the type of its binder, type e in this case.

The second significant part of the structure is C^0 , which is the locus for the shift to a propositional meaning (from t to $\langle s, t \rangle$) when the specification is $-WH$, as it was in (12). It is also the locus for the shift to an interrogative meaning, a set of propositions (from t to $\langle \langle s, t \rangle, t \rangle$) when the specification is $+WH$, as it is here. This is effected by introducing the propositional variable p that is later abstracted over. So, C^0 is defined as a function that takes a proposition q and yields the proposition p which is identical to it. It combines with the IP denotation, adjusted for its intensional meaning ($\langle s, t \rangle$), via intensional functional application. This forces p to be identical to the question nucleus. This, we will see, provides the essential control on the propositions that the question can denote.

The third significant part is SpecCP, which hosts the *wh* expression, interpreted as an existential quantifier. Its restriction is implicit in the case of monomorphemic expressions like *what*, or explicit in the case of full-fledged *wh* expressions like *which book*. The quantifier denotes the set of properties that some object/book has. It is a plain old existential quantifier, like *something* or *some book*, and binds the free variable inside C' . Co-indexation between the *wh* and the trace, as before, triggers abstraction over the variable in the base position, converting C' into a predicative expression (type $\langle e, t \rangle$). We assume, of course, that *wh* expressions can only occur with C^0_{+WH} and that this is enforced morpho-syntactically. There

is nothing in the semantics to ensure this result. The last step is the binding of the free propositional variable p , which creates the right type for the question: a set of propositions $\langle \langle s, t \rangle, t \rangle$.¹³

Let us see how the outlined procedure picks out the right set of propositions and how all the relevant propositions are brought into the denotation. Although there is no universal quantifier anywhere in the derivation, the question is a function from all propositions (cf: λp) to those propositions that are identical to the nucleus for some value of the free variable. This means that propositions like *It is raining* will not enter the denotation because they are not identical to the nucleus, regardless of the value assigned to the variable x : they simply do not denote the same set of worlds. Colloquially, the set of worlds in which it rains and the set of worlds in which Bill bought either book are not identical. More significantly, *Jill bought Emma* and *Bill bought Newsweek* will not be included in the denotation of *Which book did Bill buy*. $\wedge \text{Jill bought Emma} \neq \wedge \text{Bill bought } x^{\text{Emma}}$ because the set of worlds in which Jill buys *Emma* is not the same set of worlds in which Bill buys *Emma*. $\wedge \text{Bill bought Newsweek} \neq \wedge \text{Bill bought } x^{\text{Emma/Persuasion}}$ because there is no value of x that satisfies the restriction *book* and the nucleus condition identifying it with the buying of the magazine by Bill. The requisite control on propositions, we see, is enforced by the restriction on the *wh book*, as well as the identity relation with the question nucleus $\wedge \text{Bill bought } x$.

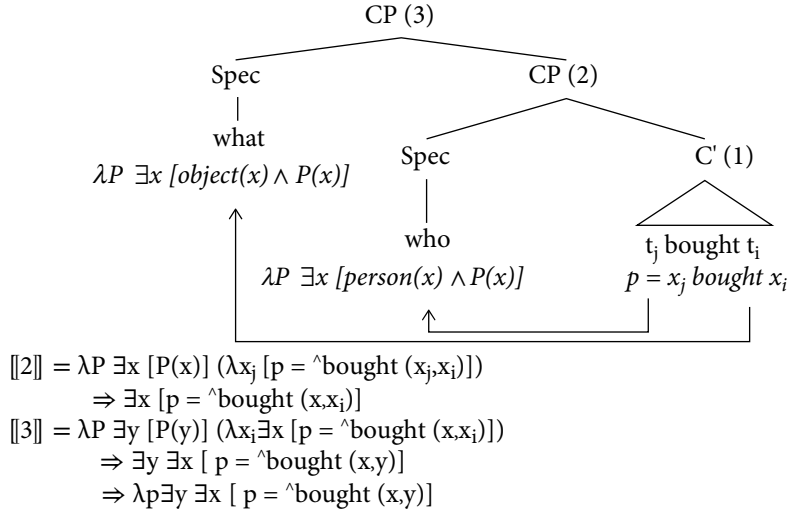
It is worth noting that the existential quantifier must take scope outside the propositional variable in order to yield the right set of propositions. If we had the *wh* taking scope inside p , we would have: $\lambda p [p = \wedge \exists x [\text{book}(x) \wedge \text{bought}(\text{Bill}, x)]]$. None of the propositions we have intuitively determined to be possible answers would pass muster. The set of worlds in which Bill bought *Emma* is a subset of the worlds in which he bought some book. The only proposition that would qualify would be the proposition *Bill bought some book*, which is a presupposition of the question, not an answer to it. This also underscores the hidden assumption in all theories of questions: if an answer to a question specifies values for a *wh* expression, that expression must have scope outside p . This, however, is not always the case (see Section 1.3.1). Still, it can be used as a rule of thumb in determining the scope of *wh* expressions.

Using answers as the guide for *wh* scope, we can extend the interpretive procedure for basic questions to multiple constituent questions, as well as to polar questions. Consider multiple constituent questions:

- (14) a. Who bought what?
 b. John bought Emma and Bill bought Persuasion.
 c. $\{\wedge \text{John bought E}, \wedge \text{John bought P}, \wedge \text{Bill bought E}, \wedge \text{Bill bought P}\}$

¹³ As we will see in Chapter 2, it is possible to define the meaning of C^0 and suitably adjust the meaning of a *wh*-DP in such a way that there is no need for this final step. However, I follow this convention here to keep the exposition simple (see also Bittner 1998).

d.



A polar question thus denotes two propositions, only one of which can be true in any situation.

As subsequent chapters will show, there is substantive discussion surrounding virtually every aspect of the semantics given here, but it nevertheless provides a reasonable starting point. Before proceeding further, however, it is worth noting some areas where the system over-generates. Assuming that *why* or *how* questions involve a simple extension of quantification to non-argumental types, one would expect multiple constituent questions to range freely over the different types. Question (16) shows that this is not the case. There is nothing implausible or problematic about (16b) as a possible answer for (16a), if (16a) were acceptable. Similarly, polar and constituent questions do not combine happily, even though quantifying an existential quantifier into a polar question would be defined. Example (17a), were it grammatical, would denote the set of propositions with each individual buying or not buying *P* (17c). Similarly, (17b) would denote a relation between Mary and the set of true propositions in (17c) if it were grammatical.¹⁵

- (16) a. *Why did Bill go how?
 b. Bill went by bus because he has no car.
- (17) a. *Did who buy Persuasion?
 b. *Mary knows whether who bought Persuasion.
 c. { $\wedge$ Bill bought *P*, $\wedge$ Bill didn't buy *P*, $\wedge$ John bought *P*, $\wedge$ John didn't buy *P*}

Note that these questions could be interpreted if one of the *wh* expressions carried echo intonation but in that case possible answers would have the form of a question. Example (16), uttered with stress on *how*, would be answered with something like *Why did Bill go by bus?* and (17a), with echo intonation on *who*, would be answered with something like *(I asked) Did Bill buy P?* We will see further discussion of such second order questions, questions whose answers are themselves questions, in Chapters 4, 7, and 9. The point to note for now is that these questions are not interpretable as regular multiple questions and that this does not follow from what we have said so far about the semantics of questions.

1.2.2 Question–answer congruence

This brief introduction underscores the methodological reliance on answers in the semantics of questions. In analyzing questions we have no direct intuitions to work with, the way we have intuitions about the meaning of declaratives. It does not make sense to ask what the truth conditions of a question are or to talk about entailment relations between questions in a pre-theoretical way. The primary methodological tool for studying the semantics and pragmatics of questions is

¹⁵ We can assume $\llbracket C' \rrbracket$ to be $[p = \wedge \text{bought}(x, P) \vee p = \wedge \neg \text{bought}(x, P)]$. After quantifying in the *wh* phrase and abstracting over *p*, we would get: $\lambda p \exists x [p = \wedge \text{bought}(x, P) \vee p = \wedge \neg \text{bought}(x, P)]$. See Section 2.3.3 for a semantic explanation.

indirect. We have intuitions about question–answer pairs which, as we have seen, can be used to probe into the meaning of questions. It is not easy, however, to determine whether all the answers we intuitively accept as appropriate responses to a question should be included in its denotation. Before looking at some of these complications, let us make precise the implicit connection between the question denotation, as laid out in the previous section, and a possible answer.

We might define an operator *Answer* that takes a question and returns the conjunction of the true propositions in the set. If it happens to be the case that Bill bought both *Emma* and *Persuasion*, the answer would include both propositions. This can be accomplished by intersecting the set of worlds denoted by each true proposition in the question denotation, using generalized intersection.^{16,17} As before, we use underlining to indicate the propositions that are true:

- (18) a. Which book did Bill buy?
 b. {^Bill bought E, ^Bill bought P}
 c. $\text{Answer}(Q) =_{df} \cap (\lambda p [p \in Q \wedge \forall p])$
 d. $\text{Answer}(18b) \Rightarrow \text{Bill bought E and Bill bought P} / \text{Bill bought both E and P.}$

It has been noted that this is not sufficient. There is also a prosodic dependence between a question and its possible answer that needs to be recognized. This is illustrated in (19), where small caps indicate focal stress:

- (19) a. What did Bill buy?
 b. Bill bought PERSUASION.
 c. #BILL bought Persuasion.

Suppose that Bill only bought *P*, the two answers in (19b) and (19c) should denote the same (true) proposition in the set denoted by (19a). The difference in focus, however, affects their suitability as answers. The felicitous answer is the one where focus in the answer corresponds to the *wh* expression in the question. This can be captured by imposing the constraint in (20a) on the *Question–Answer* relation (Rooth 1992: 86):

- (20) a. $\llbracket Q \rrbracket^o \subseteq \llbracket \text{Ans} \rrbracket^f$
 A question denotes a subset of the focus semantic value of its answer

¹⁶ Montague’s down operator gives the extension of $p: \forall p$ is evaluated true if the world of evaluation is in the set of worlds denoted by p . Similarly, in a system with explicit world variables $p(w)$ gives the value of the proposition at the actual world and is usually accepted as short for $p(w) = 1$. The actual Answerhood operator we will settle on in Section 2.2.3 is quite different from the one given here but it serves to move the discussion along.

¹⁷ The generalized intersection over a set P that is a set of sets is the set that contains elements belonging to every member of P : $\cap P =_{df} \{x: \text{for every } A \in P, x \in A\}$ (Chierchia and McConnell-Ginet 2000: 532).

$$\begin{array}{ccc}
 \text{b. } \lambda p [\exists x [p = \wedge \text{bought}(\text{bill}, x)]] & \subseteq & \llbracket \text{bought}(\text{bill}, \text{PERSUASION}) \rrbracket^f \\
 \Downarrow & & \Downarrow \\
 \{ \wedge \text{Bill bought E}, \wedge \text{Bill bought P} \} & \subseteq & \\
 & & \{ \wedge \text{Bill bought E}, \wedge \text{Bill bought P}, \wedge \text{Bill bought Newsweek} \}
 \end{array}$$

The focus semantic value of an answer in a theory like Rooth's, simplifying considerably, is the set of propositions obtained by substituting the focused expression by appropriate alternatives. The ordinary semantic value of a question will be a subset of this. We see that (19c) could be an answer to *Who bought P?* not to (19a), while (19b) could be an answer to (19a). The constraint on the question–answer relation correctly captures this intuition.

Another kind of question–answer congruence is about the form in which the answer is presented:

- (21) a. Which book did Bill buy?
 b. Bill bought Emma.
 c. #Emma was bought by Bill.

The propositional content of the two answers, again, is exactly the same but the answer in (21b) is preferred over (21c) because the syntactic form in which the proposition is expressed matches that of the question (see Kuno 1982 for relevant discussion). This intuition has not been addressed seriously in the semantic literature and, consequently, will not feature in our discussion.

There is a further relation between questions and answers that bears mentioning. The semantics given here assumes that questions have an existential presupposition and, in normal situations, there will be at least one true proposition in the set which *Answer* will pick out. But in a situation where Bill bought nothing, the only truthful and felicitous answer to (22a) is (22b):

- (22) a. What did Bill buy?
 b. Nothing.

Interestingly, the same is not true of the nearly equivalent question in (23a):

- (23) a. What is it that Bill bought?
 b. #Nothing.

The point to settle is whether negative answers are felicitous. Should the semantics of questions be adjusted to account for negative answers (Groenendijk and Stokhof 1982, 1984)? Or do negative answers implicitly deny the presupposition behind the question (Karttunen and Peters 1976; Comorovski 1996; Dayal 1996)? We will explore this further in Chapter 2 but the contrast between (22) and (23) shows that care must be taken to keep the semantics of interrogatives distinct from other context sensitive expressions they may contain, in this case the cleft.

Another case where context affects the acceptability of an answer is shown in (24):

- (24) a. Where am I?
b. You are in Helsinki

If (24a) is uttered by a passenger on a train stopped at a station, (24b) would count as a good answer. If uttered by a tourist who has lost his way while walking around Helsinki, it would not. Examples like this can be argued to require crucial reference to the goal of the question, bringing this contextual parameter into the semantics of questions (Ginzburg 1995). Alternatively, *wh* expressions can be allowed to range over entities of different granularity, the set of cities vs. the set of locations within a city, leaving it to pragmatics to determine the appropriate level for quantification (Lahiri 2002a). We will not engage seriously with this issue since it has relevance beyond questions: *John liked every place he visited* could be about every city he visited or about every part of the city he is in.

1.2.3 Types of answers

Question–answer paradigms typically involve cross-speaker dialogues, which makes the primary diagnostic available for the formal analysis of questions a particularly delicate tool to handle. We have intuitions about the appropriateness, or lack thereof, of an answer to a question. We do not have intuitions about what is included in the semantics of questions and what is a conversationally sanctioned departure from the answers that the question denotes. Determining the semantics and pragmatics of questions, then, becomes the task of the linguist.

Since the question–answer diagnostic is so dependent on conversational dynamics, let us refine our understanding of the different types of answers that we may be inclined to accept as appropriate. There are at least three distinctions that are relevant in this connection: direct vs. indirect answers, short vs. full answers, partial vs. complete answers:

- (25) a. Have you seen my book?
b. Yes, I have seen it. *Direct answer*
c. Well, Sue was reading it last night. *Indirect answer*
- (26) a. Where is Bill going?
b. Bill is going to the store. *Direct answer*
c. I think he needed cereal. *Indirect answer*

The information that (25a) or (26a) is seeking can be given in the form of a direct answer (25b)/(26b), as well as in the form of an indirect answer (25c)/(26c). A theory of questions is accountable for both possibilities. One approach that has proved fruitful is to define question meanings in such a way that the relationship to the direct answer is transparent, while using contextual information and pragmatic reasoning to relate questions to indirect answers. We might say, for example, that (26a) must denote the propositional content of the direct answer *Bill*

¹⁸ There is a small class of verbs that distinguish between polar and constituent questions. This is discussed in Chapter 5. See Chapter 7 for a brief discussion of short answers.

Response (28b), implicitly or explicitly, conveys that it is a complete answer. It would be considered an appropriate direct answer to (28a) under any theory of questions. Answer (28c) provides relevant information, but the use of hedges like *well* and accompanying intonation flag it as a partial answer. The status of (28d) and (28e) are less clear. While (28d) is a complete answer in some sense, it does not resolve the question, allowing for a follow-up: *Which ones?* Similarly, (28e) declares itself to be a partial answer but it is clearly less informative than (28b). It is a matter of debate whether such answers should have the same status as direct answers, complete or partial. For the purposes of this study I will treat them as indirect complete or indirect partial answers, and therefore not part of the core data to be explained, unless discussing specific analyses that treat them on a par with canonical direct answers.

Also relevant here are *mention-some* answers which, though partial, may be acceptable without the benefit of a hedge or other indication that partial information is being given:

- (29) a. Where can one buy an Italian newspaper?
 b. {[^]one can buy an Italian newspaper at the corner store,
 [^]one can buy an Italian newspaper at Barnes and Noble,
 [^]one can buy an Italian newspaper at the railway station}
 c. One can buy an Italian newspaper at the railway station.

If there are two places where Italian newspapers can be bought, why does (29c) suffice as an adequate response? Hintikka (1976) argues for an ambiguity in questions between *mention-all* readings of the kind we have been discussing and *mention-some* readings of the kind seen here. Groenendijk and Stokhof (1984), however, point out that such answers count as satisfactory only if it is clear that the questioner is interested in buying a newspaper and any one place will do. We will study *mention-some* questions at length in Chapter 3.

Finally, there are answers that do not conform to the propositions in the question denotation and do not, by themselves, implicate the full direct answer:

- (30) a. Who is coming to the party?
 b. Well, I know Sue had it marked on her calendar.
 c. Well, some of the invitees are out of town.

Answer (30b), with appropriate intonation, may signal that Sue, at least, is likely to come, leaving open the possibility of other attendees. Response (30c) makes it clear that not everyone can come, ruling out one possibility, without actually giving any information about those who will come. These cases of partial answers are also extremely interesting, but their relation to the direct answer relies heavily on context. By and large, they will not feature in our discussion.

Here I have presented what I consider to be a mainstream approach to the semantics of questions, spelling out some of the essential steps in the computation of a Hamblin–Karttunen analysis of questions. In Chapter 2, I will provide a more detailed account of their theories, along with that of Groenendijk and Stokhof

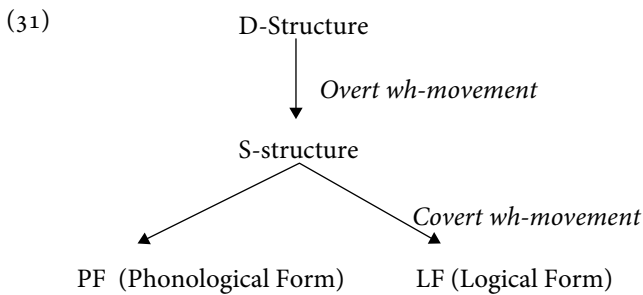
(1982). However, it is important to keep in mind that the reliance on answers in the semantics of questions, and the distinctions between different types of answers, is equally important in all current theories of questions.

1.3 Must *wh* expressions move?

I will now clarify the status of the syntactic commitments I made in Section 1.2, hoping to show that the crucial assumptions are transportable across frameworks. I also spell out the extent to which this study will focus on syntactic theories of question formation.

1.3.1 *Doing without wh movement*

My discussion of the Hamblin–Karttunen semantics assumes the Government and Binding/Principles and Parameters T-model of grammar, where interpretation is defined on LF structures (Chomsky 1981, 1986; Huang 1982; May 1985):



The semantics of questions, of course, was developed within the tradition of Montague Grammar, which did not follow the transformational model. And even within transformational grammar, recent modifications have led to different conceptions of syntactic structure. The choice of the syntactic framework adopted therefore requires some justification.

I have sketched out the mapping from syntax to semantics as involving three distinct parts, a question nucleus possibly containing a *wh* trace, a Complementizer encoding the means for a type shift to a set of propositions, and a position outside *C'* from where moved *wh* phrases bind variables inside the nucleus. Movement, crucially, occurs at two levels, S-structure for clause-initial *wh* phrases and LF for *wh* in-situ. Semantic interpretation applies to *transparent* LF, where scope relations are disambiguated. There are several other approaches to syntax that eschew *wh* movement and/or a distinction between S-structure and LF (Reinhart 1997, 1998; Ginzburg and Sag 2000; Mycock 2006; among others). To what extent do the assumptions about *wh* scope based on movement apply to approaches without movement? I sketch the connections in schematic terms in this subsection, referring the reader to the original works for complete and

Data like (33a)–(33c) have been captured by positing constraints on movement. A similar sensitivity to structure for the resolution of the slash category is built into monostratal theories to account for these data. To that extent, then, what we have to say about overt wh movement in this book is transferable to frameworks that characterize wh dependencies in alternative ways.

When we consider the scope of quantifiers and wh in-situ, the issue becomes more nuanced:

- (34) a. [_{IP} Some professor thinks [_{CP} that every student is smart]].
 b. [_{CP} Who read [_{DP}the book [_{CP} that Bill gave to who]]]?
 c. [_{CP} Who knows [_{CP} where Mary bought which book]] ?

Example (34a) does not have a reading where for each student there is a possibly different professor who considers him/her smart, the $\forall > \exists$ reading. Based on such examples, the scope of non-wh quantifiers is generally taken to be clause-bounded. In theories with LF movement, these facts have been captured by stipulating Quantifier Raising to be clause-bounded and wh movement to be island insensitive (Huang 1982). Similar assumptions are made about retrieval of stored quantifiers in non-movement approaches (Engdahl 1986).

The scope of wh in-situ in (34b)–(34c), unlike that of quantifiers, cannot be determined on the basis of direct evidence but must be mediated through intuitions about possible answers. Note that it is possible to respond to these questions by specifying the values of the matrix wh phrase as well as the wh in-situ: *Mary read the book that Bill gave to John* and *John knows where Mary bought Emma and Bill knows where she bought Persuasion*, respectively. From this, the conclusion has been drawn that the scope of wh in-situ is immune to islands.

It is, of course, possible that the fundamental generalization itself—that the specification of values is necessarily an indicator of matrix scope—may not be correct and that the scope of wh in-situ is clause-bounded like that of other quantifiers (Dayal 1996). If so, the generalization about differences between S-structure and LF movement will have to be modified, as will the distinction between storage and retrieval of different types of items. What can be learnt from this, as before, is that storage mechanisms are not syntactically blind. Any assumptions about the relationship between possible answers and the scope of wh phrases have equal resonance for non-movement accounts of scope, as they do for movement-based approaches (see Chapter 7 for discussion).

While there are non-trivial differences between the various syntactic approaches that have been proposed, the point that emerges from our discussion here is that syntactic and semantic commitments are logically independent. The real issue is the extent to which syntactic information is relevant to interpretation. To the extent that analyses depend on the diagnostic of possible answers, and the specification of values is linked to the binding of a variable in argument position by an existential quantifier with scope over the propositional variable, it is of equal relevance in frameworks employing alternative scope mechanisms to movement. However, syntactic constraints on semantic interpretation have been most extensively studied

within the tradition of transformational grammar. And it would be fair to say that the Principles and Parameters model has wider currency than non-transformational models or even its own current Minimalist incarnations. It therefore makes it the natural choice for this book, whose primary focus is on the semantics and pragmatics of questions. My brief remarks here should make it clear that the essential features of the framework are transportable and readers should have no difficulty in translating the syntactic assumptions in this book to frameworks of their choice.

1.3.2 *Delimiting syntactic discussion*

As indicated already, this book will touch upon syntactic issues as they relate to semantic concerns. It will leave out of consideration several important topics in the analysis of questions where the explanations rely purely on syntactic principles and/or where the semantic implications of syntactic constraints have not been explicitly worked out. Superiority effects and constraints on extraction, for example, are phenomena that have been much investigated, but primarily from a syntactic viewpoint. Reference to these topics will therefore be made selectively.

Superiority effects emerge in multiple constituent questions. Question (35a) is acceptable with the subject *wh* fronted but (35b) with the object *wh* fronted is not. This contrast is reduced, or even eliminated, if full *wh* expressions *which student/which book* are used (36a)–(36b):

- (35) a. Who read what?
 b. *What did who read?
- (36) a. Which student read which book?
 b. ?Which book did which student read?

A classical explanation for this contrast relies on the *Empty Category Principle* (ECP; Chomsky 1981). The ECP requires that traces be properly governed, either locally by an antecedent or by a lexical category. For (35) we have the following LF structures:

- (37) a. $[_{CP-i} \text{ what}_j [_{CP-i} \text{ who}_i [_{IP} \text{ t}_i \text{ read t}_j]]]$
 b. $[_{CP-j} \text{ who}_i [_{CP-j} \text{ what}_j [_{IP} \text{ t}_i \text{ read t}_j]]]$

In (37a), *who* moves to Spec CP at S-structure and antecedent-governs its trace. At LF, *what* adjoins to SpecCP. It cannot antecedent-govern its trace since Spec is headed by *who* and carries its index. However, the trace of *what* is lexically governed by the verb so there is no violation of the requirement that traces be properly governed. The problem with (37b) is that the trace of *what* is properly governed, both antecedent-governed from SpecCP and lexically governed by the verb, but the trace of *who* is not. The subject position is outside the maximal projection of the verb so it cannot be lexically governed by it and the Spec carries the index of *what* so it cannot be antecedent-governed. This explanation, we see, relies purely on syntactic notions and has little bearing on semantic/pragmatic

issues. However, the topic becomes relevant when we consider the absence of superiority effects in questions with full *wh* expressions.

One explanation that has been offered for the absence of superiority effects in (36) appeals to the notion of D(discourse)-linking, knowledge about the identity of the entities quantified over (Pesetsky 1987). The idea is that D-linked *wh* phrases do not enter into standard operator-variable relations, but into dependencies that are not subject to the ECP. Under this view, then, there is no problem with (36b) because it does not have to have the structure in (37b). Alternative mechanisms for assigning scope to D-linked *wh* phrases ensure full interpretability. Chapters 6 and 7 pursue this aspect of the phenomenon, probing the question of the precise connection between discourse familiarity and semantic mechanisms for interpretation. With regard to superiority *per se*, however, it stops at pointing out some semantic consequences of superiority violations in terms of possible answers but puts aside the vast syntactic literature dealing with explanations of when and why superiority effects arise.

Similarly, there are well known differences between extraction of subject vs. object *wh* phrases:

- (38) a. Who do you think (*that) came?
b. What do you think (that) he saw?

The requirement for proper government can again be used to explain the contrast between (38a) and (38b). Briefly put, the presence of the complementizer *that* interferes with antecedent government of the trace of *who* in (38a). Lack of antecedent government is irrelevant for (38b) since the trace of *what* is lexically governed. There has been no substantive discussion regarding the semantic status of these examples and we will not be concerned with them here.

The decision about what to include/exclude, then, is driven by the thematic profile of the series. While recognizing that superiority effects or constraints on extraction have played a crucial role at almost every stage in the development of syntactic theory, I have chosen not to include them, just as I have chosen not to dwell on differences between phrasal movement, feature checking, or the copy theory of movement, that have marked developments in syntactic theory from Principles and Parameters to Minimalism. At appropriate points, however, I will introduce whatever syntactic notion is needed to elucidate an approach or idea that deals with the meaning of questions. With these caveats in place I will turn now to an overview of the book, and try to give the reader a general sense of what to expect in the chapters to follow.

1.4 Roadmap to the book

I have three types of readers in mind: the linguist who is interested in questions but does not have much background in formal semantics (some of my favorite syntactician friends), the linguist who is trained in semantics but does not know much about questions (the canonical second year graduate student), and the

semanticist who has worked on questions but is curious about what has gone into this survey (those to whom I owe this book). I hope that each of you will find enough in what follows to justify your investment of time.

The survey comprises eight substantive chapters, spanning 40 years of research on the semantics and pragmatics of questions. Each chapter has a short introduction and conclusion, allowing the reader to determine if the content of that chapter is of interest to them. A glance at the table of contents will give a pretty good indication of the specific topics covered. Here I will make a few comments on the overall organization of those topics.

Chapters 2, 3, and 4 present the theoretical core of the survey. Chapter 2 presents three approaches that continue to drive much of the current thinking on the semantics of questions, as well as developments that are relatively non-controversial. Chapter 3 elaborates on the relationship between questions and answers with respect to exhaustiveness, the *mention-some/mention-all* dichotomy, and its implications for the semantics–pragmatics interface. Chapter 4 expands the discussion to include functional and pair-list answers, the latter a particularly important diagnostic in work on the syntax–semantics interface. The goal in these chapters, as in the rest of the book, is to present the reader with an explicit proposal, making clear what the motivations for particular aspects of that proposal are. The reader who is not interested in the motivations or the details of the arguments behind these developments can go directly to the snapshots provided at the end of the chapters. From this point on, the book has a loosely modular structure.

Chapter 5 focuses on indirect questions. It probes the extent to which a theory of complement selection can be derived from the lexical semantics of embedding predicates, the complement and the larger structure in which they occur. The relationship between adverbs in the matrix clause and embedded interrogatives is explored, as is the semantics of noun phrases with question-like meanings.

Chapter 6 deals with the selective extraction of *wh* expressions from weak islands: negative islands, *wh* islands, and factive/presuppositional islands. This appears to be a syntactic phenomenon at first glance but explanations that have been given show that the semantic–pragmatic profile of the extractee as well as the island is crucial, as are the presuppositions that can be independently ascribed to questions.

Chapter 7 analyzes long-distance list answers, pairing values for a matrix argument with values for a *wh* inside an island. One approach relaxes constraints for covert scope taking. Another enriches the semantics to relate the matrix *wh* to the island itself. Cross-linguistic variation in overt/covert scope taking is shown to require a more nuanced view of the diagnostic generally used to determine *wh* scope.

Chapter 8 elaborates on a focus-based semantics for questions and evaluates its role in two domains, the (in)ability of Japanese *wh* expressions to be bound from outside the local quantificational domain and the intervention effects that certain expressions have on *wh* expressions in their scope. Differences between polar and alternative questions are also discussed here.

Chapter 9, the concluding chapter, looks at negative questions, tag questions, declarative questions, echo questions, rhetorical questions, and questions that are

de facto imperatives. All of them deviate in some way from the norm of an interrogative syntactic form used to seek information. Nevertheless, they are amenable to explanations based on theories designed for canonical questions, when combined with an adequate theory of discourse.

This bird's eye view of the survey should have given my three target readers a sense of the issues that have been subjected to particular scrutiny and continue to invite discussion even today. It should also have highlighted the position which I have adopted in undertaking this study. I adjudicate, to some extent, between different perspectives on the issues under discussion, while providing enough information to allow readers to make their own determination. It has been said that "purely descriptive, wholly depersonalized history, remains what it has always been, a figment of abstract theory" (Berlin 1955). As in historical accounts, so in linguistic surveys. Not all viewpoints have been represented, or equally represented. It is just one linguist's view of the rich and varied material that our field has produced, but the references provided should allow readers to follow up on issues of interest. With so much good work to draw on, there is every reason to hope that the interested reader will come to an adequate understanding of what it is we do when we engage in the common everyday activity of asking and answering questions.

A theory of questions and answers

This chapter develops a theory of questions and answers that provides a reference point for the rest of the book. Chapter 1 introduced the rudiments of an account of questions and classified it as a Hamblin–Karttunen account. Under this view, a question denotes a set of propositions and an answer is the conjunction of the true members of that set. Here I provide the historical context for this view, clarifying in what sense it is a blend of the two theories. I refine the analysis in a number of ways, incorporating insights from later research into the semantics of questions and answers.

We start with a discussion of three papers, Hamblin (1973), Karttunen (1977), and Groenendijk and Stokhof (1982), which form the basis of most current work on the topic. In addition to the view of questions as denoting sets of propositions, the idea of questions as partitions on possible worlds is introduced. Whether an answer merely needs to denote the true propositions (weak exhaustiveness) or whether it must also rule out the false ones (strong exhaustiveness) is of particular relevance to the construction of an adequate theory of questions. The proper interpretation of the common noun inside the *wh* phrase is also significant.

We present possibilities for bridging the different positions in these three papers, providing ways of synthesizing insights from all of them. Crucial to this enterprise is the possibility of separating the question denotation from an *answerhood* operation that takes questions as argument. The possibility of more than one type of answerhood operator, capturing weak and strong exhaustiveness, is explored. The possibility of moving the truth requirement from the question denotation to the answerhood operator, and allowing answers to be restricted on the basis of properties other than truth is discussed.

We introduce the notion of maximality, something not entertained in the early theories of questions. Maximality plays a role in the interpretation of number morphology in *wh* phrases as well as in the definition of the *answerhood* operator, capturing presuppositions about existence and informativity associated with questions and answers.

We end with a snapshot of a baseline theory of questions and answers. While subsequent chapters will suggest many possible ways of refining and rethinking various aspects of the semantics and pragmatics of questions, the theory endorsed here should provide a reasonable backdrop for evaluating those proposals.

2.1 The classics

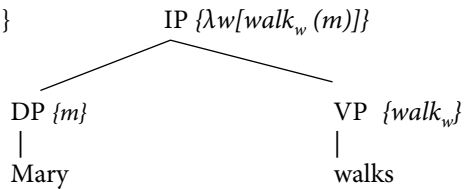
This section summarizes the three papers that can be considered classics in the field: Hamblin (1973), Karttunen (1977), and Groenendijk and Stokhof (1982), henceforth Gr&S. In discussing these works, I will use as far as possible a uniform syntactic structure for interrogatives of the kind presented in Chapter 1 (Heim 1989; Bittner 1994; von Stechow 1996). I will stay largely faithful to the original semantics, except for the uniform use of Ty2 (Gallin 1975) to facilitate comparison of theories.¹

2.1.1 Questions as sets of propositions

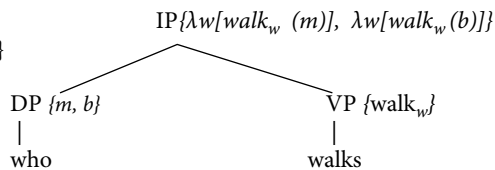
In five pages of a short paper, Hamblin (1973) laid out a semantics for questions that has proven extremely influential. He took on the challenge of showing that Montague's program for "the construction of a theory of truth . . . as the basic goal of serious syntax and semantics" was not limited to declaratives, an idea that seems obvious to current day semanticists but was still in need of defense in the early 1970s. There are two primary modifications to Montague's semantics that Hamblin proposes in order to make this extension: a shift from denotations to denotation sets for natural language expressions, and the interpretation of *wh* expressions as multi-membered sets.² He posits a recursive definition for combining denotation sets, which results in the distinction between declaratives (singleton denotation sets) and interrogatives (multi-membered denotation sets). The following derivations show how this is accomplished, assuming that Bill and Mary are the two individuals in the domain:

(1)

a. $\llbracket \text{Mary walks} \rrbracket = \{\lambda w [\text{walk}_w(m)]\}$



b. $\llbracket \text{Who walks} \rrbracket = \{\lambda w [\text{walk}_w(m)], \lambda w [\text{walk}_w(b)]\}$



¹ Ty2 allows for explicit quantification over worlds. The subscripts on predicates in these derivations indicate world variables: P_w , the extension of P at w , is equivalent to $P(w)$. See Chapter 1 fn 9 for connections with Montague's system.

² See Dowty et al. (1981) for an introduction to Montague Grammar. A noteworthy feature inherited by the papers discussed is that each rule has distinct syntactic and semantic components. To keep the presentation accessible I do not reproduce the original rules, merely showing their contribution to the syntax–semantic mapping adopted here.

Trees (1a)–(1b) show that the denotation sets of lexical items combine through point-wise functional application. The compositional procedure anticipates the account of focus in Rooth (1992).³ The fact that a proper name denotes a singleton set while a *wh* expression denotes a set with as many individuals as there are entities of the relevant type projects upwards. Thus, a declarative comes to denote a set with one proposition in it, and is identified with that proposition, while an interrogative denotes a set with as many propositions as there are entities in the domain.

The system generalizes to multiple constituent questions in the expected way ((2a)). Hamblin also defines the adverb “is it the case that” as the set with the identity function on the question nucleus and its negation for polar questions ((2b)):

- (2) a. $\llbracket \text{Who likes who?} \rrbracket = \{ \lambda w [\text{likes}_w(m, m)], \lambda w [\text{likes}_w(m, b)], \lambda w [\text{likes}_w(b, m)], \lambda w [\text{likes}_w(b, b)] \}$
 b. $\llbracket \text{Is it the case that Mary walks?} \rrbracket = \{ \lambda w [\text{walk}_w(m)], \lambda w \neg [\text{walk}_w(m)] \}$

Hamblin briefly addresses the role of the common noun in complex *wh* expressions, such as *what N*. Although he entertains the notion that the restriction to entities in the denotation of *N* may be a presupposition, he seems to lean towards the view that this membership is asserted. Mono-morphemic *wh* expressions, *who* and *what*, are treated as implicitly restricted to humans and non-humans, respectively.

Questions for Hamblin, then, denote the set of possible answers. He does not explicitly discuss what counts as an actual answer to a question. From his comment, “pragmatically speaking a question sets up a choice-situation between a set of propositions, namely, those propositions that count as answers to it” (Hamblin 1973: 254), one infers that the subset of possible answers that would count as acceptable in a given situation is contextually determined.

2.1.2 Questions as sets of true propositions

Hamblin’s theory is based on a consideration of full answers to questions. As a consequence, questions of all types, single/multiple constituent questions, questions about individuals, times or manners, polar questions, etc, all have a uniform semantic type, as opposed to a multitude of types based on the category and number of *wh* expression(s). While Hamblin only considered direct questions, this aspect of his theory is particularly attractive in the context of indirect questions, the starting point for Karttunen (1977). Question embedding predicates, for the most part, do not distinguish between various question types, so a uniform semantics for questions leads to a simpler semantics for embedding predicates (see Chapter 5 on indirect questions).

³ Rooth (1992: 84) acknowledges Dietmar Zaefferer for making the connection between his earlier work on focus (Rooth 1985) and Hamblin’s semantics for questions. See also Chapter 8.

Karttunen's interest in indirect questions leads to a significant modification of Hamblin's theory of questions, namely the restriction of question denotations to true propositions. The following verbs of communication prompt this revision: *tell, show, indicate, inform, disclose*. Take, for illustration, *tell* and *indicate*. These verbs are non-factive when embedding declaratives, but lead to veridicality when embedding interrogatives:

- (3) a. John told Mary that Bill and Susan left.
b. John indicated that Bill and Susan left.
- (4) a. John told Mary who left.
b. John indicated who left.

Sentences (3a)–(3b) can be true even if the embedded declaratives are false; (4a)–(4b) cannot be true if John told/indicated that Bill and Susan left when that is not the case. Since the restriction to truth cannot come from the embedding verb, Karttunen concludes, it must come from the question itself.

Another argument for introducing truth into the denotation of questions rests on verbs like *depend on*:

- (5) Who is elected depends on who is running.

Karttunen points out that it is simpler to define the meaning of the matrix verb on the basis of true answers to the two embedded questions than on the basis of possible answers to them.

Karttunen's modification of Hamblin's theory is cast within the general framework of natural language quantification in Montague grammar and is also informed by insights from transformational analyses in Katz and Postal (1964), Baker (1968), and Chomsky (1975). Although Karttunen follows Hamblin in taking interrogatives to denote sets of propositions, declaratives are interpreted at their normal semantic type as propositions. Wh expressions are treated as existential generalized quantifiers, adjusted to apply to interrogative level meanings. Crucial to Karttunen's proposal is the shift from declarative to interrogative meaning, mediated through the formation of *proto-questions* (Karttunen 1977:13). The output of the proto-question rule does not correspond to any natural language question but it serves two crucial purposes. It shifts the meaning to a set of propositions and it introduces the truth requirement on propositions.

In our syntactic framework, the proto-question rule can be ascribed most naturally to the $C^0_{[+WH]}$ node. Let us try to understand Karttunen's proto-question rule with reference to constituent and polar questions:

- (6) a.
- $$\begin{array}{c}
 C' \lambda q \lambda p[p_w \wedge p = q] (\lambda_w [walk_w(m/x_i)]) \\
 \Rightarrow \lambda p[p_w \wedge p = \lambda w [walk_w(m/x_i)]] \\
 \swarrow \quad \searrow \\
 C^0_{[+WH]} \quad \quad \quad IP \text{ } walk_w(m/x_i) \\
 \lambda q \lambda p[p_w \wedge p = q] \quad \quad \quad \begin{array}{c} \text{Mary}/t_i \text{ walks} \end{array}
 \end{array}$$

- b. $\llbracket C' \rrbracket = \{\lambda w[\text{walk}_w(x_i)]\}$ *for constituent questions;*
 $\{\text{walk}_w(m)\}$ or $\emptyset$ *for polar questions.*

With constituent questions, C' denotes a set of true propositions, whose determination requires the binding of the free variable inside the nucleus. With polar questions, C' denotes either the singleton set or the empty set, depending on the facts in the evaluation world. $C^0[-\text{WH}]$ would be a simple identity function on propositions: $\lambda p[p]$, as in a standard graft of Montague's semantics to this syntactic structure.⁴

Karttunen proposes further operations on proto-questions in order to generate meanings associated with actual natural language questions. Let us consider the *Yes/No Question Rule* first (Karttunen 1977:15). For our purposes, we can encode the content of the rule in a null operator in SpecCP, as in (7a). Karttunen's Y/N *Ques* rule is complicated by the truth requirement in the proto-*Ques* rule. Thus two separate cases must be considered, one in which C' denotes a set with the single true proposition (7b), and one in which the set is empty (7c):⁵

- (7) a. $\llbracket \text{OP}_{\text{Y/N}} \rrbracket = \lambda Q \lambda p [Q(p) \vee [\neg \exists q Q(q) \wedge p = \wedge \neg \exists q Q(q)]]$ *Q: type $\langle\langle s, t \rangle, t \rangle$*
 b. $\llbracket C' \rrbracket = \{\lambda w[\text{walk}(w)(m)]\}$
 $\llbracket [\text{CP OP}_{\text{Y/N}} C'] \rrbracket = \{\lambda w[\text{walk}(w)(m)]\}$
 c. $\llbracket C' \rrbracket = \emptyset$
 $\llbracket [\text{CP OP}_{\text{Y/N}} C'] \rrbracket = \{\lambda w \neg[\text{walk}(w)(m)]\}$

In (7b), the question denotes the set with the true proposition, because of the first condition in (7a). The second condition in (7a) takes care of the case where the nucleus proposition is false. The question now includes the set of worlds in which the question nucleus does not hold. For direct questions, it seems reasonable to adopt a null operator with the relevant meaning, as we have done. For indirect questions, the same could be taken as the meaning of *whether/if*. The procedure is often simplified in later accounts by interpreting $C^0_{[\text{+WH}]}$ in polar questions as $\lambda q \lambda p [p_w \wedge p = q \vee p = \lambda w \neg q_w]$.

Karttunen's treatment of constituent questions is a straightforward adaptation of Montague's quantification rule for indefinites, since he treats the *wh* phrase as an existential. Proto-questions, with an indexed free variable, feed into the *Wh Quantification Rule* and the *Wh Phrase Rule* (Karttunen 1977: 19, 24–5). We can keep the meaning of the *wh* determiner as an ordinary existential quantifier (8a), and capture quantifying into questions with the schema in (8b):

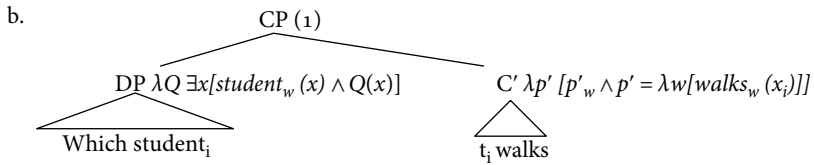
- (8) a. $\llbracket \text{which} \rrbracket = \lambda P \lambda Q \exists x [P(x) \wedge Q(x)]$ *P and Q: type $\langle e, t \rangle$*
 b. $\llbracket [\text{CP}_{[\text{DP-n}]} \text{ which } N] C' \rrbracket = \lambda p [\llbracket \text{which } N \rrbracket (\lambda x_n [\llbracket C' \rrbracket (p)])]$

⁴ Recall from Chapter 1 that IP meanings are shifted from type t to type $\langle s, t \rangle$ at C' , enabling functional application with $\llbracket C^0 \rrbracket$. See also *Intensional Functional Application* in Heim and Kratzer (1998: 308).

⁵ The relationship between the second clause of (7a) and (7c) is not transparent, but hopefully, the discussion in Section 2.2.1 will help.

The quantifying-in rule in (8b) involves the following steps. First, the $\llbracket C' \rrbracket$ is converted into type t by saturating the lambda expression denoted by C' with the variable p . Then the individual variable is abstracted over, producing an expression of type $\langle e, t \rangle$. This combines with the generalized quantifier denoted by the wh expression. By abstracting over p again, we get back a set of propositions as the meaning of the question. If Mary and Bill are the only students who walk in the actual world, though there may well be other individuals under consideration, we get sets such as (9c) for the single constituent question in (9a). Analysis (9b) provides the steps of the derivation, all involving simple functional application:⁶

(9) a. Which student walks?



$$\begin{aligned}
 \llbracket 1 \rrbracket &= \lambda p [\lambda Q \exists x [\text{student}_w(x) \wedge Q(x)] (\lambda x_i [\lambda p' [p'_w \wedge p' = \lambda w [\text{walks}_w(x_i)]](p)]] \\
 &\Rightarrow \lambda p [\lambda Q \exists x [\text{student}_w(x) \wedge Q(x)] (\lambda x_i [p_w \wedge p = \lambda w [\text{walks}_w(x_i)]])] \\
 &\Rightarrow \lambda p [\exists x [\text{student}_w(x) \wedge \lambda x_i [p_w \wedge p = \lambda w [\text{walks}_w(x_i)]](x)]] \\
 &\Rightarrow \lambda p [\exists x [\text{student}_w(x) \wedge p_w \wedge p = \lambda w [\text{walks}_w(x)]]]
 \end{aligned}$$

c. $\llbracket \text{which student walks} \rrbracket = \{\lambda w [\text{walks}_w(m)], \lambda w [\text{walks}_w(b)]\}$

Since the wh question rule is recursively defined, multiple wh questions are easily accommodated. If Mary's reading of *Emma* and Bill's reading of *Persuasion* exhaust the readings of books by students in the world of evaluation, a multiple constituent question is interpreted as (10):

- (10) a. $\llbracket \text{which student read which book} \rrbracket =$
 $\lambda p \exists y [\text{book}_w(y) \wedge \exists x [\text{student}_w(x) \wedge p_w \wedge p = \lambda w' [\text{read}_{w'}(x, y)]]]$
 b. $\{\lambda w [\text{read}_w(m, E)], \lambda w [\text{read}_w(b, P)]\}$

A standard simplification, one that was adopted in Chapter 1, is to leave the proposition p as a free variable at C^0 and abstract over it at the top node. This makes the procedure for interpreting multiple constituent questions more user friendly while achieving the same results.

The final piece of Karttunen's theory involves the embedding of questions. The question embedding rule (Karttunen 1977: 17) takes the intension of the interrogative meaning as its internal argument. The nature of the embedding predicate determines whether the intension or the extension of this argument will be relevant in the final computation. The internal argument of *wonder* is a function from worlds w' to the set of propositions p that are true at w' (11). By contrast, the internal argument of *know* takes the extension of the question (12) and yields the set of propositions that are true at the world of evaluation w :

⁶ I use bold font to indicate the key elements of the lambda expression that is to be resolved in the next step of the derivation.

- (11) a. [John [*wonders what Bill bought*]]
 b. $\llbracket \text{wonder} \rrbracket = \lambda Q_{\langle s, \langle \langle s, t \rangle, t \rangle \rangle} \lambda y [\text{wonder}_w(y, Q)]$
 c. $\llbracket \text{wonder what Bill bought} \rrbracket =$
 $\lambda Q \lambda y [\text{wonder}_w(y, Q)] (\lambda w' \lambda p \exists x [p_{w'} \wedge p = \lambda w'' [\text{bought}_{w''}(b, x)]])$
 $\Rightarrow \lambda y [\text{wonder}_w(y, \lambda w' \lambda p \exists x [p_{w'} \wedge p = \lambda w'' [\text{bought}_{w''}(b, x)]])]$
- (12) a. [John [*knows what Bill bought*]]
 b. $\llbracket \text{know} \rrbracket = \lambda Q_{\langle s, \langle \langle s, t \rangle, t \rangle \rangle} \lambda y [\text{know}_w(y, Q(w))]$
 c. $\llbracket \text{know what Bill bought} \rrbracket =$
 $\lambda Q \lambda y [\text{know}_w(y, Q(w)) (\lambda w' \lambda p \exists x [p_{w'} \wedge p = \lambda w'' [\text{bought}_{w''}(b, x)]])]$
 $\Rightarrow \lambda y [\text{know}_w(y, \lambda w' \lambda p \exists x [p_{w'} \wedge p = \lambda w'' [\text{bought}_{w''}(b, x)](w))]$
 $\Rightarrow \lambda y [\text{know}_w(y, \lambda p \exists x [p \wedge p = \lambda w'' [\text{bought}_{w''}(b, x)]])]$

Karttunen further relates question embedding verbs with their proposition embedding counterparts by a *Meaning Postulate* (Karttunen 1977: 18 ff):⁷

- (13) $\forall x \forall Q \square [\text{know}_Q(x, Q) \leftrightarrow$
 $[\forall p [Q(p) \rightarrow \text{know}_t(x, p)] \wedge [\neg \exists q Q(q) \rightarrow \text{know}_t(x, \neg \exists q Q(q))]]]$

This licenses inferences from question denotations to propositions, for verbs like *know*, *tell*, etc. It guarantees that if someone stands in the *know/tell* relation to a question, then they must stand in the *know/tell* relation to every proposition in that question. And if the question denotes the empty set, then the individual stands in that relation to the proposition that there are no true propositions of the relevant sort. In the case of (12), if Bill bought *Emma* and *Persuasion*, knowing *what he bought* is tantamount to knowing *that Bill bought Emma and Persuasion*, and, if he bought nothing, to knowing *that Bill bought nothing*. This makes it explicit that Karttunen takes an answer to a question to be the conjunction of all the propositions in the question denotation. And, in case the question denotation is empty, as the proposition that states this fact.

2.1.3 Questions as partitions

Gr&S (1982) respond to Karttunen, arguing that the basic type of a question is the same as that of a declarative, a proposition. A question is *index-dependent* so the proposition denoted varies from world to world while a declarative is *index-independent* and denotes the same proposition at every world. A question, then, determines a partition on possible worlds. A similar proposal is made in Higginbotham and May (1981) and Higginbotham (1996). I focus on Gr&S's version as an exemplar of this general perspective on questions.

The treatment of questions as propositions is motivated by the fact that it is possible to conjoin them with ordinary declaratives. If conjunction requires syntactic and semantic parallelism, an analysis of questions as propositions is appealing:

⁷ Meaning Postulates, first proposed by Carnap (1947), are used in Montague Grammar to capture language specific lexical properties by placing constraints on admissible models (Dowty, Wall and Peters 1981: 224; Chierchia and McConnell-Ginet 2000: 448–449).

- (14) a. John knows *that Peter has left for Paris*, and also *whether Mary has followed him*.
 b. Alex told Susan *that someone was waiting for her*, but not *who it was*.

As in the case of a uniform meaning for different question types, a move in the direction of uniformity between interrogatives and declaratives also allows for a simpler semantics for verbs that can embed either. It bears mentioning, though, that such co-ordination is ungrammatical under question embedding verbs: **John wondered/asked whether Peter has left for Paris and that Mary has followed him*.

Mapping Gr&S's interpretive procedure onto a GB-style syntax and adapting the original rules (Gr&S 1982: 192), we get the following for a declarative:

- (15) a. (that) Mary walks
 b.
- $$\begin{array}{c}
 C' \lambda w \text{ walk}_w(m) \quad \langle s, t \rangle \\
 \swarrow \quad \searrow \\
 C^0 [-\text{WH}] \quad \text{IP} \quad \lambda P \langle s, \langle e, t \rangle \rangle [P_w(m)] (\lambda w \text{ walk}_w) \\
 \quad \quad \quad \quad \quad \quad \quad \quad \Rightarrow [(\lambda w \text{ walk}_w)(w)(m)] \\
 \quad \quad \quad \quad \quad \quad \quad \quad \Rightarrow \text{walk}_w(m) \quad t \\
 \quad \quad \quad \swarrow \quad \searrow \\
 \text{DP} \quad \text{VP} \\
 | \quad | \\
 \text{Mary } \lambda P [P_w(m)] \quad \text{walks } \text{walk}_w
 \end{array}$$

Simplifying somewhat, *Mary* denotes the set of properties $P_{\langle s, \langle e, t \rangle \rangle}$ that characterize the individual Mary at world w , and walk_w the extension of the property *walk* at the same world. By *Intensional Functional Application* we have the formula $\text{walk}_w(m)$ which is true or false, depending on the facts at w . The specification on C^0 being $[-\text{WH}]$, the world variable is abstracted over at C' and the result is a proposition, the set of worlds where Mary walks.

A C^0 specified $[+\text{WH}]$, conversely, creates index dependence in the proposition, (16c). Instead of abstracting over the world of evaluation w , it looks for worlds w' having the same value as w with respect to the question nucleus. The relevant rule (Gr&S 1982: 192) derives the interrogative version of (15a):

- (16)
 a. Does Mary walk? / Whether Mary walks.
 b.
- $$\begin{array}{c}
 C' \lambda p \lambda w \lambda w' [p(w) = p(w')] (\lambda w'' [\text{walk}_{w''}(\text{mary})]) \\
 \swarrow \quad \searrow \\
 \Rightarrow \lambda w \lambda w'' [\lambda w'' [\text{walk}_{w''}(\text{mary})](w) = \lambda w'' [\text{walk}_{w''}(\text{mary})](w')] \\
 \Rightarrow \lambda w \lambda w' [\text{walk}_w(\text{mary}) = \text{walk}_{w'}(\text{mary})] \quad \langle s, \langle s, t \rangle \rangle \\
 \swarrow \quad \searrow \\
 C^0 [+WH] \quad \text{IP} \quad \text{walk}_w(\text{mary}) \quad t \\
 | \quad \swarrow \quad \searrow \\
 \text{Does} \quad \text{Mary} \quad \text{walk}
 \end{array}$$
- c. $\llbracket C^0_{+WH} \rrbracket = \lambda p_{\langle s, t \rangle} \lambda w \lambda w' [p(w) = p(w')]$

Let us parse (16c) to see how index-dependence is achieved. C^0 takes a proposition and creates a relation between the world of evaluation w and a proposition in the following way. Saturating the world variable in the proposition with the world of evaluation w yields, on the left of side of the equation, the extension of the proposition at w , either true or false. Saturating the world variable on the right side of the equation with a different variable w' , collects all the worlds in which the extension of the proposition on the right matches the extension of the proposition on the left, that is, at the world of evaluation. We thus get a partition that defines an equivalence relation on worlds. At each world w , the question denotes the set of worlds w' that is equivalent to w with respect to the question.

Though interrogatives and declaratives denote the same semantic type at a given world, that is, propositions (type $\langle s, t \rangle$), they are clearly different. No matter which world we interpret a declarative at, it denotes the same set of worlds. The interpretation of the interrogative varies depending on the facts at the world of evaluation. To make this concrete, consider the following model:

- (17) a. M: $W = \{w_1, w_2, w_3, w_4\}$;
 Walk = $\left[\begin{array}{l} w_1 \rightarrow \{\text{mary, sue}\} \\ w_2 \rightarrow \{\text{bill, mary}\} \\ w_3 \rightarrow \{\text{sue}\} \\ w_4 \rightarrow \emptyset \end{array} \right]$

- b. Does Mary walk?

Extension at w_2 : $\lambda w \lambda w' [\text{walk}_w(m) = \text{walk}_{w'}(m)] (w_2) = \{w_1, w_2\}$
the set of worlds in which Mary walks

Extension at w_3 : $\lambda w \lambda w' [\text{walk}_w(m) = \text{walk}_{w'}(m)] (w_3) = \{w_3, w_4\}$
the set of worlds in which Mary doesn't walk

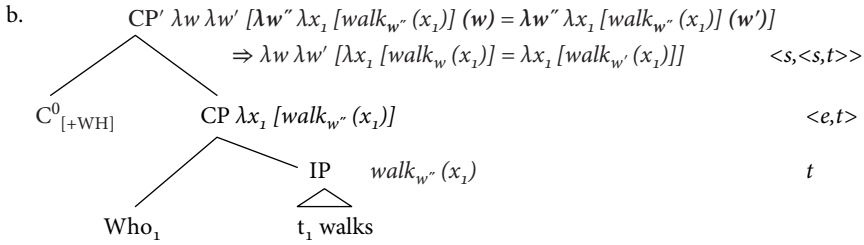
Example (17) illustrates the difference between the intension and extension of a question. The intension of a polar question denotes a partition of worlds with two cells, one where Mary walks, one where she doesn't. The extension of the question varies with worlds. At w_2 , for example, we get the proposition *that Mary walks*, at w_3 the proposition *that Mary doesn't walk*.⁸

Turning to constituent questions, an important step in interpretation is the creation of an abstract. The rules for abstract formation and abstract constituent complement formation (Gr&S 1982: 196) translate into derivations like (18). Note that this semantics requires interpreting wh phrases lower in the structure than C^0 :

⁸ The intension of a proposition: $\lambda w \lambda w' [\text{walk}(w')(m)]$ yields the same proposition at every world since the variable w does not occur inside the nucleus—the facts in the world of evaluation are not relevant to the proposition. Here, it would be the set $\{w_1, w_2\}$, whether evaluated at w_2 or w_3 .

(18)

a. Who walks?



The argument position associated with the wh expression is interpreted as an indexed variable, and the wh expression as a lambda abstractor over it. Since abstraction has to happen below the point at which index dependence is introduced, wh expressions adjoin below C^0 , at the lower CP. The only adjustment needed in the semantics is to allow index dependency to apply to categories of type $\langle s, \langle e, t \rangle \rangle$, not just type $\langle s, t \rangle$. The lower CP meaning can then feed into the meaning of C^0 , after the requisite intensionalization. This yields a function from worlds w to the set of worlds w' in which the set of walkers are the same as in w : $\lambda w \lambda w' [\lambda u [\text{walk}_w(u)] = \lambda u [\text{walk}_{w'}(u)]]$. Once again, we have an equivalence relation on worlds, this time with respect to the set of walkers. Schematically, if there are two entities in the domain, a question intension represents a partition with four cells, exhausting the space of possibilities:

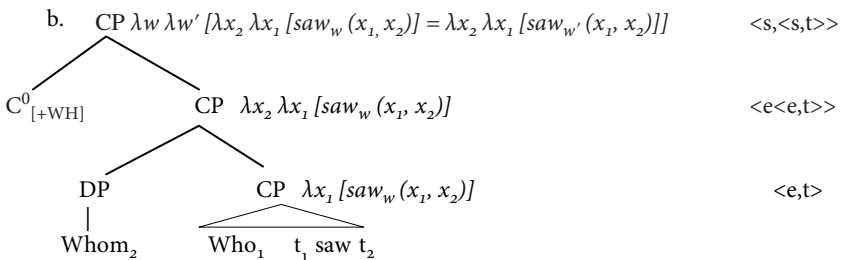
(18) c.

a	b
a,b	

At any world, the question denotes exactly one cell, the proposition that is its complete answer.

The rule for abstract formation generalizes to multiple constituent questions. Each wh expression in (19a) abstracts over one argument position (19b). The schema in (19c) represents the partition induced by this question, with a two-member domain of entities and an irreflexively interpreted predicate:

(19) a. Who saw whom?



c.

<a,b>	
<b,a>	
<a,b>	<b,a>

The rule for embedding applies without modification to interrogatives, since they have the same semantic type as declaratives (Gr&S 1982: 192). The type of the internal argument starts out intensional. As in Karttunen, embedding verbs vary along the intensional/extensional dimension. Verbs like *tell/know* are extensional, while verbs like *wonder/ask* are intensional. For completeness, I include an embedded declarative in the first class:

- (20) a. [John [*knows that Mary walks*]]
 b. $\text{know}_w(j, \lambda w \lambda w' [\text{walk}_{w'}(m)](w))$
 $\Rightarrow \text{know}_w(j, \lambda w' [\text{walk}_{w'}(m)])$
- (21) a. [John [*knows whether Mary walks*]]
 b. $\text{know}_w(j, \lambda w \lambda w' [\text{walk}_w(m) = \text{walk}_{w'}(m)](w))$
 $\Rightarrow \text{know}_w(j, \lambda w' [\text{walk}_w(m) = \text{walk}_{w'}(m)])$
- (22) a. [John [*knows who walks*]]
 b. $\text{know}_w(j, \lambda w \lambda w' [\lambda u [\text{walk}_w(u)] = \lambda u [\text{walk}_{w'}(u)]](w))$
 $\Rightarrow \text{know}_w(j, \lambda w' [\lambda u [\text{walk}_w(u)] = \lambda u [\text{walk}_{w'}(u)]])$
- (23) a. [John [*wonders who walks*]]
 b. $\text{wonder}_w(j, \lambda w \lambda w' [\lambda u [\text{walk}_w(u)] = \lambda u [\text{walk}_{w'}(u)]])$

The crucial point is that extensional verbs relate to a proposition (*type* <*s,t*>), while intensional verbs relate to a propositional concept, a function from worlds to propositions (*type* <*s*, <*s,t*>>).

2.1.4 Advantages of questions as partitions

We have seen the core of Gr&S's theory, and the points of departure from Karttunen's. We will now review those aspects that, according to Gr&S, favor their theory.

The first is the distinction between *weak* and *strong exhaustiveness*. The main point of difference can be illustrated schematically:

(24)

<u>w1</u> <i>a b c</i>	<u>w2</u> <i>a</i>	<u>w3</u> <i>b</i>	<u>w4</u> <i>c</i>
<u>w5</u> <i>a b</i>	<u>w6</u> <i>a c</i>	<u>w7</u> <i>b c</i>	<u>w8</u>

Assume we are in a world where *a* and *b* walk, say *w5*. Karttunen's theory requires that the propositions $\lambda w[a \text{ walks}_w]$ and $\lambda w[b \text{ walks}_w]$ be in the question

denotation. And knowing the question in such a world requires knowing the conjunction of these two propositions. This will include worlds in which *a* and *b* both walk, but also worlds where *c* walks as well: $\{w_1, w_5\}$. Gr&S's theory, however, singles out the set of worlds in which *a* and *b* are the only individuals who walk: $\{w_5\}$. That is, under Karttunen's theory the attitude holder may have false beliefs or no beliefs about non-walkers at w_5 . Gr&S term this *weak exhaustiveness* and argue that the *strong exhaustiveness* their theory delivers is required. Consider the following inference patterns.

- (25) a. John believes that Bill and Suzy walk

Only Bill walks

John doesn't know who walks

- b. John knows who walks

John knows who doesn't walk

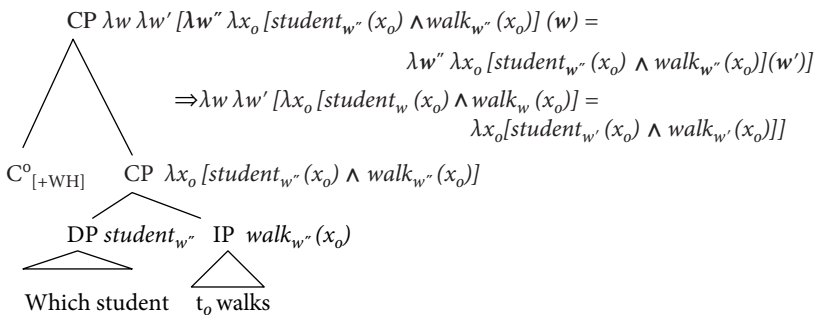
Gr&S claim that if only Bill walks, but John believes falsely of Suzy that she also walks, speakers do not judge *John knows who walks* true. Similarly, they argue that if John knows who walks, and knows who the set of individuals are, then he must know who doesn't walk. Since neither of these follows under weak exhaustiveness, they claim that questions must encode strong exhaustiveness.

A second respect in which Gr&S claim their theory to be superior to Karttunen's has to do with the interpretation of the common noun inside wh phrases. They take embedded questions like (26) to be ambiguous between a *de dicto* and a *de re* reading:

- (26) John knows which student walks.

If *a* and *b* are the two students who walk in the world of evaluation, (26) can be taken to mean that John knows that *a* and *b* are students who walk. If asked, *do you know which students walk?* he would answer in the affirmative. This is the *de dicto* reading. Alternatively, it is possible for John to know that *a* and *b* walk without knowing that they are students. If asked *do you know who walks?* he would answer affirmatively and pick out *a* and *b*, but if asked *do you know which students walk?* he might say he doesn't. This is the *de re* reading. This distinction is shown below, focusing on the direct question counterpart of (26) and skipping the obvious steps in the derivations (Gr&S 1982: 198–9):

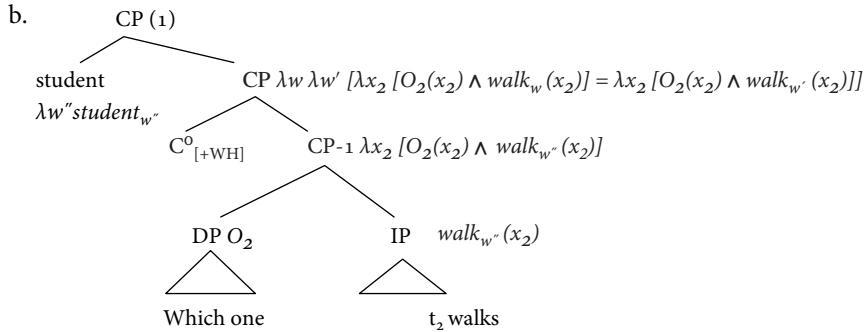
- (27) a. Which student walks?



The common noun is interpreted at the point where the abstract is formed. Thus, when the abstract combines with C^0 and the interrogative meaning is obtained, the denotation of the common noun varies on two sides of the equation. We get the set of worlds w' in which the set of student-walkers are the same as the set of student-walkers in the world of evaluation w . To know this proposition is to know the set of student-walkers in the actual world, the *de dicto* reading.

The second option differs crucially in having the common noun enter the derivation after question formation has taken place:⁹

(27)



$$\begin{aligned}
 \llbracket 1 \rrbracket &= \lambda w_{@} [\lambda O_2 [\lambda w \lambda w' [\lambda x_2 [O_2(x_2) \wedge \text{walk}_w(x_2)] = \lambda x_2 [O_2(x_2) \wedge \text{walk}_{w'}(x_2)]]] (w_{@})] \\
 &\quad (\lambda w'' \text{student}_{w''} (w_{@})) \\
 &\Rightarrow \lambda w_{@} [\lambda O_2 [\lambda w' [\lambda x_2 [O_2(x_2) \wedge \text{walk}_{w_{@}}(x_2)] = \lambda x_2 [O_2(x_2) \wedge \text{walk}_{w'}(x_2)]]] \\
 &\quad (\text{student}_{w_{@}})] \\
 &\Rightarrow \lambda w_{@} \lambda w' [\lambda x_2 [\text{student}_{w_{@}}(x_2) \wedge \text{walk}_{w_{@}}(x_2)] = \lambda x_2 [\text{student}_{w_{@}}(x_2) \wedge \text{walk}_{w'}(x_2)]]]
 \end{aligned}$$

There are two important steps here. First, there is a place-holder for the common noun at the point of abstract creation: O_2 of type $\langle e, t \rangle$. After interrogative formation, this place-holder is replaced through lambda abstraction and conversion with the denotation of the common noun in the actual world: $\text{student}_{w_{@}}$. When this happens, the property-in-extension replaces O_2 on both sides of the equation. We now have the set of worlds w' such that the set of walkers in w' are the same as the set of student-walkers in the world of evaluation, the *de re* reading. Their status as students in w' is not relevant.

It is easy enough to see how these two ways of deriving common noun meanings will yield the perceived ambiguity of questions when they are embedded under attitude verbs. Gr&S point out that Karttunen's theory only captures *de re* readings because the existential quantifier is interpreted outside the scope of the propositional variable where the question nucleus is interpreted. Their system, instead, allows for the required flexibility.

⁹ The use of $w_{@}$ in the final step has no significance and could be replaced with w to get the familiar looking question intension. The important point is that the world variable on O_2 be anchored to the same world as the predicate on the left side of the equation.

2.1.5 Section summary

In this section I have presented the key features of three seminal papers. Among the issues highlighted are the possibility of a uniform analysis of direct and indirect questions, an explicit connection between question denotations and possible answers regarding true propositions, and a distinction between weak/strong exhaustiveness. In presenting these ideas, I took liberties with the syntax but stayed faithful to the semantics, except for the use of Ty2 across the board and the simplification of certain details of Montague grammar. The goal of this section was two-fold, to provide the relevant historical background for the particular semantics we are adopting and to pave the way for later developments in the theory.

2.2 Answerhood operators

The previous discussion was restricted to arguments made by the authors themselves. In this section we will present suggestions for incorporating insights from the different approaches into a single theory of questions. We will first see how two key insights of Gr&S's theory can be incorporated into Karttunen's. We will then see how Karttunen's insights can be incorporated into Hamblin's theory. We will settle upon a blend that has broad acceptance in current work, the Hamblin–Karttunen theory of questions. Central to the enterprise is the introduction of an operator that mediates between question denotations and answers.

2.2.1 Exhaustiveness and Ans-H

Heim (1994) addresses the criticism made by Gr&S that the set of propositions approach fails to capture strong exhaustiveness. In a theory like Karttunen's, an extensional question embedding verb like *know/tell* relates individuals to the set of true propositions denoted by the question (cf. (13)). If the facts are such that John and Bill walk, then (28a) is true iff Mary stands in the *know* relation to (the conjunction of) the two propositions in (28b), underlined to indicate that they are true in the world of evaluation. But what if no one walks? Then the embedded question will denote the empty set (28c). The intersection of the empty set is the tautological proposition, which one cannot fail to believe, regardless of how ignorant one is. That is, (28a) will be automatically true if no one walks, even if Mary is totally unaware of the relevant fact since any rational person can be assumed to know a tautology:

- (28) a. Mary knows who walks.
 b. If john and bill are the individuals that walk in w ,
 $\llbracket \text{who walks} \rrbracket(w) = \{\lambda w \text{ walks}_w(j), \lambda w \text{ walks}_w(b)\}$
 c. If no one walks in w , $\llbracket \text{who walks} \rrbracket(w) = \emptyset$

Karttunen, we saw, provides a disjunctive semantics for *know* to cover the two cases. Heim taps into Karttunen's solution and proposes two notions of answerhood within his theory, which she dubs Ans_1 and Ans_2 (Heim 1994: 136). I add H to indicate authorship of the idea:

- (29) a. $\text{Ans-H}_1(\alpha, w) = \cap \llbracket \alpha \rrbracket_K(w)$
 b. $\text{Ans-H}_1(w) \text{ (who walks)} = \cap (\lambda p \exists x [p_w \wedge p = \lambda w' \text{walk}_{w'}(x)])$
 c. John and Bill walk.
- (30) a. $\text{Ans-H}_2(\alpha, w) = \lambda w' [\text{Ans-H}_1(\alpha, w') = \text{Ans-H}_1(\alpha, w)]$
 b. $\text{Ans-H}_2(w) \text{ (who walks)} = \lambda w' [\cap (\lambda p \exists x [p_w \wedge p = \lambda w' \text{walk}_{w'}(x)]) = \cap (\lambda p \exists x [p_{w'} \wedge p = \lambda w' \text{walk}_{w'}(x)])]$
 c. Only John and Bill walk.

Ans-H_1 delivers the weak exhaustiveness of Karttunen's original theory. Assuming that Bill and John are the only individuals who walk, it yields the conjunction of the two propositions denoted by the question at the world of evaluation. It admits worlds where there are no other walkers, as well as worlds with extra walkers. Ans-H_2 imports the strong exhaustiveness that Gr&S championed. It picks out all the worlds in which the answer to the question matches the answer in the world of evaluation. Effectively, proposition (30c).

Heim notes that with this enrichment, Karttunen's theory becomes a richer system since it allows for the possibility of weak as well as strong exhaustiveness, while Gr&S's is rigidly bound to strong exhaustiveness. The upshot is that the choice between the two theories must turn on empirical considerations. If there are natural language phenomena corroborating the existence of weakly exhaustive questions, Karttunen's theory, supplemented with Ans-H_2 , is to be preferred over an alternative that only has the power of Ans-H_1 . We will return to this issue in Chapter 3, but what Heim establishes is that strong exhaustiveness is not a knock-down argument against the set of propositions theory of questions.

Implicit in this discussion is also an answer to the argument in Gr&S from the conjoinability of interrogatives and declaratives. If the interpretation of an indirect question is mediated through $\text{Ans-H}_1/\text{Ans-H}_2$, which involve the intersection of the propositions in the set, the conjunction facts are no longer a challenge:

- (31)
 a. Bill knows what Mary bought and that Sue danced.
 b. $\text{know}(\text{bill}, [\underbrace{\cap [\lambda p \exists x [p(w) \wedge p = \lambda w \text{bought}(w)(\text{mary}, x)]]}_{\langle s, t \rangle}] \cap \underbrace{\lambda w [\text{danced}(w)(\text{sue})]}_{\langle s, t \rangle}])$
 c. *Bill wonders what Mary bought and Sue danced.

In fact, it has the potential advantage of accounting for (31c), on the view that question embedding verbs do not have access to this type-shift (see also Uegaki 2015).

Heim also notes the problem of incorporating *de dicto* readings in Karttunen's theory. Her suggestions are picked up by Beck and Rullman (1999) and Sharvit (2002) who capture *de dicto/de re* ambiguities using Ty2:

- (32) a. Which student walks?
 b. $\lambda p \exists x [\text{student}_w(x) \wedge p_w \wedge p = \lambda w'(\text{walk}_{w'}(x))]$
 c. $\lambda p \exists x [p_w \wedge p = \lambda w'[\text{student}_{w'/w}(x) \wedge \text{walk}(w')(x)]]$

Formula (32b) is Karttunen's original version with the common noun outside the question nucleus. Its world variable is obligatorily anchored to the actual world, resulting in the *de re* reading. The crucial move involves interpreting the common noun in the base position (32c).¹⁰ Its world variable can be bound or remain free, in the spirit of Enç (1986). When bound, the set has propositions of the form $\lambda w' [x \text{ is a student in } w' \text{ and walks in } w']$, the *de dicto* reading of the question. When free, propositions are of the form $\lambda w' [x \text{ is a student in } w \text{ and walks in } w']$, referring to those x 's who are students in the actual world, the *de re* reading. Setting aside some intricacies related to the selection of *de dicto/de re* readings of questions, this ambiguity is easily captured in this straightforward extension of Karttunen's theory.

The three arguments from Gr&S against Karttunen, we see, are tractable under reasonable modifications of the latter. The real contribution of Gr&S's paper, then, is in enriching the empirical desiderata for a theory of questions rather than in motivating a shift from questions as sets of propositions to questions as (index-dependent) propositions.

2.2.2 Truth and Ans-D_{PRELIM}

The idea that an operator mediates between question denotation and embedding predicates is also proposed by Dayal (1994). The primary argument comes from the truth requirement that Karttunen introduced into question denotations and which is hard-wired into Gr&S's theory. The proposal is to shift this requirement into the *answerhood operator*, allowing questions to denote Hamblin sets. The motivation comes from an analysis of a construction known variously in the literature as *scope marking*, *partial wh movement*, and *expletive wh construction*. Initially, this construction was thought not to exist in English, but we now know that a variant of it is found in spoken English (Dayal 1996). This makes it convenient for purposes of demonstration.

To get a feel for the construction, consider a sequence of questions like (33a):

- (33) a. Where did you go? What did you buy?
 b. What do you think? What should we buy?
 c. What do you think we should buy?

A good answer to (33a) specifies values for places and objects: *I went to the store. I bought apples, I went to the store and bought apples*, or perhaps *I went to the store to buy apples*. That is, we would interpret the sequence as a conjunction of two

¹⁰ This can be accomplished by reconstruction of the common noun at LF or, assuming the copy theory of movement, by interpreting the common noun at the tail of the movement chain, or through the use of choice functions, as favored by Beck and Rullmann.

questions. We are more likely to interpret the sequence in (33b), instead, as a single question and answer it by specifying a value for objects: *I think we should buy apples*. In that sense, it seems equivalent to (33c), which involves extraction of the embedded *wh* to matrix SpecCP. The intonational contour differentiates between a sequence interpreted as a conjunction of two questions (33a) and one interpreted as a single question (33b).

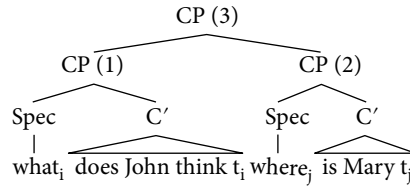
There is a vast literature on scope marking but broadly speaking it separates into two categories, dubbed *direct dependency* and *indirect dependency* in Dayal (1994). Briefly, the direct dependency approaches align scope marking with long distance extraction like (33c). I refer the reader to my earlier work as well as to recent surveys such as Fanselow (2005), Dayal (2013), Dayal and Alok (2016), and references cited therein for arguments against the direct dependency approach. We will be concerned only with sequential scope marking here which cannot be analyzed through direct dependency. This makes it possible to focus on the truth requirement in questions without getting bogged down in that controversy.¹¹

A concrete example shows the workings of the indirect dependency approach and highlights the implications for a choice between Karttunen and Hamblin:¹²

(34)

a. What does John think? Where is Mary?

b.



$$\llbracket 1 \rrbracket = \lambda p \exists p' [T(p') \wedge p = \lambda w' (\text{think}_{w'}(j, p'))]$$

$$\llbracket 2 \rrbracket = \lambda q \exists x [\text{place}_w(x) \wedge q = \lambda w'' (\text{in}_{w''}(m, x))]$$

$$\llbracket 3 \rrbracket = \lambda T_{\langle \langle s, t \rangle, t \rangle} [\llbracket \text{CP1} \rrbracket] (\llbracket \text{CP2} \rrbracket)$$

$$\Rightarrow \lambda p \exists p' [\lambda q \exists x [\text{place}_w(x) \wedge q = \lambda w'' (\text{in}_{w''}(m, x))] (p') \wedge p = \lambda w' (\text{think}_{w'}(j, p'))]$$

$$\Rightarrow \lambda p \exists p' [\exists x [\text{place}_w(x) \wedge p' = \lambda w'' (\text{in}_{w''}(m, x))] \wedge p = \lambda w' (\text{think}_{w'}(j, p'))]$$

¹¹ The absence of syntactic subordination is shown by the impossibility of variable binding across the two questions: *What does everyone_i think? What should he_{s,i} do?* Extraction cannot, therefore, apply across these clauses. Additionally, scope marking is possible with co-ordination, which is resistant to extraction. The following, based on Höhle (2000) and discussed in Dayal (2000), is relevant: *Ques: What do you think? When will Mary come and what will she bring? Ans: I think she'll come around 10 and bring pasta.* The two questions combine as follows: $\lambda p [p = \lambda w \exists q \in Q_1 \exists q' \in Q_2 [q(w) \wedge q'(w)]]$.

¹² Dayal (1994, 1996) assumes the syntactic structure given here for scope marking generally. Dayal (2000), however, argues for variation in the syntax of scope marking but a uniform semantics cross-linguistically. The structure in (34) applies to English. The intonational contour that clubs the two questions together may be taken as a reflex of the syntactic adjunction of two CPs.

Each question is interpreted as a Hamblin set. There are two points worth noting. First, since *think* takes propositional complements, the first question must quantify over propositions. Second, the variable quantified over has a (covert) restriction, just like individual variables that *what/who* and *which N* bind. We can think of this restriction as the Topic. It follows that if the quantification is over variables of type $\langle s, t \rangle$, the restriction must be of type $\langle \langle s, t \rangle, t \rangle$. Nothing special so far, but it paves the way for combining the two questions through standard functional application, as shown in (34b). The net effect is that CP₃ only denotes those propositions that John believes and furthermore are in the denotation of CP₂. That is, John's beliefs about the weather or about Sue's whereabouts are eliminated from consideration. We have sets such as (35a) for the question in (34a):

- (35) a. $\{\lambda w \text{ think}_w(j, \lambda w' \text{ in}_{w'}(m, L)), \lambda w \text{ think}_w(j, \lambda w' \text{ in}_{w'}(m, P))\}$
 b. John thinks Mary is in London.

What is the result if Mary happens to be in Paris in the world of evaluation? It turns out that (35b) is a perfectly legitimate truthful answer to (34a). This can only be if questions denote Hamblin sets (36a), not Karttunen sets (36b):

- (36) a. $\llbracket \text{CP}_2 \rrbracket = \{\lambda w' \text{ in}_{w'}(m, L), \lambda w' \text{ in}_{w'}(m, P)\}$
 b. $\llbracket \text{CP}_2 \rrbracket = \{\lambda w' \text{ in}_{w'}(m, P)\}$

But what about Karttunen's original motivation for enforcing truth inside the question denotation? Dayal's solution (Dayal 1994: 163) is to disassociate the truth requirement from the question denotation and move it to the answerhood operator in (37a). We get Hamblin sets at the CP level and Karttunen sets when a CP is directly embedded under a predicate or when a CP is a matrix question. The crucial distinctions are shown for scope marking (37b) and embedded questions (37c):

- (37) a. $\text{Ans-D}_{\text{PRELIM}}(Q) = \lambda w \lambda p [p_w \wedge p \in Q] \text{ to be revised in (48a)}$
 b. $\text{Ans-D}_{\text{PRELIM}}(\llbracket [\text{CP}_3 \text{ what does John think? Where is Mary?}] \rrbracket)$
 $\Rightarrow \text{Ans-D}_{\text{PRELIM}}(\lambda p [\exists q \in \{\lambda w' \text{ in}_{w'}(m, L), \lambda w' \text{ in}_{w'}(m, P)\} \wedge$
 $p = \text{think}_w(j, q)])$
 $\Rightarrow \text{Ans-D}_{\text{PRELIM}}(\{\lambda w \text{ think}_w(j, \lambda w' \text{ in}_{w'}(m, L)),$
 $\lambda w \text{ think}_w(j, \lambda w' \text{ in}_{w'}(m, P))\})$
 $\Rightarrow \lambda w \text{ think}_w(j, \lambda w' \text{ in}_{w'}(m, L))$
 c. John knew/told Bill where Mary is.
 $= \lambda w \text{ knew}_w / \text{told}_w(j, b, (\text{Ans-D}_{\text{PRELIM}} \llbracket \text{where Mary is} \rrbracket))$
 $\Rightarrow \lambda w \text{ knew}_w / \text{told}_w(j, b, (\text{Ans-D}_{\text{PRELIM}}(\{\lambda w' \text{ in}_{w'}(m, L), \lambda w' \text{ in}_{w'}(m, P)\})))$
 $\Rightarrow \lambda w \text{ knew}_w / \text{told}_w(j, b, \lambda w' \text{ in}_{w'}(m, P))$

What $\text{Ans-D}_{\text{PRELIM}}$ does is to sift out those propositions from the Hamblin set that are true at the world of evaluation to yield Karttunen sets. Since scope marking structures do not involve direct embedding of CP₂ under the matrix

predicate, Ans-D_{PRELIM} does not come into play in its interpretation. It only applies at the matrix level. This captures the fact that truth with respect to John's beliefs matters but not with respect to Mary's location. When the same question is embedded under extensional verbs, as in (37c), Ans-D_{PRELIM} requires truth with respect to Mary's location.¹³

An unsatisfactory aspect of Ans-D_{PRELIM} is that it still requires a shift from sets of propositions to propositions in embedded contexts. This will be addressed in revising Ans-D_{PRELIM} in Section 2.3. For now, focusing on the truth requirement, we note that it is quite easy to import the effect of Heim’s solution to the exhaustiveness problem:

- (38) a. $\text{Ans}_1\text{-D}_{\text{PRELIM}}(Q) = \lambda w \cap [\lambda p [p_w \wedge p \in Q]]$
b. $\text{Ans}_2\text{-D}/\text{H}_{\text{PRELIM}}(Q) = \lambda w \lambda w' [\text{Ans}_1\text{-D}(w)(Q) = \text{Ans}_1\text{-D}(w')(Q)]$

These versions of Ans-D_{PRELIM} apply to Hamblin sets and derive weakly/strongly exhaustive Karttunen sets, respectively.

2.2.3 Beyond truth

Another advantage of separating the truth requirement from question denotations is that it allows different embedding predicates to encode different relations to their internal argument. Although Karttunen listed a number of embedding predicates that do not show veridicality with embedded questions, he was influenced by non-factive predicates like *tell/indicate* that become veridical when they embed wh questions. From that perspective, the conclusion that the locus of veridicality is the question denotation seemed plausible enough. But as Lahiri (1991, 2000, 2002a) has discussed at length, other predicates like *be certain/agree on* do not show discernible veridicality:

- (39) a. John is certain who was at the party.
b. John and Mary agree who was at the party.

Neither (39a) nor (39b) entails that if Bill and Sue were at the party John should be certain about their presence or that he and Mary should agree about their presence. To accommodate such facts, Lahiri takes a proposition p as an answer to a question Q , iff it is in the Hamblin set denoted by the question and in the set of propositions C (Lahiri 2002a: 99–100):

- [illegible]

¹³ Dayal (1994) suggests that Ans-D_{PRELIM} is also present when embedded under intensional embedding predicates like *wonder/ask*, as functions from worlds to propositions. See Chapter 4 for scope marking with question embedding verbs and Chapter 5 for a general discussion of embedding.

C is partly determined by the lexical semantics of the embedding predicate, which can impose specific requirements based on its presuppositions (cf. 40b). The truth requirement is a default option for predicates that do not have relevant lexical presuppositions (cf. 40c).

The notion of a default, then, is a way of navigating between Hamblin and Karttunen denotations for direct questions and for some indirect questions as well. A point that is obvious enough, but still bears emphasizing, is that some restriction on the Hamblin set is required to ensure that question–answer paradigms vary across worlds.¹⁴ Such variation is essential to deriving the distinction between interrogatives as speech acts of questioning and declaratives as speech acts of assertion.

We have seen arguments here for repackaging the Hamblin and Karttunen approaches to questions. We have Hamblin sets as question denotations across the board, moving the truth requirement into an answerhood operator. To incorporate Lahiri’s insight, we would replace p_w in $\text{Ans-D}_{\text{PRELIM}}$ with $C(p)$, with C as in (40b)–(40c). Making the truth requirement a default for a contextually parameterized restriction, we create the flexibility that different question embedding predicates require. Obviously, this flexibility is not there in Karttunen, which builds the restriction to truth into the proto-question rule, nor is it there in Gr&S, where propositions are anchored to the facts of the world of evaluation. Of course, one could take Karttunen’s theory to include the variable C instead of the default truth requirement, as he may have intended it to. Note that such a move would not help with the problem of scope marking discussed in Section 2.2.2 which requires the restriction to be separate from the question denotation.

2.2.4 Section summary

In this section I discussed the developments that allow us to incorporate key insights from three different approaches to questions within a single theory. Taking Hamblin sets as question denotations and building into a separate answerhood operator a contextual restriction with a default setting for the truth requirement, we capture the variation in inferences licensed by different embedding predicates. Having a basic answerhood operator that delivers weak exhaustiveness and defining another operator that picks out worlds where the answerhood operator gives the same value, we derive strong exhaustiveness.¹⁵ Finally, interpreting the common noun inside the question nucleus and allowing for flexibility in the binding of its world variable we capture the *de dicto/de re* ambiguity in embedded contexts.

¹⁴ Keeping a fixed domain of quantification, a Hamblin denotation remains constant across worlds. If the domain varies, the Hamblin denotation will vary but not in a way that can represent a speaker’s ignorance about the nucleus proposition.

¹⁵ Some recent approaches derive Hamblin sets by optionally building in strong exhaustivity within the question nucleus (George 2011 and Nicolae 2013). They also illustrate the general point made here, that it is possible to mix and match various theories in different ways (see Chapter 3 for details).

2.3 Maximality in question–answer paradigms

We will now consider the role of number morphology in interrogative phrases, something that does not feature in the three classics discussed so far in this chapter. We will consider different proposals for importing the semantics of number into the semantics of questions and explore their impact on the relationship between questions and answers. While we will see further support for question denotations as Hamblin sets, the answerhood operator will be revised radically to capture number sensitive inferences.

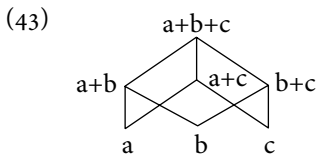
2.3.1 Number in *wh* expressions

Dayal (1991a, 1991b, 1996), draws attention to the following paradigm where singular, plural, and mono-morphemic *wh* expressions differ on the felicity of possible answers:

- (41) a. Which woman does John like?
 b. Which women does John like?
 c. Who does John like?
- (42) a. John likes Mary.
 b. John likes Mary and Sue.

Sentence (41a) can be felicitously answered only by (42a), which names a single woman, while (41b) can be answered only by (42b), which names a plurality of women. Sentence (41c) can be answered by either (42a) or (42b). These data establish that the analyses of questions and number must intersect. Intuitively, it is easy enough to describe what is at issue. The choice between a singular and a plural *wh* reflects the expectations of the speaker regarding the number of individuals who should be named in the answer. In the case of a mono-morphemic *wh* phrase, with no discernible number specification, no such expectation is in evidence. Capturing this fact, however, turns out to be non-trivial.

Dayal adopts the view, due to Sharvy (1980) and Link (1983), and by now widely accepted, that the domain of discourse includes atomic as well as plural individuals. Schematically:



The domain of individuals is an atomic semi-lattice closed under sum formation, and partially ordered by the individual part-of ($\leq$) relation: $a \leq a + b$ and $a + b \leq a + b + c$. The elements at the bottom are the atoms, those with no proper parts:

there is no x s.t. $x \leq a$ and $x \neq a$. Given this, one can treat singular common nouns as denoting in the atomic domain, and plural common nouns as denoting in the full domain.¹⁶

Applied to *wh* expressions, we get an existential quantifier which ranges over atomic individuals for *which* N_{SING} , and an existential quantifier which ranges over atomic and sum individuals for *which* N_{PL} . For mono-morphemic *wh* expressions, we can either posit a null N , which could be singular or plural, or we can treat them as having a covert restriction which makes no reference to number. These moves, in and of themselves, do not deliver the presuppositions we are interested in, regardless of which theory we adopt, Hamblin, Karttunen, or Gr&S. Let us see why.

Assume three (atomic) women in the domain of discourse, Mary, Sue, and Betty. Then, (41a) will denote the Hamblin set in (44a), and (41b) the one in (44b):

- (44) a. $\{\lambda w \text{ like}_w(j,m), \lambda w \text{ like}_w(j,s), \lambda w \text{ like}_w(j,b)\}$
 b. $\{\lambda w \text{ like}_w(j,m), \lambda w \text{ like}_w(j,s), \lambda w \text{ like}_w(j,b),$
 $\lambda w \text{ like}_w(j,m+s), \lambda w \text{ like}_w(j,m+b), \lambda w \text{ like}_w(j,s+b),$
 $\lambda w \text{ like}_w(j,m+s+b)\}$

None of the answerhood operators we have looked at so far say anything about how many propositions in these sets can be true. If John liked Mary and Sue, for example, $\text{Ans-D}_{\text{PRELIM}}$ when applied to (44a) would yield the answer in (42b) via the set: $\{\lambda w \text{ like}_w(j,m), \lambda w \text{ like}_w(j,s)\}$. Likewise, if John likes only Mary, $\text{Ans-D}_{\text{PRELIM}}$ applied to (44b) would yield $\lambda w \text{ like}_w(j,m)$, the answer in (42a). Similar problems arise with applying Ans-H_1 to Karttunen sets, which in the situations under discussion would denote subsets of (44a) and (44b) for the singular and the plural cases, respectively. Since both versions of strong answerhood operators build on the weak answerhood operator, the problem is inherited in the strongly exhaustive readings as well. And, as pointed out in Dayal (1991a, 1991b), the original theories of Karttunen and Gr&S face the same challenge. The problem is that, as things stand, the presuppositions about number that underlie our intuitions about felicitous answers to (41) do not play a role in determining the number of true propositions in the question denotation.

2.3.2 Maximality in *wh* expressions

There have been two attempts to address the problem outlined in Section 2.3.1, one using Karttunen sets, the other using Hamblin sets. I will review both but will settle on the latter, which is also in keeping with the conclusion reached in

¹⁶ The plural domain could be argued not to include atoms (Chierchia 1998) but the consensus seems to have settled in favor of including them and treating the plurality requirement as an implicature (Sauerland 2003; Spector 2007b; Zweig 2009; Chierchia 2010). Note that the plurality implicature does not arise in all contexts. *Do you have children?* is neutral with respect to the number of children expected in the answer (Schwarzschild 1996).

Section 2.2. Dayal (1991a, 1991b) builds uniqueness/maximality presuppositions into Karttunen sets by encoding definiteness into the meaning of *wh* expressions:

- (45) a. Which woman does John like?
 b. $\lambda p \exists x [x = \max y [\text{woman}(y) \wedge \text{like}(j,y)] \wedge p_w \wedge p = \lambda w' \text{like}_{w'}(j,x)]$

Quantification is over those individuals x that maximally satisfy the common noun and the question nucleus, that is, individuals who are not themselves part of any other individual that satisfies these two conditions. Since the number restricts quantification to atomic individuals, the question presupposes that only a single woman is liked by John. The question becomes, in effect, one of establishing her identity. The resulting Karttunen set is a singleton. Assuming that natural language quantification presupposes non-empty quantificational domains, the question is undefined in contexts where more than one individual meets the relevant conditions because the domain of quantification is empty.

For questions with plural *wh* expressions, the proposition must name a unique maximal individual with that property. This can be an atomic individual or a plural individual. Crucially, though, the choice of a plural over a singular *wh* expression implicates that the singleton proposition will name a plural individual. Thus even though it is semantically possible for the plural version of (45a) *Which women does John like?* to denote a propositional set naming a single woman, it is ruled out, I suggest, because the question has an existence presupposition with a plurality implicature *John likes some women*. With *who*, instead, no such implicature arises because the expression is neutral with respect to number.¹⁷ Thus the paradigm discussed in Section 2.3.1 is adequately captured. As would be obvious, this account is inspired by accounts of singular vs. plural definite descriptions (Sharvy 1980; Link 1983).

The non-standard part of the proposal is that the question nucleus contributes its meaning in two places. The primary motivation for this in Dayal's account concerns the fact that uniqueness for singular *wh* expressions is over-ridden in multiple *wh* questions: *Which man likes which woman?*, a topic taken up in Chapter 4. The results for single constituent questions in (45) can be replicated in the proposal for encoding definiteness in Rullmann (1995). And because multiple constituent questions are not considered there, Rullman's proposal works in a more straightforward way.

Following Jacobson's (1995) suggestions for extending her account of free relatives to questions, Rullmann builds maximality into Karttunen denotations inside the question nucleus. The question in (45a) would be interpreted as in (46a), with quantification restricted to atomic individuals. Its plural counterpart would be interpreted as in (46b), with quantification over atomic and plural individuals. The implicatures associated with number would be captured in a parallel fashion to the one in Dayal (1991a, 1991b):

¹⁷ We will see in Section 3.2 that positing an ambiguity between a silent N_{SING} and a silent N_{PL} will not always yield the right results, so an analysis in which number remains unspecified is to be preferred.

- (46) a. $\lambda p \exists x [\text{woman}(x) \wedge p_w \wedge p = \lambda w' [x = \max (\lambda y (\text{like}_{w'} (j, y)))]]$
 b. $\lambda p \exists x [\text{women}(x) \wedge p_w \wedge p = \lambda w' [x = \max (\lambda y (\text{like}_{w'} (j, y)))]]$

Rullmann's use of maximality in wh expressions is particularly successful in explaining negative island effects, a major focus of his work:

- (47) a. How tall is Marcus?
 b. *How tall isn't Marcus?

The contrast in (47) follows from the fact that there does not exist a unique maximal degree in the negative extension of Marcus' height but one does exist in its positive extension (see also Chapter 6).

We have seen two very similar ways of interpreting wh expressions as definites, based on Karttunen's theory of questions. Building number sensitivity into the wh phrase is an important piece of the explanation, but something more has to be added to derive the results. And as noted, the list reading of multiple constituent questions introduces further complications. This suggests the need to consider alternative solutions to the problem.

2.3.3 Maximality and Ans-D

Section 2.2 pointed out that scope marking constructions argue for Hamblin sets, with a separate answerhood operator for deriving Karttunen-like effects. The challenge of deriving the sensitivity to number morphology in question-answer paradigms using Hamblin sets is taken up in Dayal (1996). While number morphology on wh expressions is still interpreted as explicated in Section 2.3.1, number-based presuppositions are captured by defining the notion of a unique maximally informative answer. Definition (48a) replaces Ans-D_{PRELIM} (Dayal 1996: 116). As before, we adapt Heim's solution for capturing strong exhaustiveness. The operators are notated with subscript w for weak exhaustiveness and subscript s for strong exhaustiveness:

- (48) a. $\text{Ans-D}_w(Q) = \lambda w \wp [p_w \wedge p \in Q \wedge \forall p' [[p'_w \wedge p' \in Q] \rightarrow p \subseteq p']]$
 b. $\text{Ans-D/H}_s(Q) = \lambda w \lambda w' [\text{Ans-D}_w(Q)(w) = \text{Ans-D}_w(Q)(w')]$

Applying (48a) to the singular case yields (49), where each proposition names an atomic individual and none of the propositions entails any other. In any situation where John likes exactly one woman, there will be a unique maximally informative answer. Otherwise, Ans-D_w will be undefined:

- (49) a. Which woman does John like?
 b. $\lambda p \exists x [\text{woman}_w(x) \wedge p = \lambda w' \text{like}_{w'}(j, x)]$
 $\Rightarrow \{\lambda w \text{like}_w(j, m), \lambda w \text{like}_w(j, s), \lambda w \text{like}_w(j, b)\}$
 c. $\text{Ans-D}_w(49b) = \lambda w \text{like}_w(j, m)$

The plural counterpart in (50) has quantification over singular and plural individuals:

- (50) a. Which women does John like?
 b. $\lambda p \exists x [\text{women}_w(x) \wedge p_w = \lambda w' \text{like}_{w'}(j, x)]$
 $\Rightarrow \{ \frac{\lambda w \text{like}_w(j, m)}{\lambda w \text{like}_w(j, m+b)}, \quad \lambda w \text{like}_w(j, s), \quad \frac{\lambda w \text{like}_w(j, b)}{\lambda w \text{like}_w(j, b+s)},$
 $\quad \lambda w \text{like}_w(j, m+s), \quad \lambda w \text{like}_w(j, m+b+s) \}$
 c. $\text{Ans-D}_W(50b) = \lambda w \text{like}_w(j, m+b)$

If the situation has John liking Mary and Betty, there will be three true propositions in the question denotation, but only one of them will entail the other two. This is the one that Ans-D_W picks out. I assume, as before, that there is an existential presupposition behind the question, which is sensitive to the number on the *wh*: *John likes some women*. This captures the intuition that an answer naming a single individual is infelicitous.¹⁸ If the *wh* expression is monomorphemic, the existential presupposition will be neutral in this regard, allowing the maximally informative answer to name an atomic or a plural individual. The shape of the explanation, once again, is exactly what we are familiar with from analyses of definite descriptions, this time framed within the Hamblin–Karttunen blend of theories.

Bittner (1998) points out an added advantage of this approach. Karttunen (1977: 12) observed that constituent and polar questions do not combine well:¹⁹

- (51) a. *?Does John like which woman/women?
 b. $\{ \frac{\lambda w \text{like}_w(j, m)}{\lambda w \text{like}_w(j, s)}, \quad \lambda w \neg \text{like}_w(j, m)$
 $\quad \lambda w \neg \text{like}_w(j, s) \}$
 c. $\{ \lambda w \text{like}_w(j, m), \quad \frac{\lambda w \neg \text{like}_w(j, m)}{\lambda w \neg \text{like}_w(j, s)},$
 $\quad \lambda w \text{like}_w(j, s), \quad \lambda w \neg \text{like}_w(j, s),$
 $\quad \lambda w \text{like}_w(j, m+s), \quad \frac{\lambda w \neg \text{like}_w(j, m+s)}{\lambda w \neg \text{like}_w(j, m+s)} \}$

Assume that there are exactly two women in the domain of discourse, Mary and Sue. With a singular *wh* expression and the domain restricted to atomic individuals, it is clear that there will be no maximally informative answer. For every individual, either the nucleus proposition or its complement will be true and there will be no unique proposition that entails the other. Bittner does not consider the plural case but the situation does not greatly improve. In (51c), for example, the proposition $\lambda w \neg \text{like}_w(j, m+s)$ will be true since John does not like Sue, but it does not entail the two true propositions: $\lambda w \text{like}_w(j, m)$ and $\lambda w \neg \text{like}_w(j, s)$. $\text{Ans-D}_W(51c)$ is therefore undefined. $\text{Ans-D}_W(Q_{PL})$ will be defined only in situations where the nucleus holds positively for all the individuals in the domain. Since it is

¹⁸ Thanks to Mingming Liu for raising this issue and forcing me to clarify my assumptions.

¹⁹ See Chapter 1, fn 15, on how these denotations would be derived.

not defined for the general case, polar questions with *wh* expressions are deemed unanswerable, therefore deviant.

It is worth comparing the account above to a closely related, but ultimately distinct, proposal. Beck and Rullmann (1999) also start with Hamblin sets and have answerhood operators pick out maximally informative propositions. Adapting Heim's proposal, they define the following operators to capture weak and strong exhaustiveness (Beck and Rullmann 1999: 259, 268):

- (52) a. $\text{Ans-BR}_1(w)(Q) = \cap \{p: Q(w)(p) \wedge p(w)\}$
 b. $\text{Ans-BR}_2(w)(Q) = \lambda w' [\text{Ans-BR}_1(w')(Q) = \text{Ans-BR}_1(w)(Q)]$

Beck and Rullmann's starting point is the paradigm in (53), where upward scalar predicates like *be sufficient* call for the minimum number to be specified, while downward scalar predicates like *leave* call for the maximum number to be specified:

- (53) a. How many eggs are sufficient to bake this cake?
 b. {[^]one egg is sufficient, [^]two eggs are sufficient, [^]three eggs are sufficient...}
 c. How many people left?
 d. {[^]one person left, [^]two persons left, [^]three persons left...}

In a situation where two eggs are sufficient, it holds that for any *n* greater than two, *n*-many eggs are sufficient. In a situation in which three people left, it holds true for any *n* less than three, that *n*-many people left. By intersecting the true propositions, we get the proposition that entails all others.

Beck and Rullmann characterize their proposal as compatible with Dayal (1996) but there are crucial differences. Most significant is that they set aside the semantics of number morphology and do not build maximality into the definition of an answer. Example (52a) takes the subset of true propositions in a Hamblin denotation and intersects them. Consider a question with a *wh* expression over individuals, not degrees. With no lexically based entailment relations to appeal to, (52) fails in the same way as the earlier accounts discussed in Section 2.3.1 failed. If John likes Mary and Sue, regardless of whether the question *Which woman/women does John like?* has a singular *wh* expression or not, (52) will give us the proposition $\lambda w \text{ like}_w(j, m+s)$. Similarly, if he likes only Mary it will give us the proposition $\lambda w \text{ like}_w(j, m)$ even if the question uses a plural *wh* expression.

The need for combining maximality in the answerhood operator with a semantics for number in the interpretation of *wh* expressions is also seen in the following contrast:

- (54) a. John knew only one answer to the question which Dutch Olympic athletes won a medal.
 b. #John knew only one answer to the question which Dutch Olympic athlete won a medal.

Beck and Rullmann present (54a) as support for the view, originally suggested in Heim (1994), that Ans-H_1 is the lexical meaning of the common noun *answer*. The fact that it can combine with an indefinite determiner is argued by them to establish the reality of *mention-some* answers. They do not consider examples like (54b), however, which has a singular *wh* in the embedded question and is infelicitous. Chapter 3 will explore *mention-some answers*, but the relevant point here is that the shift from a plural to a singular term affects acceptability.²⁰ Once again, the vital connection between number morphology and felicity of answers is missed. Certainly, one could build number sensitivity into their account, but that would be tantamount to the proposal in Dayal (1996).

Let me mention two other phenomena where number specification on the *wh* plays a role, emphasizing that my goal in doing so is simply to show that number matters. The particular analyses for these paradigms will be discussed in Chapters 4 and 7. Examples (55a)–(55b) show that questions with universal quantifiers and definite plurals allow answers of the kind given in (55c), pairing men with women. Dayal (1992, 1996) and Krifka (1992) point out that the parallelism between universals and definite plurals breaks down when we replace *who* with a singular *wh* *which woman*. An answer to (55a) with *which woman* can still pair each man with a woman but the same does not hold for (55b) with *which woman*:

- (55) a. Who does every man love?
 b. Who do these men love?
 c. Bill loves Mary and John loves Sue.

The second data set, (56a)–(56b), shows a similar sensitivity to number:

- (56) a. Who knows where Mary bought which book?
 b. Who knows where Mary bought these books?
 c. Bill knows where Mary bought *Emma* and John knows where she bought *Persuasion*.

Baker (1968) argues that the possibility of answers specifying values for the matrix *wh* and the embedded *wh* in-situ (56c) shows that the *wh* in-situ has matrix scope. Kuno and Robinson (1972) point out that the structurally similar example (56b) also allows (56c) but is unlikely to involve matrix scope for the plural definite. Setting aside the implications for scope for now, we note an observation from Dayal (1996). Replacement of the matrix *wh* with singular *which student* does not affect the possibility of a pair-list answer for (56a) but it does for (56b). The significance of these facts in the present context is that they show the pervasive nature of number-based effects and argue for an answerhood operator like Ans-D that, in conjunction with the appropriate semantics for the *wh* phrase, delivers number sensitivity.

²⁰ See also Preuss (2001) for a more detailed discussion of Beck and Rullmann's proposal.

2.3.4 Existential presupposition and Ans-D

We have seen ample evidence that a constituent question with a singular *wh* expression is infelicitous in contexts where uniqueness is not satisfied. We will now examine the status of constituent questions in contexts where existence fails. Question-answer paradigms like (57a) are fully acceptable. This raises questions about the use of the *iota* operator in Ans-D, which would be undefined in such contexts:

- (57) a. Speaker A: Who left the party?
 Speaker B: No one.
 b. #I'm not sure whether Mary likes any student. Which student does she like?

But the anomaly of (57b), from Karttunen and Peters (1976: 355), points to the reality of the existential commitment (see also Katz 1972; Keenan and Hull 1973; Comorovski 1989, 1996; Dayal 1991a, 1991b, 1996).²¹ These data establish that existence can only ever be denied in cross-speaker exchanges. It is, of course, possible for a speaker to ask a question while overtly suspending the existential presupposition, as noted in Horn (1972):

- (58) a. I'm not sure if anyone left the party but I'd like to know who, if anyone, did.
 b. Who, if anyone, does Mary like?

Questions with clefts are interesting in this connection because they do not brook negative responses nor do they allow the speaker to suspend the existential commitment:

- (59) a. Who was it that left the party?
 #No one.
 b. *Who, if anyone, was it that left the party.

As far as I know, this distinction between clefted and regular questions has not been investigated at any length.²²

The status of the existence commitment in questions and the status of the existence commitment in clefted assertions have, however, been analyzed recently by Abusch (2010). She establishes that existence is a soft presupposition that can be canceled in the former, but it is a hard presupposition which cannot be canceled in the latter (see also Romoli 2013). In asking a question,

²¹ Karttunen (1977: 20, fn 13) recognizes that the existential presupposition is not captured in his theory. It is also not built into Comorovski's (1989, 1996) assumptions about questions and answers.

²² It is reported that many languages have obligatorily clefted questions but my preliminary and rather perfunctory investigations have not established that they are like English clefted questions in this regard.

we can assume that the speaker takes it to be answerable. That is, she expects Ans-D to be defined. The ordinary question has a soft presupposition about a positive answer to a prior polar question, making it, for all practical purposes, the conditional question: *Assuming that someone left, who did?* The hard presupposition of the cleft projects to its interrogative counterpart: *Someone left. Who did?* Schematically:

- (60) a. Who left? Presupposes: I_{SPEAKER} assume that someone left.
 b. Who is it that left? Presupposes: $We_{\text{SPEAKER+HEARER}}$ believe that someone left.

The distinction between the existence commitments of clefted and regular questions bears further investigation, of course, but the possibility of distinguishing them on the basis of soft and hard presupposition triggers is promising.

Let me end with a more general point. Answers denying the existential commitment behind the question should be carefully assessed in determining their import for the semantics of questions. For example, revisiting the inference pattern Gr&S (1982) present to argue for strong exhaustiveness, we arrive at a more nuanced picture:

- (61) a. John knows who left the party.
 No one left the party.
 John knows that no one left the party.
 b. #No one left so John knows who did/left.

While (61a) seems valid enough, it does not prepare us for the fact that (61b) is anomalous. The only difference between the two is that (61b) makes the connection between the question and the lack of existential commitment in the same sentence, that is, by one individual, the speaker.

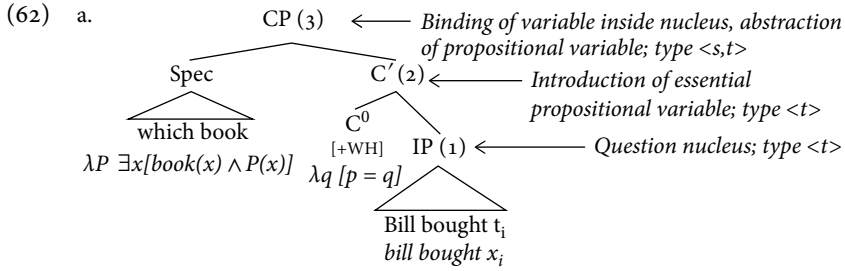
2.3.5 Section summary

Importing the semantics of number into the denotation of questions is not a controversial move but most researchers have set it aside, taking the view that it is orthogonal to the issues they are concerned with. The result of such an omission, we saw, is not always benign since number-based effects are pervasive in question-answer paradigms. We reviewed two proposals to capture number sensitivity. We settled on the one framed within Hamblin's theory since Hamblin denotations were seen in Section 2.2 to provide greater empirical coverage. Uniqueness/maximality at the propositional level, encoded in the answerhood operator Ans-D, as well as the standard semantics of number for *wh* phrases are both needed to capture the facts. The existential presupposition, also encoded in Ans-D, was discussed in relation to differences between regular and clefted questions and was shown to be an integral part of the meaning of questions.

2.4 The baseline theory

In this concluding section I will pull together the conclusions reached in previous sections, without repeating the arguments for those conclusions or noting alternatives that have been proposed. The idea is to provide an account that is explicit enough to be used as a launching pad for the discussions to follow in the rest of the book.

We begin by recalling the analysis of questions given in Section 1.2:



$\llbracket 2 \rrbracket = \lambda q [p = q] (\text{'bought}(\text{Bill}, x_i))$ *Intensional Functional Application*

$\llbracket 3 \rrbracket = \lambda P \exists x [\text{book}(x) \wedge P(x)] (\lambda x_i [p = \text{'bought}(\text{Bill}, x_i)])$ *Functional Application*

$\Rightarrow \exists x [\text{book}(x) \wedge \lambda x_i [p = \text{'bought}(\text{Bill}, x_i)](x)]$ *λ -conversion*

$\Rightarrow \exists x [\text{book}(x) \wedge [p = \text{'bought}(\text{Bill}, x)]]$ *λ -conversion*

$\Rightarrow \lambda p \exists x [\text{book}(x) \wedge p = \text{'bought}(\text{Bill}, x)]$ *Abstraction over p*

b. $\{\text{'Bill bought Emma}, \text{'Bill bought Persuasion}\}$

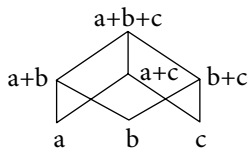
c. Set of Individuals = $\{b, j, b+j\}$ Set of objects = $\{E, P, E+P\}$

We now know the sense in which this is a blend of the Hamblin–Karttunen analyses of questions. The mapping from syntax to semantics, following the lead of Heim (1989), Bittner (1994, 1998) and von Stechow (1996), is closely modeled after Karttunen. C^0 is the point in the structure where the question nucleus is determined and the shift to interrogative meaning effected. The *wh* phrase is interpreted as an existential quantifier binding a variable from outside C^0 into a position inside the nucleus.²³ Based on arguments in Dayal (1996), it removes reference to truth from the conditions placed on propositions, resulting in the kind of Hamblin sets seen in (62b).

An addition to this system is the interpretation of number in the *wh* phrase. Following Dayal (1991a, 1991b), we adopt the ontology of individuals from Sharvy (1980) and Link (1983): singular terms denote in the atomic domain while plural terms denote atomic and plural individuals. This gives rise to different Hamblin sets for singular, plural, and neutral *wh* terms:

²³ For recent developments in the syntax–semantics mapping for questions, see Fox (2012) and its presentation in Nicolae (2013).

(63) a.



- b. $\{^{\wedge}\text{Bill saw } a, ^{\wedge}\text{Bill saw } b, ^{\wedge}\text{Bill saw } c\}$ *for which N_{SING} , for what/who*
 c. $\{^{\wedge}\text{Bill saw } a, ^{\wedge}\text{Bill saw } b, ^{\wedge}\text{Bill saw } c,$
 $^{\wedge}\text{Bill saw } a+b, ^{\wedge}\text{Bill saw } b+c, ^{\wedge}\text{Bill saw } a+c,$
 $^{\wedge}\text{Bill saw } a+b+c\}$ *for which N_{PLURAL} , for what/who*

We also adopt from Dayal (1996) the answerhood operator in (64a), and incorporate into it Heim's (1994) proposal for a second answerhood operator, in (64b). These operators introduce Karttunen's truth requirement. They are defined to apply to Hamblin sets and, when defined, to pick out a unique maximally informative true proposition from that set:

- (64) a. $\text{Ans-D}_W(Q) = \lambda w \text{ ip}[p_w \wedge p \in Q \wedge \forall p' [[p'_w \wedge p' \in Q] \rightarrow p \subseteq p']]$
 b. $\text{Ans-D}/H_S(Q) = \lambda w \lambda w' [\text{Ans-D}_W(Q)(w) = \text{Ans-D}_W(Q)(w')]$

The first operator captures weak exhaustiveness: knowing or telling a question, for example, merely requires the attitude holder to bear the relevant relation to the true propositions in the question denotation but says nothing about their relation to the false ones. The second operator captures strong exhaustiveness which requires the attitude holder to distinguish between the true and the false propositions in the set. For now, we remain neutral on whether both answerhood operators are needed in the grammar, noting only the flexibility afforded by the two operators.

The combination of questions as Hamblin sets, wh phrases restricted by the semantics of number, and answerhood operators, gives us the following question-answer paradigms. Assume a situation where Bill saw Alice but not Betty or Cal for (65), and a situation where Bill saw Alice and Betty but not Cal for (66):

- (65) a. Which woman/who did Bill see?
 b. $\text{Ans-D}_W(63b)$: Bill saw Alice.
 c. $\text{Ans-D}/H_S(63b)$ = Bill saw only Alice
- (66) a. Which women/who did Bill see?
 b. $\text{Ans-D}_W(63c)$: Bill saw Alice and Betty.
 c. $\text{Ans-D}/H_S(63c)$ = Bill saw only Alice and Betty.

If Bill saw no one or more than one woman, (65a) with a singular wh, is infelicitous because Ans-D is undefined. If Bill saw no one or only one woman, (66a) with the plural wh, is infelicitous. The source of the infelicity is different in the two cases. If he saw no one, Ans-D is undefined. If he saw only one woman, however, Ans-D is defined but the implicature of plurality that stems from the

presupposition behind the question is not satisfied. As a consequence of the theory adopted here, the response *no one* is not a direct answer to the question and must be handled separately.

We also noted differences between ordinary and clefted questions and the possibility of differentiating them on the basis of soft vs. hard presupposition triggers, following Abusch (2010). A constituent question has a soft presupposition: (*Assuming that someone left*), *who left*? The clefted question has a hard presupposition: (*Someone left*), *who was it that left*?

Turning to embedded contexts, we followed Lahiri (1991, 2000, 2002a) in allowing answerhood operators to include other types of restrictions based on the lexical semantics of the embedding verb, while keeping the truth requirement as a default. We followed Beck and Rullmann (1999) and Sharvit (2002) in deriving the difference between *de dicto* and *de re* readings of *wh* phrases by interpreting the common noun inside the question nucleus and allowing its world variable to be bound within the nucleus, yielding *de dicto* readings, or to remain free and be identified with the world of evaluation, yielding *de re* readings. In the first case, the attitude holder recognizes the membership of the relevant individuals in the common noun predicate, in the second the attitude holder need not have this knowledge.

Examples (65) and (66) show that answerhood operators mediate between question–answer dialogues across speakers. They also mediate in embedded contexts such as the one in (67a). Since *know* is extensional, we get a relation between individuals and propositions (67b):

- (67) a. [John knows [where Mary is]]
 b. $\text{know}_w(j, \lambda w' \text{ip}(p_{w'} \wedge p \in Q \wedge \forall p' [[p'_{w'} \wedge p' \in Q] \rightarrow p \subseteq p']) (w))$
 $\Rightarrow \text{know}_w(j, \text{ip}(p_w \wedge p \in Q \wedge \forall p' [[p'_{w'} \wedge p' \in Q] \rightarrow p \subseteq p'))$

We have not addressed the precise status of these operators. The following possibilities suggest themselves (thanks to Gennaro Chierchia, p.c. for discussion):

- (68) a. $\text{know}(x, Q) \leftrightarrow \text{know}(x, \text{Ans-D}(Q))$ *via a Meaning Postulate*
 b. $[\text{know}_Q] = \lambda Q \lambda x [\text{know}(x, \text{Ans-D}(Q))]$ *via lexically triggered type-shift*
 c. $[\text{know} [\text{OP}_{\text{ANS}} [\text{CP} \dots]]]$ *via a syntactically projected null OP*

We know we want a dependency between the presuppositions of the embedding predicate and the restrictions encoded in the answerhood operator. At this point, all three options appear plausible enough ways of capturing this dependency.

Let us end with two contexts where answerhood operators do not play a role. We saw that in scope marking constructions the second question must contribute pure Hamblin sets, without the mediation of *Ans-D*. The schema in (69a) has *Ans-D* associating with the Hamblin set denoted by the full structure, but not with the set denoted by *CP*₂:

- (69) a. $\text{OK Ans-D}(\text{CP}_3)$ $\text{*Ans-D}(\text{CP}_2)$
 $[\text{CP}_3 [\text{CP}_1 \text{what does John think?}] [\text{CP}_2 \text{Where is Mary?}]]$
 b. $[\text{VP wonder/ask} (? \text{Ans-D}) [\text{CP where is Mary}]]$

We might also ask whether Ans-D is implicated in the case of predicates like *wonder* or *ask* that exclusively embed questions rather than questions and propositions (69b). It may well not come into play in such cases, thereby allowing the relation to hold directly with the Hamblin set denoted by the question. Alternatively, such predicates may also involve Ans-D but the predicates would not license the shift from a propositional concept to a proposition. This would maintain the distinction between intensional and extensional predicates going back to Karttunen, but would replicate Gr&S's (1982) distinction between type $\langle\langle s, \langle s, t \rangle \rangle, \langle e, t \rangle\rangle$ for intensional predicates and type $\langle\langle s, t \rangle, \langle e, t \rangle\rangle$ for extensional ones.

This, then, is a snapshot of where we are at this point in the study of questions. If you have thought about questions at all you will most likely balk at the somewhat monolithic feel of this concluding section. But hopefully it will not stop you from reading on. If you haven't thought much about questions, consider this one possible take on a complex aspect of natural language, one that has been subjected to forty years of formal analysis. The picture presented here should help you get started on some of the rich history that will unfold in the pages to come.

Exhaustive and non-exhaustive answers

We have emphasized that our access to the meaning of questions is through possible answers. And while we have intuitions about the (in)appropriateness of question–answer paradigms, our theoretical conclusions are shaped by our assessment of what counts as a direct answer. A response classified as a direct answer reflects a reading of the question but not one classified as a conversationally relevant indirect answer. Here we consider (non)-exhaustiveness in answers to determine what they may say about the meaning or meanings of questions. Three types of answers are analyzed from this perspective: weakly exhaustive, strongly exhaustive, and non-exhaustive answers. The terms *non-exhaustive answers* and *mention-some answers* are used interchangeably, as are the terms *exhaustive answers* and *mention-all answers*.

Bearing a relation to a question, or answering a direct question, requires the relevant relation to hold with each of the true propositions in the set of propositions denoted by the question. Weak exhaustiveness is satisfied with this, but strong exhaustiveness requires the false propositions to be excluded. Does the weak–strong distinction need to be hard-wired into the semantics or is it possible to posit one kind of exhaustiveness and derive the other from it? One approach takes the variation to be encoded in answerhood operators, another takes it to be part of the question nucleus. Choosing between them brings into the discussion the broader context of conversational dynamics.

The significance of conversational dynamics increases as we turn to non-exhaustive answers. Are questions ambiguous between *mention-some* and *mention-all* readings, with different contexts or grammatical features making different readings salient? Alternatively, do they only have *mention-all* readings, though partial answers are sufficient in contexts where they satisfy conversational goals? We consider the empirical balance between semantic and pragmatic factors and the possibility that there may be more than one type of *mention-some* answer that natural language countenances.

We also consider exhaustiveness from the perspective of embedding predicates. There are predicates that seem to select exclusively for strong exhaustiveness, those that select for both strong and weak exhaustiveness, and those that select only for weak/non-exhaustiveness. Negative polarity items provide an important window into the grammatical status of these divides. Data regarding the licensing of negative polarity items in questions is also probed.

A point worth noting before we begin. A question can be posed to a single addressee who is expected to provide a satisfactory answer or it can be posed to a group, with individual responses adding up to a satisfactory answer.¹ We follow the literature in focusing on the former. With suitable adjustments, the discussion applies also to the latter.

3.1 The weak–strong distinction in exhaustive answers

This section discusses arguments against weak exhaustiveness in embedded as well as direct questions, saving discussion of arguments in favor of weak exhaustiveness for Section 3.3. Weak exhaustiveness is shown to hold its own in embedded contexts, when differences between the agent’s and the speaker’s attitudes are taken into account. Insights gained from embedded contexts are then applied to direct question contexts, taking due note of discourse dynamics. This section also engages with the formal representation of this distinction and presents a way of capturing the facts, using the answerhood operators from Chapter 2. Some recent proposals building strong exhaustiveness into the question nucleus are reviewed as well.

3.1.1 *The agent, the speaker, and the question under embedding*

Let us recall Groenendijk and Stokhof’s (1982) argument for strong exhaustiveness (Section 2.1.4):

- (1) Situation: John believes that Bill and Suzy walk
 - a. John knows who walks.
 - b. {[^]bill walks, [^]suzy walks, [^]bill+suzy walk}

Incorporating plural individuals for completeness, the embedded question in (1a) denotes the Hamblin set in (1b). If the underlined proposition is the only true one in the world of evaluation, for John to be in the *know* relation to the embedded question in (1a) requires him to know that Bill walks and Suzy does not. Speakers judge (1a) to be false in the situation under discussion. This means that John’s attitude towards the false propositions in the question denotation is relevant to the truth of (1a), not just his attitude towards the true ones. Let us probe this further.

Stepping back, note that there are four factors at play in cases like (1a): a speaker, an agent, the facts of the world, and the properties of the embedding predicate. All of them have an impact on our intuitions about exhaustiveness.

¹ Thanks to Matt Barros (p.c.) for pointing this out.

Focusing on the first three here, consider (2a) and (2b) with a slightly richer cast of characters:

- (2) a. Bill told Sue who danced at the party.
- b. Bill knows who danced at the party.
- c. {[^]john danced at the party, [^]mary danced at the party, [^]harry danced at the party...[^]john+mary+harry danced at the party}
- d. John danced at the party.

Chapter 2 established that both the non-factive *tell* and the factive *know* impose veridicality on their interrogative complements. Thus if John was the only person who danced, then of the propositions in the Hamblin set (2c), the proposition uttered by Bill to Sue and/or the proposition that Bill believes (and hence knows) is the one given in answer (2d). But who has knowledge of the relation between proposition (2d) and the fact that this proposition is the answer to the embedded question? There are exactly three possibilities.

The first possibility is that in making the reports in (2a)–(2b), the speaker is not privy to the fact that (2d) is the answer to the embedded question. This is shown by the following felicitous continuations. Here, the speaker's knowledge is incomplete and it is the agent who knows the facts of the matter:

- (3) a. Bill knows/told Sue who danced at the party but he didn't tell me.
- b. Bill knows/told Sue who danced at the party but I don't know who did.

The same report is also possible when it is the speaker who knows the facts of the matter, while the agent's knowledge is incomplete. Suppose Bill utters (2d) to Sue or gives evidence of believing this fact, but he has not given any thought to whether anyone else danced. If the speaker knows that John is the only one who danced, she can assert (2a)–(2b) truthfully even though if Bill were asked the question *Did you tell Sue?/Do you know who danced at the party?* he might declare only knowing a partial answer *all I told her/all I know is that John danced*. This is reflected in the following:

- (4) a. Bill told Sue who danced at the party but he doesn't know that he did that.
- b. Bill knows who danced at the party. He just doesn't know that he knows it.

So we see that (2a)–(2b) can be used as long as either the speaker or the agent is aware that (2d) is the exhaustive answer to the embedded question. The third option, of course, is that both the speaker and the agent are in the know. Statements (2a)–(2b) cannot be used if neither the agent nor the speaker is aware of this connection, even if both happen to know the fact reported in (2d). One way to think about this, then, is to take the interrogative to be a description of the set of propositions to which an attitude holder is related. The interpretation of an embedded interrogative must make reference to the link between the intensional object (the answerhood operator applied to a question denotation) and its

identity with a particular proposition. The ambiguity can be represented as in (5a)–(5b). The definition of the answerhood operator from Chapter 2 is repeated in (5c):²

- (5) a. $\text{know}_w(b, \lambda w [\text{identical}_w([\text{Ans-D}_w(Q)(w)], \lambda w [\text{danced}_w(j)])])$
 b. $\text{know}_w(b, \lambda w [\text{danced}_w(j)])$
 $\wedge \text{know}_w(\text{speaker}, \lambda w [\text{identical}_w([\text{Ans-D}_w(Q)(w)], \lambda w [\text{danced}_w(j)])])$
 c. $\text{Ans-D}_w(Q) = \lambda w \text{ up } [p_w \wedge p \in Q \wedge \forall p' [[p'_w \wedge p' \in Q] \rightarrow p \subseteq p']]$

The extension of $\text{Ans-D}_w(Q)$ at the world of evaluation, the maximally informative true proposition in Q , is the description denoted by the embedded interrogative and this description is identified with a particular proposition. Assuming that the identification actually holds, whether the agent or the speaker identifies the description with the true proposition is crucial to judgments involving embedded questions.

Let us apply this to the inference pattern in (1). An illustration helps:

- (6) a.

w1: b walks	w2: b walks
w3: s walks	w4: b+s walk

- b. $\text{Ans-D}_w(\text{who walks})(w_2): \lambda w [\text{walk}_w(b)] = \{w_1, w_2, w_4\}$
 c. The proposition that John believes: $\lambda w [\text{walk}_w(b+s)] = \{w_4\}$

Given this state of affairs, the agent's identification of $\text{Ans-D}_w(\text{who walks})(w_2)$ with $\lambda w [\text{walk}_w(b+s)]$, as in the logical form (5a), fails because *know* is factive.³ It also fails under a construal like (5b), where the reader/speaker in the role of the omniscient third party makes that identification, because the facts do not line up. We correctly predict the inference pattern in (1).

Locating a potential ambiguity between the agent and the speaker as the bearer of the link, as I have done here, might help explain a refinement of Gr&S's paradigm due to Spector (2005), recently discussed by George (2011). George argues that neither weak nor strong exhaustiveness quite explains the set of facts presented in (7). Assume a domain with Mary and Alice and a situation in which only Mary left the party:⁴

² The second clause of (5b), attributing identification of $\text{Ans-D}_w(Q)$ at w with the specific proposition to the speaker, would probably be better characterized as a presupposition rather than part of the at-issue content.

³ The identity would, of course, hold in John's belief worlds but that is not what is at issue here.

⁴ A version of this scenario is briefly mentioned by Gr&S. I have characterized the situation a bit differently from George (2011: 127–37). The discussion here is based on George (2011) but see also George (2013a, 2013b).

- (7) a.
- | | | |
|----------------|------|-------|
| | Mary | Alice |
| Bill's beliefs | Y | N |
| John's beliefs | Y | Y |
| Sue's beliefs | Y | ? |
| <hr/> | | |
| The facts | Y | N |
- Reported Judgments
- b. Bill knows who left. *True*
- c. John knows who left. *False*
- d. Sue knows who left. *Not quite true but not false either.*

Bill, John and Sue have different beliefs about the world. They all correctly believe, hence *know-weakly*, that Mary left. Bill is the model know-all citizen who *knows-strongly* that only Mary left. There is no doubt that (7b) will be judged true, and uncontroversially so. John, on the other hand, erroneously believes of Alice that she left. There is no doubt that (7c) will be judged false, and uncontroversially so—this is what (1) and (6) establish.

The twist comes when we look at Sue who has no beliefs about Alice. George reports that speakers hesitate to consider (7d) true but neither do they feel comfortable classifying it as false. Let us see the predictions made by strong and weak exhaustiveness here:⁵

- (8) a.
- | | |
|----------------------|------------------------|
| w1:
m left | w2:
m left |
| w3:
a left | w4:
m+a left |
- b. Sue's belief worlds: $\lambda w [\text{left}_w(m) \wedge [\text{left}_w(a) \vee \neg \text{left}_w(a)]] \Rightarrow \{w1, w2, w4\}$
- c. Ans- $D_W(\text{who left})(w2): \lambda w \text{left}_w(m) \Rightarrow \{w1, w2, w4\}$ *weak exhaustiveness*
- d. Ans- $D/H_S(\text{who left})(w2):$
 $\lambda w' [\text{Ans-}D_W(\text{who left})(w2) = \text{Ans-}D_W(\text{who left})(w')] \Rightarrow \{w1, w2\}$
strong exhaustiveness

Weak exhaustiveness predicts ready acceptance of (7d) while strong exhaustiveness suggests categorical rejection of it. So the reported intuitions do not match the predictions of either weak or strong exhaustiveness. George reports that Spector captures the nuances of this situation by enhancing the meaning of the embedding predicate with a ban against false beliefs. George himself takes this conflict as an argument against weak exhaustiveness. Here we take a slightly different view of the matter, drawing on the distinction between the speaker's

⁵ See Section 2.3.3 for discussion related to Ans- D/H_S . As its application in (8d) shows, it uses the extension of the weak answerhood operator Ans- D_W at the world of evaluation w on one side of the equation. By collecting together the worlds w' that agree with w it captures strong exhaustiveness.

vs. the agent's relation to $\text{Ans-}D_W(Q)$ as a description for the proposition $\lambda w[\text{left}_w(m)]$.

If someone were to ask Sue *do you know who left* she would not say *yes*, while both Bill and John would (though, of course, John would be misinformed). But if the speaker/reader were asked the same question, they would be justified in saying *yes*. So, this is a case where Sue's beliefs/knowledge are measured against the speaker's and judged true because it is the speaker who recognizes the link between the proposition Sue believes and its description as $\text{Ans-}D(Q)$ at the world of evaluation. The judgment changes if the speaker does not know the facts because the onus then falls on Sue, and in that case, lack of knowledge about Alice would make the sentence false. The flexibility between the agent and/or the speaker recognizing the connection between the proposition and its description as an answer to the question, illustrated in (5), seems to be the key to the problem of navigating between weak and strong exhaustiveness in such cases. The source of shiftiness in the judgment reported, I believe, is due to the fact that the presentation of the situation highlights the agent's knowledge of the connection but the inference can only go through on the basis of the speaker's knowledge. Note that this way of looking at the puzzle does not make formal reference to strong exhaustiveness. The fact that speakers do not judge (7d) as outright false is therefore not surprising.

To conclude, what I am arguing for here is an account of embedded questions in terms of the classical *de dicto-de re* ambiguity. I have suggested that there is a distinction between the role of the speaker and the role of the agent in recognizing the connection between a proposition and its description as an answer to a question. The role of the protagonists that I have emphasized here is not radical and has, in fact, been anticipated in previous work. What I have identified as the agent knowing the proposition but not its description as $\text{Ans-}D_W(Q)$, the *de re* reading of the embedded question, is what Berman (1991) terms "knowing in an objective sense." Heim (1994) argues that such knowledge is not "really knowing."⁶ I opted to present the distinction in terms of agent vs. speaker knowledge in order to articulate the idea in theory-neutral terms. I refer the reader to the articles cited to determine points of overlap.

3.1.2 *The addressee and the question posed*

When we turn to direct question-answer paradigms, we have to accommodate a changed set of factors: an addressee, a proposition, and the relation between that proposition and the question posed. Consider the following question-answer pair:

- (9) a. Which numbers between 1 and 10 are prime?
 b. 1, 2, 3, 5, 7.

⁶ See Karttunen (1977: 21–3), Berman (1991: ch. 4, sec. 3.2), and Heim (1994: 130–1).

Suppose this question were posed by a teacher to a student, and in answering it the student left out one of the prime numbers, say 1, the teacher would not consider it a satisfactory answer. If the student listed all the correct numbers but included an extra number incorrectly, say 9, the teacher would not consider that satisfactory either. A lenient teacher might give partial credit, but only if there weren't too many omissions or too many false entries. Our intuitions here are fairly clear: a satisfactory answer must list all and only those entities that satisfy the question nucleus.

Let us assume a context where the facts are not known to the questioner, making (10a) a genuine request for information:

- (10) a. Who danced at the party?
 b. John and Bill danced.
 c. John and Bill danced and Harry did not.
 d. Only John and Bill danced.

If the domain includes John, Bill, and Harry, and the addressee is well-informed about the relevant facts, the questioner would infer from (10b) that Harry did not dance. Though the addressee has not said anything about Harry, we intuitively take the omission to be a negative assertion about him. If the addressee were in doubt about Harry, she would be expected to indicate her ignorance by flagging (10b) as a partial answer with a hedge like *well, for instance* or perhaps even making it overt: *I don't know about Harry*. Absent such indication, the proposition in (10b) is taken to be the stronger statement (10c), or equivalently (10d). The issue is how exactly this strengthening happens.

This problem was brought to the fore by Gr&S (1984: 251–439), who consider several pragmatic as well as semantic solutions, which we will not review here. The account they settle on does not transfer over smoothly to the Hamblin–Karttunen semantics we have adopted.⁷ I will present instead the proposal in Spector (2007a), translating it into the terms of the discussion in Section 3.1.1 first.

The idea is that the addressee, in answering a question, does not assert just the simple proposition. Rather, the assertion is that the proposition is the complete answer to the question posed, as shown in (11a). Examples (11b)–(11c) illustrate how it works:

⁷ See van Rooij and Schulz (2004) and Schulz and van Rooij (2006) for other ways of capturing the strong exhaustiveness of answers to direct questions, inspired by Gr&S (1984).

- (11) a. $\lambda w' \text{ identical}_{w'} [\text{Ans-D}_W(\llbracket 10a \rrbracket)(w), \llbracket 10b \rrbracket]$
 b. $W = \{w_1, w_2, w_3, w_4\}$

$$\text{Dance at the party} = \left[\begin{array}{l} w_1 \rightarrow \{\text{john, bill}\} \\ w_2 \rightarrow \{\text{john, harry}\} \\ w_3 \rightarrow \{\text{john, bill}\} \\ w_4 \rightarrow \{\text{john, bill, harry}\} \end{array} \right]$$

$$\text{Ans-D}_W(10a) = \left[\begin{array}{l} w_1 \rightarrow \lambda w \text{ dance}_w(j+b) \\ w_2 \rightarrow \lambda w \text{ dance}_w(j+h) \\ w_3 \rightarrow \lambda w \text{ dance}_w(j+b) \\ w_4 \rightarrow \lambda w \text{ dance}_w(j+b+h) \end{array} \right] = \left[\begin{array}{l} w_1 \rightarrow \{w_1, w_3, w_4\} \\ w_2 \rightarrow \{w_2, w_4\} \\ w_3 \rightarrow \{w_1, w_3, w_4\} \\ w_4 \rightarrow \{w_4\} \end{array} \right]$$

- c. $\lambda w \text{ dance}_w(j+b)$ *which is:* $\{w_1, w_3, w_4\}$
 d. $\lambda w' [\text{Ans-D}_W(\llbracket 10a \rrbracket)(w') = \{w_1, w_3, w_4\}]$ *which is:* $\{w_1, w_3\}$

So, in answering (10a) with (10b), the addressee is not asserting (11c) but, rather, (11d). Now, what are the worlds in which the values for $\text{Ans-D}_W(10a)$ coincide with the proposition in (10b), that is, (11c)? They are w_1 and w_3 , the worlds where the set of dancers are all and only those named in (10b). In effect, we get a strongly exhaustive interpretation of the answer because the only available attitude holder, the addressee, asserts the relationship between a proposition p and a question, under the description $\text{Ans-D}_W(Q)$.

What we have, then, is a way of tapping into Heim's insight that strong exhaustiveness is derivative on weak exhaustiveness. The only novelty is in making explicit that the identity predicate is sensitive to discourse factors. To that extent, it can be considered a uniform account of questions as weakly exhaustive. Embedded and direct questions always compose under a single answerhood operator, namely Ans-D_W , but can give rise to strong exhaustiveness under appropriate conditions.

Similar variation with respect to the exhaustive interpretation of propositions elsewhere provides an analogy:

- (12) a. We had a party last night. John and Bill came.
 b. We invited John, Bill, Mary, and Sue to our party last night. John and Bill came.

The proposition *John and Bill came* need not have an exhaustive interpretation (*only John and Bill came*) in (12a). An exhaustive interpretation for the same proposition is at least strongly implicated in (12b), because the prior sentence

mentions other invitees explicitly and makes those possibilities salient. Not including them in the assertion is tantamount to asserting that they are false. This provides support for the position that exhaustive interpretations are sensitive to the context of use, not just in connection with question-answer paradigms but in language generally.

To complete the discussion, the proposal by Spector (2007a) just discussed is given in its original formulation in (13). Spector adopts a null operator with the semantics of *only*, from Krifka (1993). This operator, defined in (13a), takes scope over the answer and imposes the requirement that it be the strongest true proposition in the question denotation (Spector 2007a: 291). Note that Spector is working with Karttunen sets so the truth requirement enters the derivation at a different point than in our system, but the net effect is similar:

- (13) a. $\llbracket \text{OP}_Q S \rrbracket(w) = 1$ if S is the strongest member of $Q(w)$
 b. $\llbracket \text{OP}_Q \rrbracket = \lambda\phi. \lambda w. (\phi \in Q(w) \wedge \forall \Psi (\Psi \in Q(w) \rightarrow (\phi \subseteq \Psi)))$

Appealing to covert exhaustivity is part of a growing body of work that argues for the pervasive influence of covert operators in the grammar of natural language (see Chierchia 2006, 2013; Fox 2007; Chierchia et al. 2011; among others). Klinedinst and Rothschild (2011), Fox (2012), Mayr (2013, 2014), and Nicolae (2013) are other works in this category that deal specifically with questions.

3.1.3 Domain certainty and complementation

Groenendijk and Stokhof (1984) present another argument in favor of strong exhaustiveness. Sentence (14a), they claim, entails (14b):

- (14) a. Bill knows who danced at the party.
 b. Bill knows who didn't dance at the party.

- (15) dancers =

w1 a	w2 b	w3	w4 a b
w5 b	w6	w7 a	w8 a b

We can use our answerhood operators to see why. At w_1 , Ans-D_W looks for worlds in which a is a dancer for (14a) and worlds in which b is not a dancer for (14b), so we get different propositions for the two embedded questions (cf. (16)). There is no entailment relation between them, so knowing one should be independent of knowing the other. $\text{Ans-D}/H_S$, on the other hand, looks for worlds in which a is the only dancer and worlds in which b is the only non-dancer, which gives us the same set of worlds for both (cf. (17)):

- (16) a. $\text{Ans-D}_W(\text{who danced})(w_1) = \{\wedge a \text{ danced}\} \quad \{w_1, w_4, w_7, w_8\}$
 b. $\text{Ans-D}_W(\text{who didn't dance})(w_1) = \{\wedge b \text{ didn't dance}\} \quad \{w_1, w_3, w_6, w_7\}$
- (17) a. $\text{Ans-D/H}_S(\text{who danced})(w_1) = \{\wedge a \text{ danced but not b}\} \quad \{w_1, w_7\}$
 b. $\text{Ans-D/H}_S(\text{who didn't dance})(w_1) = \{\wedge b \text{ didn't dance but a did}\} \quad \{w_1, w_7\}$

The inference pattern in (14) would seem to argue for strong exhaustiveness.

But how reliable is this diagnostic? Gr&S add the important caveats that the inference between positive and negative counterparts of a question only holds if the agent is aware of the domain of quantification and this domain stays constant. There is an important example which underscores this point, albeit with a twist. Lahiri (1991: 141) credits Sylvain Bromberger for (18a), which shows that $[x \text{ know } wh \text{ not } p]$ cannot be inferred from $[x \text{ know } wh \text{ } p]$:⁸

- (18) a. Feynmann knew which elementary particles had been discovered by 1978.
 b. Feynmann knew which elementary particles had not been discovered by 1978.

Example (18a) is felicitous and true, assuming Feynmann knew his particles! Though the set of elementary particles is not known, we can restrict quantification to those that had been discovered by 1978. However, the set of particles not discovered by 1978 cannot be known and (18b) can simply not be true, at least if evaluated in 1978. So the inference from (18a) to (18b) does not go through. However, if these statements describe what Feynmann knew about elementary particles later, say in 1985, and assuming that several more elementary particles had been discovered between 1978 and 1985, the judgment for (18b) changes. If we take (18a) to be true, then (18b) follows.⁹ That is, in order for (18b) to be true, the existential presupposition, which we have encoded in Ans-D_W , must be satisfied. In general, though, since the existential presupposition of an embedded question projects up to the complex structure, there may well be contexts which only support the truth of the positive version of a question.

The relationship between positive and negative counterparts of questions is an important and frequently used diagnostic. Example (19a) illustrates Sharvit's (2002: 112) use of it for her claim that *know* can select for weak exhaustiveness

⁸ Chapter 6 discusses differences in questions with negation based on a fixed, contextually salient domain of quantification. This ties in with the reliance of negative questions on D-linking, in the sense of Pesetsky (1987). See also Section 3.2.

⁹ Even if the sentence was evaluated at a later date the domain of quantification would have to be restricted for it to be acceptable. That is, the inference can go through if the sentences are interpreted as (i)–(ii):

(i) Of the elementary particles now (in 1985) known, F knew which ones had been discovered in 1978.
 (ii) Of the elementary particles now (in 1985) known, F knew which ones had not been discovered in 1978.

(see also Heim 1994; Beck and Rullmann 1999). The variant in (19b), however, is odd:

- (19) a. I know who was admitted to the program, but I have no idea who wasn't admitted.
- b. Rupert knows which of his four students were admitted, but he doesn't know which weren't.

If *know* unambiguously selected for strong exhaustiveness, (19a) would be a contradiction. Under weak exhaustiveness the first clause simply asserts knowledge about those who were in fact admitted, leaving open knowledge about who weren't. A scenario where the agent (who in this case is the speaker) is not on the admissions committee but has seen the final list of admitted students would be one in which he or she would know the positive but not the negative extension of the predicate.

George (2011: 87) takes issue with Sharvit's conclusion. He points out that *who* in (19a) is contextually restricted to range over the set of applicants and acceptability crucially rests on the agent not knowing this set. George's point is that (19a) is consistent with the classification of *know* as strongly exhaustive. He presents (19b) to factor out domain uncertainty by restricting quantification to a small enough group that acquaintance can be guaranteed. He reports that the sentence is judged odd by many, but not all, speakers. His explanation for those who find it acceptable rests on the following two scenarios. Rupert is unaware of which of his students actually applied. Assuming that not being admitted presupposes applying, the reason Rupert cannot identify those who were not admitted is that he is uncertain about which of his students applied. As we have noted in connection with the Feynmann examples, the inference from $[V \text{ } wh \text{ } P]$ to $[V \text{ } wh \text{ } not \text{ } P]$ is contingent on knowledge about the domain of quantification. George terms this *domain uncertainty*. A second scenario in which, according to George, (19b) is accepted is when the admission process is not yet complete. Even if Rupert knows the set of applicants and the set already admitted, he cannot separate out the rest into those who have been rejected and those who are on hold. This is a case of *complementation failure*, and the inference under discussion does not go through here either.

George's arguments are not meant as arguments against weak exhaustiveness. They are arguments for keeping the relevant caveats in place when using the negation diagnostic. Note, for example, that under the *de re* scenario where it is the speaker who is the bearer of the ascription described in Section 3.1.1, *domain uncertainty* and *complementation failure* are controlled for and a weakly exhaustive ascription goes through. I should add that the literature on the properties of *know* with regard to exhaustiveness is vast and the short discussion in this and the following subsections does not do justice to it. The reader is referred to the references cited to explore this issue further. The take-home message for work on questions is to consider other embedding predicates, in addition to *know*, when drawing conclusions.

3.1.4 Alternative routes to exhaustiveness

There are other ways of capturing the weak–strong distinction in exhaustive answers. One is due to George (2011), who resurrects Gr&S’s account of *wh* expressions as restricted lambda operators.^{10,11} I simplify his definitions, which are meant to apply generally, and demonstrate for single constituent questions where abstracts yield sets of individuals:

- (20) a. Who does Anne love?
 b.

George (2011: 30) defines the question forming operator in (21), which takes a property and creates a Hamblin set. It does this by collecting the propositions obtained by existential quantification over the bearers of that property. Applied to (20b), it yields (22a)–(22b), assuming that John and Bill provide the domain of quantification:

- (21) Q: $\lambda P_{\langle s, \langle e, t \rangle \rangle} \lambda p_{\langle s, t \rangle} \exists y_{\langle e \rangle} [p = \lambda w' (P (w')(y))]$
- (22) a. $\lambda P \lambda p \exists y [p = \lambda w' (P (w')(y))] (\lambda w \lambda x [\text{love}_w(a, x)])$
 $\Rightarrow \lambda p \exists y [p = \lambda w' (\lambda w \lambda x [\text{love}_w(a, x)] (w')(y))]$
 $\Rightarrow \lambda p \exists y [p = \lambda w' (\lambda x [\text{love}_{w'}(a, x)](y))]$
 $\Rightarrow \lambda p \exists y [p = \lambda w' [\text{love}_{w'}(a, y)]]$
 b. $\{\lambda w \text{ loves}_w(a, j), \lambda w \text{ loves}_w(a, b)\}$

George (2011: 36) allows for a second operator *X* between abstract formation and question formation, shown in (23). I demonstrate only for properties, though the actual definition applies generally. Applied to (20b), we get (24), the set of individuals *S* that are the same as the individuals Ann loves in world *w*:

- (23) a. $[_{CP} Q [_{IP} X [_{IP} \text{Abstract}]]]$
 b. $X: \lambda R_{\langle e, t \rangle} \lambda S_{\langle e, t \rangle} (R = S)$
- (24) $\lambda R \lambda S (R = S) (\lambda x [\text{love}_w(a, x)])$
 $\Rightarrow \lambda S (\lambda x [\text{love}_w(a, x)] = S)$

¹⁰ Haida (2007) also takes abstract formation to be an essential step in building up interrogative meanings within a dynamic framework. However, constraints of space prevent me from presenting his analysis.

¹¹ This is the essence of George’s Chapter 2 ‘The Baseline Theory’. I set aside here modifications he entertains in his later chapters.

The question formation operation in (21), quantifying now over sets of individuals, takes (24) as input and yields (25). The set of propositions are exhaustified with respect to the individuals that Ann loves. As in Gr&S, we get a partition on the set of worlds:

- (25) a. $\lambda P \lambda p \exists S' [p = \lambda w' (P (w')(S'))] (\lambda w \lambda S (\lambda x [\text{love}_w(a, x)] = S))$
 $\Rightarrow \lambda p \exists S' [p = \lambda w' (\lambda w \lambda S (\lambda x [\text{love}_w(a, x)] = S) (w')(S'))]$
 $\Rightarrow \lambda p \exists S' [p = \lambda w' (\lambda S (\lambda x [\text{love}_{w'}(a, x)] = S) (S'))]$
 $\Rightarrow \lambda p \exists S' [p = \lambda w' \lambda x [\text{love}_{w'}(a, x)] = S']$
- b. $\{\lambda w' \lambda x [\text{love}_{w'}(a, x)] = \{\text{john}\}, \quad \text{Ann loves John not Bill}$
 $\lambda w' \lambda x [\text{love}_{w'}(a, x)] = \{\text{bill}\}, \quad \text{Ann loves Bill not John}$
 $\lambda w' \lambda x [\text{love}_{w'}(a, x)] = \{\text{john}, \text{bill}\}, \quad \text{Ann loves John and Bill}$
 $\lambda w' \lambda x [\text{love}_{w'}(a, x)] = \emptyset \} \quad \text{Ann doesn't love John or Bill}$

This is how the distinction between strongly exhaustive and non-exhaustive propositional sets is derived. George (2013a: 424) takes an answer to be some proposition in the question denotation:

- (26) x knows Q iff $\exists p$ s.t. p is in the answer set of Q and x knows that p .

When the question undergoes exhaustification the propositions in the answer set are mutually incompatible, so picking out any proposition amounts to picking out the unique strongly exhaustive answer to the question. Without exhaustification, the question denotes a standard Hamblin set and we get something weaker than the weakly exhaustive answer of Karttunen. George thus takes a question to be ambiguous between a strongly exhaustive and a *mention-some* reading. A non-trivial aspect of his proposal, then, is that weak exhaustiveness has no formal status. It is taken to be a special case of the *mention-some* answer.

Note that the issue of truth does not feature in George's analysis. In this, he follows Egré and Spector (2007) who challenge the basis for the truth requirement, citing (27) as evidence:

- (27) a. Every day the meteorologists tell us what the weather will be, but they are often wrong.
 b. The pundit told us who we should blame, but she doesn't know what she's talking about.

George recognizes that this move requires an explanation for the perceived veridicality in basic cases of questions embedded under predicates like *tell*. Examples like (28) indicate the existence of a veridical version of declarative embedding *tell*:

- (28) a. The distribution of the debris tells us that the bomb was placed under the stairs.
 b. The spacing of the puncture wounds told Anne that the victims were killed in a vampire attack.

This is an interesting tack to take, but we will not pursue it further, referring readers to George's and Egré and Spector's work.

A second approach that makes the question itself ambiguous between weakly and strongly exhaustive sets falls within what can be called the alternatives and exhaustification approach, mentioned at the end of Section 3.1.2. The theory proposed in Nicolae (2013) delivers results similar to George's at the level of the question denotation but the manner in which the sets are derived, and the implications for embedding, are substantially different. I will simplify the details of the mapping from syntax to semantics, keeping only those of immediate relevance to exhaustiveness.

The weakly exhaustive set of propositions is the basic Hamblin set and needs no elaboration. The crucial step in deriving a strongly exhaustive set of propositions is attributed to a covert operator just below C^0 ((29a)). The semantics of this operator (Nicolae 2013: 30), given in (29b), is fashioned after Rooth (1992). It takes a prejacent (the nucleus proposition p) as its argument and a contextual variable, $Alt(p)$. This set of alternatives is derived by substitutions of the individual variable denoted by the *wh* trace in the nucleus, yielding the familiar Hamblin set. The semantic contribution of the operator is to require the prejacent to be the only true member of this set. This yields a set of mutually incompatible propositions (29c):

- (29) a. $[_{CP} \text{ who } [_{C'} C^0 [_{IP} \text{ only } [_{IP} \text{ Ann loves } t_F]]]]]$
 b. $\llbracket \text{only} \rrbracket (Alt(p))(p) = \lambda w. \forall q \in Alt(p) [\neg p \subseteq q \rightarrow q(w) = 0]$ *equivalently*
 $= \lambda w. \forall q \in Alt(p) [q(w) = 1 \rightarrow p \subseteq q]$
 c. $\{\lambda w \text{ Ann loves}_w \text{ only john}, \lambda w \text{ Ann loves}_w \text{ only bill},$
 $\lambda w \text{ Ann loves}_w \text{ only john and bill}\}$

Nicolae's primary motivation for introducing covert *only* involves NPI licensing, to be discussed in Section 3.3. For now, we note that it derives the same ambiguity in questions as George. The difference between them is in embedding.

Nicolae imports $Ans-D_W$ but in a way that preserves the type of questions as sets of propositions. She defines an operator I_D with scope over the question that builds in the content of $Ans-D_W$ as its presupposition, while allowing the question denotation itself to be passed up the tree (Nicolae 2013: 51). This effectively recasts $Ans-D_W$ as a filter on question denotations:

- (30) a. $\llbracket I_D \rrbracket (Q)(w) = \exists p \in Q \wedge p = \imath p [p(w) = 1 \wedge$
 $\forall p' \in Q (p'(w) \rightarrow p \subseteq p')]$ *presupposition*
 $\lambda p. p \in Q$ *denotation*
 b. $[_{VP} \vee [_{CP} I_D [_{CP} \lambda p \dots]]]$

I_D checks for a unique maximally informative true proposition in the set. In the case of exhaustified propositional sets, this is trivial but in the case of non-exhaustified sets, I_D preserves the effects of $Ans-D_W$ with respect to

number implicatures.¹² The advantage of modulating Ans-D_W as an identity function for Nicolae has to do with selection. Instead of selecting for weak vs. strong answerhood operators, she has the embedding predicate select for strong vs. weak propositional sets.

To conclude, both George and Nicolae take questions to have both strongly exhaustive and weak or non-exhaustive interpretations, and they take embedding predicates to be sensitive to this distinction. Deferring discussion of embedding to Section 3.3, it is worth noting that something along the lines of our discussion in Section 3.1.2 is also needed under Nicolae's proposal for direct questions. Since the selectional properties of embedding predicates are not in play, the effect of strong exhaustiveness will have to be captured separately (see Nicolae 2013: 68–9). The problem with direct questions does not arise for George since the ambiguity he posits is between strong exhaustiveness and *mention-some readings*, both of which are arguably available in direct questions.

3.1.5 Section summary

Here we probed strong and weak exhaustiveness in embedded and direct questions. We noted the relevance of whether the agent or the speaker makes the connection between a proposition and its description as an exhaustive answer to the question in the case of embedded questions. The fact that direct questions seem to allow only for strongly exhaustive answers was addressed by positing that an answer is used not just to assert the proposition *simpliciter* but to assert its status as the strongest member of the question denotation. By including this conversational aspect into the analysis, we showed that it is possible to maintain a single answerhood operator that encodes weak exhaustiveness and derive the effect of strong exhaustiveness from it. The relationship between positive and negative versions of a question, a point on which strong and weak exhaustiveness differ, was considered and the conditions under which inferences are valid noted. We also presented alternative approaches that maintain an ambiguity between strong and weak/non-exhaustiveness, but at the level of the question nucleus.

3.2 Non-exhaustive answers

Exhaustive answers, our focus so far, are generally taken to represent the unmarked case. We now turn our attention to non-exhaustive answers with a view to determining whether they represent an independent reading of questions or whether they are partial answers that are considered satisfactory in specific contexts. In doing so, we review the structural as well as the pragmatic properties

¹² George does not engage with the issue of number-based effects, though he mentions the possibility of incorporating plural individuals into his system. The relevant facts will not be derived even with the proposed extension. The arguments in Section 2.3.2 will apply to it in the same way as it does to the theories critiqued there.

that have been considered necessary and/or sufficient for *mention-some answers* and, in light of that, consider various proposals.

3.2.1 *The tourist and the entrepreneur*

Consider the following paradigm, fashioned after Gr&S (1984: 458). We continue to use underlining to indicate the true propositions in the question denotation:¹³

- (31) a. Where can I buy Italian newspapers?
 b. {^you can buy Italian newspapers at the corner bookstore,
^you can buy Italian newspapers at the station,
^you can buy Italian newspapers at the co-op,
^you can buy Italian newspapers at the corner bookstore + the station,...
^you can buy Italian newspapers at the corner bookstore + the co-op+the station}
- (32) a. You can buy Italian newspapers at the corner bookstore.
 b. You can buy Italian newspapers at the corner bookstore and the railway station.
- (33) a. Well, you can buy Italian newspapers at the corner bookstore (I'm not sure about other places).
 b. You can buy Italian newspapers at the corner bookstore or at the railway station.

Gr&S point out that (31a) can be answered satisfactorily with the *mention-some* answer in (32a) if asked by an Italian tourist in Amsterdam who wants to get news of home. Any one of the places that sells such papers suffices for the tourist's purpose. If asked by an entrepreneur who wants to set up a distribution network for foreign language newspapers, however, only the *mention-all* answer in (32b) would do. If the addressee were to respond with a partial answer to the prospective distribution agent, because she does not really know every place that sells Italian newspapers for example, she would have to flag it as such with a hedge, as shown in (33a). Not to do so would make her an errant Gricean interlocutor. Along the same lines and perhaps stating the obvious, the addressee speaking to the homesick Italian could also give the *mention-all answer* in (32b), leaving it up to the tourist to decide which option would be the most convenient. The same would be true of the response in (33b), with the disjunction conveying her awareness that the tourist's goals could be met with either of the options mentioned.

This illustration highlights the importance of conversational goals in determining the appropriateness of responses. While both the tourist and the entrepreneur

¹³ As far as I know, the choice of conjunction vs. disjunction in *mention-some answers* has not been systematically studied. Disjunction in questions and answers is interesting in its own right and will be touched upon in Chapter 4.

ask the same question, the addressee understands that they have different expectations. Like all conversational factors, there is great variation in this domain as well. The homesick tourist may well want full information for future reference, even though for his immediate needs any one store would do. In that case, he might make this overt, if his dialect of English has the locution *where-all* instead of *where*. In the same vein, the entrepreneur might not really need to know the exhaustive answer. If he knows that one foreign newspaper outlet is all that the city can support, even a *mention-some answer* gives him the information he needs to abandon his plans. German *wh+alles* (Beck and Rullmann 1999) and Hindi-Urdu *wh-reduplication* are other examples of strategies to mark requests for exhaustive answers.

An important point to note is that the distinction between *mention-some/mention-all answers* also holds in embedded contexts. I use small caps to indicate that the point only holds in languages/dialects with strategies like *where all*:

- (34) a. Luigi wants to know where one can buy Italian newspapers.
 b. Luigi wants to know where ALL one can buy Italian newspapers.

Depending on whether Luigi is the homesick tourist or the entrepreneurial twin, and depending on what *wh* phrase he opts for, he is likely to get different answers.

Before going further, I would like to mention and then set aside the case of questions with indefinites. Consider a context where Prof. Smith, who has just received an award for *Mentor of the Year*, is being interviewed by a reporter. The reporter would like to talk to one of her students, and in preparation for that, asks (35a). Here, the conversational goals are in keeping with a *mention-some answer* and Prof. Smith could respond with (35b) or (35c):¹⁴

- (35) a. What topic is one of your students working on?
 b. John is working on NPI.
 c. Mary is working on FCI.

The indefinite signals the possibility for different answers depending on the choice of the student. Such questions have sometimes been called *choice questions* (Gr&S 1984; Chierchia 1993, among others). Recently, however, they have been clubbed with questions soliciting *mention-some answers* (George 2011; Fox 2012; Chierchia and Caponigro 2013; Nicolae 2013). I will maintain the old distinction and defer discussion of their properties to Chapter 4, but let me state the reason for this decision.

A *mention-some answer* typically requires a mono-morphemic *wh* expression but (35a) clearly allows for choice even with a complex *wh* phrase. Example (31a) with *what store* instead of *where* would lose its status as a *mention-some question*. Furthermore, replacing *what topic* with *what topics* or *what all* in (35a) does not

¹⁴ I find that such answers sound better if the particular student is introduced first: *Well, there's John. He's working on NPI.*

take away the flavor of choice, while we saw that such replacements shift (31a) towards a *mention-all* answer.

A second argument for keeping *choice questions* separate from *mention-some questions* is that it is possible, in certain contexts, to have *mention-some* readings for questions with universal terms. Gr&S (1984: 460) present the following as admitting a *mention-some* answer. You have to imagine that the person asking (36a) is interested in buying all of Nooteboom's books in a single visit to a store:

- (36) a. Where do they have all [the] books written by Nooteboom in stock?
- b. At Barnes and Noble. (Or at Rutgers Books . . .)

Now, note that shifting to *at which store* or *where all* has the same effect it does in (31a). Analyses that take questions with indefinites to be *mention-some* questions typically tap into differences between indefinites and universals so that the alignment of (36a) and (31a) to the exclusion of (35a) with regard to the substitutions discussed is unexpected. I take these two points as justification for keeping questions with indefinites separate from *mention-some questions*.

3.2.2 Mono-morphemic vs. complex *wh* phrases

Let us consider whether the two grammatical features typical of *mention-some* questions are strict requirements or mere propensities: the use of mono-morphemic rather than complex *wh* phrases, and the choice of possibility modals. It turns out that what counts as a restriction and what counts as a propensity seems to depend on whether we are looking at direct question-answer paradigms or embedded questions, a difference that to the best of my knowledge has not so far been systematically studied. I should emphasize that this conclusion is based on preliminary data and needs to be tested further.¹⁵

Comorovski (1996: 39) notes that *mention-some* answers are only possible with non-D-linked mono-morphemic *wh* expressions.¹⁶ In the canonical Italian tourist example, the domain of quantification is non-D-linked since the tourist is obviously unfamiliar with the city. Consider two variants of the relevant question:

- (37) a. In which store can one buy an Italian newspaper?
- b. John knows in which store one can buy an Italian newspaper.
- (38) a. In which stores can one buy an Italian newspaper?
- b. John knows in which stores one can buy an Italian newspaper.
- c. #At the corner bookstore.

¹⁵ Discussions with Gennaro Chierchia (p.c.) have helped shape my understanding of the problem but he may or may not agree with the conclusions I have drawn.

¹⁶ D(iscourse)-linking, a term due to Pesetsky (1987), refers to *wh* phrases like *which N*. They typically require a contextually salient domain of quantification, unlike *who/what* which are comfortable with domains that are not salient in the context. See also Chapters 4, 6, and 7. Many thanks to Roger Schwarzschild for very helpful discussion.

Replacing the *wh* expression with a complex *wh* expression seems to block *mention-some* answers regardless of whether the NP is singular, as in (37a), or plural, as in (38a). This holds for indirect questions as well, as in (37b) and (38b). One might argue that the singular version in (37) has a uniqueness presupposition which obfuscates the *mention-all/mention-some* distinction.¹⁷ The same, however, would not hold for the plural version in (38) where a plurality of propositions is expected to be true and a distinction between *mention-some/mention-all* potentially available. However, an answer like (38c) has to be flagged as a partial answer to be felicitous.^{18,19} Comorovski takes this to show that the plurality on *wh* is antithetical to the phenomenon.

Although the data in (37)–(38) seem to support Comorovski's generalization, let us make sure that the resistance to plurality/D-linking is absolute. Suppose a researcher needs a few people with AB blood type to test a new drug. We can reasonably assume that the study requires her to test a plurality of patients but not necessarily all the patients in the hospital. Let us also assume that the researcher has a list of patients (but not their blood types) so we know we are dealing with a D-linked context. The researcher asks (39a) of the administrator who has information mapping patients to blood types:

- (39) a. Which (of the) patients can we approach for this test?
 b. You could approach Bill and John.
 c. Or you could approach Sue and Tom.

Response (39b) is a possible *mention-some* answer, made explicit by the continuation in (39c). Thus we can say with some confidence that in the right context a D-linked plural *wh* phrase can allow for *mention-some* answers, contra Comorovski.

¹⁷ Gr&S (1984: 460) allow a *mention-some* answer to (i), which has a singular *wh*:

i. On which route to Rotterdam is there likely to be no police controls?

To me this seems possible only with a plural *wh* *which routes*. Answering the singular version with *Route 21* or *Route 25*, naming two of the routes that are free of police controls, gives the impression of denying the speaker's presupposition of uniqueness. Gr&S, of course, do not engage with the issue of number sensitivity. They present (i) to show that it is not a choice question: *no N* behaves like *every N* (cf. (36)) in this respect. Therefore, a non-exhaustive answer to (i) or (36) has to be a *mention-some* answer. Their point goes through on the plural version of (i).

¹⁸ Question (38a) is not likely to be asked by our Italian tourist; it really seems to be the prospective entrepreneur's question.

¹⁹ Roger Schwarzschild (p.c.) brought up the following scenario: A tourist (let us say Roger) is standing on a street in New Delhi in front of several stores with signs only in Devanagiri (which he could not then read, and for all I know, still cannot!). Which of the following would Roger ask?

i. Where can I buy batteries?
 ii. In which store can I buy batteries?
 iii. In which stores can I buy batteries?

(i) seems inappropriate because the stores are right in front of him, a canonical D-linked situation, (ii) presupposes that there is a unique store that sells batteries but the stores all seem to be of a similar character, (iii) seems to be over-reaching. After all, he only needs to go into one store to buy his batteries. My sense is that (i), or even (ii), might work, albeit not perfectly. I have set aside this very interesting problem in the discussion here.

Based on the above, then, it does not seem wise to think of D-linking as a factor prohibiting *mention-some answers*. What seems more likely is that D-linked singular wh expressions mask the distinction between *mention-some/mention-all* answers, while plural D-linked wh expressions require special contexts to bring out the expectation that a sub-group rather than a single individual would constitute a satisfactory non-exhaustive answer. Note that canonical cases of *mention-some answers* typically name singular entities, even though the mono-morphemic wh expression in them would allow for pluralities.

Finally, we should establish whether the canonical *mention-some answer* to a question with a mono-morphemic wh necessarily names a singularity or whether, in the right context, a plural *mention-some answer* can be given. The latter seems to be the case:

- (40) a. I need two people to help me move my things. Who can I ask?
b. You could ask Bill and Tom. (Or John and Sue...)
- (41) a. I need two students for my project on Spanish bilingualism. Who can speak Spanish fluently?
b. Bill and Sue can. (So can John and Tom).

What we can gather from the discussion so far is that while it is more common to use a mono-morphemic wh in *mention-some* contexts and that the expected response usually names a single individual, neither of these are absolute requirements. Let us now see whether this also holds in embedded contexts. The following is due to Gennaro Chierchia (p.c.):

- (42) a. Context: Harry knows that Bill and Mary are patients with AB type blood and therefore can be approached for the test. He doesn't know that Sue and John have the same blood type and could also be approached.
b. #Harry knows/found out which patients we can approach for the test.
c. Harry knows/found out who we can approach for the test.

The contrast between (39) and (42) leads to the conclusion that the reliance of *mention-some answers* on mono-morphemic wh expressions is only a propensity as far as direct question-answer paradigms go, but a hard fact when it comes to embedded contexts. It would appear that even after putting aside *choice questions*, we are still left with a multi-faceted phenomenon within the class of non-exhaustive answers.

This has important implications for the semantics of mono-morphemic wh expressions. In Chapter 2 we had noted the possibility that such expressions could either have a null $N_{\text{SING}}/N_{\text{PL}}$ or that its restriction could be implicit and hence neither singular nor plural. We now have some evidence in favor of the latter. If there were a null N_{SING} , we would predict that a *mention-some* answer would not be discernible, on a par with overtly singular wh expressions. If there were a null N_{PL} , we would predict that a *mention-some* answer would only be possible in

contexts that support the expectation of pluralities being named in the answer. That mono-morphemic *wh* expressions do not have either of these properties suggests that their restriction is implicit, neither singular nor plural, but simply neutral. Of course, the facts are also compatible with there being a null *N* that is under-determined with respect to number. What is ruled out is the possibility that mono-morphemic *wh* phrases are ambiguous between singular and plural terms.

3.2.3 *Priority modals*

When we look at examples of *mention-some questions* in the literature we notice that they typically have what Portner (2009) calls priority modals: bouletic, deontic, teleological modals. In fact, non-modal questions only admit such answers in goal-driven contexts.²⁰ It makes sense, then, to probe the extent to which this particular flavor of modality determines the availability of *mention-some answers*. Non-modal questions and questions with epistemic modality provide relevant comparisons.

At first glance, epistemic possibility seems resistant to *mention-some answers*. If the context does not favor Bill over John as a possibility, (43b) does not seem to be a possible answer. If the premise for asking the question is changed, as in (44a), the judgment changes:²¹

- (43) a. I see a light on in the office. Who might be in at this time?
b. #Bill may be in. Or John.
- (44) a. I need help. Who might be in the office at this time?
b. Bill may be in. Or John. Either of them could help you.

Using this insight, we can probe non-modal questions to see if they also admit *mention-some answers* with appropriate contextual support:²²

- (45) a. I need a ride to the party tonight. Do you know who's driving?
b. Bill is. He could give you a ride. Or John, he's driving too.

In fact, if the goal is to get a ride, the exhaustive answer in (46b) seems to over-reach. The appropriate answer is (46c), and it does not, by any means, imply that

²⁰ The only non-modal case considered so far is (36a), but the context highlights its teleological aspect.

²¹ Thanks to Gennaro Chierchia for (44).

²² In a context where the questioner is the boss and it is obvious that she is only interested in facts about people who work for her, (ii) does not rule out the possibility of people outside the office being at the party. However, this is not a *mention-some* answer but rather a *mention-all* answer to a question with implicit domain restriction. See Schulz and van Rooij (2006) and George (2011) for relevant discussion:

- i. Who was at the party?
- ii. John and Mary.

the addressee is the only one going. Most likely, the answer in (46c) would be followed by *I'll give you a ride*:

- (46) a. I need a ride. Who's going to the party?
- b. Bill, Mary, Sue, Harry ... and me.
- c. I am.

Further support for the role of goal-driven modality comes from an observation in Gr&S (1984: 544–5). *Mention-some* interpretations in embedded contexts are not possible if the matrix subject is non-human. Examples (47a)–(47b) only allow *mention-all* construals:

- (47) a. What the average grade is depends on what grade each student got.
- b. Where you can get gas depends on what day it is.

Let us see if there is a difference between priority and epistemic or non-modal embedded questions:

- (48) a. John knows/found out where to get gas.
- b. We needed help and there was a light on in the office. We asked John. He knew who might be in the office at that time.
- c. I needed a ride to the party. John told me who was driving to the party.

The facts about priority modals are clear. John does not need to know all the places to get gas for (48a) to be true. Is the same true of (48b)–(48c)? The data are not so clear, I believe. What we can say with some confidence is that the priority existential definitely survives embedding. We can conclude that a priority existential is a sufficient condition for non-exhaustive construals of questions. To the extent that epistemic/non-modal questions in embedded contexts allow for such construals, it is so only with explicit contextual support.

Strong corroboration for this comes from infinitival questions, analyzed by Bhatt (1999) as encoding covert modality. The following examples, which he notes as having non-exhaustive interpretations, are analyzed as having a covert priority existential (Bhatt 1999: 133–4):

- (49) a. Magnus knows where to get gas $\approx$ Magnus knows where he/one can get gas.
- b. Magnus knows how to solve this problem $\approx$ Magnus knows how one can solve this problem.

Infinitival questions about manner have a particular propensity for *mention-some* construals. In the following, based on Pavese (2013), the addressee need not know every way of cooking rice, just one will suffice:²³

²³ Pavese points out that Quantificational Variability (to be discussed in Chapter 5) in such questions, *partly knowing how to cook rice*, does not involve knowing some of the many ways of cooking it. Rather, it involves knowing some part of one of the ways of cooking rice.

- (50) a. Do you know how to cook rice?
 b. Sure, you can pressure cook it for 5 minutes.

I should add that neither Bhatt nor Pavese claim that infinitival questions are exclusively non-exhaustive.

To sum up, we have determined that questions with existential priority modals readily admit *mention-some* answers in direct as well as embedded contexts. Non-modal questions and questions with existential epistemic modals also allow them if the (teleological) conversational goals are made explicit. This is clear in the case of direct questions but may also hold for embedded questions.²⁴

If the facts discussed here and in the previous subsection are any indication, the phenomenon of *mention-some answers* is still in need of empirical shoring up. The framing of the issues surrounding this phenomenon will have to be refined once all the facts are in place.

3.2.4 Theoretical implications

The accounts of *mention-some questions* that have been proposed in the literature so far fall broadly into three classes: simple ambiguity approaches, scope-based approaches, and approaches based on partial answers. This classification is obviously a rough one and actual analyses typically involve a more nuanced combination of semantic and pragmatic considerations than these descriptions convey.

The following are representative of each of these three types of proposal. Beck and Rullmann (1999: 292) posit a special answerhood operator to capture *mention-some answers* (51a). George (2011) suggests that such answers arise when a scopal element interacts with the exhaustification operator (51b). Lahiri (2002a: 162) includes a contextually sensitive operator over questions, shown somewhat schematically in (51c):²⁵

- (51) a. $\text{Ans}_3(w)(Q) = \lambda P_{\langle s \langle st \rangle \rangle} [\exists p [P(w)(p) \wedge Q(w)(p) \wedge p(w)]]$
 b. $[Q [\exists_i [X [\dots t_i \dots]]]]$
 c. $[x V [\text{ENOUGH } Q]] = 1$ iff $\exists S \subseteq Q, |S| > n$, for some contextually specified large n that satisfies the speaker's purposes, and $[x V \cap S] = 1$, where x stands for the matrix subject and V for the matrix verb.

Let us briefly discuss the approaches these proposals represent, keeping in mind the empirical desiderata we have identified.

The possibility of *mention-all* and *mention-some answers* has sometimes been taken to be a simple ambiguity in questions (Hintikka 1976; Berman 1991, 1994; Beck and Rullmann 1999, for example). Beck and Rullmann's Ans_3 , a third answerhood operator in addition to those for weak and strong exhaustiveness,

²⁴ The connection between necessity priority modals and non-exhaustive answers also needs careful investigation.

²⁵ Lahiri's actual rule is couched within a theory that will be discussed in Chapter 5.

takes a Hamblin set and creates subquestions, each of which has at least one true proposition in it. Knowing or answering a question using Ans_3 involves picking out one of these subquestions, and consequently one of the true propositions from the basic question denotation. A recent variant of the ambiguity approach is George (2011), whose question denotations, we saw in Section 3.1.4, vary between sets of mutually exclusive exhaustified propositions and basic Hamblin sets.²⁶ His answerhood operator imposes simple existential quantification over these sets. When applied to exhaustified sets, the *mention-all* answer is obtained; when applied to non-exhaustified sets, the *mention-some* answer is obtained. Recall that for George, weak exhaustiveness is subsumed under the *mention-some reading* of the question and that he excludes the truth condition from his answer sets. He therefore needs to add the truth requirement specifically for *mention-some answers*.²⁷ All these proposals run the risk of over-generation, a problem the authors are well aware of. Berman tries to address the problem by positing *mention-all* as a default. Beck and Rullmann acknowledge that they do not have anything to say about pragmatic constraints on their answerhood operators. Given the nature of the differences we have noted, this seems a non-trivial gap in this general approach to non-exhaustive answers.

The second approach treats the *mention-all* and *mention-some* distinction as a structural ambiguity in questions, arising from two distinct LFs. The proposals in George (2011), Fox (2012), Chierchia and Caponigro (2013), Mayr (2013) and Nicolae (2013) fall under this category. Here I will demonstrate the approach with George's account. I should note that he includes questions with indefinites in his account of *mention-some answers* but for reasons stated in Section 3.2.1, I set those aside and focus on the modal case here. George (2011: 219) considers the possibility of deriving *mention-some* answers by optionally scoping an existential term above exhaustification but below question formation:

- (52) a. Who can fix the printer?
 b. $[Q \text{ [can } [X \text{ [who fixes the printer]]}]]$
 c. $\lambda p \exists r [p = \lambda w' \exists w'' [\text{can}(w', w'') \wedge \lambda x [\text{person}(w'')(x) \wedge \text{fixes}(w'')(\text{the-printer}(x)) = r]]]$
 d. $\{^{\wedge}\text{john is someone who in some possible world is the one who can fix the printer, } ^{\wedge}\text{bill is someone who in some possible world is the one who can fix the printer, } ^{\wedge}\text{john and bill are people who in some possible world are the ones who can fix the printer}\}$

The propositions name sets of individuals who in some accessible world are the individuals who fix the printer, leading again to a choice between several *mention-some* answers. With the modal having scope under exhaustification, we would get

²⁶ This is distinct from the scope account he proposes later in the dissertation. It is the latter which is given in (51b).

²⁷ Gr&S (1984: 461–2), who otherwise allow negative answers to questions (*no one*), do not admit them in such cases. George follows their lead on this.

the single proposition naming the set of individuals who in some world or other fix the printer, leading to the *mention-all* answer.

Note that this way of deriving *mention-some* answers requires a shift from George's baseline theory summarized in Section 3.1.4. Now, the exhaustification operation X becomes obligatory, with the distinction between strong exhaustiveness and *mention-some* readings obtained by the optionality of a term's ability to scope out. As a consequence, an existential modal is necessary and sufficient for *mention-some* readings. This has the obvious advantage of predicting their availability in direct and indirect questions with priority modals. Cases where such readings are available because the context makes conversational goals salient, could presumably be handled by positing covert modals. The details, however, would have to be worked out.²⁸

The third approach takes the view that a *mention-some answer* is a partial answer that counts as complete for purposes of the conversation at hand, an idea that goes back to Gr&S (1984). Partial answers are of course ubiquitous in question-answer paradigms, so this approach has the parsimony argument going for it. If one needs a way to derive partial answers and if one needs a way to handle conversational dynamics, putting them together seems like a reasonable option. Informally, a semantic answer to the question would be the *mention-all* answer but a cooperative interlocutor would follow the maxims of quantity and relevance in giving a partial answer, without using intonation or hedges to mark it as such. A criticism of this general approach, also going back to Gr&S and reiterated by George (2011), is that *mention-some* answers with embedded questions is problematic for such pragmatic accounts. Attempts to offset such objections can be found in Lahiri (2002a), presented in (51c), Ginzburg (1995) and van Rooij and Schulz (2004). They build contextual parameters into question denotations to encode sensitivity to the goals behind asking a question. Given that not all types of *mention-some* answers survive in indirect contexts, embedding remains an open problem for this approach until the selectiveness of embedded contexts in supporting such contextual parameters can be made precise.

To sum up, it appears that both a scope-based account and an approach in terms of partial answers, augmented by pragmatic principles, are needed to fully account for the phenomenon of *mention-some* answers.

3.2.5 Section summary

This section tried to determine which factors are necessary and/or sufficient for non-exhaustive/*mention-some* answers to count as complete. Somewhat different results were discovered in direct vs. indirect question contexts. In particular, mono-morphemic wh phrases are not a requirement for direct questions but are needed in embedded contexts. A priority possibility modal readily allows for a non-exhaustive construal in direct as well as embedded contexts. A non-modal or

²⁸ To the extent that weak exhaustiveness is a special case of *mention-some readings*, this seems to lead to an unwelcome correlation between modality and weak exhaustiveness. In fairness to George, of course, we should note that he is inclined to deny the very existence of weak exhaustiveness.

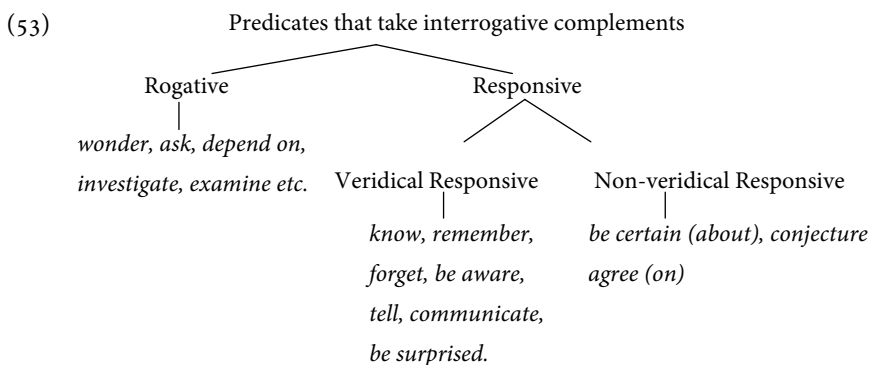
even an epistemic possibility modal requires overt contextual support, and to that extent is more evident in direct question contexts. Some of the work that has been done or is being done in this domain was reviewed, leaving it up to the reader to determine to what extent the particular approaches studied fit the modified empirical picture of *mention-some answers* presented here.

3.3 Selecting for weak, strong, or non-exhaustiveness

The proposals for dealing with strong, weak, and non-exhaustive answers to questions that we have looked at vary in the range of ambiguity they ascribe to questions and, consequently, the formal status they accord to the different types of answers. We will now approach this issue from the perspective of embedding predicates and determine the extent to which these distinctions matter in complement selection. Negative polarity items in questions will also be considered as they provide a strong argument in favor of a grammatical distinction between strong exhaustiveness and weak/non-exhaustiveness.

3.3.1 Embedding predicates

We have looked at embedded questions but we have not paid much attention to the grammatical properties of embedding predicates so far. We can start with the classification of interrogative selecting predicates from Lahiri (2002a: 287). We will restrict ourselves here to issues related to the selection of strong vs. weak exhaustiveness, deferring a broader discussion to Chapter 5:²⁹



Heim (1994) noted that the existence of predicates selecting for weak exhaustiveness would be an argument in favor of theories that allow for weak exhaustiveness, such as Karttunen. The absence of such predicates would be an argument for

²⁹ This classification subsumes the one in Karttunen (1977) and incorporates the distinction between intensional/extensional predicates from Gr&S (1982). We do not include infinitival complements (see Bhatt 1999).

theories that rule out this possibility, such as Gr&S. Most current theories that we have considered allow for both weak and strong readings. This is partly due to the discovery of a class of embedding predicates that do not take strongly exhaustive interrogative complements. This is the class of factive/veridical predicates to which *be surprised* and *be amazed* belong. Let us review some of the relevant arguments.

Heim, drawing on Berman (1991), uses the following scenario to establish that *be surprised* does not embed strongly exhaustive questions (see also Lahiri 1991, 2000, 2002a; Beck and Rullmann 1999; Sharvit 2002; Guerzoni and Sharvit 2007):

(54)	individuals:	John	Bill	Sue
	my prior expectations:	Y	Y	Y
	callers in <i>w</i> :	Y	Y	N

- a. It surprised me who called.
- b. It surprised me who didn't call.

Assume that John and Bill called as I had expected them to but Sue, who I had also expected would call, did not. According to Berman, (54a) would be judged false and (54b) true in this situation. Let us ignore plural individuals here to keep things simple:

- (55) a. $\llbracket \text{who called} \rrbracket = \{^{\wedge}\text{john called}, ^{\wedge}\text{bill called}, ^{\wedge}\text{sue called}\}$
- b. $\llbracket \text{who didn't call} \rrbracket = \{^{\wedge}\text{john didn't call}, ^{\wedge}\text{bill didn't call}, ^{\wedge}\text{sue didn't call}\}$

Under weak exhaustiveness, (55a) denotes the set of worlds in which both John and Bill called, and this includes some worlds in which Sue also called and some in which she did not. Under the weakly exhaustive reading of (55b), we would have the set of worlds in which Sue didn't call, some of which would include Bill and John as callers/non-callers. In contrast, under the strongly exhaustive interpretation, both would denote the same set of worlds, those in which there are two callers, John and Bill, and one non-caller, Sue.

In order for (54a) to be true, the lexical semantics of *be surprised* requires that the answer to the question *who called* must be a proposition that goes counter to expectations. Since I expected both John and Bill to call, the weakly exhaustive answer *John and Bill called* does not run counter to expectation, so the judgment that (54a) is false is predicted. The weakly exhaustive answer to (54b), instead, *Sue didn't call* does go counter to expectations, and is therefore predicted to be true. Since the strongly exhaustive answer to both (54a) and (54b) is *John and Bill called but Sue didn't*, the answers to both run counter to expectation. Both are predicted to be true, contrary to fact. This establishes that predicates of this class take weakly exhaustive readings of questions.

A second argument for *be surprised* to select weak exhaustiveness is based on Heim's (1994) observation that *de dicto* readings of the *wh* restrictor are only possible under strong exhaustiveness. Recall from Section 2.2 Beck and Rullmann's (1999) account of the *de dicto/de re* ambiguity. In embedded questions,

the common noun is interpreted inside the nucleus, with the world variable varying across worlds or being anchored to the world of evaluation:

- (56) a. John knows which students called.
 b. $\text{know}_w(j, \text{Ans}_{w/s}(\lambda p \exists x [p = \lambda w' [\text{student}(w)(x) \wedge \text{called}_{(w')}(x)]]))$
 c. $\text{know}_w(j, \text{Ans}_{w/s}(\lambda p \exists x [p = \lambda w' [\text{student}(w')(x) \wedge \text{called}_{(w')}(x)]]))$

In principle, (56a) can be interpreted in four ways, weak *de re*, strong *de re*, weak *de dicto* and strong *de dicto*. However, Heim observes that *de dicto* readings correlate with strong readings. That is, there is no weak *de dicto* reading.

Sharvit (2002) elaborates on this, adopting Beck and Rullmann's account of *de dicto/de re* ambiguities.³⁰ She compares predicates like *know*, which show the relevant ambiguity, to predicates like *be surprised*, which do not:

- (57) a. Scenario 1. Student Mary and student Sally left. John knows that they are students and that they left. Fred, a non-student, also left. John knows that he left and thinks he is a student.
 b. John doesn't REALLY know which students left. Indeed, he knows that Mary and Sally are students who left, but he also thinks that Fred is a student who left.
- (58) a. Scenario 2. Student Mary and student Sally left. Student Bill didn't leave. John believes that Mary and Sally are the students who left. John doesn't know that Bill didn't leave.
 b. John doesn't REALLY know which students left, because Bill is one student who didn't leave and John doesn't know that Bill didn't leave.

Let us see under what readings *John knows which students left* can be false in scenario (57a). Since John can correctly identify the members of the set of students *{Mary, Sally}* that left, the *de re* readings are true. The weak *de dicto* reading is true as well, since he correctly knows of the two student leavers that they are students and they left. Crucially, it is false under the strong *de dicto* reading since he incorrectly includes Fred as a student who left.³¹ Turning to the scenario in (58a), we find that *John knows which students left* is false on the strong *de re* reading because he can't separate the leavers and the non-leavers in the set of students *{Mary, Sally, Bill}*. It is true on the strong *de dicto* reading: John can correctly divide the student-leavers *{Mary, Sally}* from the set of individuals who are non-leavers or non-students *{Bill}*. The acceptability of (58b) is based on the strong *de*

³⁰ See Section 2.2 on this. Crucially, here we are talking of a *de dicto/de re* ambiguity with reference to the *wh* phrase. In Section 3.1 we discussed *de dicto/de re* ambiguity with respect to the whole interrogative clause.

³¹ The weak *de re* reading merely requires John to have an attitude to the proposition: *Mary and Sally left*. Their status as students is not relevant, neither is the student-hood of Fred or his leaving. The strong *de dicto* reading requires him to have an attitude to the proposition: *the students who left are Mary and Sally*.

re reading of the embedded question. To sum up, we have evidence for a three-way distinction with predicates selecting for strong exhaustiveness: weak *de re*, strong *de re*, and strong *de dicto* readings. We do not have evidence of weak *de dicto* readings.

Returning to *be surprised*, if it can be shown not to have *de dicto* readings, there will be added support to its status as selecting exclusively for weak exhaustivity:

- (59) a. It surprised John which students left.
 b. Scenario 3. Student Mary and student Sally, the only students, left. John is surprised that Mary and Sally left, but he doesn't know that they are students.
 c. #It didn't REALLY surprise John which students left. Indeed, he didn't expect Mary and Sally—the only students who left—to leave, but he wasn't aware that they are students.

The scenario in (59b) shows that (59a) is true on its *de re* reading. John would admit to not expecting Mary and Sally to leave, but their status as students wouldn't enter into it. Sharvit presents the oddity of (59c) as evidence that (59a) doesn't have a *de dicto* reading. She further shows that it is impossible to construct a scenario in which a *de dicto* reading would be true and a *de re* reading false. This is consistent with the predicate selecting exclusively for weak exhaustiveness, on the view that there is a single source for strong exhaustivity and *de dicto* readings. Sharvit's discussion of the *de dicto/de re* ambiguities is incisive but ends on an intriguing note. The absence of non-veridical weakly exhaustive verbs is noted as an accidental gap, not predicted by any theory.³²

Our concern in the rest of this section will be on predicates that exclusively select for weak exhaustiveness but I would like to touch upon one non-veridical predicate *agree on*, and its relation to exhaustiveness and *de dicto/de re* ambiguities. Sharvit (2002: 114) adopts Beck and Rullmann's (1999) semantics for *agree on* but not their contention that the weak/strong distinction is only relevant for veridical predicates (see also Klinedinst and Rothschild 2011):

- (60) a. $\llbracket \text{agree-on}^{\text{B\&R}} \rrbracket (Q)(w)(A) = 1$ iff
 (a) for all x in A and all p in $Q(w)$, if x believes in w that p is true, then for any y in A , y believes in w that p is true; and
 (b) for all x in A and all p in $Q(w)$, if x believes in w that p is false, then for any y in A , y believes in w that p is false;
 b. individuals: John Bill Sue *Harry*
 Mary's beliefs: Y Y N N
 Jane's beliefs: Y Y N —
 c. Jane and Mary agree on who called.
 d. $Q = \{\text{john called, bill called, sue called, } \textit{harry called}\}$; $A = \text{mary+jane}$

³² See Nicolae (2013) for discussion. Thanks also to Andreea Nicolae (p.c.) for helpful comments.

The lexical semantics of the verb requires that the only propositions that matter are those that the agents have views about. In the scenario (60b), ignoring Harry for now, (60c) would be true because Jane and Mary's beliefs coincide for all the individuals that they have beliefs about. The predicate being non-veridical, the facts are irrelevant. *Agree on* passes muster as selecting for strong exhaustiveness.

Now, throwing Harry into the mix, we find that Mary has a belief about Harry but Jane does not. If *agree on* optionally allows for a weak reading, then its semantics, under this reading, would only consider the first clause in (60a). Example (60c) would still be judged true under its weak reading, though false under its strong reading. Sharvit (2002: 112–15) presents the following to argue against this possibility:

- (61) a. Bill and Mary do not REALLY agree on who cheated on the exam. For example, Mary thinks that Bob didn't cheat and Bill has no opinion about Bob.
 b. John doesn't REALLY know who left. For example, he doesn't know that Sally didn't leave.
 c. It didn't REALLY surprise Bill who cheated on the exam. # For example, it didn't surprise him that Mary didn't cheat.

Sharvit concludes that *agree on* can select strong or weak exhaustiveness, analogously to *know* and distinct from *be surprised*.

Guerzoni and Sharvit (2007: 387) round off the discussion by considering rogative predicates. They treat verbs like *wonder* as internal counterparts of direct questions and provide the following semantics for them (Guerzoni and Sharvit 2007: 387):

- (62) $\llbracket \text{wonder-strong} \rrbracket^w(Q)(x)$ is defined only if it is not the case that for all $p \in Q$ s.t. $p(w)$, x believes p in w .
 When defined, $\llbracket \text{wonder-strong} \rrbracket^w(Q)(x) = \text{True}$ iff for every p in Q , x wants to know whether p is true or not.

We will see below that their characterization is related to their views on NPI licensing.

3.3.2 NPI licensing and exhaustiveness

We have seen evidence from embedding predicates that both strong and weak exhaustiveness need to be recognized in the semantics of questions. But if such a distinction exists, we may rightly expect it to be manifested elsewhere in the grammar. Guerzoni and Sharvit (2007) argue for precisely this. The fact that questions allow negative polarity items (NPI) poses a challenge. NPIs are licensed in contexts that are (Strawson) downward entailing:

- (63) a. John bought a car / *any car.
 b. John didn't buy a car / any car.
 c. Only John bought a car / any car.

Affirmative sentences are upward rather than downward entailing: *buying a car* entails *buying something* but not *buying a red car*. Consequently, they do not allow NPIs like *any N* (63a). Negative sentences reverse entailments and create hospitable contexts for NPIs. More interesting is the case of *only*, which is not downward entailing (*only John bought a car* does not entail *only John bought a red car*) and yet licenses *any*. von Stechow (1999) accounts for this by arguing that the entailment pattern has to be tested in contexts in which the presuppositions of the sentences tested are satisfied. If we confine ourselves, for example, to contexts in which John bought a red car, then it does follow from *only John bought a car* (63c) that *only John bought a red car*. This makes the proposition in the scope of *only* a Strawson downward entailing context and predicts the acceptability of *any*.

Turning to questions, it is hard to apply the notion of entailment to them (see Ladusaw 1979). We might, for example, want to say that a question A entails a question B if answering A completely answers B. But by this measure, questions do not appear to be downward entailing. Answering *did John buy a car?* does not provide an answer to *did John buy a red car?* And yet, NPIs are possible in questions. Setting that aside, Guerzoni and Sharvit (2007) observe a correlation that ties in with the concerns of this chapter:

- (64) a. Did you eat anything?
 b. Who cooked anything?
- (65) a. Claire wonders whether Frank has any books on Negative Polarity.
 b. Claire knows whether Frank has any books on Negative Polarity.
- (66) a. *It surprised Bill which students had ever been to Paris.
 b. Claire wonders which students have any books on Negative Polarity.
 c. %Claire knows which students have any books on Negative Polarity.

They argue that NPI licensing goes hand in hand with strong exhaustiveness. Polar questions are patently strongly exhaustive since the two propositions in their denotation represent a binary partition of the space of possibilities. To know one proposition in the Hamblin set to be true is to know that it can be the only true proposition in the set, or equivalently, to know the other to be false. Uncontroversially, NPIs in direct as well as indirect polar questions are grammatical, (64a) and (65). In the case of constituent questions as well, direct questions admit NPI (64b). Indirect questions, however, do so depending on the embedding predicate. Predicates that have been argued to select for weak exhaustiveness, such as *be surprised*, do not license NPIs (66a), while predicates like *wonder*, which arguably select for strong exhaustiveness, do so readily (66b). With *know* (66c), Guerzoni and Sharvit report cross-speaker variation. The diacritic % indicates that for some speakers it is fully acceptable, while for others it is not.

Speakers who accept NPIs, they contend, draw on the strongly exhaustive nature of *know*, while those who do not, display sensitivity to its weakly exhaustive nature.

Explanations for Guerzoni and Sharvit's generalization have been provided by Mayr (2013) and Nicolae (2013). Working independently, they both develop accounts that fall broadly within the alternatives and exhaustification approach to polarity (Chierchia 2013). Nicolae's proposal was summarized briefly in Section 3.1.4. The essential idea is that questions have an optional exhaustification operator, analogous to *only* inside the nucleus, which accounts for strong exhaustiveness. Given that *only* creates a Strawson downward entailing environment, the correlation of NPI licensing with strong exhaustiveness is expected:³³

- (67) a. *Who wrote anything?
 b. *Weak Exhaustiveness: $[\text{Who}_i [\text{t}_i \text{ wrote anything}]] = \{^{\wedge}\text{a wrote anything}, ^{\wedge}\text{b wrote anything}\}$
- (68) a. Who wrote anything?
 b. Strong Exhaustiveness: $[\text{Who}_i [\text{Only } \text{t}_i \text{ wrote something}]] = \{^{\wedge}\text{only a wrote anything}, ^{\wedge}\text{only b wrote anything}\}$

Mayr (2013) addresses this problem somewhat differently. He draws on the account of polarity licensing in Chierchia (2013) and combines a Karttunen-question semantics, alternatives introduced by NPIs, and a strong exhaustive answer operator. The resulting answers to questions are shown to be downward entailing. Due to constraints of space, however, I leave the details for the reader to pursue on their own.

Both Nicolae and Mayr predict not only that predicates like *be surprised* should not allow NPI, as observed by Guerzoni and Sharvit, but also that *mention-some answers* should not allow them. Mayr makes this explicit:

- (69) a. Where can I buy the New York Times?
 b. *Where can I buy any newspaper?

We can draw two conclusions from these analyses. The first is that questions are ambiguous between a strongly exhaustive and a weakly exhaustive or non-exhaustive reading. The second is that there are predicates that do not select for strongly exhaustive readings of questions. The issue that remains open is whether the predicates that do not select for strongly exhaustive questions select for weakly exhaustive questions or for *mention-some* questions or both. George (2011) suggests that they take *mention-some* complements and Nicolae that they take both weakly exhaustive and *mention-some* complements. Mayr seems to be neutral on this. A potential diagnostic to test this issue may be the possibility of *wh-ALL* in the complement of such predicates: *John was surprised*

³³ Note that there is a disconnect between question denotations and possible answers. Covert *only* licenses NPI in questions but a possible answer cannot have an NPI unless overt *only* is used. See Nicolae (2013: 39–41) for discussion.

at *who all showed up*. Beck and Rullmann (1999) have argued convincingly that *wh-ALL* is incompatible with a *mention-some* answer. If *be surprised* cannot take strongly exhaustive answers, the acceptability of *wh-ALL* in the complement of this predicate shows that it takes weakly exhaustive answers and suggests a grammatical distinction between *weakly exhaustive* and *mention-some* readings.

3.3.3 Problematic NPIs

The accounts presented above represent real advances in our understanding of NPI licensing in questions. At the same time, I believe the data they seek to capture is more nuanced than what we have seen. Consider the following contrasts (Lahiri 1991: 67–8):

- (70) a. *I know how he ever did it.
 b. I wonder how he ever did it.
 c. I don't know how he ever did it.
- (71) a. *I know why anyone bothers to listen to him.
 b. I wonder why anyone bothers to listen to him.
 c. I don't know why anyone bothers to listen to him.

The shift in judgments between positive and negative versions of the same predicate, here *know*, raises non-trivial questions for accounts based on selectional properties of embedding predicates.

The alignment of the negative responsive predicate like *not know* with rogative predicates like *wonder*, seen above, has also been noted to affect the possibility of inversion in certain dialects of English (McCloskey 2006). That is, they manifest the syntax of matrix questions:

- (72) a. *They know/discovered who was she dancing with at the party.
 b. They wondered/were asking who was she dancing with at the party.
 c. They wanted to know who was she dancing with at the party.
 d. They don't know who was she dancing with at the party.

If embedded clauses can lose their normal subordinate status under negation or modality, NPI licensing is predicted to follow the pattern of direct questions. Some preliminary discussion on this topic can be found in Dayal and Grimshaw (2009) and is briefly addressed in Chapter 5. As things stand, however, these data do not fall in line with the view that *know* can license NPIs because it can embed strongly exhaustive questions.³⁴

³⁴ NPIs are not acceptable in subject position in constituent questions but the problem with (71a) remains even when this is controlled for, especially if a generic interpretation is factored out: **I know why John bothered to listen to anyone* or **I know why John bothered anyone*.

Another cautionary note comes from NPI in *mention-some* answers. While it is true that they are not perfect, they are not ruled out either. I note the following, based on very preliminary investigation. The contexts are set up to highlight the teleological conversational goals, making the expectation of a *mention-some answer* prominent:

- (73) a. Context: A teacher planning a project wants to include a student who has visited Asia, even if it was a long time back.
 b. Who has ever been to Asia? I need a volunteer for my project.
 c. Bill has been to Asia. His parents were posted there.
 (or John, I know he traveled to several countries in Asia last summer).
- (74) a. Context: I am cooking something for which I need some exotic spices. I don't have a high expectation that the stores in my small town will carry such specialty items. But I'm willing to settle for some substitutes.
 b. Where do they sell any spices in this town?
 c. At Shoprite. Or even at Pathmark.

Speakers find the NPI in (73b) and (74b) either redundant or somewhat awkward but not entirely unacceptable. The resistance is certainly not of the order one finds in simple affirmative statements with NPI, even when they are uttered in response to questions with NPI:

- (75) a. *Bill has ever been to Asia.
 b. *They sell any spices at Shoprite.

While one cannot disregard the substantive results of the analyses discussed earlier, the facts listed here are worth keeping in mind as we move forward. They point to factors other than strong exhaustiveness as also being significant for NPI licensing in questions.

3.3.4 Section summary

Here we looked at embedding predicates to determine if they show sensitivity to strong vs. weak exhaustiveness. We looked at arguments from rogative and responsive predicates whose selectional properties suggest that such a distinction between predicates is needed. We also noted that the distinction correlates with the differential behavior of predicates with respect to NPI licensing. Though there remain several interesting strands for further investigation, our present understanding of the phenomenon supports the view that some distinction along the lines of strong and weak/non-exhaustiveness is called for. And there may even be an argument for a three-way distinction from particles that require exhaustiveness occurring in combination with predicates that do not select for strong exhaustiveness.

3.4 Exhaustiveness in the baseline theory

This chapter expanded the range of direct answers that questions admit to include *strongly exhaustive*, *weakly exhaustive*, and *non-exhaustive answers* and explored the implications of this for the semantics of questions. Several different proposals, as well as the empirical considerations behind those proposals, were discussed. I will now highlight the most significant ideas that emerged from our discussion, without attempting to integrate them into a uniform theory of questions.

We began with the difference between weak and strong exhaustiveness and proposed that there is only a single answerhood operator that delivers weak exhaustiveness: $\text{Ans-}D_W$. A structure like $[x \text{ V } Q]$ has a weakly exhaustive reading when the agent has an attitude to a proposition that the speaker recognizes as $\text{Ans-}D_W(Q)$ but not the agent; it has a strongly exhaustive reading when the agent recognizes the connection. In direct question–answer paradigms, the answer has a strongly exhaustive reading because the speaker asserts not just the proposition *simpliciter* but rather the proposition as being identical to $\text{Ans-}D_W(Q)$. Keeping to weak exhaustiveness as the basic building block for mediating between questions and answers makes it possible to maintain a distinction between positive and negative versions of questions. Other proposals in the literature take the question itself to be ambiguous, encoding an optional exhaustification operation inside the question nucleus. Depending on the answerhood operators assumed, such proposals yield an ambiguity between strong exhaustiveness on the one hand and weak and/or non-exhaustiveness on the other.

We then considered the role of contextual as well as structural properties in making available a non-exhaustive construal of questions. Here we identified a difference between direct questions and indirect questions. Mono-morphemic *wh* phrases combined with priority existential modals always allow for them, while complex *wh* phrases and non-modal/epistemic modal questions only do so with explicit contextual support, one where a positive answer to the question addresses the conversational goals. Such contextual support is easier to integrate into direct questions, so *mention-some* construals of non-modal or epistemic modal questions tend to be restricted to direct questions. We discussed three current approaches to *mention-some* answers but none of them capture the duality within the phenomenon. A combination of structural and pragmatic approaches appears to be best positioned to capture the facts. The discussion also addressed the issue of number implicatures. Complex noun phrases, which obligatorily come with number restrictions, seem resistant to *mention-some* construals. This is because the uniqueness presupposition of singular *wh* phrases blurs the distinction between *mention-all/mention-some*. Plural *wh* expressions, instead, require special contexts to make salient a non-exhaustive plural answer as meeting the conversational goals. Mono-morphemic *wh* expressions, we concluded, are underspecified with respect to number since they show the behavior of neither type of complex *wh* phrase.

The final section of the chapter approached the issue of exhaustiveness from the perspective of embedding predicates. The fact that some predicates do not force strong exhaustiveness and that some predicates explicitly reject strong exhaustiveness

is an argument for maintaining the distinction between strong exhaustiveness and weak/non-exhaustiveness. This divide is further supported by the differential behavior of negative polarity items in questions. While there is no evidence directly from embedding predicates for a three-way distinction between strongly exhaustive, weakly exhaustive, and non-exhaustive readings, there may be one in combination with the possibility of *wh-ALL* type phrases in the complement of predicates that reject strong exhaustiveness.

Single-pair, pair-list, and functional answers

Multiple constituent questions and questions with quantifiers admit answers expressing relations between individuals, either with a description or with a list. This means that there are now three types of answers that a theory of questions must account for: individual answers, which we have been studying, relational/functional answers, and pair-list answers. To add to the complexity, functional and list answers turn out to be a heterogeneous class. This chapter presents diagnostics for distinguishing between different types of list and functional answers, clearly identifying the factors responsible for shifts in interpretation. It also introduces readers to technical details of specific proposals.

Pair-list answers to multiple *wh* questions and questions with universal quantifiers seem to involve a functional dependency between two sets. Answers exhaustively pair all the members of one set in a one-one or many-one relation to members of the other set. Questions with quantifiers also allow functional answers. We explore possible connections between these two types of answers. Questions with plural definites seem to allow functional and list answers as well but they do not align with questions with quantifiers in crucial ways, suggesting distinct accounts for the two types of questions.

Functional answers to questions with quantifiers require quantification over Skolem functions, functions from individuals to individuals. Do Skolem functions leave complex traces that can explain syntactic restrictions on the availability of functional answers? What determines whether a quantifier can admit pair-list answers? Can an account of questions with quantifiers apply to multiple *wh* questions? Do questions with indefinites belong in this set? These are among the issues probed here.

The functional approach is shown to fall short of capturing the functionality of pair-list answers: the need to list every member of the domain set and the requirement of unique pairing for each such individual. Two solutions are considered, one which modifies the functional approach and one which uses higher order questions. Empirical and theoretical considerations are weighed in relation to both. The derivation of higher order questions is discussed in the context of pair-list answers, echo questions, and scope marking.

The last section considers three responses in the literature to the proposals for list answers. The first touches upon differences between matrix and embedded questions with respect to their availability. The second looks at the differential

behavior of *every* N and *each* N with respect to structural sensitivity. The third treats pair-list readings as quantification at the level of speech acts.

4.1 Pair-list and functional answers

This section explores challenges posed by constituent questions that have, in addition to a *wh* phrase, another quantificational expression: a second *wh* phrase or a universal quantifier. The first type of question admits individual and pair-list answers. The second type of question admits individual, pair-list, and functional answers. The discussion seeks to establish points of overlap between pair-list and functional answers as well as their relation to individual answers. These distinctions are particularly important in separating questions with plural definites from questions with universal quantifiers.

4.1.1 Lists and multiple constituent questions

Limiting the discussion to questions with two *wh* phrases, a question like (1a) can be answered with (1b) or (1c). Within our baseline theory it is analyzed as (2). If there are two students, Alice and Bill, and two professors, Prof. Carl and Prof. Dan, we get the denotation in (2b):¹

- (1) a. Which student met which professor?
 b. Bill met Professor Carl.
 c. Bill met Professor Carl and Alice met Professor Dan.
- (2) a. $\lambda p \exists x \exists y [\text{student}_w(x) \wedge \text{professor}_w(y) \wedge p = \lambda w' \text{met}_{w'}(x, y)]$
 b. $\{\lambda w \text{Bill met}_w \text{Prof. Carl}, \lambda w \text{Bill met}_w \text{Prof. Dan},$
 $\lambda w \text{Alice met}_w \text{Prof. Carl}, \lambda w \text{Alice met}_w \text{Prof. Dan}\}$
 c. $\text{Ans-D} = \lambda Q \lambda w \iota p [Q(p) \wedge p(w) \wedge \forall p' [(Q(p') \wedge p'(w)) \rightarrow p \subseteq p']]$

The two *wh* phrases are interpreted as existential quantifiers binding two argument positions inside the nucleus. The question thus denotes a Hamblin set with propositions varying on values for both variables. If only Bill met Prof. Carl, the only true proposition in the set will be the one underlined in (2b) and the answerhood operator (2c) will pick it out.² This is the single-pair answer to the multiple constituent question: (1b) is the analog of the individual answer to single constituent questions.

If, however, Bill met Prof. Carl and Alice Prof. Dan, Ans-D(2b) will be undefined. There will be two true propositions in the set, neither of which will entail the other. The absence of uniqueness will lead to infelicity. This theoretical

¹ A related construction has conjoined *wh* phrases: *where and when did you see him?* See Comorovski (1996) and Ratiu (2011a).

² I suppress reference to weak and strong versions of Ans-D when the distinction is not relevant to the point under discussion. (2c) captures weak exhaustiveness (see Chapter 3).

prediction flies in the face of ground reality: the pair-list answer (1c) is not only acceptable, it is possibly a more natural answer than the single-pair answer (1b).

Historically, it is the single-pair answer that has had to be defended at the empirical level. Pope (1976), who dubs them REF-QS, shows that single-pair answers are appropriate in contexts that establish the existence of a unique pair of individuals, but not their identity:³

- (3) a. Speaker A: That student couldn't solve the problem.
 Speaker B: Which student couldn't solve which problem?
 b. Mother breaking up a fight between her two children: OK, who hit who first?

The possibility of (1b) and (1c) as possible answers raises the issue of a potential ambiguity. Even though Ans-D delivers only one of the possible answers, suggesting the need for a second account, it does not settle the question. One could argue that there is no restriction on the number of propositions in the set that can be true in a given world. It is in contexts that do not favor the possibility of multiple pairs that the single-pair answer becomes visible.

Recall from Chapter 2, however, the reason for building uniqueness into Ans-D via the use of *iota*:

- (4) a. Which student/which students/who called?
 b. $\{\lambda w \text{ Bill called}_w, \lambda w \text{ Alice called}_w\}$
 c. $\{\lambda w \text{ Bill called}_w, \lambda w \text{ Alice called}_w, \lambda w \text{ Bill+Alice called}_w\}$

The choice of the singular term presupposes that only a single individual will be named in the answer, the choice of the plural term implies that a plurality will be named, while the choice of the neutral term leaves both options open. Adopting standard assumptions about interpreting number, (4a) with *which student* denotes propositional sets like (4b), naming atomic individuals, while (4a) with *which students* or *who* denotes propositional sets like (4c), naming both atomic and plural individuals. Ans-D applied to (4b) picks out a single true proposition if there is one, delivering uniqueness in the case of *which student*. The choice of an overtly plural wh phrase, due to the existential presupposition of the question: *some students called*, requires the proposition picked by Ans-D to name a plurality. Returning to multiple constituent questions, I submit that since Ans-D is needed for single constituent questions, the single-pair answer to multiple constituent questions comes at no extra cost to the theory. The challenge is to extend the system to derive pair-list answers.

The problem under discussion is not a consequence of the particular theory we have chosen but holds of any account that takes the shift from uniqueness in single constituent questions to multiple pairs in multiple constituent questions to

³ Wachowicz (1974, 1975) classifies them with echo questions, but Pope shows that REF-QS and echo questions are intonationally distinct. See also Bolinger (1978), Bartels (1997), and Truckenbrodt (2013). Also, both mono-morphemic and complex wh phrases admit such answers, contrary to what has been claimed for English by Bošković (2001).

be a semantic phenomenon. Higginbotham and May (1981), the first to address this problem, argues that pair-list answers represent one-to-one bijective relations (see also Dayal 1991a, 1991b).⁴ If there are strict restrictions on pairings, a semantic rather than a pragmatic solution is called for. Engdahl (1986: 154), however, questions the basis for the semantic operation of *absorption* proposed by them, noting cases that violate bijection:

- (5) a. Which table ordered which wine?
 b. Table A ordered the Ridge Zinfandel, Table B ordered the Chardonnay and Table C ordered the Rose and the Bordeaux.

Dayal (1996: 105–8) resurrects the argument for a semantic account based on a different set of facts. The relevant constraints are identifiable when quantification is restricted to a very small domain:

- (6) a. Which student talked to which professor?
 b. Alice and Bill both talked to Professor Carl.
 c. #Alice talked to Professors Carl and Dan.

- (7) a. $a \rightarrow c$ b. $a \rightarrow c$ c. $*a \rightarrow c$
 $b \rightarrow d$ $b \nearrow d$ $b \searrow d$

Example (6) establishes that the relations at issue are really functions. Pair-list answers exhibit *domain-cover*: every member of the set denoted by the first wh phrase must be connected to some member in the set denoted by the second wh phrase.⁵ And, they admit pairings that are one-to-one (7a) or many-to-one (7b) but not one-to-many (7c). Such clear constraints show that a semantic explanation for the shift from uniqueness to pair-lists is, after all, needed. Dayal suggests that counterexamples like (5) are only felicitous when most of the pairs are expected to fall in line with functions, with the deviations being exceptions rather than the norm. Although pair-lists are a semantic phenomenon, pragmatics comes into play in accounting for such semantically deviant cooperative responses.

The notion of *domain cover* has a natural connection with universal quantification. Multiple wh questions had been linked to questions with universal quantifiers by Groenendijk and Stokhof (1984, henceforth Gr&S), Comorovski (1989, 1996), and É Kiss (1993). However these authors either explicitly allow, or remain agnostic on whether an individual in the first set can be freely paired with any number of individuals in the second set, in contrast to what we have observed. Here we will take pair-list answers to multiple constituent questions to obey not only *domain cover* but also functional pairing and use *functionality* as a cover term

⁴ See Partee et al. (1990) and Chierchia and McConnell-Ginet (2000) on relations and functions.

⁵ Dayal uses the term *exhaustivity* to express this property, marking a distinction from *weak/strong exhaustiveness* of Gr&S (1982, 1984). However, *exhaustivity* is now used interchangeably with *exhaustiveness*, so I adopt the term *domain cover* from Bittner (1998).

for both properties. The challenge, then, is to formally capture functionality in multiple *wh* questions while preserving uniqueness in single constituent questions.

4.1.2 Lists and questions with quantifiers

Turning to single constituent questions with universal quantifiers and focusing on *every N* for now, we find functionality in pair-list answers to them as well. Karttunen (1977: 31–3), crediting Hull (1974), notes the existence of pair-list answers (8c) and Engdahl (1980, 1986) and Gr&S (1984) extend the paradigm to include functional answers (8d):⁶

- (8) a. Which professor does every student like?
 b. Every student likes Prof. Carl.
 c. Bill likes Prof. Carl and Alice likes Prof. Dan.
 d. Every student likes his/her advisor.

The theories considered so far only account for individual answers ((8b)). Let us see why:

- (9) a. $\lambda p \exists x [\text{professor}_w(x) \wedge p = \lambda w' \forall y [\text{student}_w(y) \rightarrow \text{like}_{w'}(y, x)]]$
 b. $\{\lambda w \text{ every student likes}_w \text{ Professor Carl,}$
 $\lambda w \text{ every student likes}_w \text{ Prof. Dan}\}$
 c. $a \rightarrow c$ d. $a \searrow c$
 $b \nearrow d$ $b \rightarrow d$

For some value of x , $x = a$ professor, (9a) admits propositions where every student stands in the *like* relation to that x . That is, it describes the situations depicted in (9c)–(9d). The only pair-list answer possible would be an elaboration of one of the propositions in (9b), with all the students paired with the same professor. And the only functional answer would be one where all the students have the same advisor. But we know that professors can vary with students (8c)–(8d).

Intuitively, we want to give the universal scope over the *wh*, as in (10a). But Karttunen shows that this move is misguided. Consider the model in (10b) and the three propositions in (10c)–(10e). If (10a) is evaluated at w_1 we want Ans-D to pick out proposition (10e):

- (10) a. $\lambda p \forall y [\text{student}_w(y) \rightarrow \exists x [\text{professor}_w(x) \wedge p = \lambda w' \text{ like}_{w'}(y, x)]]$
 b. <B, Prof.C> <A, Prof.D> <B, Prof.D> <A, Prof.C>
 w1 Y Y N N
 w2 N Y Y N
 w3 Y Y Y Y
 w4 Y N N Y

⁶ Karttunen's example uses *wh* expressions of the form *what N*, which have special properties (see Chapter 6) that are orthogonal to our current concerns.

- c. λw Bill likes_w Prof. Carl = {w₁, w₃, w₄}
- d. λw Alice likes_w Prof. Dan = {w₁, w₂, w₃}
- e. λw [Bill likes_w Prof. Carl $\wedge$ Alice likes_w Prof. Dan] = {w₁, w₃}

The proposed logical form in (10a) will let in a proposition p iff for every student y there is some professor x , and p denotes the same set of worlds as the nucleus proposition. But there is no proposition that meets these conditions if the domain has more than one student. This holds even if there are two students who both like the same professor. The set of worlds denoted by λw like_w(y, c) will not be the same when y is set to a and when y is set to b . Quantifying in a universal as in (10a) can only yield a meaningful non-empty set of propositions in worlds with just one student. However, a felicitous use of *every student* requires quite the opposite, a world with a plurality of students.⁷ We need an alternative account of pair-list answers to questions with quantifiers.

Functional answers involve the same issues and thus raise the question of their relation to pair-list answers. Are they just alternative ways of expressing a situation in which there is variation or is there a deeper semantic connection? The early syntactic literature (May 1985) does not pay much attention to functional answers, treating them as succinct ways of expressing lists. Engdahl (1980, 1986) argues, however, that functional answers have a bona fide independent status (see also Gr&S 1984; Chierchia 1991, 1993). She extends Karttunen's semantics to allow *wh* expressions to range over Skolem functions, functions from individuals to individuals. Deferring formal details of the proposal to Section 4.2, let us see what we can establish on the basis of this informal characterization of Engdahl's position.

An important argument in favor of the independence of functional answers is that they are available with quantifiers that do not support pair-list answers:

- (11) a. Who does no man like?
 b. His mother-in-law.
 c. #Bill doesn't like Alice and Bob doesn't like Sue.

Absent pair-lists as source, functional answers must represent a basic reading of the question.

In the reverse, one can ask if pair-list answers always derive from functional answers. Engdahl (1980, 1986) and Chierchia (1991, 1993) argue in favor of doing so, since pair-list answers are available in a subset of cases which allow functional answers. Gr&S (1984: 178), instead, treat the two as independent of each other:

- (12) a.

<i>son</i>		<i>mother</i>
John	→	Mary
Bill	↗	Suzy
Peter	→	Jane

⁷ Karttunen also points out that in worlds with no students (10a) will let in any old proposition since they will all satisfy the conditions vacuously. But, of course, Ans-D predicts infelicity since it will be undefined if its existence/uniqueness presuppositions are not met.

- b. Which woman does every man love?
- c. His mother (John loves Mary, Bill loves Suzy, and Peter loves Jane).
- d. John loves Mary, Bill loves Suzy and Mary, and Peter loves Jane.

Gr&S contend that in a situation like (12a), question (12b) can be satisfactorily and completely answered with the functional answer in (12c), with the elaboration given in parentheses. Their point is that there is a reading of (12b) that asks for a pair-list answer and in that case, the complete answer is (12d). This includes the pair *bill* and *mary*, a pair that would not be picked out by the *mother-of* function.

Note that (12d) is subject to the same caveats as example (5) from Section 4.1.1. The answer violates the presupposition of (12b) that each man loves exactly one woman. But Gr&S's general point can be made by substituting the neutral *wh* *who* in place of *which woman*. Under a pair list reading it would require all the pairs to be named but under a functional reading it would leave out the pair that does not fit the functional description *mother-of*. Arguably, though, even this pair list could be the extension of a function, since *who* would allow *bill* to be functionally related to a plural individual *mary+suzy*. It is just that such a function may not be a *natural* function that could be described by a canonical relational term like *mother-of* or *sister-of* (Engdahl 1986: 180). Once number morphology is controlled for, Gr&S's counterexample can be brought under the umbrella of list answers derived from functional answers.

Chierchia (1991, 1993) adds yet another argument linking list answers to functional answers. He observes that both pair list and functional answers become unavailable when the universal is in object rather than subject position. Again, deferring the explanation he provides until Section 4.2, it is the correlation itself that makes the case for a dependence between the two. It also establishes that functional answers are not derived from individual answers which are always available, regardless of the position of the quantifier:

- (13) a. Which student likes every professor?
 b. #Bill likes Prof. Smith and Alice likes Prof. Jones.
 c. #His advisee likes every professor.

The evidence presented here overwhelmingly argues for an independent status for functional answers to questions with universal quantifiers, and for list answers to be derived from functional answers.

4.1.3 Lists and questions with plural definites

Gr&S (1984: 181) identify questions with plural definites as allowing pair-list answers as well (see also Pritchett 1990). On the face of it, this seems to be the case:

- (14) a. Who do these men like?
 b. These men like Sue and Jane.

- c. John likes Sue, and Bill likes Jane.
- d. These men like their mothers

Dayal (1992, 1996) and Krifka (1992) note two crucial differences that argue for a separate explanation, however. First, the subject–object asymmetry brought out by (13) does not hold for questions with plural definites:

- (15) a. Who likes these men?
- b. Sue likes John and Jane likes Bill.
- c. Their mothers like these men.

Second, the pair list reading is not possible if the *wh* expression is singular:

- (16) a. Which woman do these men like?
- b. These men like Sue.
- c. #John likes Sue, and Bill likes Jane.
- d. These men like their mother.

Note that the functional answer in (16d) is only acceptable if the men are siblings. In other words, it is an individual answer. If pair-lists are derivative on functional answers, it is not surprising that there should be no pair-list answer. Substitution with a plural *wh*, *which women do these men like*, makes the point in a different way. The plural version still allows the same kind of pairing as the original. Its universal counterpart, *which women does every man like*, calls for a pairing of each man with more than one woman. Thus, functional and pair-list answers to questions with plural definites are distinct from functional and pair-list answers to questions with quantifiers.

Dayal and Krifka treat questions with plural definites as only allowing individual answers. Since definites denote individuals, whether singular or plural, the only kind of functional/list answers they can enter into are trivial. The domain set has only a single individual, making the functional answer equivalent to the individual answer. Individual answers, however, can be elaborated into list answers in the right circumstances. Scha (1981) shows that sentences like (17a) are true in situations where the total number of computers bought by a group of 200 Dutch firms is 600. Such cumulative readings require a relationship between every member of each group with some member of the other group ((17b)), but the pairings are otherwise unrestricted:

- (17) a. 200 Dutch firms bought 600 computers.
- b. $\forall x [200\text{-Dutch firms}(x) \rightarrow \exists y [600\text{-computers}(y) \wedge \text{bought}(x,y)]] \wedge$
 $\forall y [600\text{-computers}(y) \rightarrow \exists x [200\text{-Dutch firms}(x) \wedge \text{bought}(x,y)]]$

Dayal and Krifka treat pair-list answers to questions with plural definites as instances of the cumulative reading of individual answers with two plural terms:

- (18) a. Who do these men like?
 b. These men like Sue and Jane
 c. λw john+bill like_w sue+jane

The individual answer (18b), with the logical representation (18c), can be expressed as a “functional” answer if the term *their mothers* can be used for the plural individual *sue+jane*. Or it can be expressed in a list which clarifies how the relation distributes down to the atomic individuals. This approach successfully captures the requirement for a plural or neutral *wh* expression in the question and the absence of structural sensitivity. It further implies a greater role for pragmatics in determining the nature of the pairings. *These men like their wives* is likely to lead to a one–one pairing in monogamous societies, while *these men like their sisters* need not. Dayal (1996: 143) formalizes the conditions under which a pair-list can be taken as a cooperative spell-out of the individual answer:

- (19) *Cumulativity Constraint-1*: A list answer of the form $a_1 R b_1, \dots, a_n R b_n$ to a question *Q* is acceptable at a world *w* iff $a_1 + \dots + a_n R b_1 + \dots + b_n = \text{Ans-D}(Q)(w)$.

The analysis of pair-list answers to questions with plural definites as the cooperatively spelt out cumulative reading of an individual answer is generally accepted in current literature and, as such, it will not be probed further in this chapter.

4.1.4 Section summary

This section has highlighted the striking parallelism between multiple constituent questions and questions with quantifiers. Both types of questions allow pair-list answers that call for an extension of our baseline theory of questions. List answers are grammatically constrained to spell out pairings that suggest a functional basis for them. This correlates in an interesting way with the fact that questions with quantifiers allow functional answers that cannot be reduced to individual or pair-list answers. Questions with plural definites do not have genuine pair-list or functional answers. Apparent pair-list answers are cooperative elaborations of an individual answer with two plural terms and apparent functional answers are individual answers that use a relational noun to refer to a plural individual.

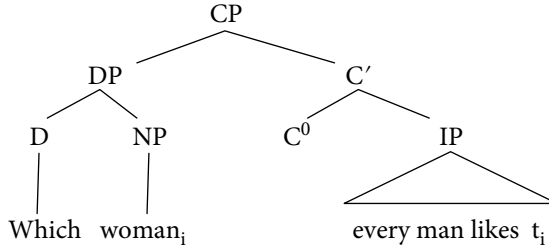
4.2 The function-based approach to pair-list answers

We now extend the baseline theory to include quantification over Skolem functions in questions with quantifiers and build structural sensitivity into the system to account for subject–object asymmetries in the availability of functional and pair-list answers. We note possible pragmatic and semantic factors that can constrain the set of quantifiers that allow list answers. This approach is also applied to multiple constituent questions. Questions with indefinite terms, which seem to offer a choice of questions to the addressee, are also discussed.

4.2.1 Quantifying over Skolem functions

Engdahl (1980, 1986) argues that functional answers provide evidence that *wh* expressions range over Skolem functions, functions from individuals to individuals, not just over individuals (see also Gr&S 1984).⁸ Adapting Engdahl's syntax to our chosen model, we find that quantifying over functions involves the same principles of interpretation as quantifying over individuals:

- (20) a. Which woman does every man like?
b.



There are two points in the derivation that we need to pay attention to. The trace of the *wh* phrase is interpreted as a variable over Skolem functions, functions whose argument can be bound by another term. In this case, the binder is *every man*:

$$\begin{aligned} \llbracket \text{IP} \rrbracket &= \wedge \forall x [\text{man}(x) \rightarrow \wedge \text{like}(x, f_i(x))] \\ \llbracket C^0 \rrbracket &= \lambda q [p = q] \\ \llbracket C' \rrbracket &= \llbracket C^0 \rrbracket (\llbracket \text{IP} \rrbracket) \Rightarrow p = \wedge \forall x [\text{man}(x) \rightarrow \text{like}(x, f_i(x))] \end{aligned}$$

The second point of note is that *which*, being an existential quantifier over Skolem functions, takes as its first argument the range of the function. The condition $\forall x [\text{woman}(f(x))]$ in the interpretation of the fronted DP restricts quantification to woman-valued functions. The universal here is not to be confused with the universal in the nucleus which is contributed by the subject term *every man*:

$$\begin{aligned} \llbracket D \rrbracket &= \lambda Q \lambda \mathcal{F} \exists f [\forall x [Q(f(x))] \wedge \mathcal{F}(f)] \\ \llbracket \text{NP} \rrbracket &= \text{woman} \\ \llbracket \text{DP} \rrbracket &= \llbracket D \rrbracket (\llbracket \text{NP} \rrbracket) \Rightarrow \lambda \mathcal{F} \exists f [\forall x [\text{woman}(f(x))] \wedge \mathcal{F}(f)] \end{aligned}$$

The final step in the derivation follows in the expected manner:

$$\begin{aligned} \llbracket \text{CP} \rrbracket &= \lambda \mathcal{F} \exists f [\forall x [\text{woman}(f(x))] \wedge \mathcal{F}(f)] (\lambda f_i [p = \wedge \forall x [\text{man}(x) \rightarrow \text{like}(x, f_i(x))]]) \\ &\Rightarrow \exists f [\forall x [\text{woman}(f(x))] \wedge \lambda f_i [p = \wedge \forall x [\text{man}(x) \rightarrow \text{like}(x, f_i(x))]](f)] \\ &\Rightarrow \exists f [\forall x [\text{woman}(f(x))] \wedge p = \wedge \forall x [\text{man}(x) \rightarrow \text{like}(x, f(x))]] \\ &\quad \lambda p \exists f [\forall x [\text{woman}(f(x))] \wedge p = \wedge \forall x [\text{man}(x) \rightarrow \text{like}(x, f(x))]] \end{aligned}$$

⁸ The functions quantified over are, in fact, of type $\langle s, \langle e, e \rangle \rangle$ but the ideas presented here can be demonstrated at the extensional level.

The interpretation of (20a) in a specific case is illustrated below:

- (21) a. Functions: *man wife-of man mother-of*
 John $\rightarrow$ Mary John $\rightarrow$ Jane
 Bill $\rightarrow$ Sue Bill $\rightarrow$ Rose
 b. $\{\wedge\forall x[\text{man}(x) \rightarrow \text{loves}(x, f_{\text{wife}}(x))], \wedge\forall x[\text{man}(x) \rightarrow \text{loves}(x, f_{\text{mother}}(x))]\}$
 c. Every man loves his wife.

In a world where every man loves his wife but not all men love their mothers, we would get (21c) as the functional answer. This demonstration showcases the advantage of using Skolem functions. Variation in the choice of women is captured without the universal taking syntactic scope over the fronted *wh*-expression. This is because the functional trace is interpreted not as an individual variable x but as a Skolemized function variable $f(x)$, with x bound by the universal. As discussed in Section 4.1, Engdahl takes pair-list answers to be a pragmatically derived graph of the function at a world, where a graph spells out the pairs in the function. She therefore posits nothing further for list answers.⁹

An aspect of Engdahl's theory that does not bear directly on list answers but is still worth mentioning is that it allows for a local interpretation of pronouns and anaphors within the *wh* phrase:

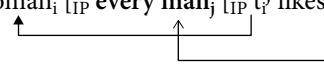
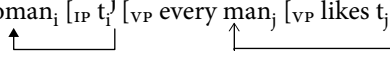
- (22) a. Which picture of himself_i/him_j does every man_i like?
 b. $\lambda p \exists f [\forall x[\text{picture-of-}x(f(x))] \wedge p = \wedge\forall x[\text{man}(x) \rightarrow \text{like}(x, f(x))]]$
 c. $\lambda p \exists f [\forall x[\text{picture-of-}y(f(x))] \wedge p = \wedge\forall x[\text{man}(x) \rightarrow \text{like}(x, f(x))]]$

Binding principles determine whether the pronoun must be bound or remain free but the binding itself is effected locally within the interpretation of the *wh* expression. In (22b), the anaphor *himself* requires quantification over functions from individuals x to pictures of x , while in (22c), the pronoun *him* requires quantification over functions from individuals x to pictures of some contextually determined individual y . These effects are obtained without appealing to syntactic reconstruction at LF to the base position where the pronoun/anaphor is c-commanded by the quantifier.

Engdahl extends the functional account to multiple constituent questions. She allows, but does not require, Skolem functions in interpreting them (Engdahl 1986: 220–34). The two options are shown in (23):

- (23) a. Which man saw which woman?
 $\lambda p \exists f_1 \exists f_2 [\text{man}(f_1) \wedge \text{woman}(f_2) \wedge p = \lambda w \text{ saw}_w(f_1, f_2)]$
 $\{\wedge\text{john saw mary}, \wedge\text{john saw sue}, \wedge\text{bill saw mary}, \wedge\text{bill saw sue}\}$

⁹ Engdahl takes quantifiers like *no N* to disallow pair-list answers for pragmatic reasons (cf. (11) above): members cannot be listed when there are no men in the *like* relation.

- (25) a. Which woman likes every man?
 b. $*[_{CP} \text{ Which woman}_i [_{IP} \text{ every man}_j [_{IP} t_i^j \text{ likes } t_j]]]$

 c. $[_{CP} \text{ Which woman}_i [_{IP} t_i^j [_{VP} \text{ every man}_j [_{VP} \text{ likes } t_j]]]]$


Example (25b) shows that the *i*-index of the *wh*-trace is bound by the *wh*-expression *which woman*. To bind the *a*-index of the trace, the universal must QR to a *c*-commanding position. This configuration, however, results in a weak cross-over (WCO) violation since the *a*-index is pronominal. There is, of course, the option of raising the quantifier to VP (25c), but it does not create the configuration needed for functional dependency. Thus, Chierchia successfully accounts for the structural asymmetry he notes with respect to functional answers. Since functional dependencies are the source of pair-list answers, the asymmetries transfer over to them (cf. (13)).

Chierchia's argument for the structure sensitivity of pair-list answers to questions with quantifiers was an alternative to the account prevalent at the time, due to May (1985). May provides structural analyses parallel to the ones in (24) and (25) for the cases under discussion. The difference is that his *wh* traces denote simple individual variables $t_{i/j}$ because he assumes quantification over individuals. The individual answer is derived in the expected way, through an LF with the scopal order in (24b). In order to derive the pair-list answer May proposes (26):

- (26) *Scope Principle*: Mutually *c*-commanding quantifiers can take scope in either order.

Since QR pulls *every man* out of the inner IP, it comes structurally close enough to the *wh* phrase to allow scope permutation.¹⁰ May does not provide a semantics for the pair-list reading. A simple-minded quantifying in of the universal into the question permitted by the Scope Principle would be open to the problems discussed in Section 4.1.2 in relation to (10). Setting that aside, May's syntactic assumptions derive the impossibility of pair-list readings with the quantifier in object position. In order to satisfy the Scope Principle, the quantifier has to raise to IP but this violates the ban on crossing dependencies (Pesetsky 1982), as evident from LF (25b). The only option is for the universal to QR to VP, as in LF (25c). But this does not meet the requirements of the Scope Principle and the resulting answer is the one where the *wh* has scope over the universal, that is, the individual answer. May does not give formal status to the functional answer.

An important difference between Chierchia's account of the asymmetry and May's is the locus of explanation, a difference that affects their applicability to multiple constituent questions (Hornstein 1995; Comorovski 1996; Dayal 1996).

¹⁰ As von Stechow (1996) notes, May's LFs are not transparent as they do not disambiguate scope relations.

Chierchia's explanation rests on the relationship of the *wh* and the quantifier at their base positions while May's explanation relies on the relationship at their landing sites. We noted in Section 4.1 that pair-list answers to multiple *wh* questions are parallel to questions with quantifiers. In particular, the subject *wh* has universal-like properties. Consider the multiple constituent question in (27a) and its possible LFs in (27b)–(27d):

- (27) a. Which man likes which woman?
 b. $[_{CP} \text{ which woman}_j [_{CP} \text{ which man}_i [_{IP} t_i \text{ likes } t_j]]]$ *satisfies Scope Principle*
 c. $[_{CP} \text{ which woman}_j [_{CP} \text{ **which man}_i [_{IP} t_i \text{ likes } t_j^i]]]]**$ *complies with WCO*
 d. * $[_{CP} \text{ **which woman}_j [_{CP} \text{ which man}_i [_{IP} t_j^j \text{ likes } t_j]]]]**$ *violates WCO*
-

May's theory predicts that since both *wh* expressions take scope at the level of CP, they will satisfy the Scope Principle, as shown in (27b). This makes the incorrect prediction that either should be able to scope over the other, assuming that wide scope translates to domain cover. If we focus on their base positions, however, the source of the asymmetry is explained under Chierchia's functional account. In (27c) the object *wh* leaves behind a functional trace and the subject *wh* binds its *a*-index. In (27d) the subject leaves behind a functional trace but the object incurs a WCO violation if it tries to bind its *a*-index. This provides the formal grounding for the intuition that the higher *wh* in a multiple *wh* interrogative behaves like a universal quantifier.

4.2.3 Tapping into witness sets

Chierchia (1993), departing from Engdahl and his own earlier account, provides a semantics for pair-list readings which builds on functional LFs of the kind shown in (24b), but with a difference:¹¹

- (28) a. $[_{CP} \text{ Which woman}_i [_{C'} \text{ does } [_{IP-2} \text{ every man}_j [_{IP-1} t_j \text{ like } t_i^j]]]]]$
 b. $\lambda P \exists A [W(\text{every man}, A) \wedge P(\lambda p \exists f \in [A \rightarrow \text{woman}] \exists x \in A [p = \wedge \text{like}(x, f(x))])]$

The interpretation of the LF for the pair-list reading is identical to the interpretation of the LF for the functional reading up until IP-1. It differs with respect to where the quantifier is interpreted. In the case of functional readings of questions, the universal is interpreted inside the question nucleus, but here it has scope outside C^0 , the point at which the nucleus is determined: $p = \wedge \text{like}(x, f(x))$. The

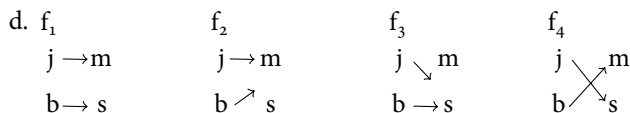
¹¹ *P* is of type $\langle \langle \langle s, t \rangle, t \rangle, t \rangle$, a set of sets of propositions, *W* stands for minimal witness sets. Barwise and Cooper (1981) define witness sets of a generalized quantifier as subsets of the common noun that are members of the quantifier. A minimal witness set is one that does not have a proper subset that is also a witness set.

accompanying semantics involves a shift in question denotations from a set of propositions to a set of sets of propositions/a family of questions.

Note though that this semantics involves a breach of strict compositionality. If we assume that quantifier raising can only be to IP and the nucleus is determined at C^0 , pulling the semantic contribution of the universal outside the nucleus is non-trivial. Chierchia (1993: 210–11) describes it as a form of *absorption*, a restructuring option: $[wh NP_i [DP_j IP]] \Rightarrow [[wh NP_i DP_j] IP]$. He also entertains the possibility that the universal could raise above C^0 , in a fashion analogous to *wh* movement.¹² The primary motivation for not allowing QR above C^0 in syntactic accounts such as May's is to account for subject–object asymmetries in pair-list answers on the basis of landing sites (see Pritchett 1990). Since the locus of explanation is the configuration inside IP in Chierchia's account, raising the universal above C^0 is a viable option. And it is an option that would be preferable from the perspective of compositionality.

Chierchia's approach to pair-list answers imports Gr&S's (1984) account of list answers into the functional approach. A question under its list reading, denotes a family of questions whose members are fixed by the number of witness sets A of the quantifier. In the case of the universal, there is only one witness set, namely the set denoted by the common noun, so the family of questions is a singleton:

(28) c. $A = \{\text{John, Bill}\}$



e. $\{\wedge \text{John likes Mary, } \wedge \text{John likes Sue, } \wedge \text{Bill likes Mary, } \wedge \text{Bill likes Sue}\}$

To answer this question is to answer the only question in the set. The portion in bold in (28b), now creates a set of propositions such that for some function f , the proposition links some member of A to some member in the range of f . If John and Bill are all the men in the domain set, and Mary and Sue the women in the range set, we get the four functions in (28d). Existentially quantifying over functions and men yields the propositions inside the single member of (28e), with all the men being paired with all the women.

We noted in Section 4.1 that all quantifiers allow functional answers, but not all of them allow pair-list answers. A prime example is *no N*. This follows from Chierchia's *absorption* operation. Since the witness set of *no N* is the empty set, it cannot generate lists. In fact, reference to minimal witness sets predicts the absence of pair-list answers for all downward monotone quantifiers. A quantifier like *no more than three N/at most three N*, for example, will have

¹² Recall that *absorption* is a term due to Higginbotham and May (1981) and referred originally to their account of how two singular *wh* phrases can compose to allow for the shift from uniqueness to bijection. It is now used as a cover term for operations that interpret two or more quantifiers in a non-iterative manner.

witness sets with up to three members but the minimal witness set will be the empty set:

- (29) a. Who do no more than three/at most three men like?
 b. No more than three/at most three men like their mothers-in-law.
 c. #John likes Mary and Bill likes Sue.

Chierchia notes *most N* as a problem since any subset of *N* larger than half qualifies as a minimal witness set.¹³ Dayal (1996) appeals to unique witness sets, which accounts for upward monotone quantifiers like *most N*, in addition to quantifiers like *no N* whose unique witness set is empty, and downward monotone quantifiers like *no more than n N* which have more than one witness set. But this leads to a non-trivial difference between them on the status of list answers to questions with indefinites.

Indefinites have minimal witness sets but they do not typically have unique witness sets:

- (30) a. Book = {Sense and Sensibility, Persuasion, Emma}
 b. Two books = $\lambda P \exists X [two(X) \wedge books(X) \wedge P(X)]$
 c. Minimal witness sets = {SS, P}, {SS, E}, {E, P}

Chierchia takes questions with indefinites to denote a family of questions, one for each minimal witness set. Dayal does not take them to allow pair-list answers, and the use of unique witness sets is meant to capture this fact. The choice of what type of witness set to use for pair-list answers, then, turns on the proper determination of the facts for questions with indefinites. We turn to this next.

4.2.4 Questions with indefinites

Questions with indefinites have two related but distinct properties, the potential for choice and the potential for pair-list readings. These two features are in evidence in the following example based on Gr&S (1984: 453–4), though the original discussion goes back to Belnap and Bennett (1977) and Belnap (1982):

- (31) a. What did two of John's friends give him for Christmas?
 b. Two of John's friends gave him a watch for Christmas.
 c. $\lambda p \exists x [p = \wedge \exists y [2(y) \wedge friends-of-john(y) \wedge give(y,j,x)]]$
 $\Rightarrow \{ \wedge two\ of\ John's\ friends\ gave\ him\ a\ watch,$
 $\wedge two\ of\ john's\ friends\ gave\ him\ a\ book \}$
 d. $\{ \langle bill \rightarrow watch \rangle, \langle peter \rightarrow watch \rangle, \langle fred \rightarrow book \rangle \}$

¹³ With a three-member domain, {SS, P, E} *most N* has three minimal witness sets: {SS, P}, {SS, E}, and {E, P}. Chierchia also notes *both N* as a problem, but under the view that this is a plural definite (Brisson 2003), it should be amenable to the plurality-based account discussed in Section 4.1.3.

Response (31b) is the answer predicted by the baseline theory, in a world where the *give* relation is as shown in (31d). The question presupposes that there is exactly one thing that was given by two of John's friends and asks for its identity.

Gr&S point to another construal of the question that emerges in the situation depicted in (32a). There is now more than one possible true answer, (32b)–(32d):¹⁴

- (32) a. Bill → {ball}
 Peter → {pen}
 Fred → {book}
- b. ^Bill gave John a ball and Peter gave him a pen.
 c. ^Bill gave John a ball and Fred gave him a book,
 d. ^Peter gave John a pen and Fred gave him a book,

The choice of friends seems to be left up to the addressee. Once this choice is made, a pair-list answer seems appropriate since the two friends chosen give different presents. This is derived in Chierchia's account in the following way:

- (33) a. What did two of John's friends give him?
 b. $\lambda p \exists A [\text{Witness-Set}(\text{two of John's friends}, A) \wedge$
 $P(\lambda p \exists f \exists x \in A [p = \wedge \text{like}(x, f(x))])]$
 c. $\{\{^{\wedge}\text{Bill gave John a ball, } ^{\wedge}\text{Peter gave John a pen, } ^{\wedge}\text{Bill gave John a pen, ...}\},$
 $\{^{\wedge}\text{Bill gave John a ball, } ^{\wedge}\text{Fred gave John a book, } ^{\wedge}\text{Bill gave John a book, ...}\},$
 $\{^{\wedge}\text{Peter gave John a pen, } ^{\wedge}\text{Fred gave John a book, } ^{\wedge}\text{Fred gave John a pen, ...}\}$

Assuming Bill, Peter, and Fred are John's friends, the minimal witness sets of *two of John's friends* are {Bill, Peter}, {Bill, Fred}, and {Peter, Fred}. The question, therefore, denotes a family of three questions, one for each minimal witness set. While there is no choice in the case of questions with universals which have only one witness set, questions with indefinites typically present a choice of questions to answer.

Szabolcsi (1997a: 320–2), building on Moltman and Szabolcsi (1994) and Szabolcsi (1997b), makes some important observations about the availability of pair-list readings, focusing on questions with singular *wh* expressions. She finds that questions with universal quantifiers such as (34a) admit pair-list answers, as we have been assuming so far, though not as readily as suggested in the literature. Questions with indefinites such as (34b), instead, are marginal at best in English. She cites Gr&S (1984: 555–6) for the clear inadmissibility of its Dutch counterparts:¹⁵

¹⁴ Perhaps, the judgment is clearer if instead of the partitive, a simple indefinite is used: *What did two guests give John for his wedding?*

¹⁵ Szabolcsi's concern is with a range of quantifiers, including indefinites like *more than n N* and *at least n N* which disallow lists. Note that % stands for variation among speakers with respect to acceptability, ? for a general assessment of marginality.

- (34) a. Which boy did every dog bite? %PL
 b. Which boy did two dogs bite? ?PL

However, to the extent that pair-list answers to questions with indefinites are at all possible, they call for an explanation. Szabolcsi (1997a: 322–3, fn. 6) speculates that they may be instances of cumulative readings of individual answers, as in Dayal's and Krifka's accounts of questions with plural definites (see also Dayal 1996: 120, fn. 31):

- (35) a. Who did two dogs bite?
 b. Fido and Spotty bit Harry and John. In particular, Fido bit Harry and Spotty bit John.

This could explain the marginality of the pair-list answer for (34b) since the *wh* expression rules out the possibility for the two dogs to be related to a plurality of boys.

A problem with this line of approach, though, is that plurality-based cumulative answers are typically structure insensitive (cf. Section 4.1.3) but (36a), with the indefinite in object position, does not seem to admit the pair-list answer in (36b), even with *who* instead of *which* *N_{SING}*. Example (37) is further evidence of structure sensitivity:

- (36) a. Who bit two boys?
 b. Fido bit Harry and Spotty bit John.
- (37) a. What did two boys read? PL
 b. Who has read two novels? *PL

If this structural sensitivity is a stable fact, then it remains a problem for an analysis in terms of cumulativity, but the contrast in (34) makes it implausible to consider extending the explanation for pair-list answers to questions with quantifiers to those with indefinites.

Regardless of what the final picture for questions with indefinites turns out to be, we must address the fact that these questions always seem to leave it up to the addressee to choose the individuals the question is about. This brings it suspiciously close to the *mention-some* phenomenon in questions, which also gives the addressee free rein in choosing an answer. In Section 3.3.1, however, we argued to keep the two apart. One reason was that they place different restrictions on the *wh* phrase:

- (38) a. Where (all) can you buy gas around here?
 b. What (all) are two of your students working on?

A *mention-some* question turns into a *mention-all* question with the addition of particles like *wh-ALL* in languages/dialects that have them (cf. (38a)) but a similar move in questions with indefinites does not affect the dimension of choice. The addressee is still free to answer (38b) with respect to any two students. If we do not

account for choice readings through the family of questions approach and we do not treat them as *mention-some* questions, an explanation remains to be given.

A possible line of inquiry, one that has not been explored at any length as far as I am aware, is that choice questions arise under a specific indefinite reading of the term. Compare the well-known case of a declarative with a specific indefinite with the kind of question we are considering:

- (39) a. If two of my relatives die, I will inherit a fortune.
 $\exists f_i [\text{die}(f_i(2\text{-relatives-of-mine})) \rightarrow \text{inherit}(I, \text{a fortune})]$
 b. What topic are two of your students working on?
 $\lambda p \exists x [\text{topic}(x) \wedge p = \wedge \text{work-on}(f_i(2\text{-of-your-students}), x)]$

Specific indefinites in examples like (39a) have been the subject of much investigation and an influential approach to them uses Skolemized choice functions to capture their special referential-like properties (Reinhart 1997, 1998, among others). Here we represent the choice function variable as existentially bound from outside the island, as in Reinhart's account but for present purposes it could also remain free. Importantly, f_i is a choice function of type $\langle\langle e, t \rangle, e \rangle$. It selects an individual out of a set whose members are sums of two individual relatives of mine. In this case, we take the index on this function to be cued to the speaker. A straightforward application of this idea to (39b) would capture the effect of choice if the index on the choice function variable is cued to the addressee instead of the speaker. This would be analogous to the interrogative flip observed in languages where evidentials have a speaker-based interpretation in declaratives but an addressee-based interpretation in questions (Speas and Tenny 2003; Murray 2010). The significant point for present purposes is that by tapping into a well-established aspect of specific indefinites, we may be able to account for the intuition that such questions involve choice. A further promising feature of an approach along these lines is the observation that specific indefinites are non-distributive (Ruys 2005).¹⁶ If this is so, it may help us understand why such questions are resistant to list answers. Obviously, much work remains to be done, at the empirical as well as the analytical level.

4.2.5 Section summary

We have extended the semantics for questions to incorporate quantification over Skolem functions for functional and pair-list answers. Syntactic asymmetries in the availability of such answers were handled by interpreting traces as functional variables with a pronominal index, sensitive to WCO effects. Looking at questions with non-universal quantifiers, we noted the extent to which the use of minimal witness sets or unique (minimal) witness sets captures the set of quantifiers that admit pair-list answers. Questions with indefinites are not amenable to accounts in terms of witness sets or to plurality-based accounts. We suggested that the choice associated with such questions may depend on the terms being specific indefinites whose indexicality is cued to the addressee.

¹⁶ The use of choice functions for the interpretation of wh phrases will be taken up in Chapter 7.

4.3 Functionality in pair-list answers

Here we show that the functional account presented above does not capture domain cover and the restriction to functional pairings. Two solutions are presented. One modifies the functional approach to allow each proposition in the question denotation to be the full graph of a function. It does so by positing an ambiguity in C^0_{+WH} . The other iterates the standard question formation operation, and distributes Ans-D over the members of the resulting family of questions. Echo questions provide direct empirical support for the existence of higher order questions in natural language. The need for higher order questions for pair-list answers is evaluated with reference to scope marking.

4.3.1 Functionality through functional absorption

Dayal (1996) points out that the same problems that were identified in Section 4.1 for the baseline theory resurface in the function-based account of pair-list readings. Chierchia's answerhood operator (Chierchia 1993: 192; Lahiri 1991: 147) says that an answer to a question is any proposition that is the conjunction of some subset of the propositions in the question denotation. This allows (40a) to be answered with (40c), in a situation like (40b), violating both components of functionality:

- (40) a. Which man likes which woman? / Which woman does every man like?
 b. {[^]John likes Mary, [^]John likes Sue, [^]Bill likes Mary, [^]Bill likes Sue}
 c. John likes Mary and Sue.

It is worth emphasizing though that the fundamental problem is not the particular answerhood operator adopted. Rather, it is the fact that these question denotations are standard Hamblin–Karttunen sets. That is, the functional dependency that Engdahl and Chierchia advocate is not reflected in their final question denotations.

To capture domain-cover and point-wise uniqueness Dayal posits, in addition to an individual C^0_{+WH} , a functional C^0_{+WH} whereby each proposition spells out the graph of a function. I will call this *functional absorption*, distinguishing it from earlier proposals about absorption. Consider the results schematically, first with a multiple wh question and the set of four functions in (29d):

- (41) a. Which man likes which woman?
 b. $[_{CP-2} \text{ which woman}_i [_{CP-1} \text{ Which man}_j [_{IP} t_j \text{ likes } t_i / t_i^j]]]$?
 c. $[[CP-2]]$ with $C^0_{+WH\text{-individual}} =$
 $\{\text{[^]John likes Mary, [^]John likes, Sue, [^]Bill likes Mary, [^]Bill likes Sue}\}$
 d. $[[CP-2]]$ with $C^0_{+WH\text{-functional}} =$
 $\{\text{[^]John likes Mary and Bill likes Sue, } \textit{graph of } f_1$
 $\text{[^]John likes Mary and Bill likes Mary, } \textit{graph of } f_2$
 $\text{[^]John likes Sue and Bill likes Sue, } \textit{graph of } f_3$
 $\text{[^]John likes Sue and Bill likes Mary}\} \textit{graph of } f_4$

Depending on which denotation Ans-D applies to, the individual or the pair-list answer is derived. Applied to (41c), it gives the single-pair answer, applied to (41d) the pair-list answer. Note that in both cases, the propositions in the set are not in an entailment relation. Thus, Ans-D is defined only in worlds where exactly one proposition is true. Let us now go through the crucial steps.

Dayal posits (42) as the meaning of the functional C^0 (Dayal 1996: 117–18). There are three distinct aspects to the proposed meaning: introduction of $\exists$ quantification over functions, restrictions on the domain and range of the function, the creation of graphs for each such function. The final denotation is the standard one, a set of propositions:

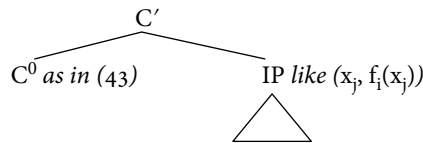
$$(42) \quad \llbracket C^0 \rrbracket = \lambda Q \lambda D \lambda R \exists f [\text{Dom}(f) = D \wedge \forall y [R(f(y))] \wedge p = \cap \lambda p' \exists y \in D [p' = Q(y)(f)]]$$

C^0 introduces, first and foremost, quantification over functions of type $\langle e, e \rangle$, with $\exists f$ binding a function variable f inside the nucleus. The type of quantification is exactly as in the Engdahl–Chierchia theory, but here it enters the computation independently of the *wh* phrase. The second contribution of C^0 is to introduce the restriction on $\exists f$. Since the quantification is over functions from individuals to individuals, there are two restrictions in place, a condition on the domain of the function $\text{dom}(f) = D$ and a condition on the range of the function $\forall y [R(f(y))]$. The third contribution is the creation of propositions that are graphs of functions. This is done by taking for each function, the intersection of the set of propositions obtained by quantifying over the members of the domain set. C^0 , because it introduces two new variables, needs to combine not just with the nucleus Q of type $\langle \langle e, e \rangle, \langle e, \langle s, t \rangle \rangle \rangle$, but also with two other arguments D and R , of type $\langle e, t \rangle$.

Let us now see how the set of propositions in (41c) is derived from an LF encoding a functional dependency. The first step, in (43), saturates the first argument of functional C^0 with the IP meaning:

$$(43) \quad \text{a. } [_{CP-2} \text{ which woman}_i [_{CP-1} \text{ Which man}_j [_{C'} C^0 +_{\text{WH-functional}} [_{\text{IP}} t_j \text{ likes } t_i^j]]]]?$$

b.



$$\llbracket C \rrbracket = \llbracket C^0 \rrbracket (\llbracket \text{IP} \rrbracket)$$

$$\Rightarrow \lambda Q \lambda D \lambda R \exists f [\text{Dom}(f) = D \wedge \forall y [R(f(y))] \wedge$$

$$p = \cap \lambda p' \exists y \in D [p' = Q(y)(f)] (\lambda f_i \text{ 'like } (x_i, f_i(x_j))$$

$$\Rightarrow \lambda D \lambda R \exists f [\text{Dom}(f) = D \wedge \forall y [R(f(y))] \wedge p =$$

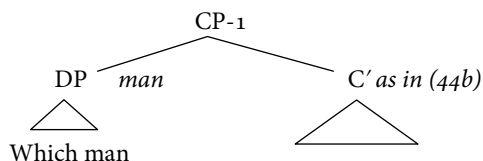
$$\cap \lambda p' \exists y \in D [p' = \lambda f_i \lambda x_i \text{ like } (x_i, f_i(x_j)) (y)(f)]$$

$$\Rightarrow \lambda D \lambda R \exists f [\text{Dom}(f) = D \wedge \forall y [R(f(y))] \wedge p =$$

$$\cap \lambda p' \exists y \in D [p' = \text{like}(y, f(y))]$$

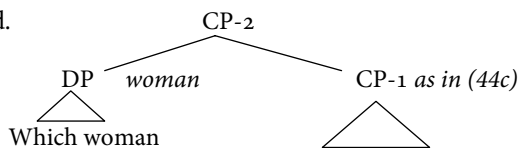
Now we saturate the domain and range arguments. Wh phrases, as restrictions on the function, have to denote sets of individuals. This can be obtained from their basic meaning as existential generalized quantifiers by applying the BE operator (Partee 1986). Alternatively, wh phrases can be treated uniformly as set denoting terms, and their generalized quantifier meaning in other contexts can be obtained through Partee's $\exists$ type shift (Bittner 1994). We follow the latter option:

(43) c.



$$\begin{aligned} \llbracket \text{CP-1} \rrbracket &= \lambda D \lambda R \exists f [\text{Dom}(f) = D \wedge \forall y [R(f(y))]] \wedge \\ &\quad p = \cap \lambda p' \exists y \in D [p' = \text{like}(y, f(y))] \text{ (man)} \\ &\Rightarrow \lambda R \exists f [\text{Dom}(f) = \text{man} \wedge \forall y [R(f(y))]] \wedge \\ &\quad p = \cap \lambda p' \exists y \in \text{man} [p' = \text{like}(y, f(y))] \end{aligned}$$

d.



$$\begin{aligned} \llbracket \text{CP-2} \rrbracket &= \lambda R \exists f [\text{Dom}(f) = \text{man} \wedge \forall y [R(f(y))]] \wedge \\ &\quad p = \cap \lambda p' \exists y \in \text{man} [p' = \text{like}(y, f(y))] \text{ (woman)} \Rightarrow \\ &\exists f [\text{Dom}(f) = \text{man} \wedge \forall y [\text{woman}(f(y))] \wedge p = \cap \lambda p' \exists y \in \text{man}(y) [p' = \text{like}(y, f(y))]] \\ &\lambda p \exists f [\text{Dom}(f) = \text{man} \wedge \forall y [\text{woman}(f(y))] \wedge p = \cap \lambda p' \exists y \text{man}(y) [p' = \text{like}(y, f(y))]] \end{aligned}$$

Note that the wh in-situ adjoins above the fronted wh. Another option would be to tuck it below the fronted wh, following Richards (1997). If the latter were adopted, the order in which the restrictions are fed in would have to be reversed but the essential results would be maintained. What is required to maintain compositionality is a match between syntactic hierarchy and semantic saturation.

The two meanings for C^0 apply straightforwardly to Chierchia's LFs for list, functional, and individual readings of questions with quantifiers. Crucially, there are two variables to be bound at C^0 in (44a), an individual and a functional variable. There is only one in (44b), either an individual variable or a functional variable:

- (44) a. $[_{CP-1} \text{ which woman}_i [_{CP-2} \text{ every man } [_C C^0_{+WH\text{-functional}} [_{IP} t_j \text{ likes } t_i^j]]]]]$
 b. $[_{CP-1} \text{ which woman}_i [_C C^0_{+WH\text{-individual}} [_{IP-2} \text{ every man } [_{IP} t_j \text{ likes } t_i/t_i^j]]]]]$

Assuming high QR for the universal and the extraction of a unique witness set from it in (44a), we get a question denotation that is the same as for multiple *wh* questions. Thus, when Ans-D applies to it we get a proposition that encodes functionality. An LF like (44b) composes with an ordinary C^0 and is interpreted exactly as in the Engdahl–Chierchia account of functional answers. That is, we get an individual or a functional answer, depending on whether the *wh* chain involves individuals or functions.

Building ambiguity into the C^0 node, then, is a way of accounting for the perceived ambiguity of multiple constituent questions and questions with quantifiers under the functional approach to pair-list answers.¹⁷ It derives the functionality of list answers, which otherwise remains elusive even with quantification over functions. The individual pieces in the complicated looking $C^0_{+WH\text{-functional}}$ are all familiar and independently motivated. Packaging them under one node, however, has not always been looked upon with favor. It is, therefore, worth emphasizing its chief merit. Functional absorption allows a single answerhood operator, Ans-D, to derive number-based effects in individual answers as well as the functionality of pair-list answers.

4.3.2 Higher order echo questions

An alternative way of capturing functionality appeals to higher order interpretations. Before we discuss this approach, however, let us try to understand the role of higher order questions in grammar by looking at an uncontroversial case. Karttunen (1977: 12) notes that constituent and polar questions cannot combine unless the *wh* phrase has echo intonation:

- (45) a. Did Mary read WHICH BOOK/*which book?
 b. Mary isn't sure whether to read WHICH BOOK/*which book?

He furthermore notes that direct questions like (45a) call for answers that are themselves questions. Echo questions raise interesting issues in their own right and will be taken up in Chapter 9. Here we present the account in Dayal (1996: 123–7), which draws on the discussion in Comorovski (1989, 1996) and in some ways anticipates the iterated C^0_{+WH} approach to be discussed in Section 4.3.3.

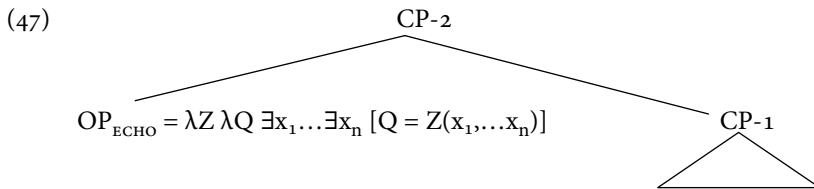
Consider the following question–answer paradigms involving at least one *wh* phrase with echo intonation:

- (46) a. Bill cooked WHAT? Bill cooked [meat]_F.
 b. WHO cooked WHAT? [Bill]_F cooked [meat]_F.
 c. Who cooked WHAT? Who cooked [meat]_F?

¹⁷ Pesetsky (2000) and Cable (2007, 2010) also posit ambiguity in C^0 to account for superiority and intervention, but the terms of the ambiguity are not quite the same.

Dayal makes the non-controversial assumption that an echo question responds to a previous utterance, at least some part of which was inaudible to the questioner. It elicits as response an utterance with focus on the inaudible part(s) of the original utterance. A single *wh* question like (46a), or a multiple *wh* question with echo intonation on both *wh* expressions like (46b), elicits a declarative as answer. However, a multiple *wh* like (46c) with echo intonation on just one *wh*, elicits an answer which is itself a question.

Dayal posits a null operator for echo questions that can combine with CPs that denote functions from (tuples of) individuals to propositions, as well as with CPs that denote functions from (tuples of) individuals to questions. In other words, it can combine with a $-_{WH}$ complement as well as a $+_{WH}$ complement, as long as there are variables associated with echo *wh* phrases inside the CP for it to bind. An answer-hood operator, similar to the one for ordinary questions accommodates the fact that echo questions relate to previous utterances, not truth (see Chapter 9 for details):



The application of the echo operator in (46) now yields the following first and second order sets:

- (48)
- a. $\text{OP}_{\text{ECHO}}(\llbracket \text{bill cooked WHAT} \rrbracket) = \text{OP}_{\text{ECHO}}(\wedge \text{bill cooked } x)$
 $\Rightarrow \{ \wedge \text{Bill cooked meat, } \wedge \text{Bill cooked pasta} \}$
 - b. $\text{OP}_{\text{ECHO}}(\llbracket \text{WHO cooked WHAT} \rrbracket) = \text{OP}_{\text{ECHO}}(\wedge y \text{ cooked } x)$
 $\Rightarrow \{ \wedge \text{Bill cooked meat, } \wedge \text{Bill cooked pasta, } \wedge \text{John cooked meat, } \wedge \text{John cooked pasta} \}$
 - c. $\text{OP}_{\text{ECHO}}(\llbracket \text{who cooked WHAT} \rrbracket) = \text{OP}_{\text{ECHO}}(\lambda p \exists y [\text{person}(y) \wedge p = \wedge y \text{ cooked } x])$
 $\Rightarrow \{ \{ \wedge \text{Bill cooked meat, } \wedge \text{John cooked meat} \}$
 $\quad \{ \wedge \text{Bill cooked pasta, } \wedge \text{John cooked pasta} \} \}$
Equivalently: {Who cooked meat? Who cooked pasta?}

At CP-1, (46a) and (46b) denote propositions. Since the non-echo *wh* in (46c) is interpreted in the normal way, CP-1 in this case denotes a question: *who cooked x?* Binding the free variable by OP_{ECHO} always shifts the denotation one level up. With this background, we turn now to the use of higher order questions to derive functionality in pair-list answers.

4.3.3 Functionality through higher order questions

Hagstrom (1998: 148) makes the same assumptions as the baseline theory for single *wh* questions: questions denote Hamblin sets, only one proposition counts as an

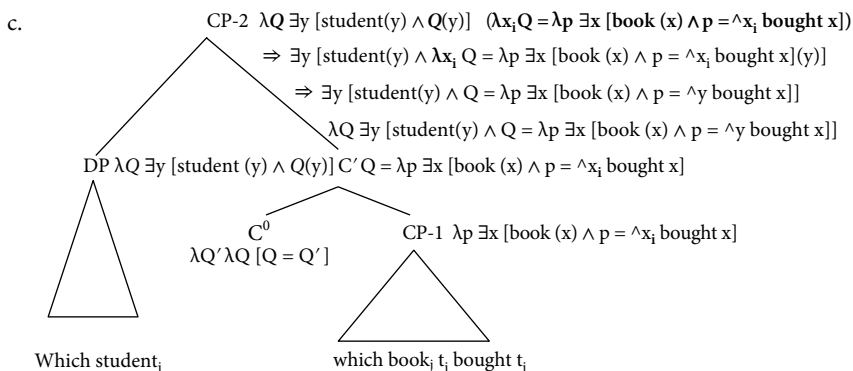
answer, a negative answer denies the uniqueness/existential presuppositions of the question. He differs on multiple *wh* questions. A multiple constituent question can denote a family of questions. To answer such a question is to answer each member of the family, with the uniqueness/existence presuppositions distributing over the subquestions. The pair-list answer is the conjunction of the individual answers to subquestions.

Hagstrom's ideas have been developed further and given a fully compositional implementation in Fox (2012).¹⁸ The key insight is to allow iteration of a type flexible version of $C^0_{+WH} \lambda p \lambda q [q = p]$. The LF in (49b) positions the *wh* in-situ below the fronted one, following Richards (1997), but places them in distinct projections of C :

(49)

a. Which student bought which book?

b. $[_{CP-2} \text{ which student}_i [_C C^0_{+WH} [_{CP-1} \text{ which book}_j [_C C^0_{+WH} [t_i \text{ bought } t_j]]]]]$



CP-1 denotes a set of propositions, though the set depends on the value assigned to the variable in subject position: *which book did x_i buy?* or equivalently: $\{\wedge x_i \text{ bought } E, \wedge x_i \text{ bought } P\}$. The higher C^0_{+WH} combines with CP-1 and the higher *wh* phrase existentially binds the free variable inside it. The abstraction over Q in the final line, the analog of abstraction over p in ordinary questions, yields a family of questions:

- (50) a. $\{\wedge \text{John bought } P, \wedge \text{John bought } E\}$, *which book did John buy?*
 $\{\wedge \text{Bill bought } P, \wedge \text{Bill bought } E\}$ *which book did Bill buy?*
 b. $\cap \{\text{Ans-D}(\{\wedge \text{John bought } P, \wedge \text{John bought } E\}),$
 $\text{Ans-D}(\{\wedge \text{Bill bought } P, \wedge \text{Bill bought } E\})\}$

¹⁸ See Nicolae (2013) for an accessible presentation of Fox's account, along with details of the syntax that have been omitted here.

Domain cover is derived by requiring that every subquestion be answered, and functional pairing by a point-wise application of Ans-D.¹⁹ The pair-list answer is the conjunction of these individual answers. The family of questions is, in effect, a plural term that has the answerhood operator distributed over it.

This approach has considerable appeal as it captures a challenging paradigm, applying independently motivated principles and basic tools of interpretation. Dayal (2016b), however, questions whether its theoretical elegance can be maintained against a wider set of facts. By eliminating functional dependencies from consideration, it leaves unaddressed the overlap between functional and pair-list readings emphasized by Engdahl and Chierchia. Of course, adding a functional core is not incompatible with building higher order questions through iterated C^0_{+WH} (Gennaro Chierchia p.c.). Questions with quantifiers also present some complications (see Nicolae 2013), but one may argue that they do so for all theories. Note that this use of higher order questions differs from Chierchia's in requiring every member of the family of questions to be answered, not just one. As it stands, then, the approach does not extend to questions with indefinites. One may argue, as we have done in Section 4.2, that list answers to such questions are a distinct phenomenon. However, those who want to bring them under the same umbrella have to make some non-trivial accommodations (see Nicolae 2013, for example). Furthermore, questions with more than two *wh* expressions may require some modification of the system as it is not clear if the truth requirement built into Ans-D should be applied at each iteration.

Dayal (2016b) also draws out the implications of this approach for the scope marking construction introduced in Section 2.2. Recall that the second CP in such constructions must denote a Hamblin rather than a Karttunen set. That is, for (51c) to be an acceptable answer to (51a), it is not necessary that Bill actually read *Emma* and John *Persuasion*. This is accomplished in the indirect dependency approach of Dayal (1994) by interpreting the two questions as in (51b) and identifying the restriction *T* in CP-1 with the denotation of CP-2. If list readings denote a family of questions, however, CP-2 would have to be shifted to a set of propositions. Ans-D cannot be used for this purpose, as in the Hagstrom–Fox approach, since it would import the truth requirement and incorrectly predict that (51c), as an answer to (51a), should entail *Bill read Emma and John read Persuasion*. Addressing the problem thus requires some adjustments in the theory.

- (51) a. What does Mary say/think/believe? Who read what?
 b. $\llbracket \text{CP-1} \rrbracket = \lambda p \exists q_{\langle s, t \rangle} [T_{\langle \langle s, t \rangle, t \rangle} (q) \wedge p = \lambda w \text{ say}_w / \text{think}_w / \text{believe}_w (m, q)]$
 $\llbracket \text{CP-2} \rrbracket = \{ \wedge b \text{ read } E \text{ and } j \text{ read } E, \wedge b \text{ read } P \text{ and } j \text{ read } P,$
 $\wedge b \text{ read } E \text{ and } j \text{ read } P, \wedge b \text{ read } P \text{ and } j \text{ read } E \}$
 c. Mary thinks Bill read *Emma* and John read *Persuasion*.

A second problem noted for the Hagstrom–Fox approach is the fact that the predicate in CP-1 of a scope marking construction must select for propositions:

¹⁹ Fox attributes this idea to Heim (2010).

- (52) a. *What does Mary ask/wonder? Who read what?
 b. $[[CP-1]] = \lambda p \exists Q_{\langle\langle s, t \rangle, t \rangle} [\mathcal{T}_{\langle\langle s, t \rangle, t \rangle, t \rangle}(Q) \wedge p = \lambda w \text{ask}_w / \text{wonder}_w(m, Q)]$

Dayal (1994) gives a type-theoretic account for this. With verbs that select propositions, the quantification can be restricted by single or multiple constituent questions, interpreted as sets of propositions, as seen in the types of q and T in (51b). Example (52a) is ruled out because CP-2, a set of propositions, cannot be identified with the restriction $\mathcal{T}$. If, on the other hand, pair-list readings could denote families of questions, one might well expect (52a) to be acceptable: *Which question Q in {What did Bill read?, What did John read?} did Mary ask?*

Dayal (2016b) draws on echo questions to further articulate this problem. Dayal (2000) shows that the only way to have predicates like *ask/wonder* in CP-1 of a scope marking construction is to interpret one of the two *wh* expressions in CP-2 with echo intonation (see also Lahiri 2002b). As discussed in Section 4.3.2, possible answers to such questions are themselves questions. Thus they are of the right type to serve as the restriction $\mathcal{T}$. Example (53a) allows for answers drawn from (53c):

- (53) a. What did Mary ask? Who bought WHAT?
 b. $[[\text{who bought } \text{WHAT}]] = \{\text{Who bought Emma?}, \text{Who bought Persuasion?}\}$
 c. $\{\wedge \text{Mary asked who bought Persuasion}, \wedge \text{Mary asked who bought Emma}\}$

Bringing echo questions into consideration has an interesting consequence for accounts of functionality when we look at questions with quantifiers. They are acceptable in scope marking under a pair-list reading with predicates like *think*:

- (54) a. What does Mary think? What did everyone buy?
 b. Mary thinks everyone bought Emma.
 c. Mary thinks John bought Emma and Bill bought Persuasion.

However, they are never acceptable with predicates like *ask*, even when the *wh* carries echo intonation:

- (55) a. *What did Mary ask? WHAT did everyone buy?
 b. $[[[_{CP} \text{everyone bought } \text{WHAT}]]] = \wedge \text{everyone bought } x$
 c. $[[[_{CP} \text{OP}_{\text{ECHO}} [_{CP} \text{everyone bought } \text{WHAT}]]]] = \{\wedge \text{Everyone bought Emma}, \wedge \text{Everyone bought Persuasion}\}$

This is because CP-2 in (55a) cannot denote a family of questions. Absent a normal *wh* expression inside the lower CP projection, we get something of type $\langle s, t \rangle$ as its meaning (55b). When OP_{ECHO} binds the echo *wh* phrase, we get a set of propositions. If list readings were derived from families of questions, one might well expect (55a) to be acceptable, given (51a), (53a), and (54a).

These data, therefore, suggest that it may be worth keeping to the lower type $\langle\langle s, t \rangle, t \rangle$ for pair-list readings of questions and reserve the higher type $\langle\langle\langle s, t \rangle, t \rangle, t \rangle$

for multiple *wh* echo questions.²⁰ To sum up, Dayal's and Hagstrom and Fox's approaches do not disagree on the existence of higher order questions or even about their derivation but rather on their role in capturing functionality in pair-list answers.

4.3.4 Section summary

It was shown that functional approaches to pair-list answers fail to capture their functionality. Functionality can be enforced through functional absorption, whereby each proposition in the set spells out the graph of a function. An alternative is to iterate question formation and build higher order questions. Functionality can then be captured by distributing Ans-D over subquestions and intersecting the resulting propositions. Higher order questions are manifested most clearly in echo questions with more than one *wh* phrase. The scope marking construction was presented as a possible empirical test for their relevance in deriving functionality in pair-list answers.

4.4 Further issues

The functional approach to pair-list answers has been extremely influential, especially Chierchia's version of it as it brought together insights from the syntactic as well as the semantic literature. There have been many responses to it, some of which we have already explored. Here we will present a few other studies that focus on the need for interpreting questions above their normal type as sets of propositions, on explanations for the structure sensitivity in pair-list readings of questions with quantifiers, and on the possibility of deriving pair-list answers by quantifying over speech acts.

4.4.1 The proper place of lifted questions

Gr&S (1984) and Chierchia (1993), we saw, both argue for lifting the meaning of questions from its basic type in order to account for pair-list and choice readings of questions. Szabolcsi (1997a) argues that lifted meanings are only justified in the complement position of extensional verbs like *know* or *find out*.

Abstracting away from the issue of functions, the pair-list reading of a matrix question with a universal quantifier can be represented with or without lifting (Szabolcsi 1997a: 312):

- (56) a. Which man did every dog bite?
 b. $\lambda P \exists W [W \text{ a witness of } \llbracket \text{every dog} \rrbracket \wedge P \text{ (which } x \in W \text{ bit which man)}]$
 c. $\lambda p \exists x \exists y [x \in W \text{ the unique witness of } \llbracket \text{every dog} \rrbracket \wedge \text{man}(y) \wedge p = x \text{ bit } y]$

²⁰ Chapter 7 will discuss the role of higher order questions in deriving list answers across islands.

Analysis (56c) treats (56a) essentially as a multiple wh question, with the witness set of the quantifier playing the role of the common noun in a wh phrase. We saw in Section 4.3 that this does not, in fact, capture functionality in pair-list answers, but let us set that aside here. Szabolcsi notes that lifted meanings are only required for choice questions. Recall from Section 4.2.4 that she does not find pair-list answers to questions with indefinites acceptable, a point that is even clearer when modified numerals like *more than n N* are used. This is evidence for staying with the normal denotation for matrix questions.

In contrast, Szabolcsi argues that lifted interpretations are needed in complements of extensional verbs. Both versions of the embedded question in (57) allow for a pair-list construal. Intuitively, *P* in (56b) can be instantiated by properties associated with embedding such as *being found out by John*, *being known by John*:

(57) John knows/found out which man *every dog/more than six dogs* bit.

Szabolcsi (1997a: 331–2) further notes that a universal, which is otherwise clause bounded ((58a)), seems to show wide scope effects in embedded pair-list contexts ((58b)). She interprets (58b) with the whole complement, interpreted at the lifted level, taking scope over the matrix clause. A possible derivation for (58c) is in (58d):

- (58) a. Some librarian found out that every student needed help. $*\forall_{ST} > \exists_L$
 b. Some librarian (or other) found out which book every student needed. $\forall_{ST} > \exists_L$
 c. [[which book every student needed]_i [some librarian found out *t_i*]]
 d. $\lambda P \exists A$ [non- $\emptyset$ witness(*A*, [every student])] $\wedge$
 $\forall x$ [*P*(which book *x* needs) iff $x \in A$]]
 $(\lambda p \exists z$ [librarian (*z*) $\wedge$ found-out(*z*, *p*))]
 $\Rightarrow \exists A$ [non- $\emptyset$ witness(*A*, [every student])] $\wedge$
 $\forall x \lambda p \exists z$ [librarian (*z*) $\wedge$ found-out(*z*, *p*)] (**which book *x* needs**) iff $x \in A$]]
 $\Rightarrow \exists A$ [non- $\emptyset$ witness(*A*, [every student])] $\wedge$
 $\forall x \exists z$ [librarian (*z*) $\wedge$ found-out(*z*, which book *x* needs)] iff $x \in A$]]

Szabolcsi notes that modified numerals, even though they allow for pair-list readings in embedded questions, do not show scope interaction with matrix arguments: *some librarian found out which book more than six students needed* does not allow librarians to vary with students. She attributes this to an independent difference between the two quantifiers: *every* allows inverse scope while *more than n N* does not.

There is another significant point related to these examples that Szabolcsi makes. Gr&S (1984) argue that quantifying in approaches cannot capture the *de dicto* readings of the common noun in pair-list construals of embedded questions. Szabolcsi, aware that her version of “quantifying in” the embedded question may also be open to this charge, challenges the generalization. She argues that there is a

subtle distinction between genuine *de dicto* readings and presupposed information that is discernible with predicates like *discover* (Szabolcsi 1997a: 339):

- (59) John has just discovered what candy every criminal craves.

Sentence (59) may be true if what John has just discovered are the candy preferences of particular individuals without discovering their criminal status. He may already be aware of their criminality. This distinction, she claims, is masked in examples with predicates like *know*.

The intuitions at play are subtle and worth probing further but constraints of space prevent me from doing so here. Let me simply note that our current understanding of the *de dicto/de re* ambiguities balance the possibilities for scope taking inside and outside opaque contexts as well as the flexibility in interpreting the world variable on the descriptive term. When it comes to pair-list readings in embedded contexts, the interactions between these two dimensions become quite complex, as we have just seen.

4.4.2 Presuppositionality and lists

Previous sections have established the relevance of Skolem functions in the grammar of natural language on the basis of questions with the quantifier *every*. We now expand our inquiry to include the quantifier *each*:

- (60) a. Which woman does every man/each man love?
 b. His mother-in-law.
 c. $\lambda p \exists f [\forall x [\text{man}(f(x))] \wedge p = \text{like}(x, f(x))]$
- (61) a. Which woman loves every man?
 b. #His mother-in-law.

Recall that Chierchia used data like these to argue that the a-index of the Skolem function is pronominal and subject to WCO. His account of structure sensitivity has been challenged on the grounds that the correlation between WCO and pair-lists is not reliable. A particularly compelling instance of this is the quantifier *each*, which aligns with *every* with respect to WCO but not with respect to structural restrictions. Karttunen and Peters (1980) discuss the pair-list reading of (62) with *each* in object position. Williams (1986) explicitly notes the lack of the expected subject–object asymmetry with respect to pair-list readings:

- (62) Which clerk is now serving each customer?

We will consider two alternatives to Chierchia's account of structure sensitivity which provide some explanation of the differential behavior of the two universal quantifiers.

Comorovski (1989, 1996) argues in connection with multiple constituent questions that pair-list answers are only possible with D-linked wh phrases, phrases

that denote contextually salient sets. This is consistent with the view that pair-list answers list members of the quantifier's witness set. Corroboration of this claim is also presented by the quantifier *any*, which like *every N*, and *each N*, allows functional answers. Unlike them, however, it does not allow pair-list answers (Dayal 1996: 121):

- (63) a. Which woman does any man / each man / every man like?
 b. His mother.
 c. Bill likes Mary and John likes Sue.

All three versions of the question in (63a) have universal force but the interpretation of the answer in (63b) varies.²¹ *Any man likes his mother* only has a generic reading, *each man likes his mother* is about a particular set of men, while *every man likes his mother* is ambiguous between the two readings. Only *each* and *every*, namely those with contextually salient domains, allow the pair-list answer (63c). But if pair-list answers always draw on terms that presuppose their domain, the difference in structure sensitivity between *each N* and *every N* cannot simply rest on that. Something more is needed.

Agüero-Bautista (2001) also pays attention to the presuppositional profile of the two universal terms but augments it with other independently motivated factors.²² The first difference has to do with their scope potential in declaratives. In (64) *every N* is trapped below negation and the indefinite, while *each N* can scope over them:

- (64) a. Someone did not meet every candidate. $\exists > \neg > \forall$; $*\forall > \exists > \neg$
 b. Someone did not meet each candidate. $\exists > \neg > \forall$; $\forall > \exists > \neg$

From this he concludes that *every N* remains below negation and the subject of the clause at LF. The second difference is between different wh phrases with respect to *there-insertion* contexts (Heim 1987):

- (65) a. *Which items are there in the fridge?
 b. What is there in the fridge?

Putting these pieces together, he proposes a constraint on the reconstruction of wh phrases (Agüero-Bautista 2001: 55) and posits the LFs in (67b)–(67c) for the questions in (67a):

- (66) Do not reconstruct a presuppositional phrase into a theta position.

²¹ The point here is independent of whether *any* is analyzed as a universal or as an existential.

²² Further support for the contrast under discussion comes from (i)–(ii) from Agüero-Bautista (2001: 82):

- i. Which boy resembles every man in your class? *PL
 ii. Which boy resembles each man in your class? PL

- (67) a. Which woman loves each man/every man?
 b. [_{CP} which woman [_{XP} **each man** [_{IP} *t_{which}* [_{VP} *t_{each}* [_{VP} *t_{which}* likes *t_{each}*]]]]]
 c. [_{CP} which woman [_{IP} *t_{which}* [_{VP} **every man** [_{VP} *t_{which}* likes *t_{every}*]]]]]

Because *each N* raises higher than *every N*, (67b) has three positions associated with *each*, while (67c) has only two positions associated with *every*. Meanwhile, *which boy* being presuppositional, can only reconstruct down until the subject, not its base position inside vP. So, *each man* c-commands the reconstructed wh phrase while *every man* does not. This according to Agüero-Bautista is why one allows pair-list readings from object position while the other does not. The explanation would extend to monomorphemic wh phrases *who loves each man?* on the view that wh phrases that support list answers are always presuppositional.

Beghelli and Stowell (1997), building on Liu (1990) and Szabolcsi (1997b), also provide an account that separates these two universal quantifiers (and others) by their potential scope sites. Unfortunately, I must leave it to the reader to consult these and other articles in Szabolcsi (1997c) on their own.

4.4.3 Quantifying into question acts

A very different view of structure sensitivity emerges from the account of pair-list readings in Krifka (2001). Krifka takes the relevant configuration to involve a topic structure. This is shown schematically in (68):

- (68) a. As for x, which dish did x make?
 b. *Topic* [_x], λ_{t₁} [*Quest* [_{which dish₂} [_{he₁} make _{t₂}]]]
 c. Which dish did x₁ (x₁ in x) make? And which dish did x₂ (x₂ in x) make? And...

Without going into the technical details, the point to note is that the question keeps its normal meaning and the topic is quantified into the speech act rather than into the standard question meaning (Krifka 2001: 22–3). We get, in effect, a conjunction of questions of the kind shown in (68c).

Krifka points out that when the universal term is focused, *which dish did EVERYONE make*, pair-list answers are ruled out. This follows on the view that focus must be interpreted inside the nucleus and cannot therefore be the topic. This is certainly compatible with Krifka's account where topics quantify over speech acts but it is also compatible with other theories that we have discussed in Sections 4.2 and 4.3. Chierchia (1993), Dayal (1996), and Fox (2012) all interpret the universal outside *C*⁰, that is, outside the domain where focused *every* must remain.

Let us probe, instead, the restriction of pair-list answers to certain quantifiers, as the explanation for it is specific to Krifka's account in terms of conjoined speech acts. Krifka agrees with Szabolcsi (1997a: 324–5) that conjunction and disjunction of questions are qualitatively different, contra Gr&S (1984). Example (69a) can be answered with a pair-list answer satisfying both requests, as in (69b). In (69c),

instead, the questioner cancels the first question in posing the second. There is no choice presented to the addressee. This is another argument against lifted meanings for matrix questions:

- (69) a. What did Mary read? And what did Judy read?
 b. Mary read Emma and Judy read Persuasion.
 c. What did Mary read? Or, what did Judy read?

Krifka concludes that quantifiers that align with conjunction (universal terms, plural definites, and conjoined proper names) are good topics while those that align with disjunction (indefinites, *most*, etc.) are not, because only conjunction is properly defined on speech acts.

Intriguing though this approach is, there are several issues that remain unresolved. One problem that Krifka himself notes is that quantifiers typically do not make good topics, and yet they lend themselves to pair-list answers. Another is the difference between *every* and *each* with respect to structure sensitivity. Krifka suggests that subjects tend to make better topics than objects, a tendency that *each* can override because it presupposes a given set of entities. We saw in Section 4.4.2, however, that presuppositionality cannot be the determining factor since even *every* participates in pair-list construals only when its domain is presupposed. Finally, using topic-hood to derive structural restrictions implies a separate explanation for similar asymmetries with functional readings of questions with quantifiers.

There are other aspects of Krifka's proposal worth probing,²³ which we cannot go into here (see Chapter 5 for a brief discussion in relation to embedded question acts). I will simply mention that this line of explanation would find interesting confirmation if list readings in languages that overtly mark topics were only available with topic-marked quantifiers. I am not aware of literature testing this prediction.

4.4.4 Section summary

The proposals discussed in this section focus primarily on pair-list readings of questions with quantifiers, independent of their functional readings. The first argued for distinctions between a matrix and a subset of embedded questions with respect to interpreting them at a higher level. The other two proposals addressed the differential behavior of the two universal terms *every* and *each* with respect to syntactic restrictions. One appealed to differences in their presuppositional profile to posit distinct LFs. The other also appealed to presuppositional differences, as they relate to the ability to function as topics that must be quantified in at the level of the speech act of questioning. Admittedly, we were

²³ For example, Krifka suggests that *which* N_{SING} is more topical than *who/which* N_{PL} and may lead to *every* N being less favored as a topic (cf. (34a)). He also differentiates how conjoined noun phrases and plural definites differ in quantifying into speech acts to account for the differences in their behavior with respect to list answers.

barely able to scratch the surface of the rich empirical and theoretical picture revealed in these and other studies in this domain.

4.5 Functions and lists in the baseline theory

We extended the semantics of questions to include quantification over Skolem functions in order to admit functional readings of questions with quantifiers. Deriving pair-list answers from functional answers explains the fact that the quantifiers that admit pair-list answers are a subset of the ones that admit functional answers. It involves extracting a set from the quantifier and merging it with the *wh* expression through an *absorption* operation. Quantifiers that can participate in absorption are restricted to those that have minimal and/or unique witness sets.

The use of minimal witness sets brings list answers to questions with indefinites under the same fold as pair-list answers to questions with quantifiers, while the use of unique witness sets keeps the two apart. The latter move treats the choice represented by such questions in terms of a specific indefinite whose indexicality is cued to the addressee. List answers to questions with plural definites were shown to be a distinct phenomenon, amenable to a pragmatic explanation as elaborations of individual answers with plural arguments.

Another property of functional questions that was discussed are apparent subject–object asymmetries. This is captured in the functional approach by positing a complex *wh* trace, with a pronominal element which is subject to the weak cross-over constraint. It applies to functional as well as list readings of questions with quantifiers. Multiple *wh* questions, which seem to privilege the subject *wh* term, are also interpreted using the same principles.

It was shown that the functionality of pair-list answers remains elusive in theories that posit standard Hamblin–Karttunen sets, even if the *wh* is interpreted functionally. One way of ensuring functionality posits *functional absorption* in C^0 . Here each proposition in the question denotation is the graph of a function. Applying Ans-D to the resulting set delivers a pair-list answer with the right properties. Applying Ans-D to denotations resulting from the ordinary meaning of C^0 derives single-pair answers to multiple *wh* questions, and individual and functional answers to questions with quantifiers.

An alternative account of functionality iterates C^0 and allows question formation to apply to questions, leading to higher order questions. Functionality is captured by intersecting the propositions obtained by distributing Ans-D over all the subquestions. Multiple *wh* echo questions provide empirical justification for higher order questions but evidence from scope marking calls into question their usefulness in deriving pair-list answers.

We also considered scope interaction between an embedded *wh* expression and quantifiers in matrix clauses. Such interaction can be explained by treating the complement as a set of questions and allowing it to permute scopally with quantifiers in the matrix clause. Comparing questions with *every N* and those with *each N*, we found that both allow pair-list answers but only the former shows

structural sensitivity. This led to proposals that posit syntactic differences between the two universal quantifiers and argue for a greater role for these syntactic differences in deriving list readings. Finally, we looked at a proposal that treats list readings of questions as conjunctions of speech acts. Quantifiers that lend themselves to list readings are treated as topics, though the quantifiers that support pair-list readings do not align with those that participate in topicalization. Differences between the two universal quantifiers are imputed to differences between them with respect to presuppositionality, but it needs to be kept in mind that presuppositionality may be a necessary condition for all quantifiers that admit list readings.

To sum up, functional and list answers are a heterogenous class. There are differences among researchers on which quantifiers allow such answers as well as on the proper account for those facts. Here we tried to make explicit where the empirical and theoretical fault lines lie. This is particularly important for list answers since they are a powerful diagnostic tool widely used to study wh scope.

Embedded questions

In the last four chapters we have moved freely between direct and indirect questions, on the assumption that the core semantics of questions remains constant across the two. Here we will take a closer look at indirect questions. While the view that direct and indirect questions form a natural class will not change, we will sharpen our understanding of the issues that apply specifically to the latter.

We compare interrogatives to relative clauses and exclamatives, constructions that share morpho-syntactic properties, and provide diagnostics to distinguish between them. We also consider *concealed questions*, noun phrases with question-like interpretations, to determine whether they belong syntactically with interrogatives or whether the two are distinct structures that simply converge in meaning. The underlying question behind both comparisons is the extent to which structural and interpretative similarities should guide formal accounts.

With diagnostics in hand, we probe the theory of selection, focusing on the view that predicates are lexically marked to take specific syntactic complements (CP/DP) and/or specific semantic arguments (propositions/questions/exclamatives). We will ask to what extent such specification can be derived from independently needed aspects of grammar. Differences in the extent to which polar vs. constituent questions can be embedded, as well as differences between different predicates with respect to the possibility of inversion in their complements are among the phenomena explored in addressing this issue.

We then look at challenges posed by concealed questions. How can the syntax of a noun phrase be aligned with the semantics of an interrogative complement? Can all noun phrases function like concealed questions or only those whose head noun can be construed as relational? Are concealed questions the same as identity questions? Do they have the full range of readings that their interrogative counterparts have? And do concealed questions have readings that identity questions lack? We look at different proposals that address these challenges.

Finally, we consider quantificational variability effects which appear to allow matrix adverbs to quantify over *wh* phrases inside a complement clause. We discuss the empirical picture in detail as well as different accounts for capturing the facts. The analyses that have been proposed have very different implications for our understanding of how an embedded question relates to the embedding predicate. In evaluating these proposals we will see that once again it is important to pay attention to the semantic contribution of embedding predicates in addition to those of the complement and the adverb.

5.1 Close kin of interrogatives

Our discussion in previous chapters has tacitly assumed that interrogatives are easily identifiable objects. This may be true in matrix contexts but not in embedded contexts. We compare interrogatives to two structures that are morpho-syntactically similar: free relatives and exclamatives.¹ We then consider a structure that does not share morpho-syntactic features with interrogatives but is semantically akin: concealed questions. The primary goal of this section is to provide diagnostics for identifying embedded interrogatives.

5.1.1 Free relatives and interrogatives

Since the semantics of interrogatives has been studied most extensively for English, the connection to relative clauses was noticed early on (Cooper 1983). There is considerable overlap in the expressions that participate in restrictive relatives, non-restrictive relatives, headless/free relatives and questions: *who*, *what*, etc. Syntactically, they pattern together in requiring *wh* fronting. This morpho-syntactic kinship makes the possibility of semantic unification attractive and Cooper shows how the various meanings can be derived from a common core of meaning.² Of direct relevance is the relationship between interrogatives and free relatives. Since they are maximally similar morpho-syntactically, it is not always obvious which one a particular embedding instantiates.

Taking the *wh* phrase to be a lambda abstractor that combines with a sentence level meaning, we have a basic predicative meaning for a relative clause (type $\langle e, t \rangle$) in (1a). We saw how question denotations can be built out of them in Chapters 2 and 3 (Groenendijk and Stokhof 1982, 1984; George 2011). Here we draw on the account of free relatives in Jacobson (1995), which agrees in spirit, if not in letter, with Cooper's account:

- (1) a. $\llbracket [_{CP} \text{ what}_i [_{IP} \text{ Mary bought } t_i]] \rrbracket = \lambda y[\text{bought}(m, y)]$
 b. $\llbracket [_{DP} [_{CP} \text{ what Mary bought}]] \rrbracket = \iota(\lambda y[\text{bought}(m, y)])$

Free relatives require an argumental type meaning. A DP structure over the free relative (1b), possibly with a null *D*, is one way to motivate this shift in meaning. The need to combine with a predicate is another. Since the free relative makes the same semantic contribution as a definite noun phrase, the *iota* type shift of Partee (1986) is posited.³

Given their morpho-syntactic similarity, we need diagnostics to distinguish between free relatives and embedded interrogatives:

¹ Wh expressions are also morphologically related to indefinites cross-linguistically and are identified semantically by Hamblin (1973) and Karttunen (1977). See Chapter 8 for further discussion.

² Syntactic differences between them have been noted in Bresnan and Grimshaw (1978) and elsewhere.

³ For the claim that free relatives are definites, even with the morpheme *-ever*, see also Dayal (1991a, 1991c, 1995, 1996, 1997). See Caponigro (2003) on indefinite free relatives.

- (2) a. I wonder what Mary saw.
 b. Mary believed what she saw.
 c. I know what Mary saw.

As Cooper points out, the underlined phrase in (2a) is an interrogative since the embedding verb *wonder* does not take noun phrase complements (**I wonder the time/John*). Similarly, we can be sure that (2b) involves a free relative since *believe* does not take questions as complements (**Mary believed whether Sue was here/who was here*). The status of (2c), however, cannot be settled on the basis of the embedding predicate's properties since it is possible for *know* to embed noun phrases (*Mary knows the time/John*) as well as questions (*Mary knows whether Sue was here/who was here*). The following can be helpful in such cases.

There are two tests that unequivocally show a *wh* complement to be an interrogative. A multiple *wh* construction has at its core, we saw in Chapter 4, a relation between individuals. Such relations can feed into question-meanings but there is no simple type shift from relations to meanings suitable for noun phrases.⁴ In administering this test, however, appropriate controls have to be used to ensure that the particular combination of *wh* expressions is independently acceptable. The awkwardness of (3c), for example, is independent of embedding. *Who* and *why* do not combine well even in direct questions:

- (3) a. *Mary believes who saw what.
 b. I wonder/know who saw what.
 c. ??I wonder/know who went why.

Another test that disambiguates in favor of interrogatives is the use of complex *wh* phrases since free relatives do not admit them:

- (4) Bill knew/wondered/*read which book Sue bought.

Of course, complex DPs become possible in free relatives if *-ever* is present. In fact, *-ever* disambiguates in favor of free relatives. It is acceptable in the complement of *believe* but not *wonder*:⁵

- (5) a. Mary believes whatever she sees.
 b. *Bill wondered whatever Mary saw.

When *-ever* occurs in interrogatives, it is restricted to mono-morphemic *wh* expressions, where it lends a somewhat rhetorical flavor to the question (6a). Note that (6c) is only acceptable with inversion in the embedded clause (see

⁴ Multiple correlatives have been argued to allow for relation-based noun phrase meanings, but multiple correlatives associate with full clauses, not arguments inside a clause. See Dayal (1991a, 1991c, 1996), as well as articles in Lipták (2009) on correlatives.

⁵ We do not include exclamation marks in the discussion here (see section 5.1.2).

Section 5.2.3 for relevant discussion). Controlling for inversion, we can say that *-ever* is not compatible with embedded interrogatives:

- (6) a. Whatever did I do to deserve this?
- b. *Whichever book was on sale?
- c. John wondered whatever did Sue do/*whatever Sue did.

Reliable though the *-ever* test is, some care must be taken in administering it. Consider predicates like *know* which take noun phrase complements. A free relative interpretation is not always available, as indicated by the somewhat marginal status of (7a). The minimally different relative clause over animate individuals in (7b), however, is fully acceptable. And (7c) shows that a free relative over propositions is equally acceptable. Similar effects can be perceived with *tell* (7d). This means that it is, after all, possible for these predicates to embed free relatives with *-ever*, as expected for predicates that admit DP arguments:

- (7) a. ?John knows whatever Mary saw.
- b. John knows whoever Mary was speaking to.
- c. John knows whatever Mary told him.
- d. John told Bill whatever he knew/?whatever he saw.

It is worth noting that even with predicates that clearly embed free relatives, there may be unacceptability with *-ever* in specific instances. Sentence (8a) is unacceptable because the conditions under which the imperative is uttered clashes with the semantic contribution of *-ever* (Dayal 1995, 1997; Jacobson 1995; von Stechow 2000; among others). With minor adjustments the sentence is redeemed in (8b). We conclude, then, that the diagnostic of *-ever* does separate free relatives from interrogatives:

- (8) a. Look at what(*ever) the cat dragged in.
- b. Look at whatever you want to look at.

The morpho-syntactic affinity between questions and relative clauses is manifested in too many languages to be an accident. As such, there have been several attempts at a unitary account of *wh* constructions: Cooper (1983), Groenendijk and Stokhof (1984), Jacobson (1995), Caponigro (2003), Haida (2007), and George (2011), among others. While there is strong empirical motivation for a unified account, the parallelism between interrogatives and free relatives is by no means perfect. Hindi, for example, has distinct *wh* series for questions and relatives (Dayal 1991a, 1996).⁶ Yiddish uses a different series for restrictive relatives from those it uses for free relatives and questions (Prince 1989). Even

⁶ Dayal (1996 and elsewhere) does not derive question meanings from basic predicative meanings that could be ascribed to relative clauses. The parallelism emphasized there is between question meanings and correlative meanings, specifically the switch from uniqueness in single *wh* constructions to functional dependencies in multiple *wh* constructions of the kind discussed in Chapter 4.

in English, restrictive relative *wh* pronouns alternate with *that*, an expression homophonous with the non-interrogative complementizer: *the book that/which Mary bought*. That said, it is clear that there are principled ways of working out a semantics for interrogatives and relative clauses based on a common core of meaning involving lambda abstraction.

5.1.2 Exclamatives and interrogatives

Interrogatives and exclamatives form a minimal pair in matrix clauses (9a)–(9b), where they are distinguished by intonation and the presence/absence of inversion. In embedded contexts, where there is no inversion in interrogatives and intonational cues are less revealing, it can be hard to determine which structure is involved. The complement in (9c) can be considered ambiguous between interrogative and exclamative:

- (9) a. How tall he is!
 b. How tall is he?
 c. She knows how tall he is.

Grimshaw (1979), building on Elliott (1971), provides some tests that distinguish between interrogatives and exclamatives. The *wh* phrase in an exclamative can include an indefinite noun phrase or the adverbial *very*, modifying the gradable adjective (10a)–(10b). These options are not available to interrogatives (10c)–(10d):

- (10) a. What a good boy he is!
 b. How very tall he is!
 c. *What a good boy is he?
 d. *How very tall is he?

Conversely, there are *wh* expressions (*which N*, *who*) that are fully acceptable in interrogatives (11a)–(11b), but not in exclamatives (11c)–(11d):

- (11) a. Which book do you like?
 b. Who do you see there?
 c. *Which book you like!
 d. *Who you see there!

These diagnostics can be used to determine the status of complements of predicates like *know* and *wonder*:

- (12) a. John knows how very tall he is.
 b. John knows what a tall boy he is.
 c. John knows which boy/who is standing there.
- (13) a. *John wonders how very tall he is.
 b. *John wonders what a tall boy he is.
 c. John wonders which boy/who is standing there.

These data establish that *know* and *wonder* can both embed interrogatives, but only the former can embed exclamatives.

Following the instinct to unify the semantics of morphologically similar types, there have been some attempts to do so for exclamatives and interrogatives. As should be clear already, *wh* based exclamatives include the dimension of degrees but there are two points on which they differ from corresponding degree questions. Exclamatives are inherently factive. Elliott and Grimshaw note that it is possible to have a first person subject expressing lack of knowledge about an interrogative but not about an exclamative (14). Also, only an exclamative entails that the relevant degree exceeds some contextually salient standard (15):

- (14) a. I don't know how tall he is.
b. *I don't know how very tall he is.
- (15) a. I know how tall he is. 6' 5" / Just 5' 6".
b. I know how very tall he is. 6' 5" / #Just 5' 6".

Gutierrez-Rexach (1996) builds exclamative meanings from question meanings, adding an emotive attitude on the part of the speaker/subject towards the true proposition that constitutes the answer to the question. Zanuttini and Portner (2003) appeal to widening, in the sense of Kadmon and Landman (1993), to capture this difference between exclamatives and interrogatives. Their basic idea is that an exclamative also denotes a set of propositions but the *wh* expression draws its values from an expanded quantificational domain:

- (16) a. D1: {5' 5", 5' 6", 5' 7", ... 6'}
b. D2: {5' 5", 5' 6", 5' 7", ... 6', 6' 1", ... 6' 5"}

If D1 is the set that the interrogative takes as its quantificational domain, the exclamative pushes the limits beyond it to D2. This, they claim, is behind the intuition that the true proposition in the set involves a high degree on the scale.

Wh exclamatives belong to a broader class of constructions expressing emotive attitudes and there is a significant body of literature on the topic, some of which we will discuss in Section 5.2. By and large, though, insights from the semantics of exclamatives have not influenced work on the semantics of questions. The flow of influence has rather been in the opposite direction. This fits in with the fact that exclamatives have an extra layer of meaning, if you will, over and above the meaning of the corresponding interrogative.

The use of polar questions as the prime exemplar of interrogatives (Grimshaw 1979) is worth flagging here. Since polar questions do not have simple noun phrase or exclamative equivalents, it is reasonable to infer from the (un)embeddability of polar questions the (un)embeddability of questions more generally. We will see in Section 5.2, however, that there are a few predicates that embed polar questions but not constituent questions and a few that display the opposite behavior.

5.1.3 Concealed questions and interrogatives

We have so far looked at clausal complements in relation to questions but noun phrases can also have question-like meanings. Baker (1968) notes that a sentence like (17a) can have an acquaintance reading which states that John knows Rome, here described as the capital of Italy. This is the standard meaning that we would expect of a noun phrase. It can also have a reading where John may not personally be acquainted with Rome but knows that it is the capital of Italy, equivalent to (17b) with an embedded interrogative.⁷ Such DPs are known as *concealed questions*:

- (17) a. John knows the capital of Italy.
b. John knows what the capital of Italy is.

Another argument for classifying such DPs with interrogatives is their behavior under predicates like *tell*. Recall that *tell* varies with respect to factivity, depending on whether the complement is a declarative or an interrogative:

- (18) a. John told Bill that it was raining.
b. John told Bill whether it was raining.

Sentence (18a) can be uttered in contexts where it is known not to be raining; (18b) requires John to have told Bill the actual state of affairs. Concealed questions pattern with interrogatives:

- (19) a. John told us that Betty had gone to Paris, but he turned out to be mistaken.
b. John told us the place Betty had gone, (#but he turned out to be mistaken).
c. John told us where Betty had gone, (#but he turned out to be mistaken).

Concealed questions look like ordinary noun phrases and to that extent might not even make the cut for inclusion in a chapter on embedding. Even semantically, the alignment is not perfect. For example, some noun phrases only allow an acquaintance reading. Sentence (20a) cannot be paraphrased as (20b):⁸

- (20) a. John knows the book.
b. John knows what the book is.

An important factor in a DP being a concealed question seems to be the presence of a relational head noun: *capital, governor, price, time*. For a city to qualify as a

⁷ Some languages distinguish between these two lexically, which can be helpful in identifying concealed questions. Hindi uses *jaan-naa* for both but *maalum ho-naa* only for factual knowledge. Spanish uses *conocer* for the acquaintance sense of knowing and *saber* for factual knowledge. Thanks to Natalie DelBusso (p.c.) for bringing this up.

⁸ Noun phrases that can have concealed question readings can also have exclamative readings (Elliott 1971; Grimshaw 1979). This does not seem to have been explored in subsequent literature.

capital, there must be a country it is related to in the relevant way but a book can be identified independently of authors or owners. Nathan (2006) gives the examples in (21) to show that this is not a low-level pragmatic effect. Even in a context where it is known that there is only one semanticist in the department, just like there is only one department head, (21b) would not be interpreted as a concealed question: *Bill told me who the semanticist is*. The minimally different (21c), however, is claimed to bring out the relevant meaning:⁹

- (21) a. John told me the department head.
 b. *John told me the semanticist.
 c. ?John told me the semanticist who teaches there.

If concealed questions are defined in terms of identity questions, as indicated above, we might well ask how such a meaning can be derived from a DP structure. Example (22a) posits an underlying identity question from which the concealed question is derived by deleting the *wh* and the copula. The underlying structure yields the desired semantic representation, (22b). This is, in essence, the proposal of Baker (1968):

- (22) a. [_{IP} John knows [_{CP} ~~what~~, [_{IP} the capital of Italy ~~is~~ t_i]]]
 b. $\lambda p \exists x[\text{capital}(x) \wedge p = \wedge x \text{ is the capital of Italy}]$

A non-trivial problem with this move is that there does not seem to be a principled way of restricting deletion to occur only when the head noun is relational.¹⁰ Another problem is that not all predicates that embed interrogatives embed concealed questions. There is no motivation for the proposed deletion that would be compatible with such selectivity. The derivational move has therefore not been pursued and concealed questions are typically taken to be basic DPs. This, then, makes the interpretation of concealed questions an active topic of research, which we will cover in Sections 5.2 and 5.3.

5.1.4 Section summary

While it is possible, and indeed tempting, to give morphologically similar structures a unified semantics, such a move needs to be balanced against two considerations. The first is the extent to which the parallelism holds cross-linguistically. The second is the extent of the overlap in meaning. In this section we focused not so much on evaluating the advisability of a unified account for interrogatives and free relatives or interrogatives and exclamatives, as on diagnostics that are helpful in distinguishing between them. In the case of DPs that only share semantic

⁹ Speakers I have consulted do not find (21a) grammatical, nor do they find a sharp contrast between (21a) and (21b)/(21c). However, *tell* does embed concealed questions in *tell the time* or *tell the temperature*.

¹⁰ Jane Grimshaw (p.c.) points out that *John knows the textbook* is markedly better, presumably because it has a relational component to it. Textbooks are typically related to courses.

similarities with interrogatives as well, we restricted ourselves to markers of identification for concealed questions.

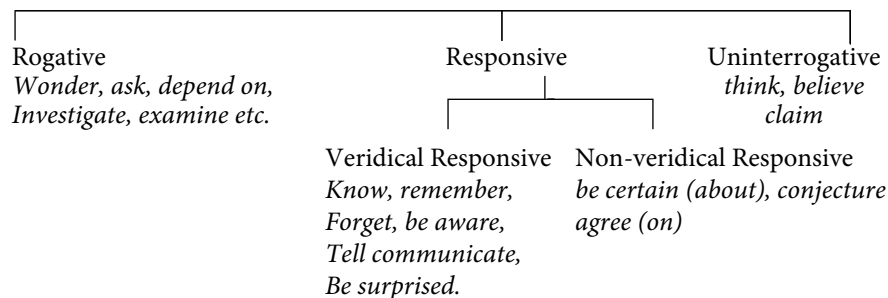
5.2 Selection

Our discussion so far has tacitly assumed a distinction between predicates that can and cannot embed interrogatives. We have also differentiated between two broad classes of predicates among those that embed interrogatives, predicates that only take interrogatives and predicates that take interrogatives as well as declaratives. We have referred to the first set as taking irreducibly intensional complements and the second as admitting their extensional counterparts. Another distinction that we have noted between interrogative embedding predicates is in terms of a choice between strongly and weakly exhaustive readings of their complement. This section probes deeper into such issues, broadly classifiable under the term *selection*.

5.2.1 C-selection and s-selection

Lahiri (2002a: 287) provides a three-way classification for interrogative selecting predicates, subsuming the classification in Karttunen (1977) and the intensional/extensional distinction in Groenendijk and Stokhof (1982). We add to it predicates that do not take interrogatives, listed under the category *uninterrogative*, completing the class of predicates that take (finite) clausal complements. We set aside non-finite complements to keep things manageable:

(23)



The basic divide between predicates that do and do not embed interrogatives is typically assumed in syntactic as well as semantic theories and is based on a fairly solid grammatical distinction between rogative and what I am calling uninterrogative predicates. In between fall the responsive predicates that can embed both interrogatives and declaratives, with a further divide between the veridical ones that entail the true answer to the interrogative complement, and the non-veridical ones which do not:

- (24) a. *John believes who will go/whether she will go.
 b. John wonders/knows/is certain who will go/whether she will go.

- (25) a. *John wonders that she will go.
 b. John believes/knows/is certain that she will go.

These divisions are often captured in terms of the $+/-_{WH}$ selecting property of embedding predicates. Rogative predicates are marked $+_{WH}$, uninterrogative predicates $-_{WH}$ and responsive predicates are ambiguous between $+/-_{WH}$. This is essentially the picture of selection in Bresnan (1972) and Chomsky (1973).

Grimshaw (1979) probes this view of selection by bringing into the discussion distinctions that are not captured by this division. Semantically similar predicates like *think* and *believe* both embed clausal declarative complements but only the latter embeds noun phrases. Related $+_{WH}$ predicates like *wonder* and *ask* differ in the same way:

- (26) a. John believed/*thought the story.
 b. John asked/*wondered the time.

Grimshaw thus argues for reference to syntactic categories in the subcategorization frames of individual verbs. $C_{category}$ -selection makes possible further distinctions within the $+/-_{WH}$ classes:

(27)	$+/-_{WH}$	$+/-_{CP}$	$+/-_{DP}$
<i>Believe</i>	–	+	+
<i>Think</i>	–	+	–
<i>Wonder</i>	+	+	–
<i>Ask</i>	+	+	+

Grimshaw further shows that the $+_{WH}$ feature is unable to distinguish between interrogatives and exclamatives, both of which involve *wh* expressions and *wh* fronting. There are $+_{WH}$ predicates that do not agree on exclamative complements:

- (28) John knew/*wondered/*asked how very tall Bill is.

This argues for differentiating complements on the basis of their meaning, that is, their $s_{semantic}$ -selectional properties. Taking Q and E to stand for the semantic objects denoted by questions and exclamatives respectively, including s-selection in the calculus leads to (29):

(29)	$+/-_{WH}$	$+/-_{CP}$	$+/-_{DP}$	Q/E
<i>Wonder</i>	+	+	–	Q
<i>Ask</i>	+	+	+	Q
<i>Know</i>	$+, -$	+	+	Q, E

Replacing the +/-WH parameter with two independent selectional parameters, c-selection and s-selection, leads to the fine-grained distinctions that natural language seems to warrant.

The relevance of c-selection has been disputed by Pesetsky (1982). He locates the difference in the embeddability of concealed questions in case theory: CPs are immune, or even perhaps resistant, to case while DPs require it. Therefore, what is at issue in examples like (26) is the case assigning property of embedding predicates rather than their c-selectional property *per se*. The ameliorating effect of prepositions is cited as evidence:¹¹

- (30) a. John wondered *(about) the governor of California.
- b. John was surprised *(at) the capital of Italy.

But note that these prepositions are not semantically innocuous. In a given context, (30a) can mean that John wondered whether the governor of California would show up, not who the governor is. Lahiri (1991, 2000, 2002a) nevertheless sides with Pesetsky, arguing that the availability of the identity reading associated with concealed questions is what counts, not the possibility of other readings.

An important point, noted by Grimshaw herself, is that predicates selecting questions always select CPs. That is, we do not find verbs that select only DPs with a concealed question reading: *[-CP, +DP, +Q]. Grimshaw (1981) suggests that this is because semantic categories have canonical syntactic realizations. The canonical syntactic form for questions and propositions is CP: any predicate that s-selects questions or propositions must allow CP complements. Pesetsky, citing Ken Hale, argues otherwise on the basis of the following:

- (31) a. *We approve (of) [_{CP} that unemployment will rise in the 1980s].
- b. We approve *(of) [_{DP} rising unemployment in the 1980s].
- c. *We paid attention (to) [_{CP} that we are departing on Thursday instead of Friday]
- d. We paid attention *(to) [_{DP} our departure on Thursday instead of Friday]

Pesetsky's point is that if it is possible to have predicates that admit [-CP, +DP, +P], where *P* stands for propositions, then it should be possible to have predicates which admit the unattested [-CP, +DP, +Q]. And once we make room for differences in case assigning properties, the distinction between predicates that do and do not take concealed questions disappears. From this perspective, c-selection is dispensable.

Nathan (2006) approaches this issue from a different angle, casting doubt on the premise that *wonder* cannot take DP complements:

¹¹ Grimshaw categorizes *be surprised/be amazed* as selecting only *E* because they do not take polar question complements. The diagnostics of multiple *wh* and complex *wh* phrases suggest, however, that they do embed interrogatives: *It surprised/amazed John who brought what, It surprised/amazed John which student had won.*

- (32) a. John wondered who left, and Sally wondered that as well/ the same thing.
 b. Who left/The same thing matters to Kim.
 c. It matters to Kim who left/ *the same thing.

In (32a) the pronominal *that* or the DP *the same thing* is anaphoric to the question *who left*. The contrast between (32b) and (32c) establishes that *the same thing* is a bona-fide DP that cannot survive the case-less extraposed position, unlike a CP. If *wonder* can c-select/case mark a DP, the reason it cannot take concealed questions, Nathan concludes, must be that concealed questions are not questions at all. That is, appearances notwithstanding, concealed questions violate the s-selectional, not the c-selectional, requirements of *wonder*.

Nathan treats concealed questions as being propositional and requiring predicates that can embed both questions and propositions.¹² Of course, there are counterexamples that he has to address. Most notably, *ask* allows concealed questions though it does not admit propositions.¹³

While the question of whether c-selection can be dispensed with as an independent parameter is not insignificant, it has proven less relevant for semantic theories of questions than the status of s-selection. However, it might be useful at this point to step back and note that the theory under discussion is one in which a lexical predicate's complement can vary arbitrarily along syntactic and semantic parameters. Thus two semantically similar predicates *P* and *P'* may differ in their ability to select a particular type of complement. While this is true of Grimshaw's view of c-selection, it does not seem to be so for s-selection, which privileges natural classes of predicates and predicts a clustering of complement types around such classes. Consequently, s-selection has proven more intriguing to semanticists.

5.2.2 Selecting between +*WH* complements

There are two generalizations that have been probed in an effort to derive the effects of s-selection. One is the observation that rogative predicates do not embed exclamatives, the other is the observation that some predicates (*be surprised/be amazed*) that embed questions discriminate between polar and constituent questions. Starting with the first generalization, although Grimshaw represents the s-selectional properties of individual predicates in subcategorization frames, her discussion makes it clear that she does not take it to be an arbitrary or idiosyncratic property of individual lexical items. She notes, for example, that rogative

¹² Concealed questions differ in this respect from regular propositional type complements which do not typically discriminate between predicates that can take only -*WH* and those that can take both +/-*WH*. See Dor (1992), who provides the basis for Nathan's account of the distribution of concealed questions, though his final proposal is significantly different.

¹³ Aloni and Roelofsen (2011) account for their distribution in terms of s-selectional properties of the embedding verb. According to them, *believe* selects for world-independent propositions, *agree on/ be certain* for world-dependent propositions or propositional concepts, *know/remember/forget* for propositions (world-dependent or independent), *be related to/depend on* for world dependent propositions, and *wonder/inquire* for propositional concepts. Thus they provide a cross-classification based on the intensionality as well as the type of the complement.

predicates cannot embed exclamatives because the factivity inherent in exclamatives is not compatible with the indeterminacy inherent in rogatives. This is the insight that D'Avis (2002 :23) casts in terms of a clash of presuppositions:

- (33) Wh-clauses that presuppose their only true answer are not allowed in question environments.

D'Avis shows how exclamatives, which presuppose the true answer to the question embedded inside them, and rogative predicates which demand ignorance of the true answer, cannot both be felicitous in any context. Thus their incompatibility is predictable.

Abels (2007, 2010) presents some potential counterexamples to the generalization that *wonder* does not select exclamatives as well as the generalization that exclamatives presuppose their true answers (see also Rett 2008; Beyssade 2009; Saebo 2010). The following are naturally occurring examples of exclamatives under *wonder*:¹⁴

- (34) a. I just wonder what a great book it might have been had there been more of a "human" look at the gangs.
 b. I can't help but wonder how very unhappy you must be in your own life to find it necessary to spread such venom about.

Through a careful study, however, Abels shows that such deviations occur in contexts where presuppositions are filtered out. As such, they turn out to be additional support for the view that semantic incompatibility drives the inability of rogative predicates to take exclamative complements. A consequence of this line of explanation is that s-selection cannot be taken as lexically specified for a predicate, or even a class of predicates. Rather, the s-selectional profile of predicates depends on the compositional semantics of the structures they occur in.

Let us now consider the incompatibility of surprise-predicates and polar question complements.¹⁵ D'Avis (2002) draws on Heim's insight that the interrogative complements of such predicates must be weakly exhaustive and expresses the restriction in (35a), seeking to derive it as shown in (35b). Note that D'Avis is working with Karttunen sets and Heim's answerhood operators discussed in Chapter 2. He therefore has to add the condition that the existential presupposition of the question be met in evaluating the relevant propositions (D'Avis 2002: 10 (for (35b)) and 26 (for (35a))):¹⁶

¹⁴ Jane Grimshaw (p.c.) notes that the acceptability of such examples is crucially tied to the presence of a modal in the complement.

¹⁵ D'Avis calls them exclamative predicates. I choose the alternative term *surprise*-predicates to avoid confusion with exclamative phrases.

¹⁶ The weak and strong answerhood operators apply to sets of true answers, not to sets of possible answers:

- (i) $\text{Ans-H}_1(\alpha, w) = \cap \llbracket \alpha \rrbracket (w)$
 (ii) $\text{Ans-H}_2(\alpha, w) = \lambda w' [\text{Ans-H}(\alpha, w') = \text{Ans-H}(\alpha, w)]$

- D'Avìs notes that a world where no one came is also in the negation of Ans₁ but facts suggest otherwise. If Heinz expected no one to come, one could not say *Heinz was amazed who came* on finding out that someone did. Rather, the declarative *Heinz was amazed that someone came* would be appropriate. In our baseline theory, the existential presupposition is built into Ans-D so this would not pose the same problem.

If the expectation was that *John will be the winner not Sue*, then the true answer *Sue is the winner* is indeed surprising. But in order to embed a question, under D'Avis' account, *norm-p* must be able to render another possible answer surprising. Since there is no such candidate here, it is incorrectly predicted to be unacceptable (see also Nicolae 2013).

Abels (2007) takes issue with a different aspect of D'Avis's proposal, namely the reliance on the strongly exhaustive answer: Ans2. He points out that *surprise* does not require knowledge of the strongly exhaustive answer, and in fact, does not even require full knowledge of the weakly exhaustive answer. If Mary, Bill, and Sue came, and John only finds out that Mary and Bill came, and knows that there may be others, he may still be surprised because he didn't expect Mary to come. Similarly, he may find out first that Bill came and not be surprised. On further finding out that Mary also came, though, he may feel surprised. This leads Abels to posit the following:

- (38) a. John entertains a proposition *norm-p* that he considers normal and which contradicts the maximal positive partial answer to Q that John knows.
 b. $\text{norm-p} \cap p = \emptyset$, where *p* is the maximally informative partial answer to the question embedded under the surprise-predicate known by the subject of the predicate.

Again, because the theory he adopts does not incorporate existential presuppositions, Abels has to rule out cases where nobody came or where John expected nobody to come.

Abels' explanation depends on a polar question being a singleton set of propositions, namely the set with just the nucleus proposition: $\llbracket \text{did Bill come?} \rrbracket = \{w: \text{came}(w)(b)\}$.¹⁷ The explanation also relies on the formal definitions of partial answers and weakly exhaustive answers in Abels' theory, as well as on the factivity and the requirement of counter-expectation inherent in the lexical meaning of surprise-predicates. Abels shows that regardless of whether the nucleus proposition is true or false in the world of evaluation, there is an irredeemable problem in satisfying the semantic requirements imposed by the various components of the structure. The impossibility of embedding polar questions under surprise-predicates is thus given a principled explanation that relies on semantic composition rather than selection.

Both D'Avis and Abels, we see, put some weight on the emotive aspects of embedding predicates. Karttunen notes that there are non-emotive predicates that are also resistant to polar questions:

- (39) a. John realized which student won.
 b. *John realized whether Bill won.

Guerzoni (2007) takes the relevant factor in the sensitivity to constituent vs. polar questions to be speaker factivity instead of the emotive nature of the embedding

¹⁷ This idea goes back to Bolinger (1978) and will be taken up in Chapter 8.

predicate. The general shape of her explanation, however, also plays on the interaction of meanings generated by the complement and the embedding predicate. Let us see how this applies to (39b). The presuppositions generated when each of the two possible answers to the polar question is embedded under *realize* are given in (40a)–(40b). They in turn generate two implicatures ((40c)). The problem comes from the speaker factivity in the matrix verb (40d), because it contradicts (40c), as shown in (40e):

- (40) a. Bill won: John didn't know before the event time that he did;
John knows at the event time that he did.
b. Bill didn't win: John didn't know before the event time that he didn't;
John knows at the event time that he didn't.
c. Speaker does not know that John realized that Bill won. $\neg K(R(j,p))$
Speaker does not know that John realized that Bill didn't win. $\neg K(R(j,\neg p))$
d. Speaker knows the truth about whether Bill won (given Speaker
factivity of V).
That is,
Speaker knows that John realized that Bill won or
Speaker knows that John realized that Bill didn't win (as the case may be).
 $K(R(j,p)) \vee K(R(j,\neg p))$
e. $K(R(j,p)) \vee K(R(j,\neg p)) \wedge \neg K(R(j,p)) \wedge \neg K(R(j,\neg p))$

Surprise-predicates are different because the presuppositions are generated by the assertion. Once that is factored in, the rest follows as above. There are, of course, open problems that Guerzoni is aware of. For example, she notes that the predicate *know* is factive, not speaker-factive. This means that the problematic implicatures would not get generated, correctly predicting the embeddability of polar questions under *know*. However, she points to self-attitude attributions as a possible argument that may suggest speaker factivity. This poses a problem for an account which uses speaker factivity to rule out polar questions:

- (41) a. I know who passed.
b. I know whether Bill passed.

Her claim is that only (41b), and not its stronger alternatives (*I know that Bill passed, I know that Bill didn't pass*), is felicitous in contexts where what is relevant is not the answer to the embedded question, but rather the identity of those who know about it. In these situations, the stronger alternatives are not relevant and so cannot give rise to the implicatures that would lead to contradiction. She also attempts to derive the strong-weak exhaustivity differences on the basis of distinctions cued to speaker's knowledge.¹⁸ While this line of inquiry is promising,

¹⁸ See the correlation made in Section 3.1 between weak-strong exhaustivity and speaker's vs. attitude holder's knowledge.

it would not be unfair to say that more work needs to be done before we can take the generalization to have been fully explained.

The general thrust of the explanations we have looked at above may well prove fruitful with respect to the opposite pattern of selection, also noted by Karttunen (1977):¹⁹

- (42) a. John doubts whether Bill will win/*who will win.
b. John doubts that Bill will win.

Interestingly, dubitative predicates like *doubt* express relations to the nucleus proposition, not its negation. John stands in the relevant relation to the positive proposition in the complement of (42a), not its negative counterpart. As noted earlier in this subsection, Abels (2010) suggests that such predicates relate to a singular proposition and a polar question denotes only the nucleus proposition (see also Chapter 8). Of course, constituent questions also have singular propositions as their true answers, but Abels gives an argument in terms of a clash of presuppositions that I leave the reader to follow up on. The general point here is that it is quite possible that the difference between a singleton and a non-singleton propositional set may be the critical factor in these cases.^{20,21}

I have laid out the proposals for deriving the selectional restrictions of surprise-predicates in some detail because they call into question the kinship between polar and constituent questions. Karttunen, who drew attention to these patterns, set the differences aside and this move has been accepted in the literature. To the extent that explanations of their differential distribution in terms of the interaction between the semantics and pragmatics of the embedding predicates and the complements are successful, they validate the intuition that polar and constituent questions belong together.

5.2.3 Selecting +/-*WH* complements

Let us now return to the fundamental divide between predicates that select propositions and those that select questions. Consider, for example, the verb *believe* which is staunchly -*WH* in the affirmative but seems to change character when it is negated. Elliott (1974) and Grimshaw (1979), for whom the possibility of polar questions is the identifier of predicates selecting questions, conclude that given the unacceptability of (43b), (43a) must embed an exclamative:

¹⁹ Note that *doubt* as a noun freely combines with constituent questions: *There is no doubt who will win*.

²⁰ This may well be the place to note the existence of predicates like *rattle off* or *list* that require plurality, broadly construed, in their complements: *She listed the names/*name; She listed which students/*student left; *She listed whether Sue had passed*. See Schwarz (1993) and Abels (2007).

²¹ Even a cursory internet search brings up counterexamples to the generalizations regarding *doubt*'s ability to take constituent questions or *realize*'s to take polar questions. Factoring out the ungrammatical examples, we are still left with enough bona-fide cases that show these generalizations to have exceptions. Close analyses would be needed to determine exactly what is at the root of such exceptions but their very existence gives credence to approaches that seek to derive s-selection through composition.

- (43) a. You won't believe who/which woman Ed has married.
 b. *You won't believe whether Ed is married.
 c. You won't believe what a jerk Ed has married.

Huddleston (1993) points out, however, that *who* and *which N* are not good in exclamatives, which establishes the status of the complement in (43a) as interrogative.²² He draws on Elliott's diagnostic of using *else* as further support:

- (44) a. Who else did you invite?
 b. *Who else you invited!
 c. You won't believe who else Ed has invited.

Huddleston concludes that *not believe* can take questions as complement, and the exclamatory force comes from the matrix predicate. It looks as if the predicate *believe* jumps ship to become a surprise-predicate in the scope of negation.

A similar shift can be observed with the $-_{WH}$ predicate *think*. In the progressive *think* begins to take on the character of *wonder* in its ability to take $+_{WH}$ complements, especially with infinitival complements:

- (45) a. *I thought whether to invite Bill/who will be invited to the party.
 b. I'm thinking whether to invite Bill.
 c. I'm thinking who to invite to the party.

I suspect that there may be some variation with regard to the acceptability of (45b)–(45c) but the contrast with (45a) is likely to hold up across speakers. One might argue here, as in the case of *not believe*, that we may be dealing with two distinct lexical items, or with lexical coercion. However, the challenge then would be to find a way to regulate the choice of the right lexical item or employ coercion in a principled enough manner to derive the observed patterns of distribution.

Setting this aside, such examples are instructive in showing that s-selection cannot be a fixed property of a predicate. They suggest a more central role for semantic composition in producing the effects of s-selection. The same conclusion is reached by McCloskey (2006) who starts from a different set of observations. He notes that rogative predicates differ from responsive predicates in allowing inversion in their interrogative complements, a root phenomenon. Although McCloskey's claim is primarily about Irish English, relative judgments hold more generally (McCloskey 2006: 87–8, 115):

- (46) a. She asked the stewards was any member of the committee in the hall.
 b. I wonder what is he like at all.
 c. Everybody wants to know did I succeed in buying chocolate for Winifred.

²² Huddleston (1993) does not entertain the possibility of these complements being relative clauses and our diagnostics support his position. The version with *which N* establishes that it cannot be a free relative.

- (47) a. *I found out how did they get into the building.
 b. *The police discovered who had they beaten up.
 c. *How many people should you invite depends on how big is your place.

It is important to keep in mind that the complement clauses in (46) are not direct quotes. The first person pronoun in (46c), which McCloskey notes is a naturally occurring example, refers to the speaker not the attitude holder. And a third person pronoun in the same context could be interpreted as varying with the universal quantifier in the matrix. The same would not, obviously, hold for direct quotes.

McCloskey initially posits distinct selectional properties for responsive and rogative predicates. The former take regular CP complements that denote questions, the latter take Force-P complements that denote question speech acts:²³

- (48) a. know [_{CP} C_{+WH}⁰ [_{TP}]]
 b. wonder [_{Force-P} QUEST [_{CP} C_{+WH}⁰ [_{TP}]]]

Formally distinguishing between these two types of predicates is not a radical move. We have seen proposals for doing so in previous chapters but, typically, those proposals tap into the intensional/extensional dimension. In Karttunen (1977) predicates that embed interrogatives take as their complement a function from worlds to sets of true propositions. Extensional predicates saturate the index of their complement and thus combine with sets of true propositions directly. The same holds for Gr&S (1982, 1984), adjusted for propositions instead of sets of propositions. In our baseline theory, questions denote the same set of propositions across indices, the Hamblin set. However, Ans-D is a function from worlds to propositions. As such, it replays the effect of the intensional/extensional distinction between rogative and responsive predicates of earlier proposals, a distinction that is not typically reflected in structural differences.²⁴

McCloskey's differentiation of the two predicate types is more fundamental. In treating inversion as a root phenomenon, he accords matrix clause status to complements of rogative predicates. The semantic underpinning comes from the theory of embedded illocutionary acts (Krifka 2001, 2014). McCloskey's explanation for (46)–(47) is that embedded clauses that show root phenomena do so because embedded Force-P makes them root clauses. However, the claim that inversion in embedded interrogatives is due to an embedded Force-P with a speech act operator faces some empirical challenges.

²³ The idea of structural differences in *wh* complementation is anticipated in discussions of Spanish. See, in particular, Suñer (1993) and Lahiri (2002a).

²⁴ One could also argue that rogative predicates embed smaller structures than responsive predicates. This would be motivated by appealing to the fact that the former select questions/sets of propositions, defined at the CP level, while the latter select propositions that are extracted from the CP denotation by an answerhood operator that occurs above it. This was mentioned as a possibility in Section 2.4 and is also briefly discussed in Lahiri (2002a: 147). Note, however, that if the difference between rogative and responsive predicates turns on intensionality, as in the baseline theory adopted in this book, a structural difference of this sort is hard to motivate.

Such clauses differ from direct questions and quotations in how they relate to modifiers that seem to operate at the level of speech acts. The questioner in (49a)–(49b) wants the addressee to give the answer quickly. Example (49c), instead, can only mean that the manner of asking was fast:²⁵

- (49) a. Quick/quickly, where did you hide the matza?
 b. Mary asked Sue, “Quick/quickly, where did you hide the matza?”
 c. Mary asked Sue quickly where she hid the matza / where did she hide the matza.

In fact, McCloskey himself introduces facts that undercut the generalizations on which his initial proposal is based. A predicate like *remember* which does not allow inversion, does so when it is in a polar question or is negated (McCloskey 2006: 112):

- (50) a. *I remember was Henry a Communist.
 b. Do you remember was Henry a Communist?
 c. ?I don’t remember was Henry a Communist.

McCloskey concludes that rogative and responsive predicates do not differ in their selectional properties. Since “the necessary discriminatory work is done by ultimately pragmatic conditions...we do not want to hardwire into the lexical entry of a resolute [responsive] predicate a constraint which forbids it to combine with a complement of the higher type” (McCloskey 2006: 116).²⁶

5.2.4 Section summary

C-selection and s-selection, we saw, are useful in talking about embedding but the terms themselves are open to interpretation. Current research makes it seem unlikely that s-selection is lexically specified for predicates in a deterministic manner. While we may continue to talk about a predicate selecting a particular type of complement, we need to be cautious about investing too much theoretical capital in this distinction. An adequate elaboration of the combinatorial possibilities, taking into account all aspects of meaning and conversational dynamics, may ultimately make a theory of s-selection redundant. Interestingly, Grimshaw herself anticipates this in her arguments for s-selection.

5.3 Concealed questions

As discussed in Sections 5.1 and 5.2, concealed questions are DPs that involve relations to questions and/or propositions derived from questions. Here we probe

²⁵ See Grimshaw (2013) and references cited there on the status of quotations as complements.

²⁶ See Dayal and Grimshaw (2009) for some discussion of the discourse conditions under which inversion in embedded clauses may be licensed.

the range of readings possible for them and the accounts that have been proposed to capture those readings. Broadly speaking, there are three possibilities that have been pursued in the literature. They have been taken to denote individual concepts, propositions, and questions. The proposals also expand the inquiry of concealed questions beyond definite noun phrases to include quantified noun phrases.

5.3.1 *Definite concealed questions*

Heim (1979) points out the interpretive complexity of concealed questions, setting the stage for current accounts of the phenomenon. Greenberg's (1977) distinction between identity questions and concealed questions provides the starting point:

- (51) a. John found out who the murderer of Smith was.
 b. John found out the murderer of Smith.
 c. $\lambda p \exists y [p = \lambda w' [y = \text{murderer-of-smith}(w')]]$

Take a situation where Jones is Smith's murderer and also happens to be the gardener. Also suppose that John has always known that Jones is Smith's murderer. What he has discovered is that the gardener and Jones are the same person. In such a situation (51a) is appropriate but (51b) is not. This casts doubts on Baker's (1968) claim that concealed questions are derived from interrogatives.

Maintaining standard assumptions about DP meanings, one would ascribe to concealed questions an individual type meaning: type *e* or type $\langle s, e \rangle$. Heim suggests working with a simple individual level meaning and building in a pragmatic component. The idea is that the description used is the contextually salient property of the individual referred to. Thus, if we are talking about John discovering something and using the description *murderer of Smith*, then that is the salient property that John has discovered about the individual Jones.

The logical form of (52a) could mean that John knows of Rome that it has a contextually salient property *P* which, given the choice of the noun phrase, would be the property of being Italy's capital:

- (52) a. John knows the capital of Italy.
 b. $\text{know}(j, x, P)$
 c. John knows the largest city in Italy.

In addition to defusing the expectation that concealed questions should be synonymous to identity questions, this move also explains why (52c) does not follow from (52a). If *P* is the *capital of Italy* the property of being Italy's largest city does not follow.

Romero (2005) critiques this line of explanation, pointing out that a logical form like (52b) suggests that it means the same as *John knows Rome*. That is, it should also have the missing acquaintance reading (see also Nathan 2006). Romero's solution is to shift to an intensional version of Heim's proposal. The

embedding predicate *know* is defined to combine with concealed questions, interpreted as individual concepts:

- (53) a. $\llbracket \text{know}_{\text{CQ}} \rrbracket = \lambda y_{\langle s, e \rangle} \lambda x \lambda w \forall w' \in \text{Dox}_x(w) [y(w') = y(w)]$
 b. $\llbracket \text{the capital of Italy} \rrbracket = \lambda w \iota x [\text{capital-of-Italy}(x, w)]$
 c. $\llbracket \text{John knows the capital of Italy} \rrbracket =$
 $\lambda w \forall w' \in \text{Dox}_j(w) [\iota x [\text{capital-of-Italy}(x, w')] = \iota x [\text{capital-of-Italy}(x, w)]]$

Formula (53c) is true in those worlds w such that every belief world of John's w' (represented as Dox_j) is such that Italy's capital in world w is the same as Italy's capital in w' . This meaning of *know* is presented as a simple generalization of the one that takes clausal questions. However, it is defined on individual concepts and builds in the identity between the extensions of the individual concept at the actual world and the worlds in John's doxastic alternatives. Thus, we no longer have the problematic acquaintance relation and we are able to get the reading that John knows the identity of Italy's capital.

Romero points out that her approach can also account for an ambiguity with nested concealed questions, discussed by Heim:

- (54)
 a. John knows the price that Fred knows.
 b. Reading A: John knows the same price that Fred knows.
 There are several relevant questions about prices:
 “How much does the milk cost?” “How much does the oil cost?”
 Fred knows the answer to exactly one of them, let's say he knows what the price of milk is. Then, John too knows the answer to that question.
 c. Reading B: John knows what price Fred knows.
 There are several relevant questions about prices:
 “How much does the milk cost?” “How much does the oil cost?”
 Fred knows the answer to exactly one of them, let's say he knows the answer to the first question. Then, John knows that Fred knows the answer to the question: “How much does the milk cost?”

These readings can be captured by minimally adjusting the meaning of *know* to take intensions as well as extensions of the individual concept:

- (55) a. Reading A: $\llbracket \text{know} \rrbracket + \text{extension of } [\text{NP the price that Fred knows}] \Rightarrow$
 $\lambda w \forall w' \in \text{Dox}_j(w)$
 $[\iota x_{\langle se \rangle} [\text{price}(x, w) \ \& \ \forall w'' \in \text{Dox}_f(w) [x(w'') = x(w)]](w') =$
 $\iota x_{\langle se \rangle} [\text{price}(x, w) \ \& \ \forall w'' \in \text{Dox}_f(w) [x(w'') = x(w)]](w)]$
 b. Reading B: $\llbracket \text{know} \rrbracket + \text{intension of } [\text{NP the price that Fred knows}] \Rightarrow$
 $\lambda w \forall w' \in \text{Dox}_j(w)$
 $[\iota x_{\langle se \rangle} [\text{price}(x, w') \ \& \ \forall w'' \in \text{Dox}_f(w') [x(w'') = x(w')]] =$
 $\iota x_{\langle se \rangle} [\text{price}(x, w) \ \& \ \forall w'' \in \text{Dox}_f(w) [x(w'') = x(w)]]]$

Although the view that concealed questions denote individual concepts originates with Heim, she herself raises concerns about the power of the system. While most

of those objections are addressed by Romero, Nathan points out that the moves that seem reasonable enough for *know* may not apply as easily to other predicates that embed concealed questions. He also notes that the restriction to relational nouns does not feature in the explanation.

Nathan's analysis of concealed questions has two goals. One is to derive their distribution, which we saw, he takes to belong with predicates that select both questions and propositions. The other is to restrict concealed questions to DPs with relational head nouns. To meet the first goal, he derives a proposition-level meaning for such DPs. Crucially, however, this proposition is built on their denotation as individual concepts, something like $\iota p \exists x_e [p = \lambda w' x = \text{the-price-of-milk}(w')]$ for *the price of milk*. He takes relational nouns to denote sets of individual concepts, sorted on the basis of the relational head noun's internal argument. *Governor*, for example, denotes the set (56a), sorted by the New England states. That is, the set of individual concepts are the relations, *governor of x*: x a New England state (Nathan 2006: 97):

(56)

- a. $\left[\begin{array}{l} [\text{CT 2002} \rightarrow \text{Rowland, 2005} \rightarrow \text{Rell}], [\text{ME 2002} \rightarrow \text{King, 2005} \rightarrow \text{Baldacci}], \\ [\text{MA 2002} \rightarrow \text{Swift, 2005} \rightarrow \text{Romney}], [\text{NH 2002} \rightarrow \text{Benson, 2005} \rightarrow \text{Lynch}], \\ [\text{RI 2002} \rightarrow \text{Almond, 2005} \rightarrow \text{Carcieri}], [\text{VT 2002} \rightarrow \text{Dean, 2005} \rightarrow \text{Douglass}] \end{array} \right]$

- b. $\llbracket \text{governor} \rrbracket_{\langle \text{se}, t \rangle} = \lambda x_{\text{se}} \exists y_e \forall w [x(w) \text{ is the governor of } y \text{ at } w]$

Nathan further takes the basic meaning of relational nouns to be type $\langle e, \langle e, t \rangle \rangle$. Drawing on Dowty (1979), he derives $\langle e, t \rangle$ and $\langle \langle s, e \rangle, t \rangle$ meanings by existentially binding the internal argument, as shown in (57):

- (57) a. $\llbracket \text{governor} \rrbracket^w = \lambda y_e \lambda x_e \llbracket \text{governor}_w(y)(x) = 1 \rrbracket \quad \langle e, \langle e, t \rangle \rangle \text{ (lexical)}$
 b. $\llbracket \text{governor} \rrbracket^w = \lambda x_e \exists y_e \llbracket \text{governor}_{\langle e, et \rangle}^w(y)(x) \rrbracket \quad \langle e, t \rangle$
 c. $\llbracket \text{governor} \rrbracket^w = \lambda x_{\text{se}} \exists y_e \forall w_1 \llbracket \text{governor}_{\langle e, et \rangle}^{w_1}(y)(x(w_1)) \rrbracket \quad \langle \text{se}, t \rangle$

Example (57a) maps its argument to a set of individuals and is used in *the governor of Vermont*. Example (57b) denotes a set and is used in *every governor voted*. The intensional meaning in (57c) is needed for cases like *the governor of Vermont changed*.

The final meanings of concealed questions are propositional but derived via question denotations (Nathan 2006: 108–9). Example (58a) shifts the meaning to a function from individuals to sets of propositions, which in his system are Karttunen sets. In a specific case, we get something like (58b). Determiners combine in the expected way. In the case of *the*, for example, *iota* would impose existence and uniqueness presuppositions on this set. Note that the restriction to relational nouns is built into the system:

- (58) a. $\lambda P_{\langle s, \langle e, et \rangle \rangle} \lambda y_{\langle e \rangle} \lambda p_{\langle s, t \rangle} [\exists x_e p = \lambda w_1 P(w_1)(y)(x)]$
 b. $\lambda p \exists x_e [p = \lambda w_1 [x \text{ is the governor of Vermont in } w_1]]$

Nathan discusses differences among relational nouns which I set aside. I also set aside discussion of how non-relational nouns can have concealed question readings when accompanied by explicit modification: *The picture on Jordan's wall*. Nathan's proposal combines the insights of Romero's development of Heim's account and, as such, captures all the readings she does. Additionally, he addresses the distributional restrictions on embedding as well as the internal restriction to relational nouns. In that sense, Nathan's synthesis has wider potential than the original versions of the account.

5.3.2 Quantified concealed questions

So far we have dealt with definite noun phrases but quantified noun phrases can also have concealed question interpretations. In fact, this is predicted since it would be hard to block any determiner from combining with a head noun once we have a way of building up a concealed question meaning from an $\langle e, t \rangle$ or $\langle \langle s, e \rangle, t \rangle$ meaning for it. The existence of quantified concealed questions is noted in Heim (1979), who points out that they are ambiguous between a "set" and a "pair-list reading":

- (59) a. Mary knows every capital city.
 b. Pair-list reading: For every country x , Mary knows what the capital of x is.
 c. Set reading: For every capital city z , Mary knows that z is a capital (of some country).

We will discuss these cases, drawing on Frana (2013). Since the pair-list reading of a quantified concealed question like (59a) entails the set reading, but not the other way around, she starts with the former. The relevant generalization is in (60):

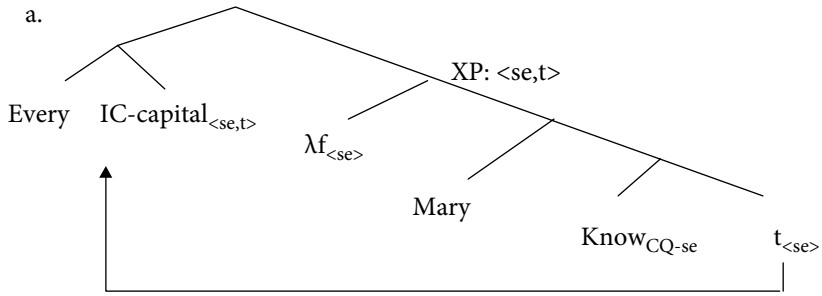
- (60) Distribution of Pair-list and Set Readings (Frana 2013:181):

Pair-list readings arise with CQs headed by relational nouns whose internal argument has not been saturated (2-place predicate NPs; semantic type $\langle e, \langle e, t \rangle \rangle$). Set readings arise with CQs headed by nouns that are not—or no longer—relational (1-place predicate NPs; semantic type $\langle e, t \rangle$).

Frana suggests that quantified concealed nouns have two incarnations, one in which the internal argument is saturated and one in which it is not. The idea is that it is possible for a lexically available argument to be existentially closed, but hard for an expression to add an internal argument.

Frana extends Romero's analysis by incorporating Nathan's idea of meaningfully sorted concepts and appealing to QR to capture the ambiguity noted above (Frana 2013: 192):

(61) a.

b. $\llbracket \text{SHIFT}(\text{capital}) \rrbracket^w = \lambda f_{\langle s,e \rangle} \forall w^* [\text{capital}_{w^*}(f(w^*))]$

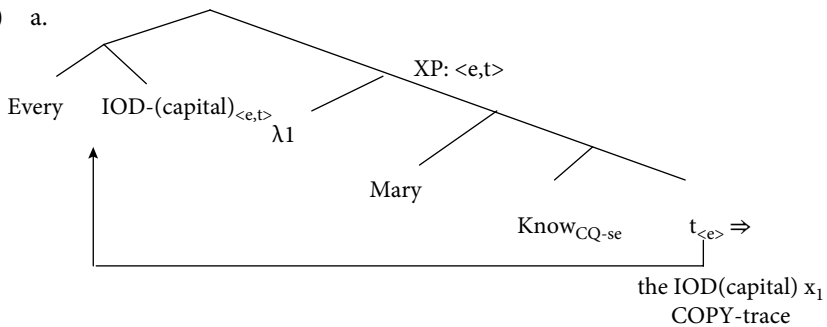
$$\left\{ \begin{bmatrix} w \rightarrow \text{Rome} \\ w' \rightarrow \text{Rome} \\ c1(\text{Italy}) \end{bmatrix}, \begin{bmatrix} w \rightarrow \text{Paris} \\ w' \rightarrow \text{Paris} \\ c2(\text{France}) \end{bmatrix} \right\}$$

c. $\llbracket \text{Mary knows every capital city} \rrbracket =$
 $\forall f_{\langle se \rangle} (\exists x_c \forall w^* [f(w^*) \text{ is the capital of } x \text{ at } w^*] \rightarrow$
 $\forall w' \in \text{Dox}_m(w) [f(w') = f(w)])$

Nathan's type shifter interprets *capital* as the set of concepts (61b), where the functions correspond to concepts individuated by country. Example (61c) is true iff every properly individuated capital concept yields the same value in the actual world w as in Mary's belief worlds w' . This is the pair-list reading.

To get the set reading of relational nouns (as well as non-relational nouns), Frana imports the operation of indefinite object deletion from Nathan (2006). Once the first set is deleted from the set of functions, the concealed question becomes functionally equivalent to a non-relational noun phrase. The trace conversion principle of Fox (2002) provides the final ingredient of the analysis (Frana 2013: 202):

(62) a.



b. Modified trace conversion.

Variable Insertion: $(\text{Det}_n) \text{Pred} \rightarrow (\text{Det}_n) [\text{Pred } \lambda y (y = x_n)]$ Determiner Replacement: $(\text{Det}_n) [\text{Pred } \lambda y (y = x_n)] \rightarrow \max [\text{Pred } \lambda y (y = x_n)]$

- c. $\forall x_e (\exists u_e [\text{capital}(u)(x) \text{ in } w] \rightarrow \forall w' \in \text{Dox}_m(w) (\llbracket \text{the IOD}(\text{capital}) x_1 \rrbracket^{w', g(1/x)} = \llbracket \text{the IOD}(\text{capital}) x_1 \rrbracket^{w, g(1/x)}))$
- d. $\forall x_e (\exists u_e [\text{capital}(u)(x) \text{ in } w] \rightarrow \forall w' \in \text{Dox}_m(w) (\exists z [\text{capital}(z)(x) \text{ at } w']))$

Example (62c), equivalent to (62d), is true iff for every x that is actually a capital of some place, Mary knows that it is the capital of some place. The trace conversion principle, by building in the descriptive content into the meaning of the trace, allows the individual concept approach to concealed questions to capture the full range of readings available to them, while retaining the restrictiveness of Nathan's system.²⁷

5.3.3 Concealed questions under covers

We have seen persuasive arguments against treating concealed questions as having the same meaning as identity questions. However, there is a version of this account, due to Roelofsen and Aloni (2008) and Aloni and Roelofsen (2011), that can withstand the criticisms leveled against earlier versions of the approach. Aloni and Roelofsen start with the idea of conceptual covers (Aloni 2001), which allows for the same set of individuals to be differentiated by different descriptions:

- (63) a. $\{x, y\}$
 b. $\{\text{the card on the left, the card on the right}\} = \text{by position}$
 c. $\{\text{the Ace of hearts, the Ace of spades}\} = \text{by suit}$
- (64) a. Conceptual Covers: Given a set of possible worlds W and a domain of individuals D , a conceptual cover CC based on (W, D) is a set of functions $W \rightarrow D$ such that:
 $\forall w \in W: \forall d \in D: \exists! c \in CC: c(w) = d$

All quantification in this system works on conceptual covers but we will demonstrate only its particular use in the interpretation of concealed questions.

Consider the set and pair-list readings of universally quantified concealed questions discussed in Section 5.3.2.²⁸ The two specific pieces of the apparatus needed are a type-shifting mechanism and a contextually driven resolution for property P .²⁹

- (65) a. John knows every phone number.
 b. $\uparrow(z, P) \alpha =_{\text{def}} ?z.P(\alpha)$

²⁷ Frana (2013) does not directly address the distribution of concealed questions under the IC approach that she adopts, but see Frana (2010) for discussion.

²⁸ Aloni and Roelofsen (2011) use the terms *specificational* for pair-list readings and *predicational* for set readings.

²⁹ I follow Aloni and Roelofsen's convention of using $?$ to represent the logical form of a question.

- c. $x \rightarrow \{\text{Ann's phone number, Bill's phone number...}\}$
 $z \rightarrow \{5403, 5431...\}$
 $P = \lambda y. y = z$ (the *identity* property)
 or the property expressed by the CQ noun phrase.

- (66) a. $\forall x (\text{phonenum}(x) \rightarrow K_j (?z. P(x)))$
 b. $\forall x (\text{phonenum}(x) \rightarrow K_j (?z. z = x))$
 c. $\forall x (\text{phonenum}(x) \rightarrow K_j (?z. \text{phonenum}(x)))$

LF (66a), the logical form of (65a), uses the type-shift schema in (65b). Resolving *P* as the identity property derives the pair-list reading (66b) and resolving it with the property expressed by the noun *phone number* derives the set reading (66c). In the latter case, the choice of a conceptual cover becomes irrelevant.

As Aloni and Roelefsen note, this analysis differs from the analyses discussed earlier in introducing the dimension of perspective, which greatly expands the range of potential readings. They argue for this widened empirical domain on the basis of examples like (67):

- (67) a. John knows every capital that Fred knows.
 b. $\forall x [(\text{capital}(x) \wedge K_f (?z_1. P_1(x))) \rightarrow K_j (?z_2. P_2(x))]$

Focusing on the reading where (67a) says something about John's knowledge in contrast/comparison to Fred's knowledge (Heim's reading A), they identify three different resolutions of the relevant variables, leading to three distinct interpretations:

- (68) a. Set reading: A
 $x \rightarrow \{\text{Rome, Berlin, Paris...}\}$
 $P_1, P_2 \rightarrow \text{CAPITAL}$
 Given this resolution: $\forall x [(\text{capital}(x) \wedge K_f (?capital(x))) \rightarrow K_j (?capital(x))]$
- b. First pair-list reading: A (*capitals given, countries asked*)
 $x \rightarrow \{\text{Rome, Berlin, Paris...}\}$
 $z_1, z_2 \rightarrow \{\text{the capital of Italy, the capital of Germany...}\}$
 $P_1 \rightarrow \lambda y. y = z_1; \quad P_2 \rightarrow \lambda y. y = z_2$
 Given this resolution: $\forall x [(\text{capital}(x) \wedge K_f (?z_1. x = z_1)) \rightarrow K_j (?z_2. x = z_2)]$
- c. Second pair-list reading: A (*countries given, capitals asked*)
 $x \rightarrow \{\text{the capital of Italy, the capital of Germany...}\}$
 $z_1, z_2 \rightarrow \{\text{Rome, Berlin, Paris...}\}$
 $P_1 \rightarrow \lambda y. y = z_1; \quad P_2 \rightarrow \lambda y. y = z_2$
 Given this resolution: $\forall x [(\text{capital}(x) \wedge K_f (?z_1. x = z_1)) \rightarrow K_j (?z_2. x = z_2)]$

In contexts where Fred and John are given a list of cities and asked to pick out those that are capitals, (68a) represents a reading in which for every city *x* such that Fred knows *x* is a capital, John also knows that *x* is a capital. In contexts where

they are given a list of cities and are asked to specify the countries they are capitals of, (68b) represents a reading in which for every x such that Fred can specify the country x is a capital of, John can also specify the country x is a capital of. In contexts where they are given a list of countries and asked to name their capitals, (68c) represents a reading in which for every country y that Fred can specify y 's capital, John can specify y 's capital as well.

Similarly, Aloni and Roelofsen note a range of readings when John's knowledge is about Fred's knowledge. This is represented by having Fred's knowledge K_f ($?capital(x)$) be part of John's knowledge K_j . This adds the possibility of the mixed readings in (69b)–(69c):

- (69) a. Set Reading: B
 $x \rightarrow \{\text{Rome, Berlin, Paris...}\}$
 $P_1 \rightarrow \text{capital}; \quad P_2 \rightarrow \lambda x. \text{CAPITAL}(x) \wedge K_f(?z_1. P_1(x))$
 Given this resolution:
 $\forall x [(\text{capital}(x) \wedge K_f(?capital(x))) \rightarrow K_j(?(\text{capital}(x) \wedge K_f(?capital(x))))]$
- b. First Mixed reading: B (*capitals given, countries asked*)
 $x \rightarrow \{\text{Rome, Berlin, Paris...}\}$
 $z_1 \rightarrow \{\text{the capital of Italy, the capital of Germany...}\}$
 $P_1 \rightarrow \lambda y. y = z_1; \quad \text{PL}$
 $P_2 \rightarrow \lambda x. \text{CAPITAL}(x) \wedge K_f(?z_1. P_1(x)) \quad \text{Set}$
 Given this resolution:
 $\forall x [(\text{capital}(x) \wedge K_f(?z_1. x = z_1)) \rightarrow K_j(?(\text{capital}(x) \wedge K_f(?z_1. x = z_1)))]$
- c. Second Mixed reading: B (*countries given, capitals asked*)
 $x \rightarrow \{\text{the capital of Italy, the capital of Germany...}\}$
 $z_1 \rightarrow \{\text{Rome, Berlin, Paris...}\}$
 $P_1 \rightarrow \lambda y. y = z_1 \quad \text{PL}$
 $P_2 \rightarrow \lambda x. \text{CAPITAL}(x) \wedge K_f(?z_1. P_1(x)) \quad \text{Set}$
 Given this resolution:
 $\forall x [(\text{capital}(x) \wedge K_f(?z_1. x = z_1)) \rightarrow K_j(?(\text{capital}(x) \wedge K_f(?z_1. x = z_1)))]$

Aloni and Roelofsen also note some exceptions to Greenberg's observation (cf. (51)). The crucial reading of (70) is one where Billy names Barak Obama, without knowing that he is Malia and Sasha's father—the content of the description:

- (70) Michelle Obama tells her daughters Malia and Sasha: Today I went to visit a primary school in the neighborhood. At some point one of the children—I think his name was Billy—was asked to name the president of some American countries. But, poor kid, he only knew your father.

While the flexibility of working with conceptual covers allows them to capture a wide range of readings, Aloni and Roelofsen are aware of the dangers of over-generation. They appeal to default resolutions for P and the domain of z to address this issue:

- (71) a. P is by default resolved to the property of being identical to z or to the property denoted by the head noun of CQ.
 b. z is by default taken to range over the *naming* cover, the rigid cover (if the individuals in the domain can be identified by ostension), a derived cover based on a functional CQ noun, if salient.

Deviation is licensed only if necessary to avoid contradictory/trivial/irrelevant meanings (maxims of *quantity*, *quality*, *relevance*) and only if the meaning obtained by shifting to a non-default resolution could not have been expressed by a more perspicuous/effective form (*manner* as blocking). Thus, a strong pragmatic component is required for a principled account within this system.

5.3.4 Section summary

Concealed questions, we saw, present the opposite situation to free relatives in terms of the mismatch between syntax and semantics: noun phrase syntax with clausal semantics versus clausal syntax with noun phrase meaning. Noun phrases with a range of determiners were shown to have the potential for concealed question interpretations. To keep concealed questions separate from full blown identity questions, they are interpreted at the level of DPs with appropriate type shifts deriving question meanings. While concealed questions do not have some readings that full identity questions do, they are ambiguous in other ways. We saw that in order to make serious headway in this domain, careful navigation between semantic and contextually sensitive aspects of meaning is needed.

5.4 Quantificational Variability Effects

The term *Quantificational Variability Effects* (QVE) in the context of interrogatives refers to a phenomenon in which an adverb of quantification in the matrix clause seems to quantify over wh expressions inside a complement clause. Proposals to capture QVE include direct quantification over individuals, quantification over propositions, quantification over (sub)questions, as well as over algebraic structures built on answers to questions. We explore these proposals in this section.

5.4.1 Quantifying over propositions

The following, from Berman (1991, 1994), illustrates QVE:

- (72) a. Sue mostly remembers what she got for her birthday.
 b. Bill knows, for the most part, what they serve for breakfast at Tiffany's.

Sentence (72a) has a salient reading in which Sue remembers most of the presents she got for her birthday and (72b) has the reading that Bill knows most of the items they serve for breakfast at Tiffany's. Examples like (72a) also have a

frequency reading that describes what most of Sue's memories are about. In this case, the adverb does not interact with the embedded question, modifying the matrix verb instead. This is not the reading of interest here.³⁰ In many cases the only, or the most plausible, reading available is the one where the adverb seems to quantify over the *wh* expression(s), the QVE reading. This calls for an explanation.

The embedding predicates that participate in QVE are the responsive predicates which select both questions and propositions. The adverbs in (73a)–(73b), which have rogative embedding predicates, modify the extent of the wondering/asking. They do not relate to the objects denoted by the *wh* expressions in the embedded clause:

- (73) a. Sue mostly wonders what she got for her birthday.
b. For the most part, Bill asks what they serve for breakfast at Tiffany's.

While this generalization will be modified to some extent, it provides a starting point for our discussion.

Berman's account brings together two lines of inquiry. One follows the Lewis–Kamp–Heim analysis of variation between existential and generic construals of indefinites (74)–(75). The other follows Hintikka's (1976) account of questions as open sentences with variable operators ((76)). This captures the *mention-some* and *mention-all* interpretations discussed in Chapter 3:

- (74) a. A student is working hard/ Students are working hard.
b. $\exists x$ [student(*x*) $\wedge$ working-hard(*x*)]
(75) a. A student (usually) works hard/ Students (usually) work hard.
b. $\forall x$ / Most *x* [student(*x*)] [work-hard (*x*)]
(76) a. John knows where to buy gas.
b. $\exists x$ [one can get gas at *x*] [John knows one can get gas at *x*]
c. $\forall x$ [one can get gas at *x*] [John knows one can get gas at *x*]

Combining them, Berman proposes logical forms like (77a) for the cases under discussion. The challenge, of course, is to block something similar for rogative predicates ((77b)). Berman locates this difference in the ability of the two predicates to embed propositions:

³⁰ The choice of adverb is important. Berman notes that they come from three distinct classes: adverbs of frequency (*seldom, usually*), adverbs of quantity (*mostly, for the most part, partly*) and adverbs of precision (*exactly, precisely, in detail*). Adverbs of quantity are the most reliable for testing QVE since they can embed interrogatives with episodic interpretations and still maintain the relevant effects. Episodicity is significant as it eliminates the possibility of quantifying over events or situations related to the embedded verb.

- (77) a. Most (x) [they serve x for breakfast at Tiffany's]
 [Bill knows that they serve x for breakfast at Tiffany's]
 b. *Most (x) [they serve x for breakfast at Tiffany's]
 [Bill wonders that they serve x for breakfast at Tiffany's]

A second challenge is to introduce the content of the embedded question into the restriction of the adverb ((77a)). Here Berman appeals to the factivity of the embedding predicate *know/remember/forget*. The logical form results from the projection of the presupposition into the restriction. Berman's account, we see, relies on an alternative to the analysis of interrogatives as denoting sets of propositions. QVE, in turn, provides motivation for adopting that alternative.

Berman's account, though intuitively appealing, has been challenged by Lahiri (1991, 2000, 2002a). Here is one empirical argument from Lahiri (2000: 377–81). The two sentences (78a)–(78b) come out equivalent under Berman's approach, (78c). The right truth conditions for them are (79a)–(79b) instead:³¹

- (78) a. John knows which French people are philosophers.
 b. John knows which philosophers are French.
 c. $\forall x[\text{French}(x) \wedge \text{philosopher}(x)]$ [John knows that $[\text{French}(x) \wedge \text{philosopher}(x)]$]
- (79) a. $\forall x[\text{French}(x) \wedge \text{philosopher}(x)]$ [John knows that $[\text{philosopher}(x)]$]
 b. $\forall x[\text{French}(x) \wedge \text{philosopher}(x)]$ [John knows that $[\text{French}(x)]$]

As Lahiri notes, the difference follows under an analysis of questions as denoting sets of propositions:

- (80) a. $\lambda p \exists x [\text{French}(x) \wedge p = \wedge \text{philosopher}(x)] \Rightarrow$
 $\{\wedge \text{Sartre is a philosopher}, \wedge \text{Proust is a philosopher}\}$
 b. $\lambda p \exists x [\text{philosopher}(x) \wedge p = \wedge \text{French}(x)] \Rightarrow$
 $\{\wedge \text{Sartre is French}, \wedge \text{Russell is French}\}$

Another problem, pointed out by Williams (2000), has to do with non-interrogative complements with bare plurals:

- (81) a. For the most part, Al knows that undergraduates got drunk.
 b. *Most x [undergraduate (x) $\wedge$ got drunk (x)] [Al knows that x got drunk]

Sentence (81a) has a factive verb taking a proposition with a bare plural. Under Berman's assumptions, this should be a prime candidate for quantificational variability (cf. (74)–(75)) but QVE is unavailable. The quantificational domain

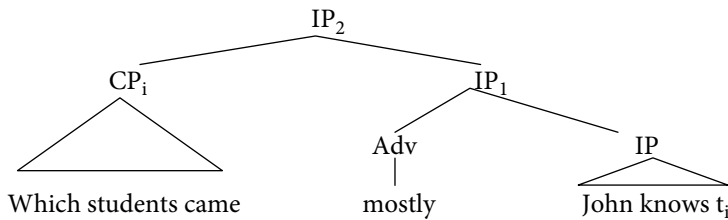
³¹ The difference is even clearer with predicates like *found out*.

of the adverb, clearly, cannot be satisfied by individuals drawn from expressions inside the complement.

Berman's account of QVE is notable for drawing attention to a phenomenon that had the potential of fundamentally altering our view of question denotations. His explanation has the appealing feature of using ingredients that are independently motivated. Even so, the account has not been accepted because of the problems we have noted here.

Lahiri shows that QVE can be explained within the standard view of questions as denoting propositional sets. He approaches the phenomenon in two ways, through quantification over propositions and through amount quantification over an algebraic structure defined over answers. We will discuss the first version here, deferring the second to Section 5.4.3. Taking Berman's separation of rogative and responsive predicates as his starting point, Lahiri gives a type-theoretic explanation that rests on the view that interrogatives always denote sets of propositions. While this matches the s-selectional properties of rogative predicates, it clashes with the s-selectional properties of responsive predicates, which are defined on propositions. In order to repair this type mismatch, Lahiri appeals to a rule of Interrogative Raising (IR), analogous to standard QR. **Most** is the propositional level counterpart of the standard quantifier *most*:

(82) a.



b. **most** p [$p \in \lambda p' \exists x$ [student(x) $\wedge$ $p = \wedge$ came(x)] $\wedge$ C(p)] [know (j , p)]

The domain of quantification is the set of propositions in the question denotation, restricted by the variable C . In this case C is the truth predicate. This yields the same results as in Berman, where quantification is over individuals. It also accounts for the differential distribution of QVE in rogative vs. responsive predicates without extra stipulation. While *knowing* a question amounts to knowing the propositions in its denotation, *wondering* about a question does not reduce to wondering about the individual propositions in its denotation.

Lahiri expands Berman's paradigm to include non-veridical predicates and shows that they align with factive responsive predicates in relevant respects:

- (83) a. John partly guessed who came to the party.
b. John told Bill, in part, who came to the party.

- (84) a. John is certain, for the most part, about who Mary loves.
b. John is convinced, in part, about who Mary's ex-lovers are.
c. John and Bill agree, for the most part, on who Mary's ex-lovers are.

The predicates in (83) arguably have both factive and non-factive uses (Kiparsky and Kiparsky 1970). More specifically, *tell* has been the canonical argument for building in factivity somewhere within the interrogative meaning itself. The predicates in (84), however, undercut the reliance on factivity in Berman's approach since they remain non-veridical even when they embed interrogatives. The relevant readings are as follows:

- (85) a. Of all the people that John thinks Mary is likely to love, he is certain about most of them.
 b. Of all the people John thinks may be Mary's ex-lovers, he is convinced about some of them.
 c. Of those people who both John and Bill think are Mary's ex-lovers, they agree on most of them.

Since Mary is not required to actually love or have loved the relevant individuals, there is no factive presupposition of the embedding predicate that the domain restriction can be traced to. However, the predicates have other lexical presuppositions that serve to carve out the domain of quantification in the relevant way. Note that these facts do not argue against this aspect of Berman's account. Rather, they argue for a greater role for the process of presupposition projection by establishing that factivity is not the only source for it. Lahiri's real contribution is to offset the challenge to a view of questions as denoting propositional sets posed by Berman's account.

5.4.2 *Quantifying over subquestions*

Lahiri's arguments against quantifying over individuals are generally accepted in the literature. Beck and Sharvit (2002) extend the discussion further by modifying the generalization about rogative predicates. They claim that all predicates that can embed a question can admit QVE, but in order to do so the adverb must have a well-defined quantificational domain. Considered out of the blue, only the structure itself can establish the existence of the domain of quantification. This works for predicates that presuppose that the embedded question has a complete true answer, namely responsive predicates and rogative predicates like *depend on*. Canonical rogative predicates, instead, are dependent on external factors for this. Beck and Sharvit point to examples like (86a) and the QVE reading (86b) in support:

- (86) a. Who will be admitted depends for the most part (exclusively) on this committee.
 b. For most people, it depends exclusively on this committee whether they will be admitted.
 c. This committee is an important factor in deciding who will be admitted.

Beck and Sharvit (2002: 116) define an operation *Part* on question denotations, which yields a set of polar subquestions for a constituent question:³³

- The QVE reading (86b) for (86a) comes from quantifying over the relevant subquestions:

- The move to quantifying over subquestions has many implications, discussed in the paper, that merit a closer examination. However, I will confine myself here to the primary motivation for their account, namely the possibility of QVE with predicates that do not select propositions, such as *depend on*.

(89) a. A: Has John found out which students cheated?
B: No. For the most part, he's still wondering.
b. For most x, John is still wondering whether x cheated.
c. Most Q [Q ∈ Part(⟦which students cheated⟧)](w)
[John wonders about Q in w]

- ³² Although predicates like *depend on* are rogative in the sense that they cannot take declarative complements, they can be separated from canonical rogative predicates like *wonder/inquire*. See fn 13 for one way of doing so.

³³ The definition given is a preliminary version that works for the phenomenon we are focusing on here. There are other cases that prompt the authors to refine the definition.

The crucial factor in (89)–(90), they argue, is that focus has been removed from the embedded question, making it “topical.”³⁴ They note that responsive predicates as well as *depend on* (though not *decide*) can have QVE readings even when the question is in focus.

Beck and Sharvit further argue that in order for the QVE reading to emerge, an existence presupposition about the complete true answer to the embedded question has to be satisfied. The canonical QVE predicates *know/tell/agree on/be certain* all presuppose this. They present examples like (91) as justification:

- (91) a. Mary is certain who ate the strawberries
(but there were no strawberries/#but she doesn’t believe that there were strawberries).
b. Mary and Joe agree on who ate the strawberries
(but there were no strawberries/#but they don’t/#Mary doesn’t believe that there were any strawberries).

Note that this presupposition is not about the truth of the complement. The predicates in (91a)–(91b) are non-veridical, but the impossibility of the continuations reveals that they have existential presuppositions consistent with the predicates’ lexical semantics. Beck and Sharvit note, furthermore, that such a presupposition is missing from *wonder* and *ask*. Sentence (92a) seems coherent. Example (92b), instead, requires the typical metalinguistic effort associated with the negation of presuppositions (Beck and Sharvit 2002: 145). Sentence (92c) completes the paradigm. Although rogative predicates do not have presuppositions about the existence of an actual true answer qua speaker (92a), they have such a presupposition qua the attitude holder (92c):

- (92) a. Mary isn’t wondering who ate the strawberries because there were no strawberries.
b. Mary doesn’t know who ate the strawberries
(?? because there were no strawberries).
c. Mary is wondering who ate the strawberries
(#but she doesn’t believe there were strawberries).

If what is required for QVE is a well-defined quantificational domain, whereby the subquestions together provide the actual answer to the question, the (im)possibility of QVE with rogative predicates turns on the level of contextual support available. Topicality amounts to the question being part of the common ground, and thus implying the existence of a complete true answer.

Beck and Sharvit refine their proposal to capture a wide range of facts that are otherwise elusive. Nevertheless, there remain some unresolved issues. One such

³⁴ See also Lahiri (2000, 2002a) on what he terms focus affected readings and QVE.

issue is the unavailability of QVE with singular embedded wh (Beck and Sharvit 2002: 129–31):

- (93) a. #Jill partly/mostly knows which boy cheated.
 b. $\text{Part}(\llbracket \text{which boy cheated} \rrbracket(w)) = \{\text{Did Bill cheat? Did John cheat?}\}$

The problem is partly addressed by requiring each answer to a subquestion to be strictly partial, and only in conjunction with answers to the other subquestions to provide the complete answer to the complement. Given the singularity presupposition of *which boy cheated* only one of the subquestions can yield a true answer and this violates the requirement of strict partiality. However, Beck and Sharvit explicitly argue for allowing QVE with strong exhaustivity. As they note, this now makes it possible for each subquestion in *which boy cheated* to be strictly partial: *Bill cheated* and *John didn't cheat* together provide the strongly exhaustive answer. This leads to the incorrect prediction that (93a) should be acceptable.

Surprise-predicates also raise some issues:

- (94) a. Jill was, for the most part, surprised who showed up.
 b. $\text{Part}(\llbracket \text{Who showed up} \rrbracket(w)) = \{\text{Did Bill show up? Did Sue show up? Did John show up?}\}$
 c. #Most Q [$Q \in \text{Part}(\llbracket \text{who showed up} \rrbracket(w))$] [Jill was surprised Q]

Since surprise-predicates do not select polar questions, (94c) cannot be the right analysis for (94a). One possibility is to shift the question to a proposition in order to combine with *surprise*. The standard move of using an answerhood operator does not seem to give the right result. If Jill expected Bill, Sue, and John to show up but only John did, (94a) would be judged false. However, the answer to the first two questions would be *Bill didn't show up* and *Mary didn't show up*. Since both propositions run counter to Jill's expectations, the sentence would be predicted to be true. Beck and Sharvit (2002: 142) are aware of the problem and take a different tack, arguing that in this case the subquestions are not polar questions but wh questions: *{which French student showed up? Which tall student showed up?}* which *surprise* can combine with.

5.4.3 Propositional conjunction algebras

We now return to Lahiri's account of QVE. We reviewed his initial proposal for quantifying over propositions in Section 5.4.1. Ultimately, Lahiri does not endorse that proposal, arguing instead for amount quantification over propositional conjunction algebras (PCAs). The two yield equivalent results but for two cases, which prompts Lahiri to shift to what he admits is a more complicated semantics. Let us first introduce the PCA-based account before evaluating the motivations for it, using the version in Lahiri (2000), which is essentially the account in Chapter 3 of Lahiri (2002a).

Lahiri starts with a Hamblin denotation for questions built out of atomic individuals. The PCA is defined on possible answers, where a possible answer is

the intersection of any subset of propositions in the question denotation (95a). A PCA is therefore a set of propositions, each of which is a conjunction of some subset of possible answers. Lahiri defines the supremum of a set of propositions as the proposition that entails all other propositions in the set, if there is one, and is undefined otherwise (95b).³⁵ The final piece of the analysis rests on defining sets of atoms of the algebra of answers (95c), where the members of the Hamblin set constitute the atoms:

- (95) a. $\text{Ans}(p, Q)$ iff $\exists S \in \text{Pow}(Q)[p = \cap S]$ (Lahiri 2000: 331)
 b. If φ is an open formula containing the propositional variable p , then the
 $\sigma_Q(p) [\varphi(p)] = \sup_Q \{p: \varphi(p)\} = \cap \{p: \varphi(p)\}$ (Lahiri 2000: 333)
 c. $\psi(a) = \{b: b \text{ is an atom} \wedge b \leq a\}$ (Lahiri 2000: 355)
 d. $\text{most}^+(P)(S) = 1$ iff $|\psi \sup_p(P) \otimes \sup_p(P \cap S)| > |\psi \sup_p(P) \otimes (\sup_p(P \cap S))^-|$ (Lahiri 2000: 358)

With these pieces in place, most^+ can be defined for PCAs ((95d)). Like normal quantifiers, it relates two sets. Here P is the set of true propositions (setting C to its default value), S the set of those propositions that the attitude holder stands in the appropriate relation to, and S^- the complement of that set.³⁶ The intersection of P and S is compared with the intersection of P and S^- according to an appropriate measurement function ψ and evaluated as true if the first exceeds the second. Let us unpack this with reference to a concrete example (Lahiri 2000: 358):

- (96) a. John, for the most part, found out which students came to the party.
 b. $\llbracket \text{which students came} \rrbracket = \{p_1, p_2, p_3\}$
where $p_1 = a \text{ came}, p_2 = b \text{ came}, p_3 = c \text{ came}$
 c. $\text{PCA} = \{p_1, p_2, p_3, p_1 \cap p_2, p_1 \cap p_3, p_2 \cap p_3, p_1 \cap p_2 \cap p_3\}$
 d. $\text{most}^+([\lambda p [\text{Ans}(p, Q) \wedge \vee p]])([\lambda p [\text{knows}(p)(j)]] = 1$ iff
 $|\psi \sup_p(A)| \geq |\psi \sup_p(B)|$, where
 A is the set of true propositions in $\text{Ans}(96b)$ that John knows and
 B is the set of true propositions in $\text{Ans}(96b)$ that he does not.

This yields the right truth conditions. If a , b , and c are all and only the individuals who came to the party and John knows of at least two of them that they did, (96a) will be true.

Let us now consider the two arguments Lahiri gives for moving to the PCA-based account, one from surprise-predicates and one from pair-list answers. Let us take each in turn. Consider a situation where John and Bill's being at the party together may be surprising (because they dislike each other's company, for

³⁵ That is, the supremum is what we get when we apply Ans-D to a set of propositions.

³⁶ I am simplifying considerably here. See Lahiri (2000, 2002a) for details on defining complementation on the algebra of answers. I am also ignoring the $\emptyset$ in computing the PCA.

example) but neither John's nor Bill's being there on their own is surprising. In such a situation, (97a) would be judged true (Lahiri 2000: 363):

- (97) a. It is surprising who came to the party.
 b. $\forall p [p \in Q \wedge p(w)] [\text{surprising}(p)]$
 c. $\llbracket \text{Who came to the party} \rrbracket = \{^{\wedge}j \text{ came}, ^{\wedge}b \text{ came}\}$

Lahiri argues that quantifying over propositions would make it false since the two propositions in the set (97c) are not surprising. The non-distributivity of such predicates, then, is empirical justification for moving away from quantifying over propositions.

It is not clear, however, that the conclusion Lahiri draws is warranted. The problem demonstrated in (97) is intractable only if plurality is not included in the system. If the denotation of the embedded question is as given in (97c), the only way to access both Bill and John's coming to the party is through the two propositions about them as individuals. But if number is factored into the calculus, the embedded question would include a distinct proposition referring to the group of John and Bill, hence to their joint appearance. I elaborate the conditions slightly to highlight the point under discussion:³⁷

- (98) a. $\llbracket \text{Who came to the party} \rrbracket = \{^{\wedge}\uparrow(j+b) \text{ came}, ^{\wedge}j \text{ came}, ^{\wedge}b \text{ came}\}$
 b. $\forall p [p \in Q \wedge p(w) \wedge \text{found-out}(\text{speaker}, p)] [\neg \text{expected}(\text{speaker}, p)]$
 c. $\exists p [p \in Q \wedge \text{found-out}(\text{speaker}, p) \wedge \neg \text{expected}(\text{speaker}, p)]$

All three propositions in (98a) are true in the situation under discussion, and in finding out that the group of John and Bill came to party, the speaker no doubt finds out that the atomic individuals John and Bill came to the party. That is, all three propositions will be in the domain of quantification. A default universal incorrectly predicts the sentence to come out false. There may, however, be a confound to this way of articulating the facts.

One possibility is that *surprise* may have a weak semantics ((98c)), with an existential as its default quantifier (cf. Section 5.2.2). If so, (97a) would be correctly predicted to be true. Another possibility is that the context under discussion highlights John and Bill qua group, not qua individuals, so that the propositions $^{\wedge}j \text{ came}$ and $^{\wedge}b \text{ came}$ are not included in the quantificational domain. If so, (98b) or (98c) would both make the correct prediction for (97a).

Some support that the domain of quantification in such contexts is a singleton comes from the behavior of adverbs of quantification. Speakers find (99a) infelicitous in the situation under consideration.³⁸ This may be due to the domain of quantification being restricted to a single proposition referring to the group of John and Bill:

³⁷ Groups are treated here as in Link (1983) and Landman (1989). I am not including the possibility of representing the plurality through sums as they would admit distributive inferences.

³⁸ Thanks to Bill McClure and Hazel Pearson for illuminating discussion of relevant cases.

³⁹ Lahiri (2002) does look at other phenomena where plurality and distributivity are in play. For example, QVE occurs when the embedded question has plural definites: *John for the most part knows where the kids are hiding* (Williams 2000). See also Beck and Sharvit (2002). Other cases involve interaction with plural definites in the matrix: *the witnesses know which Klansmen were present at the Lynching*. Lahiri's analysis of these cases relates to our discussion of questions with plural definites in Chapter 4 of this book (see also Chapter 7).

should have read a book and John should know for a majority of them, the book they read. Statement (101a) would be infelicitous in the given context.⁴⁰

It is worth noting that the source of the problem is not the account of QVE in terms of propositional sets but the interpretation of the embedded question as an ordinary Hamblin set based on atomic individuals. Lahiri is aware of this and discusses the alternative account of pair-list answers in Dayal (1996), presented in Chapter 4. Recall that under that view, the embedded question denotes the set (102) instead of (101c), with each proposition giving the graph of a function:

- (102) $\{^{\wedge}a \text{ read } 1 \text{ and } b \text{ read } 2 \text{ and } c \text{ read } 3,$
 $\quad \quad \quad \underline{^{\wedge}a \text{ read } 2 \text{ and } b \text{ read } 3 \text{ and } c \text{ read } 1, \dots}\}$

Lahiri correctly notes that though this, in combination with Ans-D, captures the functionality of pair-list answers, it cannot provide the domain of quantification in QVE structures. Assume that the underlined proposition is the answer to the embedded question in (101a). Intuitively, John should know two of the three atomic propositions that make up this answer. However, the atomic propositions are no longer accessible at the embedded CP node, which is where the domain of quantification is determined. This problem obviously calls for a modification of the approach and has been addressed in Dayal (2016b). The modification involves a certain amount of complexity that would take us too far afield to introduce here. Interestingly, Lahiri's PCA-based account as presented in this section is also insufficient to address the problem of QVE with pair-list answers. He explores a refinement that interprets such questions as sets of sets of sets of propositions (Lahiri 2002: 125–31). I refer the reader to the original work for details, merely flagging the added complexity of including third order questions into the system.

In the version of the theory that we have presented here, Lahiri explicitly rules out propositions based on plural individuals, restricting question denotations to logically independent propositions. If questions denote propositions built on atomic individuals and PCAs are based on these denotations, we do not expect the singular version of (103), earlier given as (93a), to be unacceptable:⁴¹

- (103) Jill partly knows which boys / #which boy cheated.

Lahiri (2002: 198–203) revises his theory to incorporate plurality, showing that it is possible to preserve the account of the plural version of (103) in terms of atomic propositions. He does not specifically discuss singular wh phrases but such cases remain recalcitrant even under these revisions. The problem, I believe, lies in the

⁴⁰ Lahiri also draws on the discussion in Moltmann and Szabolcsi (1994) and Szabolcsi (1994) to show that these cases suffer from the proportion problem.

⁴¹ Lahiri points out that singular terms that can lend themselves to a natural part-whole structure can be quantified over in the relevant way. There are some obvious points of comparison with the type of plurality based pair-list readings discussed in fn 39:

- (i) John knows, for the most part, Beethoven's fifth symphony.
- (ii) John knows, for the most part, the boys/*the boy who live there.

notion of answerhood that Lahiri subscribes to. Allowing an answer to be the conjunction of any subset of the question denotation yields extremely weak results even when the singular–plural distinction is factored in. This is so for constituent questions with one *wh* expression, multiple constituent questions, and constituent questions with quantifiers, as we saw in Chapter 4. Here we see that it also has unwelcome consequences for QVE.

5.4.4 Section summary

We looked at the interaction of adverbs in the matrix clause with *wh* expressions inside its complement. Berman’s analysis allows the adverb to quantify directly over individuals denoted by the *wh* expression(s), requiring a treatment of questions as open propositions rather than sets of propositions. The alternative proposed by Lahiri maintains the analysis of questions as sets of propositions, with adverbs quantifying over propositions. We also probed the difference between rogative and responsive predicates with respect to QVE. Beck and Sharvit posit quantification over subquestions, with special attention to presuppositions in creating the appropriate quantificational domain for the adverb. Finally, we discussed Lahiri’s account in terms of amount quantification over an algebra of answers. We looked at data motivating these proposals as well as data presenting challenges for them.

5.5 Embedding indirect and concealed questions

This chapter fleshed out issues specific to indirect questions. It started by providing diagnostics to distinguish embedded interrogatives from the morpho-syntactically similar free relatives and exclamatives. Multiple *wh* questions, polar questions, and questions with *which N* are among the properties that identify a complement as interrogative. The presence of *-ever* marks it as a free relative while *wh* expressions with intensifiers *how very adj* and indefinite noun phrases *what a(n) adj N* mark it as exclamative. We also considered concealed questions, which pose a different kind of challenge. They are noun phrases that overlap in meaning with identity questions.

The distinction between interrogatives, exclamatives, and concealed questions was essential in examining the other side of the embedding equation, the theory of selection. The classical view as lexical specification of $\pm\text{WH}$ on predicates, while useful in many ways, cannot capture more fine grained differences in the selection of interrogatives, exclamatives, and concealed questions. A two-dimensional system based on category (c-selection) and meaning (s-selection) was shown to handle such distinctions. Roughly speaking, the choice between interrogatives and concealed questions is attributable to the c-selectional properties of interrogative selecting predicates, the choice between interrogatives and exclamatives to their s-selectional properties.

One of the interesting developments in this area has been the attempt to derive s-selection from compositional semantics. There are two generalizations in

particular that have been subjected to close scrutiny in this connection: the inability of rogative predicates to embed exclamatives and the inability of surprise-predicates to embed polar questions. Evidence that complementation options can shift depending on the larger structure is an added argument against lexically specifying s-selectional properties on predicates. The generalization that rogative predicates (*ask*, *wonder*) cannot take exclamatives, and uninterrogatives (*think*, *believe*) cannot take interrogatives does not always hold. Similarly, there are shifts in the predicates that allow embedded interrogatives to behave like root clauses, arguing against the view that some predicates are lexically marked to select speech acts as complements. The fact that selectional options for a single predicate can shift depending on properties of the larger structure clearly points to a more fluid view of selection than standardly assumed. However, much work remains to be done before the theory of s-selection can be dispensed with. It therefore makes sense for now to continue using the terms c-selection, s-selection, and the features $+/-WH$ in talking about embedding.

Concealed questions were shown to raise a host of issues, independently of selection. Their complexity lies in the fact that a noun phrase with fairly unremarkable structural properties has to be associated with interrogative semantics. The head needs to be restricted to relational nouns, and definites as well as quantified noun phrases need to be recognized as concealed questions. Semantically, they are akin to, but not the same as, identificational questions. And their interpretation must be flexible enough to admit a range of subtly distinct readings. One approach that attempts to separate them from identificational questions treats them as denoting individual concepts, with the overt description naming the most salient property in the context. This allows for nested concealed question meanings and requires some modification of the lexical meanings of embedding predicates. Another view builds on this approach, but taps into the relational nature of the head noun. It derives propositional level meanings for them. Quantified concealed questions, which reveal an ambiguity between so-called set and pair-list readings, are captured with reference to the relational component in the head noun. Finally, we reviewed an account of concealed questions that treats them as identity questions, but under a particular perspective, a *conceptual cover*. The various readings of concealed questions are argued to arise from various ways of pragmatically resolving their *cover*. By giving greater weight to pragmatics, this account also expands the range of meanings that concealed questions can be associated with.

The final section dealt with quantificational variability effects, which in its simplest form can be understood with reference to schemas like [$CP...Adverb...[CP...wh...]$]. The adverb, instead of modifying the matrix verb, seems to quantify over the set denoted by the embedded *wh* expression. Appearances notwithstanding, such direct quantification is empirically untenable. Instead, QVE is better captured in accounts interpreting the *wh* with scope within the complement. Three different proposals were considered. In one, quantification is over the propositions in the question denotation. In another, the question is reanalyzed as a set of subquestions and quantification is over answers to subquestions. A third proposal posits quantification over an algebra of answers to the question. The role

of the embedding predicate (rogative vs. responsive) in admitting quantificational variability and the significance of its lexical presuppositions in restricting the domain of quantification are among the themes that these proposals engage with. At the end of Chapter 4 the issue of whether expressions inside complements can have semantic presence outside their local domain was raised. The discussion of QVE elaborates on that and anticipates the issues to be discussed in the next two chapters.

It should be obvious that I am no longer attempting to incorporate the insights of the various proposals discussed into our baseline theory. This is because we are well past the point where this could be done without doing serious disservice to the many complex issues that emerge from these discussions. I hope, however, that the baseline theory continues to provide a convenient reference point for evaluating the multiple viewpoints we touch upon as the scope of our investigation continues to expand.

Weak islands and extraction

So far we have focused primarily on *wh* expressions involving quantification over individuals, or extensions to individual-based functions: *who*, *what*, *which N*. The assumption is that the same would apply to *wh* expressions like *what N*, *when*, *where*, *how*, and *why*, and to a large extent they do but there are also contexts where different types of *wh* display distinct patterns of behavior. The phenomenon of weak islands is one such context, distinguishing between *wh* expressions ranging over individuals as opposed to those ranging over manners, reasons, and amounts.

Three constructions that are representative of weak islands are presented: negative islands, *wh* islands, and factive/presuppositional islands. All of them allow some *wh* expressions to be extracted but not others. A canonical example of a good extractee is *which N* and a canonical example of a bad extractee is *how Adj*. The phenomenon raises the following questions. How do weak islands differ from strong islands, such as complex noun phrases and coordinate structures, which do not allow extraction of any *wh* expression? Do weak island effects require a two-pronged explanation, focusing on the nature of the island as well as the nature of the *wh* expression? Does the division between arguments as good and adjuncts as bad extractees hold up under scrutiny? Or is a better handle on the data afforded by classifications based on the notions of referentiality and D-linking?

With answers to these basic questions in place, we explore the semantic underpinnings of the phenomenon. One proposal is that there are two radically different types of domains of quantification, one that is suitable for individuals and another that is suitable for amounts, manners, or degrees. This difference results in a difference in grammaticality when the question in the scope of the *wh* embeds a weak island. Another proposal treats negative island effects in terms of maximality within Karttunen's semantics for questions. The proposal claims that negative islands are fundamentally different from *wh* and factive/presuppositional islands, for which a syntactic account may be more suitable. Finally, the role of presupposition in creating the weak island effect is considered, and the extent to which a shift in context may effect a change in the paradigm is probed.

This last account in terms of presuppositions shares the assumptions of our baseline theory and is elaborated upon by the next set of proposals. They all take questions to denote Hamblin sets and appeal to the answerhood operator to pick out the maximally informative proposition from it. Weak island effects arise because the semantics of certain types of questions interacts with the requirement of informativity in ways leading to semantic deviance. These proposals enlarge the

empirical picture by showing that an appropriately placed modal repairs potential weak island violations. Together, they present a uniform account of all types of weak island violations and do so within a theory of questions and answers that has independent justification.

6.1 Weak islands, referentiality, and D-linking

Here we illustrate the phenomenon of islands, identifying the properties that distinguish weak and strong islands. We also review the early syntactic literature on the topic, which focuses on referentiality and D-linking as the relevant properties of extracted phrases. These notions, though related, are nevertheless distinct. This section seeks to shed some light on what these terms mean, in addition to articulating their role in accounts of weak islands.

6.1.1 Weak islands

Ross (1967), in his seminal work on syntactic transformations, noted the impossibility of questioning into certain structures:

- (1) a. *What/which book did you meet the person who wrote?
 b. $[_{CP} \text{ what/which book}_i [_{IP} \text{ you met } [_{DP} \text{ the } [_{NP} \text{ person } [_{CP} \text{ who wrote } t_i]]]]]$
 c. What/which book is such that you met the person who wrote it?
- (2) a. *What/which book did you buy the CD and?
 b. $[_{CP} \text{ what/which book}_i [_{IP} \text{ you bought } [_{DP} [_{DP} \text{ the CD}] \text{ and } [_{DP} t_i]]]]]$
 c. What/which book is such that you bought the CD and that book?

The unacceptability of (1a) and (2a) is ascribed to syntactic constraints induced by the type of constituent from within which the *wh* expression is extracted. This is corroborated by the fact that minimally different constructions with the intended meaning, as in (1c) and (2c), are acceptable, if perhaps a bit clunky. It is further corroborated by the fact that the same effects show up with topicalization. The non-*wh* phrase *this book* aligns with *what/which book* with respect to movement:

- (3) a. $*[_{CP} \text{ This book}_i [_{IP} \text{ I met } [_{DP} \text{ the } [_{NP} \text{ person who wrote } t_i]]]]]$
 b. $*[_{CP} \text{ This book}_i [_{IP} \text{ I bought } [_{DP} [_{DP} \text{ the CD}] \text{ and } [_{DP} t_i]]]]]$

It thus seems reasonable to attribute the observed restriction to a general constraint on movement, rather than to a specific aspect of question formation. Complex noun phrases and coordinate structures, we conclude, are simply islands for movement.

A division within the class of islands emerges when we consider constituents that allow the extraction of only certain types of *wh* expressions. The following seem to distinguish between extraction of arguments vs. adjuncts:

- (4) a. ?_{[CP Who_i did [IP you find out [CP that Mary had talked to t_i]]]}?
 b. *_{[CP How_i did [IP you find out [CP that Mary had behaved t_i]]]}?
 c. *_{[IP Why_i did [IP you find out [CP that Mary had left t_i]]]}?
- (5) a. ?_{[CP Who_i did [IP you wonder [CP whether Mary had talked to t_i]]]}?
 b. *_{[CP How_i did [IP you wonder [CP whether Mary had behaved t_i]]]}?
 c. *_{[CP Why_i did [IP you wonder [CP whether Mary had left t_i]]]}?

The first exemplifies a factive/presuppositional island, the second a *wh* island. Argument phrases originating inside the complement clause can be fronted relatively easily, but not adjunct phrases.¹ Note that (4b) is acceptable but only as asking for the method by which you found out about Mary's behaving (well). It is unacceptable under an embedded construal in which the question is about Mary's behavior. Example (4c) is an acceptable question but only if it is construed as asking for the reason for wondering rather than the reason for Mary's leaving. Similar comments apply to (5). To complete the comparison, (6a)–(6b) show that an embedded construal is available when there is no island involved—*think* does not implicate factivity and there is no *wh* expression in the embedded CP:

- (6) a. _{[CP How_i do [IP you think [CP Mary will behave t_i]]]}?
 b. _{[CP Why_i do [IP you think [CP Mary left t_i]]]}?

Factive and *wh* islands are called weak because they block extraction of adjuncts but not arguments. They differ from complex NPs and coordinate structures which are considered strong because they block extraction of arguments as well as adjuncts.

The same contrast shows up in questions with negation, which justifies classifying negative questions as weak islands (see Rizzi 1990; Cinque 1990):

- (7) a. Who_i didn't you talk to t_i?
 b. *How_i didn't you behave t_i?
 c. *Why didn't you behave t_i?

Example (7c) is acceptable as a question about the reason for not behaving: *I didn't behave and it was because I was angry*. It is not acceptable as a question about reasons that did not motivate good behavior: *I behaved but it was not because I was afraid (I behaved because I am polite)*.

The presentation of strong and weak islands given so far has idealized the data to some extent since the lines of division are known to be murky. For example, adjunct clauses show selectivity of the kind associated with weak islands.

¹ Judgments in these cases tend to vary in the literature. Relative judgments, i.e. the contrast between (4a)/(5a) and (4b)–(4c)/(5b)–(5c), are robust. The same holds for (8a)/(9a) and (8b)–(8c)/(9b)–(9c).

Nevertheless, they are classified as strong because all extraction is blocked when the adjunct is tensed (8c):

- (8) a. ?Which politician_i did you go to England [after meeting t_i]?
 b. *How much water_i did you make the pasta [after boiling t_i]
 c. *Which politician_i did you go to England [after you met t_i]?

Note that *how much water* is an argument of the verb *boil* in (8b), but we already know that argument-hood is not relevant to extraction out of strong islands. One factor that has been shown to be relevant to them is the contrast between DP and PP traces (see Obenauer 1984; Cinque 1990; among others):

- (9) a. ?Which topic_i did you leave [without talking about t_i]?
 b. *About which topic_i did you leave [without talking t_i]?

Our discussion will be limited to negative, *wh*, and factive/presuppositional islands. Following standard practice, we classify adjunct islands as strong and defer discussion to Chapter 7 where we consider other strong islands as well.

6.1.2 Referentiality and D-linking

Some of the early accounts of weak islands are cast within syntactic theories that take movement to be regulated by bounding nodes or barriers (Huang 1982; Lasnik and Saito 1984; Chomsky 1986). Strong islands are islands because they involve more than one such node in the path from the base position to the clause-initial position. From this perspective, weak islands differ in the strength of the barrier they pose for movement. This, however, does not explain the difference between arguments and adjuncts noted in Section 6.1.1. Huang argued that two different principles are needed to explain weak islands: Subjacency, which keeps track of the number of barriers crossed, and the Empty Category Principle (ECP), which regulates traces. An argument extracted from a weak island incurs a subjacency violation but its trace is properly governed by the lexical verb. Since an adjunct is not lexically selected by the verb, its trace cannot be lexically governed by it. Its only option is to be governed by the fronted *wh* expression, which is blocked by the barrier posed by the island. The trace and the fronted *wh* do not meet the locality clause of the ECP. That is, an adjunct not only violates subjacency, it also violates the ECP. This explains why argument extraction may not be perfect but is generally acceptable, while adjunct extraction is ungrammatical. An adjunct is fully acceptable only when it can proceed through intermediate stops in the lower CPs, ensuring that each link of the chain is locally governed, hence ECP compliant.

Describing the selectivity of weak islands in terms of arguments and adjuncts is not quite accurate as there are cases where arguments cannot escape *wh* islands:

- (10) a. *What did John ask whether these pearls cost t ? (Ross 1984)
 b. *These pearls cost.

- (11) a. *How did John ask whether to behave ?
 b. *John behaved.

Rizzi (1990) therefore argued that weak island effects are sensitive to a semantic property of the extractee, independently of its status as argument or adjunct, namely the property of *referentiality*. Wh expressions ranging over amounts, manners, reasons, etc. are not referential while those ranging over individuals are. The theory of *Relativized Minimality* blocks X from governing Y if a similar operator Z intervenes between them and can be put to service. Here, for example, the wh expression in the lower clause blocks binding of the trace by the antecedent because it is a potential binder for it: [how/why_{antecedent} ... [wh_{intervener} ... t_{bindee}]]. Obenauer (1984) is an early proposal along the same lines.

Rizzi's account of the weak island phenomenon raises a non-trivial question about the applicability of the term *referentiality* to wh expressions. Quantifiers like *every N* or *most N* can be distinguished from *a N*, *the N*, or proper names on the basis of types ($\langle\langle e, t \rangle, t \rangle$ or e) and/or their ability to support discourse anaphora. It is unclear what criteria can be used to make similar distinctions within the class of wh expressions, which are indeterminate by definition.

Cinque (1990), partly influenced by Kroch (1998) and Comorovski (1989), adopts instead the notion of D-linking, due to Pesetsky (1987).² Pesetsky argues for a distinguished role for wh expressions that range over a set of entities salient in the immediate context. He makes a tripartite distinction between wh expressions that are inherently D-linked (*which N*), those that are not normally D-linked but may become D-linked in particular contexts (*who*, *what*, and, to a lesser degree *how*, *why*, etc.) and those that resist D-linking altogether (*who the hell*/*what the hell*, etc.). The following from Kroch provides a good illustration of the distinction at the heart of Pesetsky's classification. The difference between (12b) and (12c) is in the use of *who* vs. *which N*:

- (12) a. Context: A man walks into an apartment building in front of two women who are conversing on the sidewalk. One woman says to the other:
 b. Who just went into the building?
 c. Which man just went into the building?

Question (12b), in the context of (12a), can be asked without the speaker having any expectations about the identity of the individual. Any adult male named in the answer would be acceptable. In asking (12c) in the same context, instead, the speaker expects the addressee to name an individual familiar to both of them. D-linking, then, is a semantic/pragmatic notion, like referentiality, but one that can be applied to the domain of quantification over which a wh expression ranges. Furthermore, it has a morpho-syntactic reflex in favoring wh expressions with a determiner and a head noun.

² Kroch (1998) was written in 1989 and widely circulated before being published.

The following paradigm from Pesetsky (1987) illustrates how contextual support shifts a non-inherently D-linked *wh* to become D-linked:

- (13) a. I figured out what goes where.
 b. ?I figured out where what goes.
 c. I know that we need to install Transistor A, Transistor B and Transistor C, and I know that these three holes are for transistors but I'll be damned if I can figure out from the instructions *where what* goes.

Pesetsky points out that superiority effects of the kind seen in (13a)–(13b) do not hold of questions with *which N*. They allow either order (*which transistor goes into which hole? Which hole does which transistor go into?*), leading to the generalization that superiority violations are ameliorated under D-linking. With appropriate context, as in (13c), superiority violations with *what/where* also become acceptable. Put differently, *what/where* can become D-linked with contextual priming and when they do, show the same behavior as other D-linked expressions with respect to superiority.

By adopting D-linking over referentiality, Cinque accounts for the following contrasts:³

- (14) a. ?What were you wondering how to fix *t*?
 b. Which car were you wondering how to fix *t*?
 (15) a. *How many points are you wondering whether to earn *t*?
 b. How many points are the jurors debating whether to take off *t*?

The contrast in (14) is detectable in out of the blue contexts: *which N* appears acceptable because it is inherently D-linked, *what* does not because there is no contextual priming. Example (15), from Kroch (1998), shows that even amount *wh* expressions can be D-linked and without overt contextual support. Talk of jurors makes salient contexts of games and it becomes possible to think of a limited range of points that may be deducted for a particular infraction. D-linking has the expected ameliorating effect. Note that this elastic view of D-linking and its role in weak islands suggests that the phenomenon would have a gradient quality, subject to the ease or difficulty with which an extractee can become D-linked.

D-linking, however, has its limits. The non-inherently D-linked *wh* expression in (16) can be extracted without the benefit of contextual cues (Kroch 1998: 29). No specific group of friends/relatives need to be known to speaker and hearer:

- (16) Who were you wondering whether to visit on your vacation?

³ D-linking does not dispense with all argument-adjunct asymmetries. The following, from Rizzi, is shown by Kroch to contrast even though both have D-linked *wh* expressions:

- (i) *For which reason_i don't you know if we can say that Gianni was fired *t_i*?
 (ii) Which reason_i don't you know if we can give *t_i* for Gianni's firing?

Szabolcsi and Zwarts (1993) show that even a so-called aggressively non-D-linked *wh* can be extracted, casting further doubt on the role of D-linking:

- (17) a. Who the hell saw John?
 b. ??Who the hell are you wondering whether to invite t?
 c. What the hell do you still not know how to spell t?

Wh-the-hell expressions can be used, they claim, as information seeking questions, but only if there is unquestionable evidence for the existential presupposition behind the question. Example (17a) requires that it be patently obvious to speaker and hearer that someone saw John. They argue that it is the difficulty in fulfilling this requirement of unquestionable evidence that is responsible for the apparent unacceptability of extracting *wh-the-hell* expressions from weak islands. It is easier to imagine the right context for some questions, such as (17c), than for others, like (17b).

There remain several mysteries about D-linking which we will return to in this chapter and the next. For now, note that even though neither referentiality nor D-linking appear to provide completely satisfactory solutions to the weak island problem, they have proven useful in highlighting the importance of semantic/pragmatic considerations in tackling what was initially considered a purely syntactic phenomenon.

6.1.3 Section summary

This section distinguished weak islands from strong islands as structures that allow the selective extraction of *wh* expressions. Negative, factive/presuppositional, and *wh* islands were identified as exemplars of such islands. The basic separation between good and bad extractees seems to be between *wh* expressions like *which N* and *who/what* vs. *how/why*. D-linking, though fundamentally a pragmatic notion, is noted to have a morpho-syntactic manifestation that broadly cuts across the same lines, with *which N* being more open to D-linking than *who/what* or *how/why*. We now turn to proposals that elaborate on the semantic underpinnings of the accounts discussed above.

6.2 Alternatives to Relativized Minimality

Here we will survey three early responses to the Relativized Minimality account of weak islands, each of which addresses the role of D-linking/referentiality in some way. As indicated earlier, there are two related but distinct ways of classifying D-linking. There is a morpho-syntactic classification that privileges *which N* and there is a pragmatic one which lets in other expressions under appropriate contextualization. It is important to keep both in mind as we look into these proposals and see how other aspects of the semantics of questions may affect explanations of the weak island phenomenon.

6.2.1 *Weak islands through quantificational domains*

Szabolcsi and Zwarts (1993) take the basis of weak island effects to be semantic. They locate it in the distinction between the types of domain that individuals and non-individuals denote in, and the interaction between these domains and questions with weak islands. That is, there are two key aspects to their proposal: the difference in denotation domains and the type of operations needed to compute particular structures.

To get a feel for the first, compare the question–answer possibilities in (18)–(19):

- (18) a. How did everyone behave?
 b. Everyone behaved badly.
 c. a behaved badly, b behaved stupidly, and c behaved deviously.
 d. #Everyone behaved badly and additionally, a behaved stupidly, b cleverly and c deviously.
- (19) a. What did everyone read?
 b. Everyone read W&P.
 c. a read W&P, b read E, and c read PP.
 d. Everyone read W&P, and additionally, a read E, b read P, and c read AK.

Statement (18b) is appropriate in situations where everyone has behaved uniformly, as in context (20a). Example (18c) is appropriate in situations where everyone's behavior is distinct, as in (20b). Example (18d) is inappropriate as an answer, even in a context like (20c) where there is a common core to everyone's behavior. It is as if each person's behavior encompassed both aspects and merged them into one indivisible whole. To represent this situation, a pair-list answer would have to specify for each person what their composite behavior was. This is clearly not so when the *wh* ranges over individuals. The paradigm in (19) and (21) is parallel, but (21c) is described by answer (19d):

- | | | | | | | |
|------|----|-----------|----|---------------|----|-----------------------|
| (20) | a. | a → badly | b. | a → badly | c. | a → badly & stupidly |
| | | b → badly | | b → stupidly | | b → badly & cleverly |
| | | c → badly | | c → deviously | | c → badly & deviously |
-
- | | | | | | | |
|------|----|---------|----|---------|----|--------------|
| (21) | a. | a → W&P | b. | a → W&P | c. | a → W&P & E |
| | | b → W&P | | b → E | | b → W&P & P |
| | | c → W&P | | c → P | | c → W&P & AK |

Szabolcsi and Zwarts provide many other arguments to buttress their claims, from different languages, which I omit here. They establish that good extractees (*who/what*, etc.) range over unordered sets of individuals while bad extractees (*how, how many*, etc.) range over partially ordered sets. This raises two questions. Why should this have an effect on weak islands? How does D-linking ameliorate potential weak island violations?

To answer the first question, a guiding principle about scopal elements is provided:

- (22) Scope and Operations: Each scopal element SE is associated with certain operations (e.g. *not* with complements). For a *wh*-phrase to take scope over some SE means that the operations associated with SE need to be performed in the *wh*-phrase's denotation domain. If the *wh*-phrase denotes in a domain for which the requisite operation is not defined, it cannot scope over SE.

Taking negative islands for illustration, assume a set of books $\{T, E, P, J\}$ and the question *what didn't a read?* Suppose the set of books *a* read is $\{T\}$. Then, the set of books *a* didn't read is its complement set: $\{E, P, J\}$. We cannot perform a similar complementation operation over manners for *how didn't John behave?* because complementation is not defined for partially ordered sets. As Szabolcsi and Zwarts note, their proposal is a truly relativized minimality effect: the meaning of each extractee determines what interveners it is sensitive to. Though they intend their proposal to extend to other weak islands, they do not actually elaborate on those (see Chapter 8 for further discussion).

I will turn now to the issue of D-linking. As we see, there is no particular role for it in the theory as presented above and, in fact, Szabolcsi and Zwarts welcome this result. Recall that they dismiss the notion that the so-called aggressively non-D-linked *wh* *wh-the-hell* is a bad extractee (23a)–(23b):

- (23) a. What the hell do you still not know how to spell?
b. What the hell are you upset that you took?

Example (23a) can be used in contexts where the speaker sees someone madly searching through the dictionary and (23b) if a thief sees his fellow thief trying to smuggle an item back into a house they have just robbed. Their general point is that such expressions do not just carry the standard existential presupposition that can be implicitly denied by an addressee with an answer *no one/nothing*. The presupposition is stronger in that the basis for it must be manifest in the context, making deniability impossible. Their unacceptability in weak islands, they claim, is for independent reasons, though they do not actually elaborate on what those reasons might be.

Of course, Szabolcsi and Zwarts recognize the role that D-linking seems to play in the amelioration of weak island violations but they claim that the effect is indirect. D-linking can force a domain that does not denote an unordered set to do so. Thus, (24a) becomes acceptable in contexts where a list of potential scores and receivers' names are posted. They also suggest that contextualization/D-linking aids in an alternative verification procedure called "look-up" that circumvents the normal Boolean operations. Thus, if (24b) is asked with reference to individuals already under discussion, the "look-up" procedure can check for each candidate whether or not everybody supports him/her:

- (24) a. How many scores did no one receive? (Answer: 2–22 and 27)
 = Which of the figures on the blackboard has no name next to it?
 b. Who did everybody support? (Answer: The candidate from Ohio)

What is not explicitly addressed, but can be gleaned from the discussion, is why there should be a difference between *what/who* and *which N* with respect to facilitating “look-up.” What is it about inherently D-linked whs and non-inherently D-linked whs that results in this variation in behavior? One difference is the presence of a head noun that sets the granularity of the question, noted by Szabolcsi and Zwarts in relation to contextual priming of granularity. Note though that *what N* and *which N* overlap on granularity but do not have the same properties (see, for example, Kochovska (2010) on Macedonian clitics that distinguish between them; see also Heim (1987) and Barros (2013)). The second difference between *which N* and *what/who* (or even *what N*) goes back to Comorovski (1989, 1996), who points out that *which N* is inherently partitive, suggesting that there is an inner definite DP over which some property can be distributed. This is consistent with Szabolcsi and Zwarts’ take on D-linking, where the definite could be taken to provide the peg for the look-up procedure. What remains elusive is the fact, also noted in Comorovski, that extraction out of weak islands in Romanian requires D-linking even for individual-denoting wh expressions. Szabolcsi and Zwarts’ explanation of extraction crucially rests on the distinction between denotation domains as they relate to Boolean operations, not on the availability of “look-up” procedures.

6.2.2 Weak islands through maximality

Rullmann (1995) takes questions to denote Karttunen sets but with a uniqueness presupposition built into the nucleus (cf. Chapter 2):

- (25) a. Which books did(n’t) John read?
 b. $\lambda p \exists x [\text{book}(x) \wedge p(w) \wedge p = \lambda w' x = \text{MAX}(\lambda y [\text{read}(w')(j,y)])]$
 c. $\lambda p \exists x [\text{book}(x) \wedge p(w) \wedge p = \lambda w' x = \text{MAX}(\lambda y [\neg \text{read}(w')(j,y)])]$

With positive questions, maximality is defined over the domain of books, so MAX picks out the largest plurality in the set of books John read. With negative questions, there is a potential problem. If the question ranges over the set of all possible books, the computation is problematic, but if the set is cut down to size by D-linking, the computation becomes manageable. What the question requires, then, is not a D-linked domain *per se* but a reasonable sized domain of quantification, which D-linking makes available. Note that Rullmann’s use of D-linking is similar to Comorovski’s in equating the role of D-linking with the kind of domain narrowing that is ubiquitous in natural language: *John read every book* is typically interpreted with respect to a contextually restricted domain of books.

Rullman's appeal to maximality is especially relevant to degree and amount questions, which, according to him, show sensitivity to negative islands because they typically range over sets lacking a maximal element. Schematically:

- (26) a. How tall is/*isn't John?
 b. $?n[n = \text{MAX}(\lambda d [\text{John is } d\text{-tall}])]$
 c. $?n[n = \text{MAX}(\lambda d [\text{John is not } d\text{-tall}])]$

If John is 6' tall, the positive question denotes the proposition that 6 is the n such that it is the maximal degree of John's heights. The negative question is undefined because there is no degree that constitutes the maximal element of the degrees above 6'.⁴ D-linking, we can infer, would solve the problem. A plausible context with a limited set of degrees is needed to ask the question.

Rullmann follows Kroch (see Section 6.2.3) in treating amount questions as potentially ambiguous, but not in weak island contexts. He argues that this is due to the problem with defining maximality over an open set, at least for negative islands:

- (27) a. How many books did John want to read?
 b. $?n[n = \text{MAX}(\lambda n' \exists x[\text{books}(w)(x, n') \wedge \text{want-to-read}(w)(j, x)])]$
 c. $?n[n = \text{MAX}(\text{want-to-read}(w)(j, \lambda w \lambda P \exists x[\text{books}(w)(x, n') \wedge P(w)(x)]))]$

John may either have a desire to read a certain set of books, and the speaker asks the addressee what the cardinality of that set is. This may be if John is too young to know his numbers, for example. This is the referential reading (27b). Or John may know his numbers and has a desire to read ten books, maybe as a way of delaying his bedtime. This is the amount reading (27c).

The negative case also has two potential readings:

- (28) a. How many books did John not want to read?
 b. $?n[n = \text{MAX}(\lambda n' \exists x[\text{books}(w)(x, n') \wedge \neg \text{want-to-read}(w)(j, x)])]$
 c. $?n[n = \text{MAX}(\neg \text{want-to-read}(w)(j, \lambda w \lambda P \exists x[\text{books}(w)(x, n') \wedge P(w)(x)]))]$ *undefined*

John the pre-numerical child may have a desire not to read the books he finds scary. The addressee is being asked about the cardinality of this set, an available reading. The amount reading is the one where John has a desire not to read a particular number of books. Let's say he is sleepy and he only wants to read 2 books. Then 3 is a number such that he doesn't want to read a set of that many books, and 4 is a number such that he doesn't want to read a set of that many books and so on. MAX is undefined since for any number n for which it holds that John doesn't want to read a set of books of that cardinality, it also holds for numbers bigger than n .

⁴ This assumes the relevant meaning to be *at least d-tall* while Rullmann assumes an *exactly d-tall* meaning. This does not affect the essentials of the proposal.

Now consider contexts with a D-linked set of numbers (Rullmann 1995: 15):

- (29) a. Rob is a super athlete who has run international races of almost any distance.
How far has Rob never run?
b. He has never run a 25 mile race and he's never run a 100 mile race (though he has run all others, including 50 a fifty mile race).

Rullmann notes that in such cases the degrees (number of points, distances) are not viewed as being ordered on a linear scale. Instead, they are treated as if they were ordinary entities structured by a Linkian part-of relation. This allows for an answer like (29b) which picks out a maximal element 25+100 to the exclusion of 50. This is consistent with Szabolcsi and Zwarts' view of the role of D-linking on non-individual denoting expressions.

A final point. Rullmann's account of negative islands does not readily extend to other weak islands. He addresses this, noting that it is as it should be since the two phenomena are distinct. For example, shifting to a D-linked *wh* expression merely reduces the unacceptability in *wh* and factive islands whereas it leads to complete acceptability in negative islands:

- (30) a. ?Which man did you wonder whether Bill invited t?
b. ?Which man did you regret that Bill invited?
c. Which man do you think Bill didn't invite?

Rullmann also notes that there is cross-linguistic variation in the case of *wh* and factive islands while negative island effects are robustly attested across languages. He sees these as arguments for treating *wh* and factive islands in syntactic terms while reserving the maximality-based semantic account for negative islands.

6.2.3 Weak islands through presuppositions

Comorovski (1989, 1996), using cross-linguistic evidence, argues that *wh* island violations are acceptable only if the extractee is D-linked, a conclusion that we saw is accepted by Cinque. Comorovski's account also seeks to explain the putative relativized minimality effects from the interaction of D-linking with the presupposition behind questions. Romanian (31a) for example, is unacceptable with a non-D-linked *wh* but would be acceptable with a D-linked fronted *wh* (Comorovski 1996: 169):⁵

⁵ For some Romanian speakers extraction remains bad even with D-linked *wh*. Interestingly, extraction becomes acceptable for them if the matrix clause has a *wh*. That is, [*wh*_i [Subj V [*wh*_j [*t*_j V *t*_j]]]] is bad but [*wh*_k *wh*_i [*t*_k V [*wh*_j [*t*_j V *t*_j]]]] is good. We will discuss multiple fronting languages further in Chapter 7. Thanks to Anamaria Falaus, Andreea Nicolae, and Dafina Ratiu for discussion.

- (31) a. $*[_{CP} Ce_i [_{IP} pro \quad \text{\textit{\text{ştii}}} \quad [_{CP} cine_j [_{IP} t_j a \quad \text{\textit{distrus}} t_i]]]]$
 what you-know who has destroyed
 “What do you know who has destroyed?”
- b. Answerability:
 A constituent question is answerable iff the truth of its presupposition can be checked.
- c. $\forall x [thing(x) \rightarrow \exists y [person(y) \wedge destroy(y,x)]]$

Comorovski, recall from Chapter 2, takes a constituent question to have an existential presupposition. Example (31a) is unacceptable because a person who answers it tacitly accepts that for all things there is someone who destroyed it, but given that the set is not D-linked, this is impossible. By making the set manageable, D-linking makes the question answerable. There are several points of note here. First, Comorovski’s use of D-linking seems to serve the same purpose as domain selection normally does. Second, Comorovski uses a morphological classification of D-linking to derive ungrammaticality. She does not address whether (31a) would improve if the context made salient a particular set of objects that the question could be about.

Kroch (1998) accepts the significance Comorovski places on the existential presupposition but explains weak islands somewhat differently. To appreciate his point, consider amount questions again:

- (32) a. How much did the book cost?
 b. Presupposition:
 There is an amount of money such that the book cost that amount.
 c. Answer: \$10.

Question (32a) can be answered with (32c), consistent with the presupposition in (32b). Amount questions become interesting when they involve extraction from thematic positions that are equally compatible with amounts and entities. Unlike *cost* which can only take amounts, *publish* can take entities like *these books* or amounts like *5000 books*:

- (33) a. How many books did the editor publish this year?
 b. ‘Amount’ Presupposition:
 There is an amount of books such that the editor published that amount this year.
 c. ‘Referential’ Presupposition:
 There is a set of books such that the editor published that set this year.
 d. Answer: 5000 books.

Under one reading, the question is about the amount of books, under the other it is about a set of books described in terms of its cardinality. The answer under either reading is the same.

Kroch taps into this distinction to shed light on the role of D-linking/referentiality. He notes that extraction out of weak islands is typically compatible with the referential reading of amount questions:

- (34) a. How many books did Bill ask whether the company was interested in publishing?
- b. ‘Referential’ Presupposition: There was a set of books for which Bill asked whether the company was interested in publishing them.

When a referential reading is not possible, we get pragmatic oddity:

- (35) a. *How much money was John wondering whether to pay?
- b. ‘Amount’ Presupposition: There was a sum of money about which John was wondering whether to pay it.

We can easily think of contexts in which John’s thoughts could be of the form: *should I pay \$5 or \$6 or \$7 for this?* But this is not what (35a) is attempting to ask. The person who would answer the question would be saying, of \$5, \$6, \$7, what John is wondering about is whether he should pay \$5 or not. He is not wondering about whether he should pay \$6 and he’s not wondering about whether he should pay \$7. Now going back to (15b), repeated in (36a), we see why it works:

- (36) a. How many points are the jurors debating whether to take off ?
- b. Answer: 50 points (the jurors are debating whether to take off 50 points for the infraction or not, they are not debating about taking off 100 points and they’re not debating about taking off 150 points).

Kroch’s claim is that (36a) is acceptable because it is easy for the presupposition of a unique amount under debate to be satisfied in the context of jurors and points. Of course, the jurors could also debate whether to take off 50, 100, or 150 points, but (36a) is not the question that can be used to capture that meaning.

Kroch concludes that the impossibility of extracting a “non-referential” amount quantifier is not due to semantic non-referentiality. Weak island effects arise when the presupposition of uniqueness cannot be satisfied. In general, this is harder to satisfy for manners and amounts than for individuals. Crucial to Kroch’s explanation is the addition of the dimension of uniqueness to Comorovski’s existential presupposition for questions. This brings it in line with the baseline theory of questions that we have adopted in this book. In Section 6.3 we will see this connection made explicit.

6.2.4 Section summary

We looked at proposals that re-interpret the Relativized Minimality account of weak islands in semantic/pragmatic terms. Essentially, they flesh out the semantic underpinnings of the reliance on referentiality/D-linking, and in doing so, they

reshape the empirical contour in subtle ways. The semantic reworking makes reference to unordered vs. partially ordered quantificational domains, “small enough”/finite domains for the maximality operator to work on, and/or to the existential and uniqueness presuppositions of questions. In each case, D-linking (or lack thereof) can be a facilitating (or hindering) factor but is not held directly responsible for the weak island effect. This is considered desirable as the data do not line up neatly along the D-linked/non-D-linked divide.

6.3 Maximal informativity and modal obviation

Section 6.2 set the stage for the next set of proposals, those that adopt the core assumptions of our baseline theory: that questions denote Hamblin sets and an answerhood operator picks out the unique proposition in the set that is maximally informative. If no such proposition exists, the operator is undefined and the question infelicitous. Weak island effects arise because the interaction of certain wh expressions with these aspects of question meaning leads to semantic/pragmatic deviance. A uniform account of all three types of weak islands is provided and the paradigm extended to include modal configurations that repair weak island violations, reinforcing the view that weak islands are a semantic/pragmatic phenomenon, not a syntactic one.

6.3.1 Negative islands, degree questions, and dense scales

Although the maximality requirement in Rullmann (1995) derives the negative island effect for amount questions, Beck and Rullmann (1999) present examples that pose a problem for that particular way of building in maximality (see also Chapter 2). Underlining indicates truth at the world of evaluation:

- (37) a. How many eggs are sufficient to bake a cake?
 b. 3 eggs are sufficient, and so are 4 eggs and so are 5 eggs.
 c. Ans-D($\{\overset{\wedge}{2}$ eggs are sufficient, $\overset{\wedge}{3}$ eggs are sufficient, $\overset{\wedge}{4}$ eggs are sufficient...})

If 3 eggs are needed in the recipe, there is no unique number that would satisfy the condition $\text{MAX}(\lambda n' \exists x[\text{eggs}(w)(x, n') \wedge \text{sufficient-to-bake-a-cake}(w)(x)])$. However, there is a true proposition, namely $\overset{\wedge}{3}$ eggs are sufficient to bake a cake, that entails all the other true propositions in the set. That is, examples like (37a), along with the number-based felicity requirements on questions, prompt the move we have made in opting for Hamblin sets and an answerhood operator that builds in the uniqueness presupposition:⁶

⁶ Beck and Rullmann do not incorporate the semantics of number morphology or uniqueness and existence presuppositions. See Chapter 2 for problems this gives rise to. Anticipating the accounts to be reviewed here, I use Ans-D to demonstrate the point under discussion.

- (38) $\text{Ans-D}(Q) = \lambda w \text{ } \wp \in Q[p(w) \wedge \forall p' \in Q[p'(w) \rightarrow p \subseteq p']]$

As Beck and Rullmann note, an unfortunate consequence of this move is that the maximality-based account of negative islands is lost:

- (39) a. How tall is/*isn't John?
 b. $\{\dots \wedge \text{John is tall to degree } 5'8'', \wedge \text{John is tall to degree } 5'9'', \wedge \text{John is tall to degree } 5'10'' \dots\}$
 c. $\{\dots \wedge \text{John isn't tall to degree } 5'9'', \wedge \text{John isn't tall to degree } 5'10'', \dots \wedge \text{John isn't tall to degree } 6' \dots\}$

If John is exactly 5'9'' tall, Ans-D, applied to the positive version of (39a), will pick out $\wedge \text{John is } 5'9'' \text{ tall}$ as the maximally informative true answer. Ans-D applied to the negative version of (39a) will pick out $\wedge \text{John is not } 5'10'' \text{ tall}$ since it entails all the other true propositions: $\wedge \text{John is not } 5'11'' \text{ tall}$, $\wedge \text{John is not } 6' \text{ tall}$ and so on. The unacceptability of negative degree questions remains unexplained.

Fox and Hackl (2006) correct this problem by combining the maximal informativity of Ans-D with density in measurement scales:

- (40) a. MAXIMAL INFORMATIVITY PRINCIPLE (MIP): any question presupposes that it has a maximally informative answer, that is, a true answer that logically entails all the other true answers.
 b. UNIVERSAL DENSITY OF MEASUREMENTS (UDM): measurement scales needed for natural language semantics are always dense. (Fox and Hackl 2006: 542)
 c. $\forall d \forall d' [(d < d') \rightarrow \exists d'' (d < d'' < d')]$

A dense measurement scale offsets the problem posed by maximal informativity in (39) by making Ans-D undefined for degree questions. The proposition $\wedge \text{John is not } 5'10'' \text{ tall}$ is no longer the most informative proposition in the set because there now exists a more informative proposition which names a point between 5'9'' and 5'10'', say 5'9.5''. But UDM ensures that even this proposition is not maximally informative because there is another point in between 5'9'' and 5'9.5'' and so on *ad infinitum*. Since Ans-D is doomed to presupposition failure, the result is general unacceptability, assuming with Gajewski (2002) that irredeemable presupposition failure leads to ungrammaticality.

Fox and Hackl are aware that the negative island effect also holds in questions that do not intuitively range over dense scales:

- (41) How many children does(*n't) John have?

They address this problem by distinguishing between a deductive system which is contextually blind, and the actual interpretation of a question which is contextually sensitive. The logic of scales always includes density and since the presupposition of maximal informativity cannot be met for negative questions, they get

ruled out while positive questions are admitted. Once a question is admitted, context plays its usual role in determining what level of granularity the quantification should involve. For questions over individuals' heights, we may be satisfied by quantification that rounds to the closest inch or we may be interested in a more precise measurement. For quantification over numbers of children, the most likely contexts will involve whole numbers.⁷

An interesting consequence of Fox and Hackl's proposal, one that is not recognized in earlier theories, is the ameliorating effect of modals. They draw on initial observations by Kuno and Takami (1997) and show that the facts differentiate between statements with $\neg\Box/\Diamond\neg$ vs. $\Box\neg/\neg\Diamond$ order:

- (42) a. How many children are you not allowed to have?
 b. *How many children are you allowed not to have?
- (43) a. How many children are you required not to have?
 b. *How many children are you not required to have?

Fox and Hackl reason that (44) is felicitous in a context where the addressee does not know the exact weight of the vessel but is in a position to put an upper bound on its possible weight:

- (44) How much are you sure that this vessel won't weigh?

Though there can be no minimal degree d that the vessel doesn't weigh, due to density, there can be a minimal upper bound to possible weights. There can be a minimal weight such that in every world of the modal base, worlds that represent the addressee's certainty about the vessel's weight, the vessel does not have that weight. That is the amount that the maximally informative true answer in the question will name.

Thus Fox and Hackl provide an explanation in terms of the logic of modality and negation within UDM that accounts for modal obviation. Their more general point with regard to negative islands is that violations of maximal informativity lead to ungrammaticality but if maximal informativity can be preserved, apparent violations of negative islands are rendered innocuous.

A problem for this theory that Fox and Hackl are aware of concerns the role of D-linking in negative islands. The standard move of allowing D-linking to facilitate the shift from dense to discrete scales is not open to them, since all scales are considered dense for the purposes of calculating maximal informativity. The explanation for D-linking has to steer clear of this in order to preserve the explanation for examples like (39) and (41). Fox and Hackl resolve the paradox by appealing to a contextually licensed syntactic variable C , responsible for domain

⁷ Contexts that might use a more fine-grained measure would be statistical reports:

- (i) How many children does the average American family have?
 (ii) The average American family has 2.3 children.

restriction, *pace* von Stechow (1994). Crucially for them, levels of granularity enter into the determination of truth conditions quite differently from contextually licensed syntactic variables. Levels of granularity are among the parameters of the valuation function. As such, they are not represented in the syntax and are invisible to the module that checks grammaticality. The contextually licensed variable *C*, because it is present in the syntax, can save a sentence from ungrammaticality because its contribution is visible to the relevant module.

6.3.2 Negative islands, degree questions, and interval semantics

Abrusan and Spector (2011), building on Abrusan (2007), account for negative island effects and modal obviation without recourse to density of scales. Instead, they adopt an interval-based semantics for degrees (Schwarzschild and Wilkinson 2002). A gradable adjective denotes a function from intervals to sets of individuals. The adjective *tall*, for example, is a function from intervals *D* to individuals whose heights are in *D*. Example (45a) follows the convention in Heim and Kratzer (1998) of placing the presuppositions between a colon and a period. An interval, they contend, must contain a continuous set of degrees ((45b)):

- (45) a. $\llbracket \text{tall} \rrbracket = \lambda D: D \text{ is an interval. } \lambda x. x\text{'s height} \in D$
 b. Given a scale *E*, that is, an ordered set $(E, \leq)$, an interval on *E* is a subset *D* of *E* such that: $\forall d_1 \in E \forall d_2 \in E \forall d_3 \in E [((d_1 \in D) \wedge (d_3 \in D) \wedge (d_1 \leq d_2 \leq d_3)) \rightarrow d_2 \in D]$

If John is 5ft tall, we get (46b) as the interpretation for the positive question:⁸

- (46) a. How tall is John?
 b. $\{ \wedge \text{John's height is in the interval } [5, 5],$
 $\wedge \text{John's height is in the interval } [1 \dots 5],$
 $\wedge \text{John's height is in the interval } [5 \dots 11], \dots$
 $\wedge \text{John's height is in the interval } [1 \dots 4]$
 $\wedge \text{John's height is in the interval } [6 \dots 11] \}$

Any proposition built on an interval that includes 5ft in it is true, while any proposition that builds on an interval that does not include 5ft is false. Now, of the true propositions, there is one that entails all others, namely the proposition that John's height is in the smallest of the intervals, namely $[5, 5]$. This satisfies the presupposition that there is a maximally informative true answer to (46a).

The negative question, in the same situation, denotes (47b) with the last two propositions true. Since neither of them entails the other, maximal informativity fails and the question is ruled out:

⁸ In Abrusan and Spector's notation, inward pointing brackets on intervals indicate inclusion of end-points, outward pointing brackets indicate their exclusion.

- (47) a. *How tall isn't John?
 b. {[^]John's height isn't in the interval [5,5],
[^]John's height isn't in the interval [1...5],
[^]John's height isn't in the interval [5...11],...
[^]John's height isn't in the interval [1...4]
[^]John's height isn't in the interval [6...11]}

As Abrusan and Spector point out, the presupposition of the negative question can only be satisfied if the degree in question is 0. This is not plausible in the case of heights, but if we switch to speed we can imagine a car that is not in motion and whose speed will therefore be included in the interval [0, 0]. The presupposition of the negative question *How fast isn't the car going?* will then be satisfied by the maximal interval that includes all the points above 0:] 0, + ∞[. They note that this violates a fundamental requirement for questions, namely that the answer should not already be available. If a question can only have one possible answer as a matter of grammar, there is no context in which it can be felicitous.⁹ Thus, maximal informativity in combination with an interval-based semantics for degrees can account for negative islands without appealing to dense scales.

Abrusan and Spector further show how their account deals with the modal obviation facts discussed by Fox and Hackl:

- (48) a. How fast are we not allowed to drive?
 b. How fast are we required not to drive?
 c. {[^]we are not allowed to drive [65 mph, 65mph],
[^]we are not allowed to drive [66 mph, 66 mph]
[^]we are not allowed to drive [66mph, 67 mph]...}

If the speed limit is 65mph, then the first proposition in the set above is false, but the other two are true. Of these, the first entails the second as well as any proposition that is built out of an interval that includes [66, 66] while excluding [65, 65], that is, any interval whose start point is the minimal degree that crosses the limit. The same explanation extends to the equivalent case of negation scoping under necessity. Informally, (48a) says: *for what interval I, is it not allowed that our speed be in I?* Of course, the question can only be asked if it is known that there is a maximum speed limit but not what it is.

Now, consider (49) with the scope order of the modal and negation reversed. The relevant logical facts are given in (50):

- (49) a. #How fast are we allowed not /not required to drive?
 b. For what interval I, is it allowed that our speed not be in I?

⁹ Abrusan and Spector add an explicit statement to this effect: *for at least one other world w' compatible with common knowledge, [the maximally informative answer] A[to Q] is not the maximally informative answer in w'*. This also holds for most theories, albeit sometimes implicitly, in the requirement that a question should express a state of ignorance about the proposition(s) that are true in the actual world.

- (50) For any two intervals I_1 and I_2 , the two following statements are equivalent:
- a. *It is allowed that our speed not be in I_1 entails It is allowed that our speed not be in I_2 .*
 - b. I_2 is included in I_1 .

It follows that the maximally informative true answer, if it exists, is based on an interval that includes all the intervals that yield a true answer. They dub this the ‘maximally informative interval’ and demonstrate the problem for each of the following situations:

Case 1: There is no particular speed s such that our speed must be exactly s . For any speed d , it is permitted that our speed not be in $[d, d]$. Then the maximally informative interval M , if it exists, must include all the intervals I such that it is allowed that our speed not be in I : M must be the interval $[0, +\infty[$, which means that there is a permitted world in which we have no speed, not even the null speed. This is a contradiction. So there is no maximal interval M .

Case 2: There is a particular speed s , distinct from 0, such that our speed must be exactly s . Since $s \neq 0$, it is allowed that our speed not be 0 and it is also allowed that our speed not be d , for any d strictly above s . The maximally informative speed, if it exists, must include 0 and any speed above, which is again the interval $[0, +\infty[$, which leads to a contradiction.

Case 3: Our speed must be exactly 0. Here there is a maximally informative interval, $]0, +\infty[$. This, however, is problematic. If the logic of the question determines that it can only be answered if the speed is 0, the question has exactly one maximally informative answer in every world, which is to say it is not a question at all.

Under all possible scenarios, then, there is a problem. This provides an explanation of the negative island effect for degree questions. Abrusan and Spector provide an interesting argument in support of their interval-based account. They predict that (51a) could be answered by (51b) while Fox and Hackl’s account (or Beck and Rullmann’s), predicts that it should be answered by (51c):

- (51) a. How fast must one drive on this highway?
 b. One must drive between 45mph and 75 mph.
 c. One must drive at 45 mph at least.

Both answers are considered acceptable by speakers. In order to account for (51c) they modify their approach in the following way. They take adjectives to denote sets of degrees, as in standard approaches, and incorporate an operator Π , which takes two sets of degrees as arguments. The second argument has to be an interval and it presupposes that the first has a maximum. Different readings are obtained depending on the relative scope of the modal and Π :

- (52) a. $[\Pi\Pi] = \lambda P_{\langle d, t \rangle} : P \text{ has a maximum. } \lambda I_{\langle d, t \rangle} : I \text{ is an interval. } \text{Max}(P) \in I.$
 b. For what interval I , is it required that $\Pi(\lambda d. \text{we are } d\text{-fast})(I)? = \text{For what interval } I, \text{ is it required that our speed be in } I?$
 c. For what interval I , $\Pi(\lambda d. \text{is it required that we drive at least } d\text{-fast})(I)?$

Interpretation (52b) derives (51b), as in the interval-based approach, while (52c) yields (51c), as in the set of degrees approach. Abrusan and Spector show that introducing this operator does not dilute the results for negative islands. It simply allows for an alternative LF, one in which the maximality presupposition of Π plays the same role that maximality plays in the account of negative islands in Rullmann (1995).

A final argument for the interval-based approach is the following:

- (53) a. Jack and Peter are devising the perfect Republic. They argue about speed limits on highways. Jack believes that people should be required to drive at a speed between 50 and 70 mph. Peter believes that they should be required to drive at a speed between 50 and 80 mph. Therefore
- b. Jack and Peter do not agree on how fast people should be required to drive on highways.

For Jack and Peter to disagree on a given question Q , there must be at least one proposition in the denotation of the question to which they assign different truth values (see Chapter 3). This is so in context (53a). They differ on the proposition: *people should be required to drive at a speed included in interval [50, 70]*. However, they agree on the proposition: *people should be required to drive at at least 50 mph*.

Fox and Hackl's theory is meant to account for a wide range of facts, in addition to negative islands. Abrusan and Spector provide an alternative for negative island effects, noting that density of scales may be needed for other phenomena. They note that interval semantics provides a uniform account for all amount questions, including those that do not seem to involve density. Furthermore, ungrammaticality can be computed without setting aside contextual parameters such as granularity. Abrusan and Spector do not engage with the issue of D-linking, but since their approach to calculating ungrammaticality does not involve exempting granularity, they have the option of treating D-linking as simply facilitating the shift from scales to discrete objects. In fact, some of the examples discussed by them may find a ready solution under this view of D-linking.¹⁰

6.3.3 *Wh islands and degree questions*

Most semantic accounts of weak islands focus on negative islands. Abrusan (2011a, 2011b), building on Abrusan (2007) and Abrusan and Spector (2011), addresses other weak islands as well. Her account of degree questions in *wh* islands builds on the semantics we have already introduced. Let us compare a question over individuals with one over degrees:

¹⁰ I have in mind their example (33) *how many children does none of these women have?* They consider cases in which for a given number n , for every number below it there is a woman who has n number of children. They show that interval semantics, though applicable here, would only work in highly implausible contexts. However, it is possible to interpret this question in the kinds of contexts where specific numbers are being discussed and intervals would not apply.

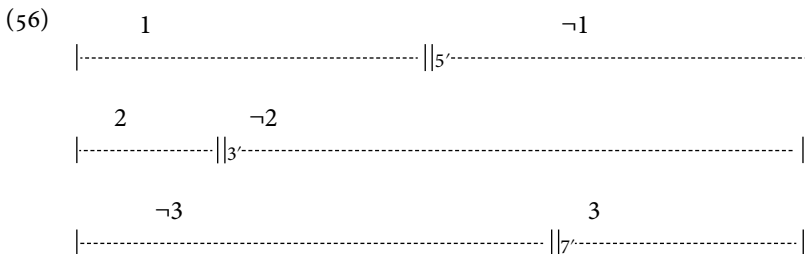
- (54) a. Who does Mary know whether she should invite?
 b. $\lambda q \exists x [\text{person}(w)(x) \wedge q = \lambda w' \text{ know}(w')(m, \lambda p [p = \lambda w'' \text{ she}_M \text{ should invite } x \text{ in } w'' \vee p = \lambda w'' \text{ she}_M \text{ should not invite } x \text{ in } w''])]$
 c. $\{^{\wedge} \text{Mary knows whether she should invite Bill,}$
 $^{\wedge} \text{Mary knows whether she should invite John,}$
 $^{\wedge} \text{Mary knows whether she should invite Fred,}$
 $^{\wedge} \text{Mary knows whether she should invite John+Bill,}$
 $^{\wedge} \text{Mary knows whether she should invite John+Fred,}$
 $^{\wedge} \text{Mary knows whether she should invite Fred+Bill,}$
 $^{\wedge} \text{Mary knows whether she should invite John+Fred+Bill}\}$

If Mary knows whether she should invite John and Bill, the maximally informative answer is: $^{\wedge} \text{Mary knows whether she should invite John+Bill}$. This is because under a distributive interpretation, it entails the two other true propositions. Furthermore, Abrusan proposes that asserting that a proposition is a complete true answer is tantamount to asserting that the other members of the Hamblin set not entailed by it are false (see also Section 3.1). In the case at hand, the negation of all the non-underlined propositions in the Hamblin set hold. Since it does not follow from $^{\wedge} \neg \text{Mary knows whether she should invite } J+F+B$ that $^{\wedge} \neg \text{Mary knows whether she should invite } J+B$, maximal informativeness is preserved and (54a) is deemed acceptable.

Choosing any of the propositions denoted by a degree question, instead, leads to incoherence:

- (55) a. *How tall does Mary know whether she should be
 (in order to join the basketball team)?
 b. $\lambda q \exists I [I \in D_I \wedge q = \lambda w' \text{ know}(w')(m, \lambda p [p = \lambda w'' \text{ her}_M \text{ height should be in } I \text{ in } w'' \vee p = \lambda w'' \neg \text{ her}_M \text{ height should be in } I \text{ in } w''])]$
 c. $\{^{\wedge} \text{Mary knows whether her height should be in } I-1,$
 $^{\wedge} \text{Mary knows whether her height should be in } I-2,$
 $^{\wedge} \text{Mary knows whether her height should be in } I-3\}$

The pictorial representation in (56), which we can take to mark 1 to 10, illustrates. Interval 2 is fully contained in interval 1 and interval 3 is fully contained in the complement of interval 1:



Suppose the maximally informative answer is *Mary knows whether her height should be in I-1*. The other two propositions in (55c) will have to be false, since neither of them are entailed. But this means something like the following, given the assumption that knowing if a particular height is in a particular Interval means knowing that it is not in its complement Interval:

- (57) #Mary knows whether her height should be between 0 & 5 or between 5 & 10
 but she does not know whether her height should be between 0 & 3 or
 between 3 & 10
 and she does not know whether her height should be between 0 & 7 or
 between 7 & 10.

The only way out of this conundrum is to choose the interval that includes all the intervals (the interval 0 to 10), but that is tantamount to saying *Mary knows whether she should have a height*. This is not informative. The felicity condition for asking the question cannot be met if only one answer is allowed by the grammar. Thus the ungrammaticality of (55a) is located in the interaction of the semantics of degrees, the semantics of the embedding predicate, and the presupposition that questions have a maximally informative answer.

Abrusan shows how this approach applies to amount questions, varying the embedding predicate, as well as switching from polar questions to constituent questions in embedded position. Here is how modal obviation in amount questions is dealt with:

- (58) a. Context: Cough syrup contains some steroids. For this reason, Olympic athletes need to inform the World Anti-Doping Agency (WADA) if they have to take more than a certain amount of it. But I forgot what this amount was, so I ask my friend, Michael Phelps:
 ?How much cough syrup does the WADA need to know whether you took t?
 b. Context: Losing a lot of weight too fast can also be suspicious when it comes to Olympic athletes. Therefore WADA has an interest in such cases as well.
 ?How many pounds does the WADA want to know whether the boxers lost t?

There are two parts to the proposed explanation. First, the logic of universal quantification resolves the contradiction that would otherwise obtain. This is discussed in depth by Fox (2007) and Abrusan adapts that general line of argument to the *wh* island case. Second, as Kroch (1998) has pointed out, finding contexts that can satisfy the presupposition of the question is key. Manipulating contexts for *wh* islands proves to be difficult, hence the slight marginality of some examples, and the improvement when the context makes available a limited set of choices.

6.3.4 *Weak islands and manner questions*

Let us see how Abrusan (2011b) deals with weak island effects in questions that quantify over manners (and reasons). The general logic of the explanation should be familiar by now: something in the nature of the quantification leads to deviance when combined with the requirement of maximal informativity. Along with the well-known resistance of manner questions to weak islands, Abrusan shows that they too are sensitive to modal obviation:

- (59) a. *How didn't John behave?
 b. How was John not allowed to behave?

Like Szabolcsi and Zwarts (1993), Abrusan appeals to a difference between the domain of manners (and reasons) and the domain of individuals. She admits structured domains for adjuncts, not unlike domains for individuals. This accounts for the conjunction of manners in the same way as for the conjunction of individuals:¹¹

- (60) a. How did John run?
 b. {[^]John ran fast, [^]John ran carelessly, [^]John ran fast+carelessly...}
 c. John ran fast and carelessly.

In a context where John's running was fast as well as careless there is a maximally informative proposition, the italicized one in (60b). We can conclude that there is no other manner that describes his running.

The crucial distinction that Abrusan (2011: 281) introduces between the domain of individuals and that of manners is to treat the latter as containing contraries (Horn 1989):

- (61) a. Two statements are contraries if they cannot be simultaneously true.
 b. Two statements are contradictories if they cannot be simultaneously true or false.
 c. P: wisely; fast; by bus
 P': unwisely; slowly; by car

The context might implicitly restrict the domain of manners to particular dimensions but within a particular dimension, a manner as well as its contraries are represented. The context might focus on speed (fast, slow) and attention (careful, careless) rather than on style (clumsy, graceful) but for each dimension there will be a pair of contraries. Some examples of contraries are given in (61c): one can drive fast, slow, or neither fast nor slow. Similarly, one can travel by bus, by car, or by neither bus nor car. Crucially, one cannot both be fast and slow at the same

¹¹ Abrusan illustrates the proposal using event semantics, but I believe the illustration provided here captures the essential features of her analysis. I make this change in the interests of space, since switching to event semantics would require introducing a certain amount of formal detail.

time and one cannot both travel by bus and by car at the same time. On this view, the denotation of (60a) is not (60b) but rather (62):

- (62) { $\wedge$ John ran fast, $\wedge$ John ran slowly, $\wedge$ John ran at medium speed,
 $\wedge$ John ran carelessly, $\wedge$ John ran carefully, $\wedge$ John ran with medium-care ...
 $\wedge$ John ran fast+carefully ... }

Ans-D applied to (62) picks out the same maximally informative answer as before. But now consider the negative question (63a), keeping the situation constant and focusing just on the dimension of speed:

- (63) a. How didn't John run?
 b. { $\wedge$ $\neg$ John ran fast, $\wedge$ $\neg$ John ran slowly, $\wedge$ $\neg$ John ran at medium speed
 $\wedge$ $\neg$ John ran slowly+medium-speed ... }

If John ran fast, there will be two speeds of which it is true that John didn't run at that speed: slow and medium. This is where the problem arises. If the two underlined propositions are both true and neither entails the other, the presupposition that there is a maximally informative true proposition is not satisfied. We could presumably take the plural manner *slow+medium-speed*, which would make the proposition $\neg$ John ran slowly+medium-speed the maximally informative statement. But here Abrusan posits a general requirement that it should, in principle, be possible for the members of a plurality to hold at the same time. So, *fast+careless* is defined because it is possible for a single event to be conducted in a fast as well as a careless manner, but *slow+medium-speed* is undefined because there is no single event which can be conducted at both speeds. That is, plural manners can only be formed across dimensions, not within them.

Let us see how modal obviation works, interpreting (64a) in a context where John is not allowed to drive fast. I use explicit quantification over worlds to illustrate:

- (64) a. How is John not allowed to drive?
 b. { $\lambda w \neg \exists w' w \leq w' \text{ drive}(w')(j, \text{slow}),$
 $\lambda w \neg \exists w' w \leq w' \text{ drive}(w')(j, \text{fast}),$
 $\lambda w \neg \exists w' w \leq w' \text{ drive}(w')(j, \text{medium-speed})$ }

Here the first and the third propositions are false. This means that the propositions in the scope of negation must be true: it is permitted that John drives slow and it is permitted that John drives at medium speed. This is not a problem since the contrary behaviors (*slow*, *medium-speed*) are distributed across worlds.¹² Now consider what happens when negation scopes over necessity:

¹² Note that there would be no problem even if the situation were a bit stricter and John was not allowed to drive fast and he was not allowed to drive at medium speed. Then in all the permissible worlds John's driving would be uniformly slow but, interestingly, the answer would not make reference to the plural individual *fast+medium-speed*: John is not allowed to drive fast or at medium-speed.

- (65) a. *How is John not required to drive?
 b. $\frac{\{\lambda w \neg \forall w' w \leq w' \text{ drive}(w')(j, \text{slow}),$
 $\lambda w \neg \forall w' w \leq w' \text{ drive}(w')(j, \text{fast}),$
 $\lambda w \neg \forall w' w \leq w' \text{ drive}(w')(j, \text{medium-speed})\}}$

The question presupposes that there is exactly one true proposition, and let us assume that it is that he is not required to drive slow. This means that the other two propositions are false, which means that in every accessible world, John's driving is *fast+medium-speed*. But we know that such plural manners are undefined. The ungrammaticality is explained.

Abrusan takes the same approach to explain factive/presuppositional islands. Along with the presence of contraries for manners we also have the projection of the factive presupposition:

- (66) a. *How does Mary regret that John drove the car?
 b. $\lambda p \exists \alpha [\alpha \in D_M \wedge p = \lambda w' \text{ Mary regrets that John drove the car in } \alpha \text{ manner in } w']$
 c. $\{^{\wedge} \text{Mary regrets that John drove the car fast},$
 $^{\wedge} \text{Mary regrets that John drove the car slowly},$
 $^{\wedge} \text{Mary regrets that John drove the car at medium-speed}\}$
 d. Projected Presupposition:
 For every manner $\alpha \in D_M$ Mary believes that John drove the car in α

Example (66a) is a question about John's speed at a particular time/event, and a question must represent a genuine choice between possible speeds, so we get the set of propositions in (66c). Maximal informativeness can be satisfied by one of these propositions. Due to the factivity of *regret*, however, the quantification is over propositions that the attitude-holder believes to be true (though her regret need only be about one of them). In this case, it would require Mary to believe that John's manner of driving was *fast+slow+medium-speed*, but of course such a plural manner does not exist. To establish that the presupposition projected is indeed universal, Abrusan presents (67):

- (67) a. Who among these ten people does Mary regret that Bill invited?
 b. Projected Presupposition: $\forall x \in \{\text{these 10 people}\}$: Mary believes that Bill invited x.

Abrusan, then, is successful in explaining a rather complex set of facts on the basis of a small set of defensible semantic principles. Let us return to the issue of D-linking. Abrusan discusses the kinds of contexts that Kroch brought into the discussion of weak islands and shows that violations are only remedied when the domain that is explicitly given rules out contraries, as in (68a) but not in (68b):

- (68) a. ?How did you not play chess?
 A: Blindfolded, B: Drunk, C: In a bathing suit.

- b. *How do you not speak French?

A: Very well, B: so-so, C: badly

Although Abrusan does not address the issue of the class of *wh* expressions that Pesetsky identified as “inherently D-linked,” that is *which N*, I note again that as with Szabolcsi and Zwarts, two factors seem to be relevant, the inherent partitivity of *which N* phrases and the role of *N* in setting the level of granularity.

6.3.5 Section summary

The explanations reviewed in this section can be seen as developing the insight in Kroch (1998) that weak island effects arise when the existence and uniqueness presuppositions of the question are not satisfiable. The proposals cashed this out in terms of the Maximal Informativity Principle, as in the baseline theory we have proposed. For degree questions we considered two proposals, one that appeals to density in measurement scales and one that appeals to an interval-based semantics. For questions over manners and reasons, the explanation rests on including contraries in the quantificational domain. Thus, these proposals also develop the insight in Szabolcsi and Zwarts that there is a difference between the quantificational domains for good and bad extractees. On the empirical side, modal obviation facts established that the problem cannot be syntactic. This approach extends beyond negative islands, to factive as well as *wh* islands. A unified semantic/pragmatic explanation is possible for the three structures that we took as exemplars of weak islands.

6.4 Weakness in islands

This chapter looked at the phenomenon of weak islands (negative, *wh*, and factive/presuppositional), structures which selectively allow extraction of *wh* phrases:

- (69) a. [_{CP} *wh*_α [TP ... [ISLAND ... *wh*_α ...]]]
 b. *[_{CP} *wh*_β [TP ... [ISLAND ... *wh*_β ...]]]

The syntactic literature has focused on determining what properties of α and β might allow or disallow a dependency across an island. There are three proposals for the values of α and β : they stand for arguments vs. adjuncts, for referential versus non-referential entities, or for D-linked versus non-D-linked expressions. The relativized minimality approach of Rizzi (1990) and Cinque (1990) is the most widely accepted syntactic account at this point, with D-linking being the most likely candidate separating α and β . D-linking refers to the requirement of a contextually salient set of entities in the discourse (Pesetsky 1987). *Which N* is generally considered inherently D-linked, *who/what* can be D-linked in particular contexts, and *how adj/how many N/how/why* are essentially non-D-linked, though in very specific contexts they too may get D-linked. They differ from *wh-the-hell* which is classified as aggressively non-D-linked (but see Szabolcsi and

Zwarts 1993). The propensity for expressions to get D-linked results in a certain amount of variation in the data, but the generalization in (69) is essentially reliable.

Weak islands have been approached from a semantic perspective as well. The locus of explanation is the mode of composition, not the syntactic nature of the wh dependency. Rullmann (1995) argues that a question denotes the maximal element that the wh expression ranges over. For *which N/who*, a positive or negative question can pick out such an entity: *which is the maximal entity that left/didn't leave*. For degrees, however, things are different: *which is the maximal degree to which she is tall* is defined but *which is the maximal degree to which she is not tall* is not. D-linking can help by shifting the quantificational domain to one where maximality holds. Szabolcsi and Zwarts (1993) argue that some wh expressions range over unordered sets for which the Boolean operations required for computing islands are well defined; others range over partially ordered sets for which the relevant operations are unsuitable. D-linking is a facilitating factor but does not provide an entirely reliable way to carve out the empirical generalization. Under both accounts, technically speaking, there is no island. It is just the nature of the wh expression that does or does not result in a well-defined question meaning. There are two points worth noting here. One is that D-linking is not part of the explanation, though it does play an indirect role in it. The second point is that these accounts work best for negative islands. A somewhat different approach to weak islands appeals to the existence and uniqueness presuppositions of a question (Comorovski 1989, 1996; Kroch 1998). Kroch argues that the contexts most readily available for amounts, for example, do not satisfy these presuppositions when there is a weak island in the nucleus. Thus the locus of explanation shifts to aspects of question meaning that have been argued for independently in this book.

The next set of proposals takes this further and explains weak island violations by appealing to the requirement for every felicitous question to have a maximally informative answer. Fox and Hackl (2007) consider degree questions as ranging over dense scales. The positive question satisfies the requirement of maximal informativity because the answerhood operator picks out the proposition naming the maximal degree. The negative question does not satisfy it because there can be no maximally informative statement with dense scales. An important consequence of this line of explanation is that an appropriately placed modal can distribute the degrees over worlds in such a way that the question becomes answerable. It further shows that weak island effects are not absolute, even for a given wh expression and a given weak island. Abrusan and Spector (2011) work within the same general approach to weak islands but instead of a dense measurement scale they adopt an interval-based semantics for degree questions. They argue that this provides a better account of non-degree questions, while preserving the modal obviation facts along the lines proposed by Fox and Hackl. This builds on earlier work by Abrusan (2007, 2011a, 2011b) where the behavior of wh expressions ranging over degrees and manners in presuppositional, negative, and wh islands is addressed. The notion of maximal informativity plays a crucial role again, as does the claim that manners denote contraries (pairs that can be false at the same time, but not true at the same time). The result is a theory rich enough to tackle a rather complex and disparate-seeming set of weak island effects.

Wh inside islands

This chapter continues to explore issues related to scope taking out of islands, but shifts the focus from *wh* expressions that have been extracted to those that remain inside the island. The primary diagnostic for determining matrix scope is the possibility of specifying values for the embedded *wh* in the answer. The central theoretical issue of concern is the role of syntactic constraints on the scope of such *wh* expressions. If *wh* expressions can take extraordinary wide scope, the locus of explanation lies in the syntactic module. If they do not actually take scope out of islands but merely appear to do so, the burden shifts to the semantic component. The issue that lies at the heart of the phenomena covered in this chapter turns on the following question: how reliable is the diagnostic of using specification of values in the answer in the determination of *wh* scope?

Nominal islands, for example noun phrases and prepositional phrases, are strong islands which require overt pied piping when they contain *wh* expressions. Pied piping raises significant challenges. Should the non-*wh* material be reconstructed into the base, leaving only the *wh* expression in matrix scope position at transparent LF? Or should the *wh* be interpreted with scope over the island, but in ways that yield the effect of a direct question? *Wh* in-situ languages as well as fronting languages are discussed in this connection.

Wh islands, introduced in Chapter 6, are also probed but this time from the perspective of *wh* expressions that, while remaining inside the island, seem to interact at the matrix level. Long-distance list answers, pairing values for a matrix *wh* and an embedded *wh*, suggest that the *wh* takes scope at the matrix level. This implies a fundamental syntactic difference between overt and covert scope taking in their sensitivity to islands. But is there sufficient motivation to posit differences in island sensitivity between overt and covert scope? Is it possible to interpret the embedded *wh* within the island and reinterpret the long distance list as pairing the matrix *wh* with the island rather than an expression within the island? These questions are explored, taking into account cross-linguistic differences between multiple *wh* fronting, single *wh* fronting, and *wh* in-situ languages.

Long-distance lists raise similar questions in the case of *wh* inside adjunct islands and give rise to similar responses. Do we need mechanisms for giving wide scope to *wh* expressions inside adjuncts? Or can we assign scope to *wh* phrases at the site of the island and capture long-distance lists through relations with the adjunct island? Evidence from single-pair answers, multiple-pair answers, and so-called trapped list answers are discussed from both perspectives.

This chapter, we will see, extends the discussion to include strong islands but one type of strong island that is omitted is coordinate structures. This is because

they have not featured much in discussions of covert scope taking, even though their relevance is self-evident. D-linking is discussed at various points, especially in relation to the perception that D-linked *wh* phrases can violate syntactic constraints on scope. In this sense too this chapter continues to probe themes introduced in the last chapter.

7.1 Pied piping

Here we consider the phenomenon of nominal pied piping, in which a *wh* expression moves to the scope position along with the containing DP or PP, a constituent that is otherwise an island for movement. Arguments for such pied piping are straightforward in the case of *wh* fronting languages but less clear for *wh* in-situ languages. We look at the challenge pied piping poses for interpretation and elaborate on two approaches that have been pursued in this regard. In one, the embedded *wh*, and only the *wh*, is in scope position at transparent LF. In the other, the larger phrase containing the *wh* is interpreted in that position.

7.1.1 Subjacency and reconstruction

A simple question like (1a) is not tractable under our working assumptions about the syntax–semantics mapping in interrogatives:

- (1) a. Whose book did Bill read?
 b. $[_{CP} [_{DP} \text{whose}_j \text{book}]_i [_{C'} [_{TP} \text{bill read } t_i]]]$
 c. Bill read Tolstoy's book.

If (1b) is the transparent LF, to use von Stechow's term for referring to the syntactic structure that undergoes interpretation, we get semantically undesirable results. In quantifying the *wh* expression in Spec CP into C', we have to decide whether to focus on the properties of books or on the properties of book-owners/authors. If the *wh* expression is an existential over books, based on the head NP, we must treat the author/owner as existentially bound within the *wh* phrase: $\lambda P \exists y [\exists x (\text{person}(x)) \wedge y = \text{the-book-of-}x \wedge P(y)]$. This is the property set of books that are in some way unique among their authors'/owners' books. It leads to the derivation in (2). Though the propositions in the set are reasonable enough they do not give the information the question is after. The propositions should name books, but via their owners/authors:¹

¹ Suppose Tolstoy, Austen, and Dostoyevsky are the authors on a reading list with the books as listed in (2b). Assume that the books listed for the same authors on a previous reading list were *Anna Karenina*, *Persuasion*, and *The Idiot*. Sentence (1c) can be continued with *just like last time* even though *War and Peace* was not an option the last time. The response is valid if Bill read *Anna Karenina*, the book of Tolstoy that was on the previous list. In other words, we need the authors to stay fixed but books to vary across indices, while remaining anchored to the authors.

- (2) a. $\lambda p \exists x \exists y [person(x) \wedge y = \iota z [book-of-x(z)] \wedge p = \wedge read(b,y)]$
 b. $\{\wedge Bill \text{ read War and Peace}, \wedge Bill \text{ read Emma}, \wedge Bill \text{ read Crime and Punishment}\}$

Alternatively, if the *wh* expression is interpreted as the property set of owners/authors with unique books, taking the possessive *wh* as semantic head, we get: $\lambda P \exists x [person(x) \wedge \exists y [y = the-book-of-x] \wedge P(x)]$. But now we get the set of undefined propositions in (3), with the objects of the reading activity being people, not books:

- (3) a. $\lambda p \exists x \exists y [person(x) \wedge y = \iota z [(book-of-x(z) \wedge p = \wedge read(b,x))]]$
 b. $\#\{\wedge Bill \text{ read Tolstoy}, \wedge Bill \text{ read Austen}, \wedge Bill \text{ read Dostoyevsky}\}$

Reconstructing everything in the DP except the *wh* phrase into base position gives the desired results (Engdahl 1986; Cresti 1995; von Stechow 1996; among others):

- (4) a. $[_{CP} [_{DP} \text{who}_j [_{IP} \text{bill read } t_j\text{'s book}]]]$
 b. $\lambda p \exists x [person(x) \wedge p = \wedge \exists y [y = \iota z (book-of-x(z)) \wedge read(b,y)]]$
 c. $\{\wedge Bill \text{ read Tolstoy's book}, \wedge Bill \text{ read Austen's book}, \wedge Bill \text{ read Dostoyevsky's book}\}$

A further advantage of adopting reconstruction is that it explains the available binding theoretic possibilities (see Engdahl 1986; Jacobson 1998; Sharvit 1998; Sportiche 2005; among others, for relevant discussion):

- (5) a. Which picture of himself_i did John_i give to Mary?
 b. Which picture of him_{*i/j} did John_i give to Mary?
 c. Which picture of John_i did he_{*i/j} give to Mary?

The interpretation of the reflexive, the pronoun, and the proper name in the pied piped phrase is constrained by the base position. Furthermore, the reflexive and the pronoun can get bound variable readings, again as dictated by the base position:

- (6) a. Which picture of himself_i did every student_i give to Mary?
 b. Which of his_{i/j} pictures did every student_i give to Mary?

Note that English syntax prevents the extraction that would lead to the kind of structure that the semantics seems to require:

- (7) a. $*[who_i \text{ did you read } [t_i\text{'s book}]]$
 b. $*[whose_i \text{ did you read } [t_i \text{ book}]]$

Giving the *wh* expression matrix scope violates syntactic constraints while moving the island itself is syntax compliant. Reconstruction at LF solves the problem of interpretation, even though it creates exactly the same structure that is unacceptable at S-structure.

One way to think of the problem is in terms of subadjacency constraints holding at S-structure but not at LF, a claim made forcefully by Huang (1982) on independent grounds. We can illustrate this with the following examples from English and Chinese, deferring discussion of further paradigms to Sections 7.2 and 7.3. Extracting a non-D-linked *wh* phrase out of a *wh* island is marginal in English (8a), as we saw in Chapter 6. Extraction out of a complex noun phrase is unacceptable, regardless of whether the *wh* is D-linked or not (8b). Huang notes that the Chinese counterparts in (9) are fully acceptable:

- (8) a. ??/*What_i does John wonder [who bought t_i]?
 b. *Who_i/which politician_i do you like [books that criticize t_i]?
 (9) a. ni xiang-zhidao [shei mai-le sheme]
 you wonder who buy-ASP what
 “Who is such that you wonder what he/she bought?”
 “What is such that you wonder who bought it?”
 b. ni zui xihuan [piping shei de shu]
 you most like criticize who REL book
 “Who is such that you like books that criticize him/her?”

As indicated, (9a)–(9b) can be interpreted as direct questions, seemingly in violation of islands.² Huang claims that syntactic constraints only apply to expressions scoping overtly, not to those scoping covertly. In other words, islands only hold at S-structure.

In related languages like Korean and Japanese, Nishigauchi (1986, 1990) notes a difference between the two types of islands. The following Japanese examples illustrate:

- (10) a. tanaka-kun-wa [Mary-ga nani-o kat-ta ka] sitte-imasu
 Tanaka Mary what bought Q knows
 “Tanaka knows what Mary bought.”
 b. tanaka-kun-wa [Mary-ga nani-o kat-ta koto-o] sitte-imasu ka
 Tanaka Mary what bought Comp knows Q
 “What does Tanaka know that Mary bought?”
 c. tanaka-kun-wa [Mary-ga doko-de nani-o kat-ta ka] sitte-imasu ka
 Tanaka Mary where what bought Q knows Q
 “Does Tanaka know where Mary bought what?”
 NOT: “What is such that Tanaka knows where Mary bought it?”

Japanese has a question morpheme *ka* (and *no*) that indicates interrogative force. In (10a), with *ka* in embedded C, the *wh* in-situ is interpreted with embedded

² The *wh* island facts for Chinese are somewhat controversial (see Nishigauchi 1990: 32, fn 13; and Dayal 1996: 228) but the complex NP facts hold up. See Tsai (2013) for a more nuanced description of cases like (9b).

scope. In (10b), with *ka* in matrix C, the embedded *wh* in-situ gets matrix scope. This covert scope taking does not cross an island since the embedded clause does not have a Q morpheme. The interesting case is (10c), where both the embedded and the matrix C host Q morphemes. Nishigauchi emphasizes that the embedded *wh* only has local scope, with the matrix Q forcing a polar interpretation for the matrix clause. That is, Japanese *wh* in-situ cannot covertly scope out of a *wh* island, contra Huang.

Nishigauchi notes that in the case of complex noun phrases, however, a direct question interpretation for an embedded *wh* is readily available (11a). The same is true for *wh* expressions inside adjunct phrases. This presents an interesting state of affairs. *Wh* islands, which are weak islands, block matrix scope for embedded *wh* in-situ while nominal and adjunct islands, which are strong islands, seem to allow such scope. Nishigauchi resolves this puzzle by arguing for a pied piping analysis for complex noun phrases (11b). Crucially, the *wh* expression moves from within the noun phrase to adjoin to it, and its quantificational properties get transferred to the containing phrase. This is followed by LF pied piping of the whole DP to Spec CP:³

- (11) a. Kimi-wa [DP [CP dare-ga kai-ta] hon-o] yomi-masi-ta-ka
 You-TOP who-NOM wrote book-ACC read Q
 ‘Who is such that you read a book that he/she wrote?’
 b. [CP [DP who_j [DP [CP t_j wrote] book]]_i [IP you read t_i] Q]

Nishigauchi’s point is that *wh* scope is constrained by islands and any apparent wide scope effects are due to pied piping. *Wh* islands do not lend themselves to pied piping and therefore *wh* expressions inside them do not manifest the so-called wide scope effect.

Covert pied piping has been claimed for a range of languages (see Heck 2008 and references cited therein). It should be obvious that it poses the same problems for interpretation that we articulated in connection with overt pied piping. Appeal to covert pied piping loses much of its force, however, if it has to be accompanied by (covert) reconstruction. A more desirable alternative would be one that derives the appropriate set of propositions while interpreting the pied piped phrase in scope position.

7.1.2 Alternatives to reconstruction

Nishigauchi (1986, 1990) interprets the pied piped structure (12a), treating the Q morpheme *ka* as an unselective binder (Lewis 1975; Heim 1982; Kamp 1981):

³ See Cable (2007, 2010) and references cited therein for somewhat different ways of construing pied piping. I represent the island in (11) as DP, though depending on the analysis of Japanese case marking, it could be a PP.

- (12) a. $[_{CP} [_{DP} \text{who}_j [[_{t_j} \text{wrote }] \text{book}]_i [_{IP} \text{you read } t_{i/j}] Q_{i/j}]]$
 b. $\lambda p \exists x \exists y [\text{book-that-}y\text{-wrote}(x) \wedge p = \wedge \text{read}(\text{you}, x)]$
 c. $\{ \wedge \text{you read W\&P}, \wedge \text{you read PP} \}$

von Stechow (1996) points out that this would lead to (12b)–(12c), an undesirable result. The problem is the familiar one, namely that the variable corresponding to the owner/author does not feature in the nucleus. von Stechow presents a number of other arguments against the unselective binding approach, some of which we anticipated in our earlier discussion. Although von Stechow himself opts for a reconstruction-based solution, Sternefeld (2001, 2002) presents an alternative that preserves the integrity of Nishigauchi's syntax. His account, which adopts the choice functional approach to quantification in Reinhart (1997, 1998), applies to structures like (12a). It avoids the problems that the unselective binding approach faces.

Reinhart argues that LF movement of *wh* in-situ is incompatible with Minimalist assumptions. She interprets *wh* in-situ in their base position using choice functions, functions from properties to individuals.⁴ Her proposal can be illustrated with English multiple *wh* questions:

- (13) a. Which student read which book?
 b. $[\text{which student}_i [t_i \text{ read which book}]]$
 c. $\lambda p \exists f \exists x [\text{student}(x) \wedge CF(f) \wedge p = \wedge \text{read}(x, f(\text{book}))]$

While the fronted *wh* quantifies over individual students in the standard way, the *wh* in-situ is interpreted with a functional variable in argument position. This variable is bound from above C^0 , where the propositional core of the question is determined. That is, from the position where normal *wh* expressions take scope. The choice function takes the set of books as its argument and returns an arbitrarily chosen member of that set. Consequently, there are as many distinct choice functions as there are books. Quantification over choice functions thus delivers the same result as quantification over individuals, but without requiring covert movement and/or reconstruction.

Returning to covert pied piping, Sternefeld (2001, 2002) agrees with von Stechow's criticism of Nishigauchi's account but provides a choice-function-based alternative. Cable (2007, 2010) proposes essentially the same account of the semantics of pied piping, though his claims about their syntax are substantively different.⁵ Cable also explores a wider range of facts, some of which we will take up in the next chapter, where we will also consider a third proposal for interpreting such structures, due to Kratzer and Shimoyama (2002) and Shimoyama (2001, 2006). Here, I will present Sternefeld's (and

⁴ Reinhart argues specifically against the unselective binding approach (see Section 7.3).

⁵ Cable as well as Engdahl, whose account we will review in Section 7.1.3, use intensional versions of choice functions and Skolem functions respectively. For the purposes at hand, we can work with extensions. For discussion of contexts requiring functions in intension, see Engdahl (1986: 167–8). See also fn 1.

Cable's) insight about the interpretation of a pied piped noun phrase in slightly modified packaging. For ease of exposition I will often use individual variables x, y instead of choice function variables when there are no islands at issue. This has no substantive effect on the interpretation, but is meant to flag the distinction between island sensitive and island insensitive scope taking:

(14)

- a. $[_{CP-1} [_{DP-i} \text{whose book}] [_C \text{did John read } t_i]]$
 b. $\llbracket \text{whose book} \rrbracket =$
 $\lambda P_{\langle e, t \rangle} \exists x_e \exists f_{\langle \langle e, t \rangle, e \rangle} [\text{human}(x) \wedge CF(f) \wedge P(f(\lambda z [z = i_y (\text{book-of-} x(y))]))]$
 or equivalently
 $\lambda P_{\langle e, t \rangle} \exists g_{\langle \langle e, t \rangle, e \rangle} \exists f_{\langle \langle e, t \rangle, e \rangle} [CF(g) \wedge CF(f) \wedge$
 $P(f(\lambda z [z = i_y (\text{book-of-} g(\text{human}))(y)))]$

The pied piped phrase denotes an existential generalized quantifier over books (14b). The way in which it is built up, however, makes crucial reference to the author/owner argument in defining the properties of books. In the innermost part we recognize the inherent definiteness of a possessive noun phrase by taking the set of books to be uniquely instantiated for an individual owner/author x : $\lambda z [z = i_y (\text{book-of-} x(y))]$. We now feed this set in as the argument of a choice function, resulting in the book in that set. The generalized quantifier denotes the set of properties P of this book. Since we get to this book via the author/owner, there will be as many books as there are authors/owners.

When the pied piped phrase combines with C' and the question nucleus is substituted for P through lambda conversion, we get propositions where the argument of *read* is a book. The propositions in the question denotation vary on the choice of authors/owners because x is existentially bound from outside the nucleus:

(14)

- c. $\llbracket C' \rrbracket = [p = \wedge \text{read}(j, x_i)]$
 d. $\llbracket [14a] \rrbracket = \llbracket \text{whose book} \rrbracket (\lambda x_i \llbracket C' \rrbracket)$
 $\Rightarrow \exists x \exists f [\text{human}(x) \wedge CF(f) \wedge \lambda x_i [p = \wedge \text{read}(j, x_i)] (f(\lambda z [z = i_y (\text{book-of-} x(y))]))]$
 $\Rightarrow \exists x \exists f [\text{human}(x) \wedge CF(f) \wedge p = \wedge \text{read}(j, f(\lambda z [z = i_y (\text{book-of-} x(y))]))]$
 $\Rightarrow \lambda p \exists x \exists f [\text{human}(x) \wedge CF(f) \wedge p = \wedge \text{read}(j, f(\lambda z [z = i_y (\text{book-of-} x(y))]))]$
 e. $\{ \wedge \text{John read the book of Austen/Austen's book,}$
 $\wedge \text{John read the book of Tolstoy/Tolstoy's book} \dots \}$

This procedure can be freely iterated. Example (15c) gives a partial derivation for a single-pair or REF-Q reading (see Chapter 4) of (15a) under LF (15b):

- (15) a. Whose mother read from whose book?
 b. $[\text{whose mother}_j [\text{whose book}_i [t_j \text{ read from } t_i]]]$

- b. $\llbracket \text{whose mother} \rrbracket =$
 $\lambda P_{\langle e, t \rangle} \exists u_e \exists f'_{\langle \langle e, t \rangle, e \rangle} [\text{human}(u) \wedge CF(f') \wedge P(f'(\lambda z [z = \iota y (\text{mother-of-}u(y))])]$
- c. $\llbracket \text{whose book } [t_j \text{ read } t_i] \rrbracket =$
 $\exists x \exists f [\text{human}(x) \wedge CF(f) \wedge p = \wedge \text{read}(x_j, f(\lambda z [z = \iota y (\text{book-of-}x(y))])]$
- d. $\llbracket 15a \rrbracket = \llbracket 15b \rrbracket (\lambda x_j \llbracket 15c \rrbracket)$
 $\Rightarrow \lambda p \exists u \exists f' [\text{human}(u) \wedge CF(f') \wedge \exists x \exists f [\text{human}(x) \wedge CF(f) \wedge p =$
 $\wedge \text{read-from}(f'(\lambda z [z = \iota y (\text{mother-of-}u(y))]), f(\lambda z [z = \iota y (\text{book-of-}x(y))])]]]$

The multiple-pair reading of (15a) involves creating a dependency between the set of choice functions leading to mothers and the set of choice functions leading to books. Expression (16d) gives the nucleus proposition in schematic form (see Chapter 4 for details):⁶

- (16) a. $f_{CF}: \{f_{CF-1}(\text{john}, \text{bill}) \rightarrow \text{john's mother/alice},$
 $f_{CF-2}(\text{john}, \text{bill}) \rightarrow \text{bill's mother/betty}\}$
- b. $g_{CF}: \{g_{CF-1}(\text{Austen}, \text{Tolstoy}) \rightarrow \text{A's book/Emma},$
 $g_{CF-2}(\text{Austen}, \text{Tolstoy}) \rightarrow \text{T's book/AK}\}$
- c. $\lambda p \exists F_{\langle e, e \rangle} [\text{Dom}(F) = f \wedge \text{Range}(F) = g \wedge p = \cap \lambda p' \exists f' \in f [p' = \wedge \text{read}(f', F(f'))]]]$
- d. $\lambda p \exists F_{\langle e, e \rangle} [F: \llbracket \text{whose mother} \rrbracket \rightarrow \llbracket \text{whose book} \rrbracket \wedge$
 $p = \wedge \text{read}(\text{graph-of-some-}f \in F)]]$
- e. $\{\wedge \text{John's mother read Austen's book and Bill's mother read Tolstoy's book},$
 $\wedge \text{John's mother read Tolstoy's book and Bill's mother read Austen's book},$
 $\wedge \text{John's mother read Austen's book and Bill's mother read Austen's book},$
 $\wedge \text{John's mother read Tolstoy's book and Bill's mother read Tolstoy's book}\}$

There are two main distinctions between choice functions and Skolem functions worth noting. Choice functions are functions from sets of individuals to individuals and can be generalized to functions from any set of type $\langle \alpha, t \rangle$ to an arbitrarily chosen member of type α . This operation can apply to *wh* in-situ and by extension to larger phrases inside the nucleus and be existentially bound from outside to yield well defined question denotations.

Skolem functions are functions from individuals to individuals that can be used to capture functional dependencies between them. They are useful for modeling pair-list answers between individuals because they manifest functionality: domain cover whereby every individual in the domain set is mapped on to some individual in the range set, and point-wise uniqueness whereby they are mapped to only one such individual. Generalizations of this approach beyond individuals can be used

⁶ Assuming the set of individuals whose mothers and books we are interested in are Bill and Sue, we can apply the tests from Chapter 4 to establish the functional nature of pair-list answers:

- (i) Bill's mother and Sue's mother both read Austen's book.
(ii) #Bill's mother read Tolstoy's book and Austen's book.

However, as we will see, pair-list answers involving larger structures often tend to favor one-one pairings.

to represent functional readings involving larger structures. For perspicuity, I abbreviate where convenient, as in (16d). An alternative way to capture the functionality of list answers was discussed in Chapter 4, using higher order questions and distributing Ans-D over its members (Hagstrom 1998; Fox 2012). The working hypothesis I adopt is that natural language has higher order questions, but they are not essential to basic pair-list readings. Pair-list readings are available for two set denoting terms, regardless of types, as long as they are in a local configuration. The reader is encouraged to weigh the implications of the issues explored here and decide for themselves whether the crucial distinctions that emerge can be maintained in the alternative approach, which derives pair-list answers through higher order questions. Higher order interpretations will play a role in long-distance list answers, as we will see later in this chapter. The alternative approach, as far as I can tell, is not committed to using higher order questions for long-distance lists (see Dayal 2016b for discussion).

It bears emphasizing that appealing to choice functions does not mark a shift from the view that overt and covert scope taking have different sensitivities to syntactic islands but it just casts the distinction in different terms. Overt scope taking involves movement and respects islands, while covert scope taking taps into choice functions which can bind across islands.⁷ It is similarly true that the choice of Skolem functions in interpreting *wh* expressions is independent of the issue about locality in scope taking. Engdahl (1986), for example, argues for functional *wh* but takes the scope of *wh* in-situ to be insensitive to islands.

Let us return to the issue of pied piping and reconstruction. Engdahl (1986: 211–16) observes an ambiguity when a fronted *wh* phrase includes a pronoun. We represent the different readings, using Chierchia's syntax for functional dependencies:

- (17) a. [[which of his_k books]_i [did every author_j recommend t_i^j]]
 b. $\lambda p \exists f_{\langle e, e \rangle} [\forall x [\text{book-of-}x(f(x))] \wedge p = \wedge \forall z [\text{author}(z) \rightarrow \text{recommend}(z, f(z))]]$
 c. $\lambda p \exists f_{\langle e, e \rangle} [\forall x [\text{book-of-}x(f(x))] \wedge p = \wedge \forall z [\text{author}(z) \rightarrow \text{recommend}(z, f(z))]]$

In (17b) the pronoun *y* in the fronted phrase is interpreted deictically. The closure operation on the function restricts quantification so that for any function *f* and individual *x*, the output will be a book authored by *y*. If Tolstoy is the value assigned to the free variable *y*, the restriction on the function is that its output should be a book by Tolstoy. If the functional trace inside the nucleus is bound by the subject we get a reading in which every author chooses a (possibly different)

⁷ A similar dichotomy between island sensitive overt scope taking vs. island insensitive alternative binding mechanism is posited by Pesetsky (1987) based on D-linking. See Charlow (2014) for a recent discussion of issues related to locality and scope.

book by Tolstoy.⁸ If the pronoun inside the *wh* is interpreted as a bound variable (cf. *which of his own books*), we get reading (17c) where authors are linked to their own books. There is a third option, namely one in which there is no functional dependency. The pronoun inside the *wh* would be free and the propositions in the set would name different books by some individual, Tolstoy for example. In this case, Ans-D would pick one such proposition, naming a single book that every author recommends.

The key insight is that the morpho-syntax and the structural configuration of the base determines whether the pronoun is an anaphor subject to Principle A (which must have a syntactic antecedent) or a non-anaphoric pronoun subject to Principle B (which must be disjoint from relevant antecedents). The closure operation on the function is sensitive to this in a language specific way. If instead of the pronoun *his*, a fronted *wh* has an anaphor like *himself/herself* that is licensed at the base position, the closure operation would have to enforce the type of binding seen in (17c).

While Engdahl herself uses a system equivalent to reconstruction for basic cases of pied piping (Engdahl 1986: 208–10), the procedures she discusses can be used to dispense with reconstruction for intricate patterns of pronominal reference and bound variable readings:

- (18) a. $[_{CP} \text{ Whose book}_i [_C \text{ did every author read } t_i]] ?$
 b. $\llbracket \text{whose book} \rrbracket = \lambda F \exists f \exists u [\text{person}(u) \wedge \forall x [f(x) = \iota z [\text{book-of-}u(z)]] \wedge F(f)]$
 c. $\llbracket [C'] \rrbracket = [p = \wedge \forall y [\text{author}(y) \rightarrow \text{read}(y, f_i(y))]]$
 d. $\lambda p \exists f \exists u [\text{person}(u) \wedge \forall x [f(x) = \iota z [\text{book-of-}u(z)]] \wedge$
 $p = \wedge \forall y [\text{author}(y) \rightarrow \text{read}(y, f(y))]]$
 e. Every author read a book of Tolstoy's.

The fronted *wh* is interpreted as the property set of a function, a function from individuals to some contextually relevant book of theirs ((18b)). Quantifying in derives a set of propositions which vary on books (assuming that uniqueness inside the possessive is relativized appropriately), but the books are referred to via their authors/owners.

There are, then, viable alternatives to reconstruction for interpreting pied piped structures. Note that the reconstruction approach resolves in favor of syntactic complexity (we need to make room for a syntactic operation with different properties than standard fronting) but maintains a simple semantics (nothing beyond what we have said in our baseline theory for interpreting *wh* expressions in Spec CP). The function-based approaches, on the other hand, resolve in favor of

⁸ Under our assumptions this is the functional reading of the question, not the pair-list reading. This distinction would be evident if the quantifier binding the functional trace was *no author* as it only supports functional readings (see Chapter 4). We would now get an answer like *No author recommended his favorite Tolstoy novel*, allowing Tolstoy's novels to vary with the authors in the current context. We are following Chierchia (1993) in treating pair-list answers as being based on functional answers but with a distinct syntax and semantics.

semantic complexity (we need to incorporate the role of another argument in building up a suitable functional meaning) but maintain the integrity of syntactic operations (the semantics interprets whatever the syntax places in scope position). This tension between a simple semantics and a simple syntax will also characterize the proposals we consider in relation to *wh* inside other islands in Sections 7.2 and 7.3.

7.1.3 *Pied piping multiple wh*

One of the diagnostics that has been used in arguments for and against pied piping vs. island insensitive movement as well as in arguments for and against pied piping with and without reconstruction has to do with the status of short answers. Consider Nishigauchi's analysis of Japanese covert pied piping and von Stechow's critique of it:⁹

- (19) a. Kimi-wa [PP [CP dare-ga kai-ta] hon-o] yomi-masi-ta-ka
 You-TOP who-NOM wrote book-ACC read Q
 "Who is such that you read a book that he/she wrote?"
 b. Austen desu
 It's Austen.
 c. [PP [CP Austen-ga kai-ta] hon] desu
 "It's a book that Austen wrote."

Nishigauchi's argument is that the acceptability of (19c) is not predicted by a theory in which only the *wh* has matrix scope, but is expected if the *wh* and its containing phrase are in SpecCP. The idea, briefly, is that the short answer can either spell out the *wh* phrase *dare-ga*, "who" with *Austen*, or the pied phrase *dare-ga kai-ta hon-o*, "a book that who wrote" with *a book that Austen wrote*. The force of this argument holds equally in both current views of short answers, the view that they are derived by movement of a constituent to the left periphery of CP followed by ellipsis of TP (Merchant 2004; Weir 2014) and the view that they are complete syntactic objects unto themselves (Stainton 1998; Ginzburg and Sag 2000; Jacobson 2013).

Von Stechow notes that the unselective binding view Nishigauchi relies on leads to incorrect results when the complex noun phrase has more than one *wh* expression. He attributes (20a) to Shin-Sook Kim:

- (20) a. Kimi-wa [PP [CP dare-ga kai-ta] dono hon-o] yomi-masi-ta-ka
 You-TOP who-NOM wrote which book-ACC read Q
 "Which book that which person wrote did you read?"

⁹ Pesetsky (1987: 113) notes that many Japanese speakers prefer the short answer (19c) which repeats the whole island. Some find (19b) unacceptable, some marginal. The contrast disappears for such speakers when a D-linked phrase is used and a set of individuals explicitly given, from which the value of the *wh* is to be drawn.

- b. $Q_i j$ [who_i wrote which book_j] [you read t_j]
 c. For which x , y , x a book, y a person, did you read x ?

The unselective binding view of the Q morpheme predicts that the owner/author will not feature in the answer, making it equivalent to (19a). Question (20a), to the extent that is possible, calls for an answer naming a single book and a single author: *The Children of Darkness by Wolf von Niebelschütz*. Similar facts hold for English (21a):

- (21) a. Which mountain in which country did you climb?
 b. The Dom, which is in Switzerland.

von Stechow emphasizes that this is not a multiple wh question which asks for a list of mountains and countries. Rather, it seems to ask for the identity of the mountain the addressee is presupposed to have climbed, with the further requirement that the country in which it is located also be named. The relative clause in the answer has a non-restrictive or appositive interpretation.¹⁰

Sternefeld uses higher order interpretations involving choice functions to address this problem. A slightly modified presentation is given in (22):

- (22) a. $[_{DP} [_{DP} \text{which mountain}] [_{PP} \text{in which country}]]_i$
 b. $\llbracket 22a \rrbracket = \lambda P \exists f \exists g [CF(f) \wedge CF(g) \wedge P(f(\lambda x [\text{mountain}(x) \wedge \text{in}(x, g(\text{country}))]))]$
 c. $\llbracket [C' \text{you climbed } t_i] \rrbracket = [p = \wedge \text{climb}(\text{you}, x_i)]$
 d. $\llbracket 21a \rrbracket = \llbracket 22a \rrbracket (\lambda x_i \llbracket 22c \rrbracket)$
 $\Rightarrow \lambda p \exists f \exists g [CF(f) \wedge CF(g) \wedge p = \wedge \text{climb}(\text{you}, f(\lambda x [\text{mountain}(x) \wedge \text{in}(x, g(\text{country}))]))]$

For any given mountain, there is exactly one country in which it can be located. When Ans-D is applied to (22d) we get a proposition naming a unique pair of mountain and country. Note, of course, that the proposition only refers to the mountain. We have to assume some extra-semantic requirement that the form of the answer and the form of the pied piped phrase be isomorphic.¹¹

There are further nuances to pied piping with multiple wh expressions that show up when embedding occurs under responsive predicates (Engdahl 1986: 231–3):

- (23) a. Who remembers whose recording of which Beethoven symphony Mary prefers?
 b. John does.

¹⁰ It is not clear to me how von Stechow derives the interpretations for (20) and (21).

¹¹ This may well be due to requirements on the focus structure of questions and answers (see Chapter 8 for related discussion).

- Answer (23b) is not problematic but answer (23c) bears discussion. Engdahl provides the following S-structure for (23a). I have added a trace in the argument position of the embedded CP-2 and underlined the expressions whose values feature in the pair-list. Double underlining indicates the *wh* whose values do not feature in the answer:

- Let us take the unacceptability of (23d) to establish that the larger DP-j is not in matrix Spec. But, then, how can values for the more deeply embedded wh be paired with the matrix wh? Engdahl argues, in effect, for reconstruction of everything but the higher wh in CP-2, something akin to (24b). Now, *which Beethoven symphony* is in-situ and can move to the matrix clause. She follows the line of argument that subjacency does not hold for the covert scope taking that would be involved here.¹²

(25)

a. [_{CP-1} Vem minns [_{CP-2} [vilken inspelning av vilken symfoni] [_{TP} Maja
Who remembers which recording of which symphony Maja
ville veta [_{CP-3} vem some hade t_j]]]]
wanted know who that had

b. [_{CP-1} who... [_{TP-1} [_{CP-2} [_{DP-j} which recording] [_{TP-2}...
[_{CP-3} who [_{TP-3} [t_j [_{PP} of which symphony]]]]]]]]]]

¹² Anticipating Section 7.2, a non-reconstruction approach would have to build in a functional dependency between the matrix *who* and the complement interpreted as: *{whose recording of the 5th symphony does Mary prefer? Whose recording of the 9th symphony does Mary prefer?}*.

vilken symfoni is put in store inside CP-3 there is no occurrence of the relevant variable in the meaning of CP-3. It is not entirely clear to me, however, that there is a principled difference between (24b) and (25b) in this respect. One way to think about these facts, perhaps, is that *vilken symfoni* does not have an interpretation independently of *vilken inspelning*, ‘which recording.’ But this takes us in a different direction from Engdahl’s reconstruction account.

Engdahl does not devote much space to these examples and they have not been explored further in the literature, as far as I know. I have included them here because they show properties that are of relevance to the issues discussed in this chapter. First, note that the pair-list readings under discussion instantiate long-distance dependencies across wh islands, the topic of Section 7.2. Second, they seem to suggest extraordinary scope of wh expressions out of islands that are already in scope position. Third, they seem to require that the island have multiple wh in it. Example (26a), with just one wh expression, does not seem to admit the pair-list construal. Question (26b), with a predicate designed to facilitate the intended reading, still does not admit the pair-list answer (26c) and (27) follows the same pattern:

- (26) a. Who remembers [_{CP} [_{DP} whose recording [_{PP} of the 5th symphony]]_j Mary prefers t_j]?
- b. Who remembers [_{CP} [_{DP} whose recording [_{PP} of the 5th symphony]]_j Mary has played t_j]?
- c. *John remembers that Mary has played Karajan’s recording of the 5th symphony and Bill remembers that she has played Kleiber’s recording of the 5th symphony.
- (27) a. Who remembers [_{CP} [_{DP} Karajan’s recording [_{PP} of which symphony]]_j Mary prefers t_j]?
- b. Who remembers [_{CP} [_{DP} Karajan’s recording [_{PP} of which symphony]]_j Mary has played t_j]?
- c. *John remembers that Mary has played Karajan’s recording of the 5th symphony and Bill remembers that she has played Karajan’s recording of the 9th symphony.

We will put further discussion of these structures on hold for now, briefly returning to them at the end of Section 7.2.2, but their significance to our exploration of wh in-situ and scope should be clear.

7.1.4 Section summary

We looked at proposals for interpreting pied piped noun phrases and prepositional phrases and noted the need for either reconstruction of the non-wh material or alternative function-based interpretations of the pied piped phrase. The two approaches place the burden of explanation in different domains. On the reconstruction approach, wh movement has to be undone, leaving only the wh in

scope position. This allows for the simple semantics of our baseline theory. On the function-based approaches, we need to appeal to a more complex semantics for the larger phrase, making room for one argument to be referred to through another. This complexity, however, is offset by the fact that the integrity of syntactic movement is preserved. Though hard to separate empirically in the basic cases, consideration of a wider range of facts having to do with pair-list readings can help determine the choice.

7.2 The wh triangle

We now take up the issue of list answers across clauses, starting with classic cases of lists specifying values for wh in-situ inside wh islands. One view is to allow the wh to take scope at the matrix clause level, in violation of the island. The alternative view is that such list answers are due to a functional dependency between the matrix wh and the island as a whole, interpreted as a set of questions. Empirical justification for this move as well as the mechanism by which this dependency leads to pair-list answers are discussed. The consequences of these views are also laid out. Among the issues touched upon are the status of higher order questions and the diagnostics used for determining wh scope.

7.2.1 Long-distance lists in one fell swoop

Baker (1970) pointed out that questions like (28a) admit two types of answers, giving values for the matrix wh expression (28b) or pairing values for the matrix wh with values for the embedded wh in-situ (28c). He analyzed this as a scope ambiguity. In a movement-based account of scope, Baker's analysis can be rendered into the distinct LFs in (29) and (30):¹³

- (28) a. Which student knows where Mary bought which book?
 b. John knows where Mary bought which book.
 c. John knows where Mary bought AK and Bill knows where she bought W&P.
- (29) a. $[_{CP} \text{ which student}_i [_{IP} t_i \text{ knows } [_{CP} \text{ where}_j [_{CP} \text{ which book}_k [_{IP} \text{ mary bought } t_k \text{ at } t_j]]]]]$
 b. $\lambda p \exists x [\text{student}(x) \wedge p = \wedge \text{know}(x, \text{Ans}(\lambda p' \exists y \exists z [\text{place}(y) \wedge \text{book}(z) \wedge p' = \text{bought-at}(m, z, y)]))]$
 c. $\{ \wedge \text{John knows where Mary bought which book, } \wedge \text{Bill knows where Mary bought which book} \}$

¹³ I do not show the functional dependency in representing the pair-list reading of (28). We will return to this issue later. Here I simply focus on evidence for and against matrix scope for the wh in-situ, as provided by specification of values in the answer. The issues discussed in this section and the next are also discussed in Dayal (2016a, 2016b).

- (30) a. $[_{CP} \text{ which student}_i [_{CP} \text{ which book}_k [_{IP} t_i \text{ knows } [_{CP} \text{ where}_j [_{IP} \text{ mary bought } t_k \text{ at } t_j]]]]]]]$
 b. $\lambda p \exists x \exists z [\text{student}(x) \wedge \text{book}(z) \wedge p = \wedge \text{know}(x, \text{Ans}(\lambda p' \exists y [\text{place}(y) \wedge p' = \text{bought-at}(m, z, y)])])]$
 c. $\{^{\wedge} \text{John knows where Mary bought AK and Bill knows where Mary bought W\&P, } ^{\wedge} \text{John knows where Mary bought W\&P and Bill knows where Mary bought AK} \dots \}$

The scopal account of this ambiguity goes against the generalization discussed in Chapter 6 that *wh* islands do not readily permit extraction of non-D-linked *wh* expressions. The facts above, however, do not change if *which N* is changed to *who/what*. Thus, we now also have evidence from a fronting language that standard syntactic constraints do not apply to the scope of *wh* in-situ, as claimed originally by Huang (1982). The appeal of this approach is that once we allow *wh* islands to be violated at LF, nothing further is needed to derive the pair-list answer.

There is, however, reason for caution. Full disclosure: I am not a disinterested party in this debate. The arguments against unconstrained scope that I give below draw heavily on my own previous work. An early argument against giving *wh* extraordinary wide scope, due to Kuno and Robinson (1972), questions the validity of the diagnostic used. They point to questions like (31a) which admit pair-list answers but do not contain an embedded *wh* in-situ:

- (31) a. Who knows where Mary bought these books?
 b. John knows where Mary bought AK and Bill knows where she bought W&P.

Such list answers are acceptable, they claim, in contexts where an answer specifying values for the matrix *wh* is unsatisfactory. If the information that the questioner is interested in happens to be distributed over several individuals, a cooperative addressee might proffer a response that provides this elaboration. The list does not count as a semantic answer and no conclusions about the syntax of the question should be based on it. If there exists a pragmatic account for questions with embedded definites, they reason, the same should account for examples with embedded *wh*.

While the point raised by Kuno and Robinson in relation to (31) has merit, it does not transfer over to (28a). Dayal (1996: 145–9) draws attention to grammatical differences between questions with embedded definites and those with embedded *wh* in-situ. List answers to the former but not the latter are dependent on plurality. A singular *wh* in the matrix (32a), or a singular *wh* in the complement (32b), blocks the possibility of list answers:

- (32) a. Which student knows where Mary bought these books?
 b. Who knows in which shop Mary bought these books?

The explanation for these effects builds on the explanation for pair-list answers to monoclausal questions with plural definites in Dayal (1992, 1996; Krifka 1992), namely that they represent a cumulative reading of an individual answer (see Chapter 4). Ans-D applied to the question in (32a) is defined at a world *w* if and only if one atomic individual, say *bill*, knows where Mary bought the plurality of books but this cannot be expanded into a list. Turning to (32b), Ans-D applied to *in which shop did Mary buy these books?* is only defined if there is exactly one shop in which Mary bought the plurality of books. Any addressee who accepts the singularity premise would simply name the individual(s) who is in the *know* relation to the singular proposition identifying the shop.

The key insight is that semantic answers with plural arguments can have distributive or cumulative readings. This applies equally to simple questions that are about relations between individuals *who bought these books*, as to complex questions that relate individuals and sets of propositions *who knows where Mary bought these books*.¹⁴ Adverbials like *each* or *jointly* disambiguate between distributive/cumulative readings for both types of questions. While this account does not support the conclusions Kuno and Robinson draw about the Baker example, it does underscore their contention that list answers are not a foolproof diagnostic of scope. That is, it confirms that a semantic account of the pair-list answer to the Baker example is needed but it does not entail that such answers are to be derived by giving the embedded *wh* matrix scope.

Dayal (1996) observes that a question must have a “*wh* triangle” configuration in order to allow a long-distance list answer. The first part of this triangle is the need for a multiple *wh* complement. The following Hindi examples are instructive as Hindi finite complements are known to be scope islands for *wh* (Davison 1984; Mahajan 1990; Dayal 1991a, 1996; among others):

- (33) a. kaun jaantaa hai ki meri-ne kyaa khariidaa
 Who knows that Mary-ERG what bought
 “Who knows what Mary bought?”
 b. kaun jaantaa hai ki meri-ne kahaaN kyaa khariidaa
 Who knows that Mary-ERG where what bought
 “Who knows where Mary bought what?”

Example (33a) only admits answers specifying values for the matrix *wh*; (33b) readily admits list answers of the kind seen in English. The source of such answers in Hindi, the data suggests, cannot be covert *wh* movement since that option is not available in the language. The contrast between a single *wh* complement and a multiple *wh* complement with respect to pair-list readings can also be detected in English:

- (33) c. ?Which student knows that Mary bought which book?

¹⁴ In Dayal (1991a, 1991b), all questions with pair-list readings were treated as involving plurality. In Dayal (1992, 1996) plurality-based lists were distinguished from function-based lists. A distinction between multiple-pair and single-pair dependencies across islands was posited only in Dayal (2002).

Example (33c) is somewhat marginal since the *wh* in the embedded clause is not fronted but to the extent that it is good, it does not allow a multiple-pair answer. This is in opposition to the Baker example which has a multiple *wh* complement.

The second relevant property is the presence of a *wh* expression in the matrix clause. Recall that the Japanese example in (10c) does not allow scoping out of a *wh* island. Replacing the matrix subject with a *wh* makes a long-distance list answer possible (Dayal 1996: 98):¹⁵

- (34) a. Tanaka-kun-wa [Mary-ga doko-de dono hon-o kat-ta ka]
 Tanaka Mary where which book bought Q
 sitte-imasu ka
 knows A
 “Does Tanaka know where Mary bought which book?”
- b. dono sensei-ga [Mary-ga doko-de dono hon-o kat-ta ka]
 which professor Mary where which book bought Q
 sitte-imasu ka
 knows Q
 “Which professor knows where Mary bought which book?”

The third and final part of the triangle is that the multiple *wh* question has to be the complement of the matrix verb. Mahajan (1990) attributes examples like (35a) to David Pesetsky:¹⁶

- (35) a. [Which student believes [(that) John knows
 [where Mary bought which book]]]?
 b. [Which book does [Bill believe [(that) John knows
 [(that) Mary bought]]]]?

Here there are no islands involved, as witnessed by the acceptability of overt movement in (35b). Yet a clause intervening between the matrix and the multiple *wh* complement seems to block the list answer to (35a). This fact is intractable under an approach that posits unconstrained covert movement. It remains so even under Richards’ (1997) *Principle of Minimal Compliance*, according to which if one *wh* expression in a clause obeys a syntactic constraint, others get a free pass. If unconstrained movement of a *wh* in-situ were at issue, the presence of *where* in the lowest clause of (35a) would free up *which book* to take scope outside the

¹⁵ Hindi (33b) also loses the list reading if the matrix subject is replaced with a referential term. And the same holds for the original Baker example. The Japanese paradigm illustrates the relevance of the *wh* triangle more dramatically since it shows that even a Q morpheme in matrix Spec does not result in wide scope for the embedded *wh*.

¹⁶ Mahajan (1990) and Kim (1991) take covert *wh* movement to be an instance of Quantifier Raising. I agree with them as far as locality of *wh* scope is concerned, but distinguish between the semantics of *wh* and quantifier scope. See also Charlow (2014) for a semantics for local scope taking with the appearance of extraordinary scope.

island. But what we observe here is that there is a locality constraint that holds beyond the island proper.

7.2.2 Long-distance lists through higher order questions

If a specific configuration of (at least) three *wh* expressions, $[\dots wh_1 \dots V [wh_2 \dots wh_3]]$, is critical for long-distance lists, any account of the phenomenon must make crucial reference to it. Dayal (1996) posits the following two LFs:¹⁷

- (36) a. Who knows where Mary bought what?
 b. $[_{CP-1} \text{ who}_i [_{IP} t_i \text{ knows } [_{CP-2} \text{ where}_k \text{ what}_j [_{IP} \text{ mary bought } t_j \text{ at } t_k]]]]$
 c. $[_{CP-1} \text{ who}_i [_{CP-2} \text{ where}_k \text{ what}_j [_{IP} \text{ mary bought } t_j \text{ at } t_k]] [_{IP} t_i \text{ knows } t_{CP-2}]]$

LF (36b) is a question about the matrix *wh* and thus admits an individual answer. A plurality-based list answer is also possible but only if the conditions discussed in Section 7.2.1 hold. LF (36c) captures the long-distance list reading. The central idea is that this is due to a functional dependency between the matrix *wh* and the multiple *wh* complement, not between the matrix *wh* and a *wh* inside the complement. It is a case of a dependency with a covertly pied piped clausal structure, if you will.

The dependency rests on the complement being interpreted as a family of questions. The function relates this set to the set of individuals who know answers to the questions in it:

- (37) a. $[[CP-2]] = \{\text{where Mary bought the pen, where Mary bought the book}\}$
 b. $\lambda p \exists x \exists f_{\langle \langle s, t \rangle, t \rangle, e \rangle} [\text{Dom}(f) = [[CP-2]] \wedge \text{Range}(f) = \text{person} \wedge p = \cap \lambda p' \exists q \in [[CP-2]] [p' = \text{know}(f(q), \text{Ans}(q))]]$
 c. $f_1 [\text{where M bought the pen} \rightarrow \text{John, where M bought the book} \rightarrow \text{Bill}]$
 $f_2 [\text{where M bought the pen} \rightarrow \text{John, where M bought the book} \rightarrow \text{John}]$
 $f_3 [\text{where M bought the pen} \rightarrow \text{Bill, where M bought the book} \rightarrow \text{John}]$
 $f_4 [\text{where M bought the pen} \rightarrow \text{Bill, where M bought the book} \rightarrow \text{Bill}]$
 d. $\{^{\wedge} \text{John knows where M bought the pen and Bill knows where M bought the book,}$
 $^{\wedge} \text{John knows where M bought the pen and John knows where M bought the book,}$
 $^{\wedge} \text{Bill knows where M bought the pen and John knows where M bought the book,}$
 $^{\wedge} \text{Bill knows where M bought the pen and Bill knows where M bought the book} \}$

¹⁷ I follow Richards (1997) in utilizing “tucking in” to derive the order of the *wh* expressions in embedded SpecCP in (36b). This has no substantive effect on the analysis, except for the mechanical shift in the order in which arguments are fed into the operator in C^0 (see Chapter 4). As regards (36c), we will see arguments for a somewhat different analysis. Section 7.2.3 will provide the justification.

If Mary bought a book and a pen and John and Bill are the individuals in the domain, we have four functions from the family of questions to individuals, demonstrated in (37c). As with atomic questions with two *wh* expressions, the dependency is translated into propositions that for each such function gives the graph of that function. This derives the set of propositions in (37d). Two of these propositions, those based on f_2 and f_4 , are indistinguishable from the individual answer since the same individual relates to both questions. But the propositions based on f_1 and f_3 yield genuine multiple-pair answers.

There are two key aspects of the extension from the atomic to the complex case worth noting. The first is the interpretation of CP-2 as a family of questions, type $\langle\langle\langle s,t \rangle, t \rangle, t \rangle$, a set of sets of propositions. The idea that multiple *wh* interrogatives can permit higher order questions was discussed in Chapter 4 in relation to echo questions. Small caps indicate echo intonation:

- (38) a. Bill cooked WHAT? Bill cooked [meat]_F.
 b. WHO cooked WHAT? [Bill]_F cooked [meat]_F.
 c. Who cooked WHAT? Who cooked [meat]_F?

Question (38a) elicits in response a declarative (type $\langle s, t \rangle$) with focus on the expression that was inaudible in the previous utterance. A multiple *wh* interrogative like (38b), with echo intonation on both *wh* expressions, also elicits a declarative answer, this time with focus on both expressions that were inaudible. However, a multiple *wh* like (38c), with echo intonation on just one *wh*, elicits an interrogative answer (type $\langle\langle s, t \rangle, t \rangle$). Given the view that a question denotes the set of its possible answers, (38c) denotes a set of questions. We have direct evidence of the potential that a multiple *wh* interrogative has for denoting a family of questions. While this demonstration is most plausibly made for echo questions, it underscores a general point. Multiple *wh* interrogatives, in principle, can have second order interpretations whereas single *wh* interrogatives cannot.

The syntactic structure invoked in Dayal (1996) is reproduced in (39a) and a somewhat modified semantics in (39b). The *wh* already in SpecCP at S-structure contributes to the denotation of the lower CP as a set of propositions (the standard question denotation). The *wh* in-situ has two options. It can take scope at lower SpecCP and be interpreted within this CP in ways we have discussed in Chapter 4. We would get either the REF-Q/single-pair reading, or a functional/ multiple-pair reading. Alternatively, if the *wh* in-situ is interpreted through a choice functional analysis and is bound by the operator in the higher CP*, we get a family of questions cued to the *wh* in-situ. Crucially, binding from CP* cannot yield a family of questions unless the lower CP is a question:

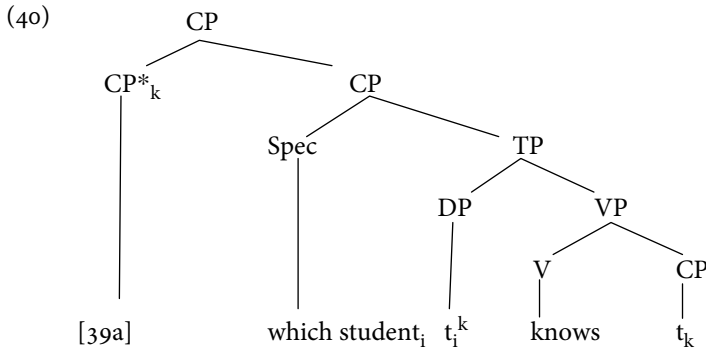
- (39) a.
-
- ```

graph TD
 CP_star[CP*] --- Spec
 CP_star --- C_star[C*]
 Spec --- OP_j[OPj]
 C_star --- C0_star[C0*]
 C_star --- CP[CP]
 C0_star --- Where[Where]
 CP --- mary_bought[mary bought which bookj]

```

- b.  $\llbracket \text{CP} \rrbracket = \lambda p \exists x [\text{place}(x) \wedge p = \wedge \text{Mary bought } f_j(\text{book}) \text{ at } x]$   
 $\llbracket \text{C}^{0*} \rrbracket = \lambda Z [Q = Z]$   
 $\llbracket \text{C}'^* \rrbracket = \llbracket \text{C}^{0*} \rrbracket (\llbracket \text{CP} \rrbracket) \Rightarrow [Q = \lambda p \exists x [\text{place}(x) \wedge$   
 $p = \wedge \text{Mary bought } f_j(\text{book}) \text{ at } x]]$   
 $\llbracket \text{OP}_j \rrbracket = \lambda P \exists f [\text{CF}(f) \wedge P(f)]$   
 $\llbracket \text{CP}^* \rrbracket = \lambda P \exists f [\text{CF}(f) \wedge P(f)] (\lambda f_j [Q = \lambda p \exists x [\text{place}(x) \wedge$   
 $p = \wedge \text{Mary bought } f_j(\text{book}) \text{ at } x]])$   
 $\Rightarrow \lambda Q \exists f [\text{CF}(f) \wedge Q = \lambda p \exists x [\text{place}(x) \wedge p = \wedge \text{Mary bought } f(\text{book}) \text{ at } x]]$
- c.  $\{Q_1: \text{where did Mary buy Emma}, Q_2: \text{where did Mary buy Persuasion}\}$

The final piece of the puzzle is the relation between the two arguments of the matrix verb, the matrix subject and the multiple wh interrogative. Dayal adopts the rule of Interrogative Raising (Lahiri 2002a: 76–95). If responsive predicates like *know* are of type  $\langle \langle \langle s, t \rangle, t \rangle, \langle e, t \rangle \rangle$ , that is a relation between individuals and questions, or type  $\langle \langle s, t \rangle, \langle e, t \rangle \rangle$ , that is a relation between individuals and propositions, there will be a type mismatch when its complement denotes a family of questions. This mismatch is resolved by raising CP and adjoining it to matrix CP, in a fashion analogous to QR. The raised CP leaves behind a trace of type  $\langle \langle s, t \rangle, t \rangle$ , something the predicate can combine with, possibly through Ans-D:



The pair-list answer is derived through a functional dependency between  $\text{CP}^*$  and *which student*, that is, between  $\{Q_1, Q_2\}$  and  $\{student_1, student_2\}$ . For each question in the Q-set, the matrix question looks for the individual in the student-set who knows the answer to it.

Note that this procedure provides a way of approaching list readings of examples like (23) discussed in Section 7.1: *who remembers whose recording of which symphony Maria prefers*. It does so without appealing to reconstruction followed by long-distance movement of a wh expression that was already in the embedded SpecCP. Instead, it involves projecting a  $\text{CP}^*$  at the embedded clause level, with the rest as given in (40).



The IR account, because it rests on a higher order interpretation of the complement CP, not only explains why this CP must have more than one wh expression, it also explains why locality above the island is necessary for list answers:

- (41) a. [<sub>CP-1</sub> which student<sub>i</sub> [<sub>IP</sub> t<sub>i</sub> said [<sub>CP-2</sub> that John knows  
[<sub>CP-3</sub> where Mary bought which book]]]]?  
b. [<sub>CP-1</sub> which student [<sub>IP</sub> t<sub>i</sub> said [<sub>CP-2</sub>  
[<sub>CP-3</sub> where Mary bought which book] John knows t<sub>CP-3</sub>]]]

IR is a raising operation like QR, which is known to be a local operation. CP-3 can only move up to Spec of CP-2, as in (41b). This does not bring it close enough to the matrix subject to enter into a functional dependency with it.<sup>18</sup>

The explanation for the wh triangle, then, is argued to be a generalization of the procedure for interpreting functional dependencies in multiple wh questions relating individuals. It is also an extension of those approaches to pied piping where the constituent containing a wh expression is interpreted as the scopal expression. By weakening the diagnostic of using specification of values for the embedded wh as an indicator of matrix scope for that wh, we provide an explanation that does justice to the empirical observation about the wh triangle.

### 7.2.3 A little housekeeping

The alert reader will have noticed one respect in which long-distance list answers do not follow the pattern of list answers to monoclausal questions. In the latter case, it is the subject that is assumed to set the domain of the function and the object its range. In the long-distance case, however, it seems to be the opposite. Note that this problem is not limited to the IR-based account given in Section 7.2.2. It also holds for accounts that posit extraordinary scope for wh in-situ. Dayal (1996, 2002) notes it as an open problem for both approaches. However, there may be a way of framing the issue that sheds some light on possible directions to take.

Note first of all that unlike nominal/prepositional pied piping, the kind of clausal pied piping posited in Section 7.2.2 does not by itself lead to a direct question interpretation: *John knows where Mary bought which book* is a declarative with an embedded question, not a direct question about books.<sup>19</sup> One might draw the analogy with a universal quantifier which cannot on its own generate a question interpretation ((42a)). In combination with a wh as in (42b), however, it

<sup>18</sup> I set aside the interpretation of (41b), as it is orthogonal to the point under discussion.

<sup>19</sup> That is, it is not the kind of clausal pied-piping that has been proposed for Basque (Ortiz de Urbina 1993).

can transform a single *wh* question into a *de facto* multiple interrogative like (42c), by providing the domain argument of a functional dependency with the *wh*:

- (42) a. [each student / every student] likes John.  
 b. Which professor does [each student / every student] like?  
 c. Which student likes which professor?

When a universal is in object position, there is a breakdown in parallelism between the two exponents *each N* and *every N*:

- (43) a. Which professor likes [each student / every student]  
 b. Who knows [<sub>CP\*</sub> OP<sub>j</sub> [where Mary bought *which book<sub>j</sub>*]]

The object *every N* does not lead to list answers, but the same is not true of the object *each N*, the inherently D-linked universal (Section 4.4.2). We could say, then, that the higher order multiple *wh* CP discussed in Section 7.2.2 is not an outlier, but rather, functions like *each N*. What requires further elaboration is the correlation between D-linking and the possibility of reversals in setting the domain of the functional dependency.

This may also be the place to correct an error in reported judgments about the *wh* triangle in multiple fronting languages like Bulgarian:

- (44) [<sub>CP</sub> koj<sub>i</sub> [<sub>TP</sub> t<sub>i</sub> znae [<sub>CP</sub> koja kniga<sub>j</sub> kade<sub>k</sub> [<sub>TP</sub> e kupi  
 Who knows which book where has bought  
 Mariya t<sub>j</sub> t<sub>k</sub>]]]]  
 Maria  
 “Who knows where Maria has bought which book?”

Dayal (1996: 91) had reported such sentences as admitting long-distance lists pairing individuals and books but that is not the case. For some, but not all speakers, it is possible to list values for all three *wh* expressions. Since I was not tuned in to this possibility at the time, I may have misconstrued such judgments for list answers pairing two of the three *wh* expressions.<sup>20</sup> The facts seem to be replicated in other multiple fronting languages, such as Romanian.<sup>21</sup> I therefore conclude that a further fact to be incorporated into analyses of pair-list answers is that higher order questions are only possible when the embedded multiple *wh* question has a *wh* in-situ. This is manifested more dramatically in Hungarian which allows two options: the option in (45a) which has an embedded *wh* in-situ, and the one in (45b) which has multiple fronting in the embedded clause:

<sup>20</sup> I am unclear how such answers are derived. See Section 7.3.3 for related facts in connection with adjunct islands. I am indebted to discussions with Catherine Rudin, Elena Dimova, and Boris Harizanov for clarification.

<sup>21</sup> Thanks to Dafina Ratiu-Gasparin, Andreea Nicolae, and Anamaria Falaus for clarification and discussion regarding the Romanian facts.

- (45) a. Melyik versenyző dicsekedett, hogy hol végzett melyik versenyen?  
Which sportsman boasted, that where finished which competition-on  
Melyik versenyző dicsekedett, hogy melyik versenyen  
Which sportsman boasted that which competition-on  
hol végzett?  
where finished  
 “Which sportsman boasted about where he finished in which competition?”

Surányi (2006) notes that (45a) allows for answers pairing the matrix wh (set of sportsmen) with the embedded wh in-situ (set of competitions), as in English. Question (45b), with both embedded wh phrases fronted, can only be answered by specifying a value for the matrix wh. A long-distance list answer is unavailable (Balázs Surányi p.c.).<sup>22</sup>

A priori, this would seem to be an argument in favor of unconstrained LF movement but this conclusion would be hasty. The facts are equally compatible with the approach spelt out in Section 7.2.2. Given that we have posited a double CP structure, we might take multiple fronting structures to be those in which all wh expressions move to the standard scope position in the lower CP ((46a)). This is how we get REF-Q/single-pair or function-based pair-list readings of the embedded CP in these languages:

- (46) a. [<sub>CP</sub>\* [<sub>CP</sub> wh<sub>1</sub> wh<sub>2</sub> [<sub>TP</sub>... t<sub>1</sub>... t<sub>2</sub>... ]]] <<s,t>,t>  
 b. [<sub>CP</sub>\* OP<sub>2</sub> [<sub>CP</sub> wh<sub>1</sub> [<sub>TP</sub>... t<sub>1</sub>... wh<sub>2</sub>]]] <<<s,t>,t>,t>

If a language allows a second wh to remain in-situ at S-structure, as in (46b), that wh can be bound from CP\*. This leads to an interpretation as a family of questions, triggering IR of the multiple wh complement and paving the way for long-distance list answers.<sup>23</sup>

The following facts are also worth considering in this connection. Hirschbuhler (1978) points out that questions like (47) obligatorily require long-distance list answers. He explains this by noting that the wh expressions *what* and *whether*, in (47a) for example, cannot combine at the embedded clause level. This forces *what* to move to matrix Spec:

- (47) a. Who knows whether Mary bought WHAT/which book?  
 b. Who remembers how Mary solved WHAT/which problem?  
 c. Who knows why Mary bought WHAT/which book?

<sup>22</sup> There are many subtle differences between a clause with multiple fronting and a corresponding version with wh in-situ in Hungarian discussed in Surányi (2002, 2006), which I have set aside here. Balázs Surányi (p.c.) informs me that minor modifications, such as subjunctive mood in the complement or wh expressions with similar types of domains, can enhance the contrast.

<sup>23</sup> We might want to generalize this to wh expressions whose denotation includes a functional variable not bound at the level of CP. This would differentiate multiple fronting of the kind shown in (46b) from multiple fronting of the kind found in the Engdahl examples, such as (23). While the wh in Spec in (45b) has to bind an argument position inside TP, the one in (23) does not.

An alternative explanation is that the *wh* is forced to be interpreted in CP\* and because this leads to a higher order interpretation, the embedded clause is forced to IR and form a dependency with the matrix *wh*.<sup>24</sup> Interestingly, Hirschbuhler notes that the embedded *wh* must be stressed. This is not so surprising given the connection between echo intonation and higher order denotations that we have drawn here.<sup>25</sup>

Finally, consider a paradigm from Dayal (1996: 130) which notes further consequences of this account:

- (48) a. Which student thinks/believes which teacher is nice?  
 b. Johnny thinks/believes Ms. A is nice and Billy thinks/believes Ms. B is nice.  
 c. \*[<sub>CP1</sub> Which student<sub>i</sub> [<sub>TP</sub> t<sub>i</sub> thinks (that) [<sub>CP2</sub> Bill believes [<sub>CP3</sub> which teacher is nice]]]]?
- (49) a. Which students knows which teacher is nice?  
 b. \*Johnny knows Ms. A is nice and Billy knows Ms. B is nice.

The predicates *think/believe* select propositions, not sets of propositions. Example (48a), however, can be rescued by interpreting the question *which teacher is nice* as a set of propositions and pairing it with the set of students after IR. Note, that an intervening clause, as shown in (48c), has the familiar blocking effect on the list answer. In this case, it leads to ungrammaticality because of the selectional properties of the embedding predicate. With a predicate like *know*, which can take propositions or sets of propositions (49), we have to restrict IR to only occur when necessary or we would incorrectly predict (49a) to allow the answer in (49b).

#### 7.2.4 Section summary

Just as in the pied piping cases discussed in Section 7.1, *wh* in-situ inside *wh* islands have been given two analyses. In one, the *wh* expression itself escapes the island and forms a multiple *wh* interrogative at the matrix level. In the second, the *wh* expression is bound locally from just outside the island but in a way that forms a second order question. This family of questions plays a role in long-distance lists analogous to questions with a D-linked universal quantifier. This approach was argued to do justice to the generalization that long-distance list answers are only available when the configuration involves a *wh* triangle: a multiple *wh* complement, another *wh* argument in the immediately higher clause, and an embedding

<sup>24</sup> Another interesting observation is made by Lasnik and Saito (1992: 118) who take the following as evidence of obligatory long-distance movement: *Who wonders what who bought?* Here the embedded clause shows a superiority violation and obligatorily has a long-distance list reading pairing the two underlined phrases.

<sup>25</sup> Hirschbuhler's observations connect these cases to echo questions in an interesting way. The correlation between the intonation of echo questions and the D-linked *wh* expressions that seem to participate in long-distance lists remains under-explored at this point (see also Comorovski 1996 and Dayal 1996 for relevant discussion).

predicate that selects for sets of propositions. A generalization of this is the case of a complement that denotes a set of propositions, another *wh* argument in the immediately higher clause, and an uninterrogative embedding predicate that selects only propositions.

### 7.3 Long-distance lists across adjunct islands

The same puzzles that we found with nominal and *wh* islands resurface with adjunct islands. *Wh* expressions inside adjunct phrases cannot be extracted, but nevertheless can have their values specified in answers. One option is to allow the *wh* phrase to take matrix scope, the other is to interpret it locally with scope just over the adjunct island. In other words, list answers can be derived either by relating two *wh* expressions in violation of the island or by relating a matrix *wh* argument with the island as a whole. As in the other island cases, the former requires a relaxation of syntactic constraints on scope taking, the latter requires an enhanced semantics for the adjunct phrase. The importance of single-pair, multiple-pair, and trapped pair-list readings is also discussed.

#### 7.3.1 Single-pair answers via choice functions

Adjunct islands are used by Reinhart (1997) to argue against the unselective binding approach to *wh* in-situ and in favor of choice functions (cf. Section 7.1). Example (50a) has a *wh* in-situ in the antecedent, which is a strong island (50c):

- (50) a. Which linguist will be offended if we invite which philosopher?  
 b. Prof. Smith will be offended if we invite Professor Brown.  
 c. \*[Which philosopher<sub>i</sub> [will Prof. Smith be offended [if we invite t<sub>i</sub>]]]?

Reinhart argues against covert movement but makes the standard assumption that an answer spells out the values of all and only the *wh* that have matrix scope. She interprets the *wh* in-situ with a function variable in its base position but existentially bound at the matrix level, as part of the interrogative formation rule:

- (51) a.  $\lambda p \exists x \exists f [\text{linguist}(x) \wedge \text{CF}(f) \wedge$   
 $p = \wedge x \text{ will be offended if we invite } f(\text{philosopher})]$   
 b.  $\{\wedge \text{Prof. Smith will be offended if we invite Prof. Brown,}$   
 $\wedge \text{Prof. Jones will be offended if we invite Prof. White...}\}$

Reinhart's use of choice functions, we see, gives exactly the same results as would be derived if the *wh* in-situ moved to take scope at the matrix clause level. It has theoretical appeal because there is no movement involved, so the island insensitivity has a principled basis. In fact, the view that only non-movement-based scope mechanisms should be insensitive to islands had been proposed earlier by Pesetsky (1987) who argued for unselective binding into islands. Reinhart's

argument for choice functions is a response to the particular type of binding proposed by Pesetsky.

Reinhart shows the flaw in the unselective binding approach with reference to (52a):<sup>26</sup>

- (52) a.  $\lambda p \exists x \exists y [\text{linguist}(x) \wedge$   
            $p = \wedge \text{philosopher}(y) \wedge \text{we invite } y \rightarrow x \text{ will be offended}]$   
       b. Prof. Smith will be offended if we invite Donald Duck.

Given the logic of conditionals, any non-philosopher is sufficient to make the antecedent false and the conditional true, allowing for answers like (52b). This problem is avoided with the use of choice functions whose value is restricted to members of the argument set, here the set of philosophers.

While Reinhart extends the reach of pair-list answers to conditionals, she does not make a distinction between single-pair and multiple-pair answers. Dayal (2002) notes that the examples she discusses, while requiring specification of values for both *wh* expressions, do not lend themselves to pair-list answers:

- (53) a. \*Prof. Smith will be offended if we invite which philosopher.  
       b. #Prof. Smith will be offended if we invite Prof. Brown and Prof. Jones will be offended if we invite Prof. White.

The explanation proffered is that multiple-pair answers require two *wh* expressions to enter into a functional dependency, with one providing the domain and the other the range of the function. Each member of the question denotation builds on the graph of a function relating the two sets. As such, when *Ans-D(Q)* applies to such a set we get a multiple-pair list answer. But if the denotation of the question has atomic propositions only, as in (51b), *Ans-D* applied to it will necessarily yield a single-pair answer. That is, Reinhart's analysis, in conjunction with the presupposition that there be a maximally informative true proposition in the set, guarantees a single-pair reading in these cases.

Surányi (2006) makes a similar observation in relation to Hungarian examples involving a matrix *wh* and a *wh* inside a temporal adjunct:

- (54) Melyik fiú let ideges miután felhívta melyik lányt?  
       Which boy became angry after up-called which girl-ACC  
       ‘Which boy got angry after he phoned which girl?’

Surányi's observation and the core of the explanation is similar to the one mentioned above. In essence, multiple-pair answers require functional absorption of two *wh* expressions adjacent to each other. Long-distance binding of a choice

<sup>26</sup> Choice functions were also introduced to account for the apparent island insensitivity of specific/referential indefinites, discussed in Fodor and Sag (1982) and many others since then (see Ruys 2006 for a survey). On choice functions, see also Winter (1997).

function variable may allow the *wh* inside the island to take wide scope but will not deliver the kind of adjacency needed for multiple-pair answers.

It seems though that multiple-pair answers are not always ruled out, as was pointed out to me by Kate Davis (p.c.) and Satoshi Tomioka (p.c.). It has also been noted by Cheng and Demirdache (2010: 466), who attribute the following to Chris Tancredi:

- (55) Context: each of two philosophers will be offended if we invite one of two linguists.  
 What I want to know is,  
 Q: Which philosopher will be offended if we invite which linguist?  
 A: Quine will be offended if we invite Chomsky and Lewis will be offended if we invite Kayne.  
 A': #Quine will be offended if we invite Chomsky.

The approach of Dayal (2002) and Surányi (2006) faces the most fundamental of problems, at least as far as English is concerned. The facts it is supposed to account for turn out not to hold. Before we discard this approach, however, let us keep in mind that multiple-pair answers become available only with explicit contextual support of the kind shown in (55), whereas monoclausal multiple *wh* questions (*which linguist offended which philosopher?*) admit multiple-pair answers quite readily. If anything, it is the single-pair answers to these questions that have needed justification in the past. It may be worthwhile highlighting here a distinction between D-linking and contextual priming with regard to scope. A claim one finds fairly often in the literature is that D-linked *wh* expressions can escape islands. Some of the examples used to show this involve single-pair answers of the type we have seen. To derive the multiple-pair answer, the contexts that have to be provided seem to require an explicit statement about the distributive nature of the relation between two sets. I mention this here as something that may be worth probing in future studies of scope and D-linking. In Section 7.3.3 we will see that such contextual priming does not appear to be necessary when the island contains more than one *wh* expression.

### 7.3.2 Multiple-pair answers via higher order choice functions

Just as in the case of nominal and *wh* islands, it is possible to give a semantics for adjunct phrases with the *wh* inside them. I draw on Sternefeld (2001, 2002) for this demonstration, starting first with a simple declarative to lay out the core assumptions about the way adjuncts combine with the main clause:

- (56) a. Mary left because John arrived.  
 b.  $\llbracket \text{because John arrived} \rrbracket = \lambda p \llbracket \text{because}(\wedge \text{john arrived}, p) \rrbracket$   
 c.  $\llbracket 56a \rrbracket = \lambda p \llbracket \text{because}(\wedge \text{john arrived}, p) \rrbracket (\wedge \text{mary left})$   
 $\Rightarrow \text{because}(\wedge \text{john arrived}, \wedge \text{mary left})$

The adjunct phrase denotes a set of propositions, specifically, those propositions that hold in virtue of (are caused by) the adjunct proposition. After functional application, we have a sentence that is true if an appropriate causal relation holds between John's arriving and Mary's leaving.

The issue to settle is whether a reasonable semantics can be given for the adjunct if there is a *wh* expression in it. Sternefeld proposes a solution that he characterizes as non-standard question formation for interpreting such phrases. The construction that Sternefeld is interested in involves complications having to do with Hungarian scope marking, which would take us too far afield to introduce here. Let me therefore go to a simpler construction, starting first with (57a), which is obviously not acceptable on its own. Setting that non-trivial point aside for the moment, we can derive a set of sets of propositions, each of which is the set of propositions that are caused by a different boy's arrival:

- (57) a. \*because which boy arrived  
 b.  $\llbracket \text{because which boy arrived} \rrbracket =$   
 $\lambda R \exists x [\text{boy}(x) \wedge R = \lambda p [\text{because } (^x \text{ arrived}, p)]]$   
 $\Rightarrow \{\lambda p [\text{because } (^{\text{john}} \text{ arrived}, p)], \lambda p [\text{because } (^{\text{bill}} \text{ arrived}, p)]\}$

Embedding this inside a clausal structure may be marginal as a non-echo question but to the extent that it is acceptable, it is interpreted as (58b):<sup>27</sup>

- (58) a. ??Mary left because which boy arrived?  
 b.  $\llbracket 58a \rrbracket = \lambda p \exists f [\text{CF}(f) \wedge p = f(\llbracket 57b \rrbracket)] (\llbracket \text{Mary left} \rrbracket)]$   
 $\Rightarrow \frac{\{\text{because } (^{\text{john}} \text{ arrived}, ^{\text{mary left}})\}}{\text{because } (^{\text{bill}} \text{ arrived}, ^{\text{mary left}})}$   
 c. Mary left because John arrived.

The important step in the computation is the use of a choice function variable to pick out the members of the set denoted by (57a) when composing the adjunct with the matrix clause. Since there will be as many distinct choice functions as individuals, we end up with a standard question denotation. Assuming we are in a world where John's arrival causes Mary to leave, the answerhood operator will pick out the proposition naming him.

The problem, of course, is that such a question is not acceptable in English but there is considerable cross-linguistic variation on this.<sup>28</sup> That aside, it does set the

<sup>27</sup> In a context like the following, we would get a non-echo question:

- (i) Speaker A: Mary left because that boy arrived.  
 (ii) Speaker B: Mary left because which boy arrived?

The reference of the indexical is what is at issue in B's response, as in the REF-Q cases discussed in Chapter 4.

<sup>28</sup> Shimoyama (2006: 142) provides the following as an acceptable corresponding question in Japanese:

- (i) Taro-wa [dare-ga kita-kara] kaerimasita ka  
 Taro-TOP who-NOM came-because left Q  
 "Who is such that Taro left because he/she came?"





### 7.3.3 Lists trapped inside adjunct wh triangles

An interesting new twist enters into the discussion of list answers and islands with the introduction of trapped pair-lists in Romanian multiple wh questions (Ratiu 2007, 2011a, 2011b). Here we will draw on Cheng and Demirdache (2010) who extend the discussion to include English. The following paradigm illustrates the phenomenon of trapped pair-list answers. The question has a wh in the matrix and a multiple wh in the adjunct, setting up a wh triangle configuration:

- (61) a. Which teacher got upset because which boy had a fight with which girl?  
 b. *Single triple answer*  
 The history teacher got upset because John had a fight with Sue.  
 c. *Trapped pair-list*  
 The history teacher got upset because John had a fight with Sue and Bill had a fight with Jane.  
 d. *Lists of trapped pairs*  
 The history teacher got upset because John had a fight with Sue and Bill had a fight with Jane, and the math teacher got upset because Terry had a fight with Mary and Chris had a fight with Clare.

Cheng and Demirdache establish that an answer pairing the matrix wh with one of the embedded wh phrases, to the exclusion of the other, is not possible. There are a number of other subtle points they discuss with regard to multiple wh questions that I will set aside here, and instead focus on their explanation for the data in (61).

Cheng and Demirdache derive the trapped readings by interpreting both wh phrases inside the island, adopting Sternefeld's semantics for wh adjunct phrases and the functional dependency account of Dayal (1996):

- (62)
- [<sub>CP</sub> which teacher<sub>1</sub> [<sub>t</sub><sub>1</sub> got upset [<sub>ISLAND</sub> which girl<sub>3</sub> which boy<sub>2</sub> [because <sub>t</sub><sub>2</sub> fight <sub>t</sub><sub>3</sub>]]]]
- 

In (62) the adjunct clause has a functional dependency between the two wh phrases inside it. Furthermore, the whole adjunct clause enters into a functional dependency with the matrix wh, in a procedure analogous to the one we discussed in relation to the wh triangle in Section 7.2, albeit adjusted for the type of the adjunct clause. Since the facts are quite complex, I will try to present the account schematically. Let us assume that the individuals at issue are

matrix position at LF. It is unclear, however, how that approach would deal with the trapped pair-list readings to be presented in Section 7.3.3.



Let us briefly review Cheng and Demirdache's motivation for adopting the view that *wh* in-situ can escape islands. They note that multiple fronting languages like Romanian rule out the possibility of answers pairing matrix *wh* with an embedded *wh* in the classic Baker type example, while single fronting languages like English allow it (cf Section 7.2.3). I repeat the contrast from Hungarian and the explanation I offered in Section 7.2:

- (65) a. Melyik versenyző dicsekedett, hogy hol végzett  
 Which sportsman boasted that where finished  
melyik versenyen?  
 which competition-on OK<sub>PL</sub>
- b. Melyik versenyző dicsekedett, hogy melyik versenyen hol  
 Which sportsman boasted that which competition-on where  
 végzett ?  
 finished \*PL  
 "Which sportsman boasted about where he finished in which  
 competition?"
- (66) a. [CP\* OP<sub>2</sub> [CP wh<sub>1</sub> [TP ... t<sub>1</sub> ... wh<sub>2</sub>]]] <<<s,t>,t>,t>  
 b. [CP\* [CP wh<sub>1</sub> wh<sub>2</sub> [TP ... t<sub>1</sub> ... t<sub>2</sub> ...]]] <<s,t>,t>

Under the view that only *wh* in-situ can be bound by the OP in CP\*, we can handle these examples without giving the *wh* in-situ matrix scope. We take list answers across *wh* islands to instantiate a dependency between a matrix *wh* and a multiple *wh* complement interpreted as a family of questions. If this view of *wh* in-situ is adopted, the claim that *wh* in-situ need not obey subadjacency can be given up and Cheng and Demirdache's proposal for trapped pair-list readings maintained. *Wh* in-situ will always take local scope, either by moving to CP or by being bound by the operator in CP\*. In the latter case, they lead to higher order interpretations for the island due to another *wh* being bound at CP. In such cases, they give the appearance of interacting scopally with the matrix *wh* (see Dayal 2016b for elaboration and further discussion).

To sum up, recent work on multiple fronting languages has established that there is a difference between overt vs. covert scope taking such that in some cases covert scope taking seems to obey subadjacency (leading to trapped pair-list readings) and in some cases seems to defy subadjacency (the *wh* island type example). Cheng and Demirdache's proposal allows island insensitive covert movement for one case while forcing locality in the other case. Another option is to maintain locality in the interpretation of *wh* in-situ but allow it to take scope above the island, as I have suggested here. The role of intervening clauses as well as of trapped pair-list readings, as we saw, is critical in choosing between these two options.

### 7.3.4 Section summary

We looked at *wh* in-situ inside adjunct islands and saw that choice functions can be used to give them wide scope without actually moving them out. This approach

is theoretically appealing in that the absence of syntactic constraints on scope is linked to a system that does not involve movement. Another approach uses choice functions and interprets the *wh* in-situ at the site of the adjunct island, yielding higher order interpretations. Differences between single-pair and pair-list answers across adjunct islands, as well as the possibility of trapped pair-list answers in cases when a *wh* triangle can be created over the adjunct, provide empirical justification for recognizing both options in the grammar of natural language.

## 7.4 Scoping out or staying local

This chapter was concerned with a phenomenon that can be represented schematically in the following way:

- (67) a.  $[_{CP} [_{TP} \dots [_{ISLAND} \dots wh \dots ]]]$   
       b.  $[_{CP} wh_i [_{TP} \dots [_{ISLAND} \dots t_i \dots ]]]$   
       c.  $[_{CP} [_{ISLAND} \dots wh_i \dots ]_j [_{TP} \dots t_j]]$
- 

Suppose that a question that has the form of (67a), with a *wh* expression inside an island, allows answers that specify values for that *wh* expression. Does this indicate a structure like (67b) where the interpretation is defined on the unpronounced head of the *wh* chain at the matrix scope position? Or does this indicate a structure like (67c) where the island itself has matrix scope? Proposals along both lines of inquiry were presented and their empirical and theoretical implications discussed.

We first looked at nominal islands, with *wh* expressions inside DPs and PPs, and noted that if questions were restricted to quantifying over individuals, something like (67b) would be required for the right set of propositions to be generated. Overt pied piping, where the island is clearly in scope position, requires the island minus the *wh* to be reconstructed back into its base position. When there is no overt pied piping, however, the argument for reconstruction has less appeal. There are two accounts that seem plausible for these cases. One is to relax the syntactic constraints on scope to apply only to overt movement. The other is to enrich the semantics to quantify over more complex semantic objects, thereby allowing for a compositional semantics for the island as a whole. The status of short answers as well as complexities introduced by having more than one *wh* inside the island were also discussed.

Similar issues arise in the case of *wh* islands, where a list answer provides values for the matrix *wh* and an embedded *wh* in-situ. The argument for giving matrix scope to a *wh* in-situ inside such islands is straightforward but we looked at some data that is problematic for this approach. In particular, this effect is only observed when the structure forms a *wh* triangle:  $[_{CP-1} \dots wh_1 \dots V [_{CP-2} \dots wh_2 \dots wh_3 \dots ]]$ . It was shown that a multiple *wh* question can be interpreted as a family of

questions if the wh in-situ is bound by an operator from above the position at which the normal question is formed. The island, CP-2 in the triangle, has the structure in (68a). When one wh is interpreted in CP (resulting in a standard question meaning, type  $\langle\langle s, t \rangle, t \rangle$ ) and the other is bound by the operator in CP\* we get a meaning of type  $\langle\langle\langle s, t \rangle, t \rangle, t \rangle$  for the complement:

- (68) a.  $[_{CP^*} OP_3 [_{ISLAND} wh_2 [TP \dots t_2 \dots wh_3 \dots ]]]$   
 b.  $[_{CP} [_{CP^*} OP_3 [_{ISLAND} wh_2 [TP \dots t_2 \dots wh_3 \dots ]]]]_k [_{CP} wh_1 [t_1 V t_k]]$

This allows CP\* to raise to the matrix Spec and interact with the matrix wh ((68b)). The list answer pairs questions in the set denoted by CP\* with individuals denoted by wh<sub>1</sub>. This interaction is facilitated by interrogative raising, which analogously to quantifier raising, respects locality. It is predicted that such long-distance answers cannot occur across intervening clauses. It is also predicted that a direct question meaning will not be available unless there is a wh expression in the matrix clause. Finally, taking overt movement to be necessarily to Spec of CP, it is also predicted that multiple fronting structures will not allow long-distance answers of the relevant kind.

The discussion of adjunct islands introduces distinctions between single-pair, multiple-pair, and trapped pair-list readings. This leads to the following possibilities for interpreting a *wh* inside an adjunct:

- (69) a. [<sub>CP</sub> [<sub>TP</sub> [<sub>ISLAND</sub>... wh...]]]  
 b. [<sub>CP</sub> wh<sub>1</sub> [<sub>TP</sub> t<sub>1</sub> [<sub>ISLAND</sub> wh<sub>2</sub> ]]]  
 c. [<sub>CP</sub> wh<sub>1</sub> [<sub>TP</sub> t<sub>1</sub> [<sub>ISLAND</sub> wh<sub>2</sub>... wh<sub>3</sub>...]]]

Example (69a) is ungrammatical in English but acceptable with a direct question interpretation in Japanese. Such structures become acceptable even in English if there is an additional *wh* expression in the matrix clause ((69b)). We explored the option of allowing binding of the embedded *wh*, interpreted as a choice function, from across the island. This leads to single-pair answers. We also explored the option of interpreting the island itself at a higher level, a set of propositions, and having that interact with the matrix *wh*. This leads to list answers pairing propositions with individuals. Structures like (69b), which may otherwise lend themselves to a single-pair answer, admit multiple-pair answers only if the context establishing a distributive relation between two sets is explicitly given. D-linking of the *wh*, where the set of individuals denoted is established in the context, proves inadequate for this. In the case of (69c) multiple-pair answers are readily available, but here the two embedded *wh* expressions must enter into a functional dependency before entering into one with the matrix *wh*, leading to what is known as trapped list readings.

One type of island that we did not discuss but which is clearly relevant to the issue of wh in-situ and scope is the one instantiated by coordinate structures. It is unclear whether the same considerations apply to overt and covert scope taking

out of coordinate structures as to the islands studied here. Coordinate structures are known to pose some very special challenges and are certainly worth probing further.<sup>30</sup>

Let me end on a more general note. We have direct evidence that some amount of pied piping occurs in natural language. Although the principles involved in interpreting pied piping structures are universal, there is considerable cross-linguistic variation in the overt manifestation of such structures. The idea that syntactic constraints on overt and covert scope taking may vary is based on theoretical and empirical considerations, but two diametrically opposed conclusions have been drawn from this distinction which this chapter has tried to navigate between. One view is that *wh in-situ* is impervious to syntactic constraints. The other is that the scope of such expressions is strictly local. The first requires a relaxation of standard syntactic constraints that fall under the rubric of subadjacency. The other requires an enriched semantics for the island. The kind of evidence that this debate rests on is indirect and relies very much on different emphases that can be placed on the specification of values in an acceptable answer. Our discussion sought to highlight the issues that remain open in this area of research.

<sup>30</sup> For relevant discussion, see Munn (1999), Chaves (2012), den Dikken and Lahne (2013), among others.

## Focus and alternatives

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This chapter picks up on the discussion in Chapter 5 regarding morpho-syntactic kinship that *wh* expressions and question formation may share with other phrases and constructions. Here we look at cross-linguistic connections between indefinites and *wh* phrases and between strategies for questioning and marking focus. Should these correlations lead to a shared semantics for *wh* phrases and indefinites? Do they argue for a semantics of questions in terms of the semantics of focus?

We discuss current versions of a semantics for questions in which *wh* expressions and indefinites introduce multi-membered sets of individuals but separate semantically at the clausal level, reviving in essence Hamblin's original theory of questions. This approach agrees with the baseline theory in grouping indefinites and *wh* phrases with each other but differs on the need for *wh* expressions to be existentially bound. It also differs on the meaning ascribed to  $C^0_{+WH}$ . We discuss proposals within this approach with a view to answering the following questions: Do the properties of Japanese indefinites/*wh* phrases necessitate such shifts? How well does this approach account for the scope of *wh* in-situ, discussed in Chapter 7? We also look at the theory of inquisitive semantics, which puts together statements with indefinites and questions in a natural class, but with different assumptions about the syntax-semantics divide. And in doing so, casts the connection between indefinites and questions within a broader framework of conversational dynamics.

A second concern of this chapter is the phenomenon that goes by the name of intervention effects. The basic generalization is that the presence of certain expressions between a *wh* in-situ and its scope position leads to ungrammaticality. One generalization is that these interveners are focus sensitive operators. By tapping into the connection between a theory of questions and a theory of focus, ungrammaticality can be tracked to the competing demands imposed by the intervener and the *wh*. But how solid is the generalization that interveners are focus sensitive operators? We consider alternative explanations based on different characterizations of the empirical generalization regarding intervention.

Finally, we consider polar questions, polar alternative questions, and alternative questions. Do polar questions denote a set of two possible answers or do they, in some crucial way, privilege the nucleus proposition? Differences between polar and polar alternative questions, both grammatical and pragmatic, are explored in answering this question. The ambiguity of questions with disjunction is also investigated. The role of prosody in distinguishing the polar question reading



from the alternative question reading and the proper analysis of those alternatives is explicated, as is the status of possible answers to alternative questions.

## 8.1 Focus-based semantics for questions

This section presents accounts that capitalize on morpho-syntactic affinities between *wh* phrases and indefinites in relation to focus. The first represents something of a return to Hamblin's original analysis of questions. The second casts Hamblin's question semantics in a bi-dimensional framework, where ordinary and focus semantic values operate at distinct though related levels. Implications for the mapping between the syntax of interrogatives and the semantics of questions are articulated. The problem of island effects within focus-based approaches to question meanings is also considered, with particular reference to Japanese. A third account takes the affinity between indefinites and questions further, arguing for a logic that allows declaratives with indefinites and interrogatives to behave as a natural class in conversational exchanges.

### 8.1.1 *Wh phrases, indefinites, and focus*

In Chapter 5 we noted the morpho-syntactic affinity between interrogatives, free relatives, and exclamatives, and considered proposals that sought to unify them semantically. Here we return to the issue of morpho-syntactic affinities, this time probing the connection between interrogatives and indefinites. The two are claimed to have a common morphological component in many languages (Haspelmith 1997; Bhat 2000; Haida 2007). The underlined phrase in the German question in (1), for example, can be interpreted either as a *wh* expression or an indefinite (Haida 2007: 180–2). In the first case, we get a multiple constituent question, in the second a simple question:

- (1) Wer mag was  
 Who likes what  
 “Who likes what?”/“Who likes something?”

Crucially, however, this alternation is dependent on prosody. If the in-situ phrase has intonational focus we get the interrogative meaning, otherwise we get the indefinite meaning. This does not apply to the fronted phrase, however, which is necessarily interpreted as a *wh* phrase. Setting this aside, there are good grounds for connecting questions and focus. Haida (2007) describes in some detail how the expression of focus varies across languages and establishes that the strategy for question formation, while also variable across languages, tracks the strategy for focus within individual languages. I give two examples discussed by Haida to illustrate this.

Hungarian manifests the correlation between interrogative and focused phrases in the syntax. Both must occur in the preverbal position (2a)–(2b), with any expression preceding this position being interpreted as topic. When the *wh* phrase

occurs in topic rather than focus position (2c), it leads to ungrammaticality (Haida 2007: 171–3):

- (2) a. János        haza        kísérte        Marit.  
          John-NOM home escorted Mary-ACC  
          “John escorted Mary home.” = “It was home that John escorted Mary.”
- b. János        kit            kísért        haza  
          John-NOM who-ACC escorted home  
          “Who did John escort home?” = “As for John, who did he escort home?”
- c. \*mit            Marinak       adott János?  
          What-ACC Mary-DAT gave John-NOM  
          Intended: “What did John give to [Mary]<sub>FOCUS</sub>?”

Another type of correlation between focus and questions is seen in Gurune. Examples (3a)–(3b) show this for direct object. The focus-sensitive operator *maʔa*, “only” adjoins to a focused associate. The complex can stay in-situ or be fronted. When it appears in-situ, it must be accompanied by the particle *la* but not otherwise. Examples (4a)–(4b) show the same pattern for interrogatives (Haida 2007: 176):

- (3) a. AdɔNɔ        zaã            nyɛ        la        Atia        maʔa  
          Adongo yesterday see *la* Atia only
- b. Atia maʔa        tɪ        AdɔNɔ        zaã            nyɛ  
          Atia only *ti* Adongo yesterday see  
          “Adongo saw only [Atia]<sub>FOCUS</sub> yesterday.”
- (4) a. AdɔNɔ        zaã            nyɛ        la        anɪ  
          Adongo yesterday see *la* who
- b. anɪ        ta        AdɔNɔ        zaã            nyɛ  
          who *ta* Adongo yesterday see  
          “Who did Adongo see yesterday?”

These illustrations are not meant to be exhaustive. They are simply representative of general tendencies across languages. In what follows I will assume, as is commonly done in the literature, that there is a non-trivial correlation between *wh* phrases and indefinites and between question formation and focus. The two proposals discussed in Section 8.1.2 can be seen as capitalizing on these correlations.

### 8.1.2 Hamblin *redux*

Rooth (1985, 1992) is a standard starting point for semantic studies of focus. The proposal is that every expression has, in addition to its normal semantic value ( $\llbracket \cdot \rrbracket^0$ ), a focus semantic value ( $\llbracket \cdot \rrbracket^f$ ) as well. The latter is obtained by substituting the ordinary semantic value of focused expressions with alternatives of the same

type. The focus semantic value of complex expressions is built up compositionally through point-wise functional application of the focus semantic values of its parts:

- (5) a. John talked to Bill<sub>F</sub>  
 b.  $\llbracket \text{Bill} \rrbracket^o = \text{bill}$ ;  $\llbracket \text{Bill} \rrbracket^f = \{x: x \in D\}$   
 c.  $\llbracket \text{John talked to Bill} \rrbracket^o = \wedge \text{John talked to Bill}$ .  
 d.  $\llbracket \text{John talked to Bill} \rrbracket^f =$   
 $\{\wedge \text{John talked to Bill}, \wedge \text{John talked to Sue}, \wedge \text{John talked to Harry}\}$

This bi-dimensional semantics provides an elegant way to model question–answer congruence: the ordinary semantic value of the question must be a subset of the focus semantic value of the answer (Rooth 1992: 86).<sup>1</sup>

- (6) In a question–answer pair  $\langle \Psi, \alpha \rangle$ ,  $\llbracket \Psi \rrbracket^o \subseteq \llbracket \alpha \rrbracket^f$

Thus, (5a) is a felicitous answer to (7a) but not to (8a). This accords well with our intuitions:<sup>2</sup>

- (7) a. Which man/who did John talk to?  
 b.  $\{\wedge \text{John talked to Bill}, \wedge \text{John talked to Harry}\}$   
 b'.  $\{\wedge \text{John talked to Bill}, \wedge \text{John talked to Harry}, \wedge \text{John talked to Sue}\}$
- (8) a. Which man/who talked to Bill?  
 b.  $\{\wedge \text{John talked to Bill}, \wedge \text{Harry talked to Bill}\}$   
 b'.  $\{\wedge \text{John talked to Bill}, \wedge \text{Sue talked to Bill}, \wedge \text{Harry talked to Bill}\}$

Rooth's theory does not directly address the role of focus in questions. The principle in (6) is equally compatible with Hamblin (1973), Karttunen (1977), or the blend of Hamblin/Karttunen that we have adopted. There is, however, an obvious parallel between the way focus semantic values are built up in Rooth's theory of focus and the way question meanings are derived in Hamblin's theory (see Section 2.1.1).<sup>3</sup> This parallelism has been made explicit by a number of authors (Ramchand 1997; Hagstrom 1998; Shimoyama 2001, 2006; Kratzer and Shimoyama 2002; Beck 2006). Here we will review two of them.

Recall that Hamblin's and Karttunen's theories both treat *wh* expressions and indefinites as a natural class. They differ, however, in the interpretations accorded to them. Karttunen treats them both as existential quantifiers. Hamblin, instead, takes them to denote sets of individuals. These sets combine with other expressions through pointwise functional application, resulting in a set of propositions

<sup>1</sup> The relation must be subset rather than identity because the focus semantic value does not include the restriction imposed by the *wh* phrase: (5d) includes women as alternatives to *Bill*, but the *wh* phrase restricts the alternatives in (7b) to men.

<sup>2</sup> Note that (5a) is a congruent answer to (7a) even in situations where it is false.

<sup>3</sup> Rooth (1992: 84) credits Dietmar Zaefferer with pointing out the connection between Hamblin's question semantics and Rooth's (1985) focus semantics.

at the level of the clause. This is the desired result when the source of the alternatives is a *wh* phrase. When the source of the alternatives is an ordinary indefinite, a type shift to a single proposition is required. Kratzer and Shimoyama (2002) propose a set of propositional operators that act upon the set of alternatives. Of the ones given in (9), of immediate relevance are *Q* and  $\exists$ :

- (9) Where *A* is a set of propositions:
- a.  $[Q](A) = A$
  - b.  $[\exists](A) = \{\text{the proposition that is true in all worlds in which some proposition in } A \text{ is true}\}$
  - c.  $[\forall](A) = \{\text{the proposition that is true in all worlds in which all propositions in } A \text{ are true}\}$

Kratzer and Shimoyama show the workings of this system in relation to Japanese indeterminate pronouns. This is a term coined by Kuroda (1965) to capture the fact that the same base form can display a variety of readings, depending on the closest operator in the structure. The overlap in meaning between Japanese *wh* phrases and indefinites is illustrated below:

- (10) a. *dare-ga odotta no*  
           Who-NOM called Q  
           ‘Who called?’
- b. *dareka-ga odotta*  
           someone-NOM called  
           ‘Someone called.’

Applying (9a)–(9b) to the alternatives generated by (10a)–(10b), we get:

- (11) a.  $\llbracket 10a \rrbracket = [Q](\{\hat{a} \text{ called}, \hat{b} \text{ called}\}) \Rightarrow \{\hat{a} \text{ called}, \hat{b} \text{ called}\} \quad \langle \langle s, t \rangle, t \rangle$   
       b.  $\llbracket 10b \rrbracket = [\exists](\{\hat{a} \text{ called}, \hat{b} \text{ called}\}) \Rightarrow \hat{a} \text{ called} \vee \hat{b} \text{ called} \quad \langle s, t \rangle$

Under this view of the syntax–semantic map, note that the Japanese *Q* morpheme *no* (or *ka*) is semantically vacuous. The same holds for the null operator in  $C^0$  in languages lacking an overt *Q*-morpheme. Note also that *wh* expressions and indefinites make the same contribution to meaning. The difference between them, noted in Section 8.1.1, cannot be located in their semantics directly. It has to be located in the relationship they have to different sentence level operators, as in (9).

The idea that *wh* expressions have the same semantics as focused phrases is taken further by Beck (2006) who proposes that *wh* expressions only have focus semantic values. This goes hand in hand with her analysis of the *Q* operator as taking the focus semantic value of its TP argument and converting it into the ordinary semantic value, a requirement for grammaticality:<sup>4</sup>

<sup>4</sup> I use the version in Rooth (1992) in presenting this and other focus-based accounts in this chapter. The reader is referred to the original articles for formulations using distinguished variables to capture the semantic contribution of focus.

- (12) a.  $\llbracket \text{who} \rrbracket^o$  is undefined;  $\llbracket \text{who} \rrbracket^f = \{x: x \in D\}$   
 b.  $\llbracket [\text{TP who left}] \rrbracket^o = \text{undefined}$ ;  $\llbracket [\text{TP who left}] \rrbracket^f = \{^a \text{left}, ^b \text{left}\}$   
 c.  $\llbracket [\text{CP } C^o_{+WH} [\text{TP who left}]] \rrbracket^o = \llbracket [\text{TP who left}] \rrbracket^f = \{^a \text{left}, ^b \text{left}\}$

Kratzer and Shimoyama's and Beck's reworkings of Hamblin's theory are embedded within broader discussions related to other aspects of wh phrases and indefinites. Under both accounts, wh expressions have to be interpreted inside  $C^o$ . We saw in Chapter 1 that this poses a problem when wh expressions are existentially bound within the question nucleus. However, this is not an issue for Hamblin semantics since no existential binding is ever involved in the interpretation of wh expressions. Note that we still need the principle in (6) to ensure the felicity of question–answer pairs. Focus on wh expressions is used to generate the set of alternatives but in the final output that set of alternatives is the ordinary semantic value of the question, which is what (6) refers to in capturing question–answer congruence.

### 8.1.3 The inquisitive lens

Let us now turn to a rather different perspective on how statements with indefinites and questions are related. We have so far taken declaratives, whether they include indefinites or not, to denote propositions, sets of possible worlds. And we have taken interrogatives to denote sets of propositions:

| (13) | <i>Example</i>    | <i>Sentence Type</i> | <i>Speech Act</i> |
|------|-------------------|----------------------|-------------------|
| a.   | John left         | declarative          | assertion         |
| b.   | Someone left      | declarative          | assertion         |
| c.   | Did someone leave | interrogative        | question          |

Inquisitive Semantics presents a way of thinking about these sentence types that cross-classifies them on the basis of two separate properties (Groenendijk and Roelofsen 2009):<sup>5</sup>

| (14) | <i>Example</i>    | <i>Informative</i> | <i>Inquisitive</i> | <i>Speech Act</i>      |
|------|-------------------|--------------------|--------------------|------------------------|
| a.   | John left         | +                  | –                  | assertion              |
| b.   | Someone left      | +                  | +                  | assertion and question |
| c.   | Did someone leave | –                  | +                  | question               |

To understand the two key notions, informativeness and inquisitiveness, let us step back and consider the now standard view of discourse dynamics, going back to the seminal work of Stalnaker (1972, 1978) and Karttunen (1974). On the dynamic semantics view, an utterance is interpreted against a body of information, the common ground.<sup>6</sup> This is a set of propositions that the discourse

<sup>5</sup> We will set aside here types with negative values for both, contradictions and tautologies.

<sup>6</sup> See Chierchia and McConnell-Ginet (2000) for an introduction to dynamic semantics. See Murray (2014) on how to incorporate the insights of inquisitive semantics into the original dynamic

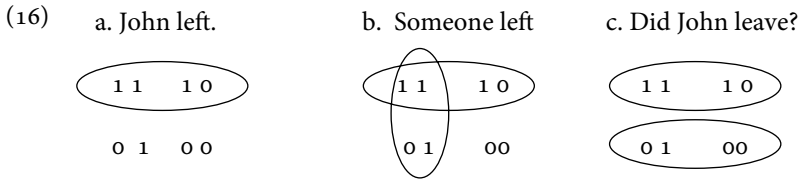
participants are in agreement about, the facts that can be taken as settled between them. The common ground describes a set of worlds, the context set, which are those worlds in which all of the mutually accepted propositions hold. An assertion updates the common ground by eliminating those worlds from the context set where the asserted proposition is false. As information grows, the context set shrinks. It is obvious that declarative sentences have the potential to enrich the common ground in this way. It thus makes sense to think of declarative meaning not as a proposition, a set of worlds, but in terms of context change potential. An interrogative clearly does not enrich the common ground. What it does instead is raise an issue by making salient a set of alternatives. Depending on which alternatives are asserted in response to the question, the common ground is enhanced. Thus their impact on the common ground is indirect, at best.

Turning now to inquisitive semantics, the values under the first column *+/- informative* essentially refers to the context change potential of declaratives in dynamic semantics. The second column, *+/- inquisitive* gives formal status to the discourse contribution of interrogatives, recognizing it as an equally important building block of conversational exchange. Crucially, thinking in terms of the conversational exchange potential of interrogatives in this way opens up the possibility of distinguishing between declaratives along these lines as well. Consider the following paradigm:

- |      |                               |                           |
|------|-------------------------------|---------------------------|
| (15) | a. Speaker A: Did John leave? | Speaker B: Yes, he did.   |
|      | b. Speaker A: Someone left.   | Speaker B: Yes, John did. |
|      | c. Speaker A: John left.      | Speaker B: Yes, he did.   |

The interrogative in (15a) must be followed by a response in order for the dialogue to be considered successful. While the same is not required for the declarative in (15b), a response naming John as the one who left is perfectly natural and moves the conversation forward in an incremental way. This is not so for (15c). The common ground is enriched based on Speaker A's contribution as long as Speaker B does not object to the statement. The positive response is, at best, an overt acceptance of Speaker A's assertion. This, then, provides empirical motivation for characterizing indefinite statements as hybrid, inquisitive as well as informative.

The differences in (15) can be represented pictorially. Assume a domain with two individuals, John and Mary. Within each pair of numerals, take the one on the left to stand for the truth or falsity of *John left*, the one on the right to stand for the truth or falsity of *Mary left*. Example (16a) denotes one possibility, and eliminates one; (16b) denotes two possibilities and eliminates one; (16c) also denotes two possibilities but does not eliminate any:



The two sentences that agree on informativity, shrink the context set by eliminating worlds in which either John doesn't leave (16a), or worlds in which neither John nor Mary leaves (16b). The non-informative sentence in (16c) does not eliminate any worlds. The first sentence is uninquisitive in that it presents only one alternative. The addressee can choose to accept or reject it, but there isn't anything more that she can do. The other two, however, raise two possibilities and now the addressee has a role to play in choosing between those options.

There are some obvious parallels here with Kratzer and Shimoyama's version of alternative semantics presented in Section 8.1.2. Apart from the cases discussed so far, they also agree on disjunction. In alternative semantics disjunction is an alternative generator (Alonso-Ovalle 2006); in the inquisitive semantic approach it is a hybrid. One way to think about the two approaches, as far as denotations go, is that they agree on meanings until just before the clause-level closure operation takes effect in alternative semantics. Inquisitive semantics does not have such an operation cued to a syntactic node. By allowing alternatives continued existence, it makes it possible for a single sentence type to make different contributions to dialogue. Hybrids can function as questions based on their inquisitive meanings or they can type shift to proposition level meanings and function as assertions. In terms of the conversational potential of the two pure types, uninformative inquisitive sentences and uninquisitive informative sentences, inquisitive semantics does not depart from the dynamic semantic view in any significant way, at least for the atomic cases. I refer the reader to Groenendijk and Roelofsen (2009) and references cited there for details of the logic and application to some semantic and pragmatic natural language phenomena. There has not been much work on constituent questions, the bulk of our investigations so far, within inquisitive semantics (but see Anderbois 2011; and Theiler 2014). There have been several inquisitive semantic analyses of polar and alternative questions, however, and we will have occasion to look at some of them later in this chapter.

#### 8.1.4 Islands and indeterminate pronouns

Let us return now to the alternative semantic account in Kratzer and Shimoyama (2002) and Shimoyama (2001, 2006). In order to appreciate the richness of their proposal, let us expand our discussion to include universal readings of Japanese indeterminate pronouns:

- (17) a. dareka-mo      odotta  
           everyone-NOM    called  
           "Everyone called."

- b. [[Dono gakusei-ga syootaisita] sensei]-mo odotta  
Which student-NOM invited teacher-MO danced  
“For every student  $x$ , the teacher(s) that  $x$  had invited danced.”
- c. Yoko-wa [[[Taro-ga nan-nen-ni nani-nituite kaita] ronbun]-mo  
Yoko-TOP Taro-NOM what-year-in what-about wrote paper-MO  
yuu-datta ka] siritagatteiru  
A-was Q want-to-know  
“Yoko wonders whether for every topic  $x$ , every year  $y$ , that paper that Taro wrote on  $x$  in  $y$  got an A”

Apart from showing the universal force of indeterminates in the presence of *mo*, (17a)–(17c) also show properties that were in evidence in relation to the Q morpheme *ka* in Chapter 7. The universal seems to bind an indeterminate across the island formed by a complex noun phrase (17b), but binding cannot happen across another binder. The Q morpheme *ka* in (17c) forces the matrix into a polar question, while the embedded indeterminate has universal force due to the *mo* immediately above it.

*Mo* is analyzed as a standard universal quantifier denoting the set of properties  $P$  that every member in the denotation of its sister constituent has (Shimoyama 2006: 154–5):

$$(18) \quad \text{For } \llbracket \alpha \rrbracket \subseteq D_e, \llbracket \alpha \text{ mo} \rrbracket = \{\lambda P [\forall x [x \in \llbracket \alpha \rrbracket \rightarrow P(x)]]\}$$

Here we interpret a structure where *mo* combines with a complex noun phrase:

- (19) a. [[Dono gakusei-ga teisyutusita] syukudai]-mo yuu-datta  
which student-NOM submitted hw-assignment-MO A-was  
“Every homework assignment that a student had handed in got an A.”
- b.  $[_{TP} [_{DP} \text{ which student submitted hw-assignment}]-\text{mo got-A}]$
- c.  $\llbracket \text{which student submitted hw-assignment} \rrbracket =$   
 $\{\iota y [\text{hw}(y) \wedge \text{submitted}(z, y)]: \text{student}(z)\}$   
 $\Rightarrow \{\iota y [\text{hw}(y) \wedge \text{submitted}(\text{Taro}, y), \iota y [\text{hw}(y) \wedge \text{submitted}(\text{Ziro}, y)]]\}$   
 $\llbracket \text{which student submitted hw-mo} \rrbracket =$   
 $\{\lambda P [\forall x \in \llbracket \text{wh st submitted hw} \rrbracket \rightarrow P(x)]\}$   
 $\Rightarrow \{\lambda P [\forall x \in \{\iota y [\text{hw}(y) \wedge \text{submitted}(z, y)]: \text{student}(z)\} \rightarrow P(x)]\}$
- d.  $\llbracket 19a \rrbracket = \forall x \in \{\iota y [\text{hw}(y) \wedge \text{submitted}(z, y)]: \text{student}(z)\} \rightarrow \text{got-A}(x)$

Since the indeterminate denotes a set of individuals, the constituent *which student submitted hw-assignment* denotes the set of homework assignments  $y$ , such that each  $y$  is the assignment submitted by some student. When *mo* applies to this, we get universal quantification over the set. However, the force of universality rests on the indeterminate ensuring that the complex noun phrase is not a singleton. These effects are achieved without the indeterminate moving covertly, even within the island. Shimoyama calls this the “direct restrictor” account, as opposed to the “embedded restrictor” approaches of Nishigauchi (1990) and von Stechow (1996).



There are other more complex aspects to the data that Shimoyama brings into the discussion. The observation that aligns it with the discussion of islands in Chapter 7 is that indeterminates cannot be bound from outside a *mo/ka* phrase:

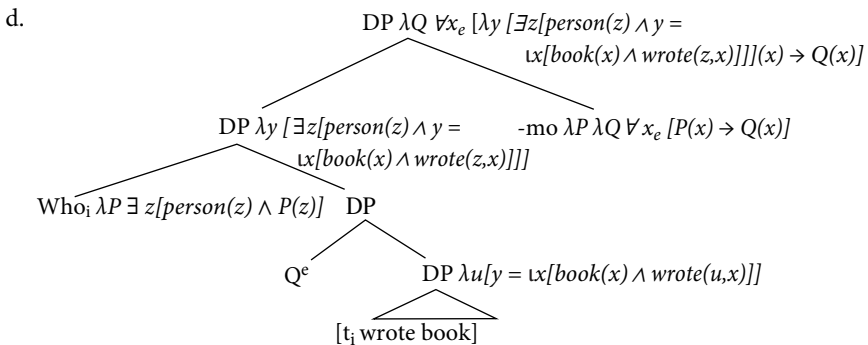
- (20) a.  $*[ \dots [ \dots \text{indeterminate}_i \dots ] -ka/mo \dots ] -ka/mo_i$   
 b.  $[ \dots [ \dots \text{indeterminate}_i \dots ] -CNP/Adjunct \dots ] -ka/mo_i$

Shimoyama's explanation is that both *ka* and *mo* are defined on sets of alternatives. Once a set of alternatives combines with one of these operators and is factored into its meaning, the alternatives are no longer available for a similar operator higher up. The appeal of the system lies in explaining apparent scope effects as a by-product of the mode of composition.

Although the direct restrictor view of *mo* and *ka* is couched within Hamblin's semantics, it turns out that it is not dependent upon it. Heim (2010) shows that it is possible to get equivalent results within an analysis of indeterminates as existential generalized quantifiers, as in Karttunen. Heim generalizes the interpretation procedure across categories so there is uniformity across *ka* and *mo* phrases. I show below how it applies to *mo* phrases. It should be clear that it would also apply to *ka* phrases with appropriate adjustments in the syntax:

(21)

- a. Family of Q morphemes:  $\llbracket Q^\sigma \rrbracket = \lambda v_\sigma \{v\}$   
 b. Wh Phrase Quantification Rule: If  $\alpha$  has daughters  $\beta$  and  $\gamma$ , where  $\beta$  is of type  $\langle \langle e, t \rangle \rangle$  and  $\gamma$  of some type  $\langle \sigma, t \rangle$ , then  $\llbracket \alpha \rrbracket = \{v_\sigma . \llbracket \beta \rrbracket (\lambda x_e v \in \llbracket \gamma \rrbracket (x)) = 1\}$   
 c.  $\llbracket mo \rrbracket = \lambda P \lambda Q \forall x_e [P(x) \rightarrow Q(x)]$  (i.e.  $\llbracket mo \rrbracket = \llbracket every \rrbracket$ )  
 Syntax: *mo* subcategorizes for a sister headed by a Q morpheme.



The essential insight of Shimoyama and Kratzer (2002) and Shimoyama (2001, 2006) is that the restriction of Japanese particles is provided by its immediate sister, however complex. And, as we see from (21), this insight is as compatible with a focus-based view of indeterminates as with their treatment as existential quantifiers. That is, indeterminates do not force a choice between Hamblin's and Karttunen's theories or the blend of Hamblin and Karttunen that we have adopted.

A question that arises for the direct restrictor view, worth noting here, is with respect to the scope of indeterminates in *wh* triangle configurations. Recall from Chapter 7 that such cases admit long-distance list readings between the matrix and an embedded *wh* phrase:

- (22) a. dono sensei-ga [Mary-ga doko-de dono hon-o kat-ta ka] sitte-imasu ka  
       which professor Mary where which book bought Q knows Q  
       ‘Which professor knows where Mary bought which book?’  
       b. [<sub>CP</sub> [<sub>TP</sub> which professor<sub>i</sub> [<sub>CP</sub> [<sub>TP</sub> ... where<sub>j</sub> ... **which book<sub>k</sub>** ... ] Q<sub>j</sub> ] ... ]  
       Q<sub>i, k</sub>]  
       c. [<sub>CP</sub> [<sub>TP</sub> which professor<sub>i</sub> [<sub>CP</sub>\* [<sub>CP</sub> [<sub>TP</sub> ... where<sub>j</sub> ... **which book<sub>k</sub>** ... ] Q<sub>j</sub> ]  
       OP<sub>k</sub> ]<sub>1</sub> ... ] Q<sub>i 1</sub> ]

One view is that such pair-list readings arise when an embedded *wh* in-situ takes scope at the matrix level, disregarding islands. The other is that they arise due to a functional dependency between a matrix *wh* and a *wh* complement interpreted as a family of questions. The two approaches are represented schematically in (22b) and (22c) respectively. Neither view blends readily with the mode of composition we have considered in this section. It raises the following question: how can the contribution of a *wh* indeterminate get past the first operator so that it can be bound by a higher operator, whether that higher operator is in the next higher  $C^0$  or in the  $CP^*$  projection immediately above it? As far as I know, the challenge posed by such cases has not been addressed within this approach.<sup>7</sup>

### 8.1.5 Section summary

We noted cross-linguistic morphological affinities between *wh* phrases and indefinites, and between *wh* phrases and focus. This affinity is directly captured in Hamblin (1973) and its more recent incarnations, in which a *wh* expression generates alternatives that percolate up to the level of the clause. A particular advantage of this approach is that a compositional account can be given for the binding of Japanese indeterminates from across nominal and adjunct islands but not across *wh* or other quantificational islands. In this connection we saw that the insights of this approach can also be replicated within the Hamblin/Karttunen account where *wh* expressions are existentially quantified. We noted, however, the challenge posed by long-distance list answers that seem to require a *wh* phrase to escape the first available binder in order to be bound by matrix  $C^0$  or by an operator in the  $CP^*$  immediately dominating the island. We also introduced Inquisitive Semantics, a theory of conversational dynamics that allows statements with indefinites to share formal properties with interrogatives.

<sup>7</sup> Kotek (2014) proposes a hybrid system in which *wh* expressions can either be interpreted via focus or via *wh* movement. She furthermore assumes, following Cheng and Demirdache (2010), that the scope of *wh* in-situ is syntactically unconstrained. If both options are freely available, it is not clear to me how the locality effects discussed here could be derived in a principled manner. See Chapter 7 for problems with this aspect of Cheng and Demirdache’s proposal. See also Dayal (2016a, 2016b).

We will now look at intervention effects, a phenomenon in which a question becomes unacceptable if a *wh* phrase appears in the scope of certain operators at S-structure. Although the initial exploration of intervention effects was within the syntactic literature, it has also intrigued semanticists and at present there are several semantic explanations available. The biggest issue remaining open is the precise characterization of the empirical terrain. It is hard to evaluate the different proposals definitively since they often differ on the generalizations they seek to capture. The proposals we survey will convey a sense of the empirical and theoretical complexities involved.

Hoji (1985), and later Beck (1996a, 1996b) and Beck and Kim (1997), note a contrast in grammaticality based on the relative position of a *wh* in-situ and a quantifier. Examples (23) and (24) are from German, but similar facts hold in other languages that allow *wh* in-situ and scrambling. In single *wh* fronting languages these effects only show up in multiple *wh* interrogatives because only they allow *wh* in-situ:

- Example (23a), with a negative quantifier preceding a *wh*, is unacceptable, but it is acceptable with a proper name in the same position. Example (23b) shows that scrambling the *wh* expression to the left of the quantifier repairs the problem. Both orders are acceptable with the universal quantifier but there is a difference in interpretation. The base order forces a pair-list reading. That is, the universal in (24a) cannot be interpreted inside the question nucleus, it must take scope over the *wh* phrases. Recall from Chapter 4 that negative quantifiers do not allow wide scope/pair-list readings, so (23a) does not have this option. Not all quantifiers, we will see, give rise to these effects. Much of the work in analyzing intervention has therefore focused on determining the precise set of interveners.

Beck (1996a, 1996b) and Beck and Kim (1997) explain intervention effects in terms of a constraint on LF movement, positing a filter that can be represented schematically as in (26a). English glosses are used for ease of exposition:

- (26) a.  $*[\dots X_i \dots [Q \dots [t_i^{LF} \dots]]]$   
 b.  $*[\text{whom}_i \text{ where}_j [t_i^{SS} \text{ nobody seen } t_j^{LF}]]$   
 c.  $[\text{whom}_i \text{ where}_j [t_i^{SS} t_i^{LF} \text{ nobody seen } t_j^{SS}]]$

In (26b), the LF for (23a), “where” moves at LF across the quantifier, violating the filter. In (26c), the LF for (23b), “where” scrambles across the quantifier at S-structure, where the filter does not apply. Subsequent movement of “where” to Spec CP creates an LF trace but no crossing of the quantifier is involved in this segment. This proposal argues against an alternative explanation in terms of relativized minimality (Cinque 1990; Rizzi 1990), where intervention effects arise because QP is a closer potential binder for the trace than the antecedent in Spec CP.<sup>8</sup> While (26a) captures the core generalization, it is not entirely satisfactory as it stipulates an otherwise unmotivated dichotomy between traces created at S-Structure and LF.<sup>9</sup> Furthermore, it does not take into account the nature of the quantifiers that inhibit such movement.

An approach to intervention effects that discriminates between potential interveners is the dynamic approach of Honcoop (1996, 1998) and Haida (2007). They note the overlap with weak island effects discussed in Chapter 6. Honcoop shows that several of the scopal elements identified by Szabolcsi and Zwarts (1993) as creating weak island effects are also at play in intervention. These quantifiers (negation, *nobody*, *no N*, *less than n N/fewer than n N*, *exactly n N*, etc.) share the property of creating inaccessible domains for anaphora. Schematically:

- (27) a.  $\dots Q_i [\varphi \dots [\sigma X \dots \text{indefinite}_i \dots] \dots] \dots$   
 b. *The Intervention Generalization:  $* \dots Q_i [\varphi \dots [\text{Weak Island Operator} \dots \text{indefinite}_i \dots] \dots] \dots$*

In (27a), the indefinite must be dynamically bound by Q but if X is an operator that creates an inaccessible domain, this binding is blocked, leading to the generalization in (27b).

Haida (2007) elaborates on this explanation within his approach to wh expressions as focused indefinites. I give the essence of his analysis of intervention effects

<sup>8</sup> Other explanations, apart from the ones to be discussed below, include Hagstrom (1998), Pesetsky (2000), Kim (2002), Cable (2007, 2010), Kotek (2014), among many others.

<sup>9</sup> Note that Huang’s (1982) distinction between S-structure and LF movement goes in the other direction, with the latter being more permissive than the former.

in Hindi (Haida 2007: 263–4), though the Hindi facts have been somewhat inaccurately presented in the literature and need to be corrected for future work on this topic:<sup>10</sup>

- (28) a. ??kisii-ne nahiiN kyaa paRhaa?  
 Someone-ERG not what read  
 “What did no one read?”  
 b.  $[_{CP} C^{[+Q]} [_{TP} \text{no one } [_{VP} \text{what}_2^{[+F]} \text{read}]]]$   
 c.  $Q^i (\lambda i. \neg \exists u_1 \exists u_2 (u_2 = \sigma v. \text{read}(i)(u_1, v)))$

LF (28b) is the intervention configuration. It presupposes *there is no pair of individuals and things such that the individual read that thing* (28c). In any context where this presupposition is satisfied, the question (in effect) *what is it* cannot be asked. When the *wh* is moved over the intervener (29a), things change:

- (29) a.  $[_{CP} C^{[+Q]} [_{XP} \text{what}_2^{[+F]} [\lambda 2 [_{TP} \text{no one } [_{VP} \text{what}_2^{[+F]} \text{read}]]]]]$   
 b.  $Q^i (\lambda i. \exists u_2 (u_2 = \sigma v. \neg \exists u_1. \text{read}(i)(u_1, v)))$

The presupposition now is *there is something that no one read* and in a context that satisfies this presupposition, the question *what is it* can be felicitously asked.

Haida also discusses multiple *wh* questions which I cannot go into here. I note though that many of the points he makes have already been covered in our discussion in Chapter 4. I also note the point made by Mayr (2014) that this line of approach merges negative island and intervention effects, though the two have been shown to be distinct (see also Beck 2006). The former blocks overt

<sup>10</sup> Note that Hindi doesn’t have a quantifier corresponding to *nobody*. That meaning is expressed through a combination of an indefinite and negation. Also, I have added the ergative marker on the subject in (28a). Beck (2006: 2) and, following her, Haida (2007: 262–3) give the paradigm in (i)–(ii). I have left the error in case-marking in place in (i)–(ii); (iii)–(iv) highlight the fact that the effects are not really about *wh* phrases but about the position of negation with respect to the verb:

- (i) ??koi nahiiN kyaa paRhaa?  
 anyone not what read-PERF.M  
 (ii) kyaa koi nahiiN paRhaa  
 what anyone not read-PERF.M  
 “What did no one read?”

Sentential negation in Hindi occurs left adjacent to the verb, which is responsible for the improvement in (ii). That the *wh* is not critical is shown by the fact that the contrast between (i) and (ii) also holds when *kyaa* is replaced by a referential term as in (iii)–(iv):

- (iii) ??kisii-ne nahiiN yeh kitaab paRhii  
 someone not this book read-PERF.M  
 (iv) yeh kitaab kisii-ne nahiiN paRhii  
 this book someone not read-PERF.M  
 “No one read this book.”

This is so when (iii) is read with neutral intonation. It may be acceptable if the intonation suggests some kind of scrambled word order. Thanks to Rajesh Bhatt (p.c.) for discussion of these points.

movement while the latter blocks covert scope. The former manifests modal obviation, something that has not been noted for the latter. The other proposals we will consider in this section keep the two apart.

### 8.2.2 *Tapping into focus semantics*

Beck (2006) takes a radically different position from her earlier account. Adopting generalization (30) from Kim (2002), she shifts the locus of explanation from the nature of the trace to the relationship between the relevant components of the offending structure: the *wh* in-situ, the intervener, and the *Q* morpheme:

- (30) a. A quantificational or focusing element may not intervene between a *wh*-phrase and its licensing complementizer.  
 b.  $*[Q_i [\dots [\text{intervener} [\dots \text{wh-phrase}_i \dots ]]]]$   
 c.  $*[Q_i [\dots [\text{FocP} [\dots \text{wh-phrase}_i \dots ]]]]$   
 A focused phrase (e.g. ‘only’+NP) may not intervene between a *wh*-phrase and its licensing complementizer.

Beck treats *wh* expressions as obligatorily focused, as in Hamblin’s original account. Translated into Rooth’s account of focus, this means that *wh* expressions only have a focus semantic value, not an ordinary semantic value. A corollary to this is the claim that the *Q* morpheme is defined only on focus semantic values. We have already seen the basic account in Section 8.1.2, repeated as (31) for convenience:

- (31) a.  $[Q_i [\text{who}_i \text{ left}]]$   
 b.  $\llbracket \text{who} \rrbracket^f = \{x \mid x \in D_e\}$   
 $\llbracket \text{left} \rrbracket^f = \{\text{left}\}$   
 $\llbracket \text{who left} \rrbracket^f = \{\lambda w \text{ left}(w)(x) \mid x \in D_e\}$   
 $\llbracket Q \text{ XP} \rrbracket^o = \llbracket \text{XP} \rrbracket^f$   
 c.  $\llbracket \text{who} \rrbracket^o = \text{undefined}$   
 $\llbracket \text{left} \rrbracket^o = \{\text{left}\}$   
 $\llbracket \text{who left} \rrbracket^o = \text{undefined}$

Crucially, the *Q* morpheme is defined on the focus semantic value of its argument and is impervious to its ordinary semantic value.

Now, consider a case in which a focus-sensitive operator like *only* intervenes between the *Q* morpheme and a *wh* expression (Beck 2006: 16–17). English is used for expository purposes. A structure like (32a) would be unacceptable in any language that otherwise allows *wh* phrases to remain in-situ:<sup>11</sup>

- (32) a.  $*\text{Only John saw who?}$   
 b.  $[_{CP} Q_2 [_{IP_3} \text{only}_C [_{IP_2} \sim C [_{IP_1} \text{John}_{F1} \text{saw who}_2]]]]]$

<sup>11</sup> The squiggle operator  $\sim$  is defined in Rooth (1992: 86–90) to introduce constraints on the interpretation of focus.

$[[IP_1]]^\circ$  is undefined because of the *wh* in-situ, and so  $[[IP_2]]^\circ$  is undefined. The meaning of *only* uses both focus and ordinary semantic values of its preajacent, combining them to output a single ordinary semantic value. This means that the ordinary semantic value and the focus semantic value for  $[[IP_3]]$  are undefined and consequently *Q* is undefined. Beck posits the following constraint to explain the unacceptability of such CPs:

- (33) *Principle of Interpretability*: An LF must have an ordinary semantic value.

Movement has a redeeming effect (Beck 2006: 21). The crucial category in (34b) is the one labeled *X*, where the contribution of focus is fully calculated:

- (34) a. *Who* has only John seen?  
 b.  $[_Z Q_3 [_Y \text{who}_3 [_1 [_X \text{only}_C [\sim C [[John]_{F_2} \text{has seen } t_i]]]]]]]$

The trace of a moved *wh* expression has an ordinary semantic value so the structure it occurs in has an ordinary semantic value. A focus semantic value is generated by *John*, so the preajacent of *only* has both values. At *X*, the ordinary and focus semantic values are reset so *X* has an ordinary semantic value and there remains a variable at the position of the trace. This is bound by the moved *wh* operator in *Y*. When the *Q* morpheme combines with the category *Y*, we again have a focus semantic value due to the moved *wh* expression. The *Q* operator converts it into the ordinary semantic value and we obtain a standard question meaning. Scrambling the *wh*, in languages that allow it, has the same effect. The contrast between overt and covert *wh* movement with respect to intervention is derived as a semantically driven relativized minimality effect based on focus semantic values.

Li and Law (2016) refine the generalization regarding focus related intervention, drawing on Chinese data, due to Aoun and Li (1993) and Yang (2012). A relevant contrast is shown in (35):

- (35) a. ?\*Ta *zhi* yaoqing-le [Libai]<sub>F</sub> chuxi *shenme huodong*?  
 He only invite-Asp Libai attend what activity  
 Intended ‘‘What was the activity *x* such that he only invited [Libai]<sub>F</sub> to attend *x*?’’  
 b. Libai *zhi* chuxi-le *shenme huodong*?  
 Libai only attend-Asp what activity  
 ‘‘What was the activity *x* such that Libai only attended *x*?’’

Li and Law’s generalization is in (36a), where it is the  $XP_F$  that creates the problem. Sentence (35b) has the structure  $[Q \dots [Focus \text{ sensitive operator } [\dots WH \dots ]]]$  and is clearly exempted:

- (36) a. ?\*[*Q*...[Focus sensitive operator [ $XP_F$ ...*WH*...]]]  
 b. \*[...[Focus sensitive operator [focus alternatives...ordinary alternatives...]]]

The intervention effect, they claim, arises not because the focus sensitive operator cannot relate to the meaning of the *wh* phrase, but rather because the focus associate and the *wh* phrase together create an untenable quantificational domain. Briefly, they take focused phrases and *wh* phrases to give rise to alternatives via distinct means. In (35a), for example, the focused phrase contributes focus alternatives to Libai:  $\{Libai, Dufu\}$  and the *wh* contributes alternatives based on events:  $\{dinner, dance\}$ . Composing the two yields (37) as the focus semantic value of the VP:

(37)

$$\left\{ \left\{ \lambda y \lambda w \text{invite-to-attend}_w(y, \text{Libai}, \text{dinner}), \lambda y \lambda w \text{invite-to-attend}_w(y, \text{Libai}, \text{dance}) \right\} \right\} \\ \left\{ \left\{ \lambda y \lambda w \text{invite-to-attend}_w(y, \text{Dufu}, \text{dinner}), \lambda y \lambda w \text{invite-to-attend}_w(y, \text{Dufu}, \text{dance}) \right\} \right\}$$

Question (35a) is unacceptable because *zhi*, “only” requires its prejacent to be of type  $\langle\langle e, \langle s, t \rangle \rangle, t \rangle$ , not  $\langle\langle\langle e, \langle s, t \rangle \rangle, t \rangle, t \rangle$ . The VP in (35b), instead, has only one set of alternatives, those generated by the *wh* phrase. It therefore has the right type to function as the quantificational domain of *zhi*. They further show that the configuration *only* > *WH* >  $XP_F$  is also unacceptable with association between *only* and *XP*, for the same reasons.

Li and Law’s approach in terms of quantificational domains has clear merit, given the facts they cite. As they point out, however, it does not explain cases where the offending interveners are quantifiers like *everybody* or *nobody*.

### 8.2.3 Appealing to anti-topicality and anti-additivity

Tomioka (2007) presents a rather different perspective on intervention. He notes that LF-based approaches, such as Beck (1996b) and Beck and Kim (1997), do not account for the following observations: judgments of unacceptability are fragile, ranging from merely marginal to unacceptable, the set of interveners within and across languages is not homogeneous, and intervention effects do not obtain in embedded contexts.

Tomioka argues for a pragmatic account, within the *Information Packaging* framework (Chafe 1976; Prince 1981; Vallduvi 1992, 1995). The crucial distinction is between *focus* and *ground*, with the latter being divided into *link* and *tail*. Roughly speaking, *focus* is the new information, while *link* connects an utterance to the previous context by setting up the theme of the utterance. The *tail* is then what is left of the old information. Japanese (38a) has topic marking on the subject. It is the *link*, the *wh* is the *focus*, and the verb the *tail*:

- (38) a. John-wa *nani-o* yon-da-no  
           John-TOP what-ACC read-past-Q  
           “What did John read?”
- b. \**daremo nani-o* yom-ana-katta-no  
           anybody what-ACC read-neg-past-Q  
           “What did no one read?”
- c. \**daremo-wa*,            \**dareka-wa*,    \*[John-ka Bill]-wa,    \*John-mo-wa  
           anybody/everyone,    someone       John or Bill       John-also



Tomioka's explanation for intervention effects starts with the view that the pre wh expression must be a topic but not all quantifiers can be topic marked. Example (38c) lists some Anti Topic Items (ATI). Ungrammaticality is predicted when an ATI occurs to the left of the wh. This raises the question of why scrambling should redeem intervention effects. Tomioka argues that scrambling places the ATI into the *tail* portion of the interrogative, with which it is fully compatible. Scrambling also has the effect of phonologically reducing the material to the right of the scrambled phrase, a further confirmation of its information structure status as *tail*.

This line of approach, linking intervention effects with (anti)-topicality, provides a simple explanation for a contrast that the accounts discussed in Sections 8.2.1 and 8.2.2 do not engage with. The pre wh subject of a matrix clause must be topic marked (39a), but not the subject of an embedded clause (39b). Interestingly, there is no discernible intervention effect in (39c) with a quantifier as the embedded pre wh subject. This is as expected since there is no pressure in the embedded position for an ATI that precedes the wh to function as topic:

- (39) a. ??John-ga *nani-o* yon-da-no  
           John-NOM what-ACC read-Q  
           ‘What did John read?’  
       b. Kimi-wa [<sub>CP</sub> John-ga *nani-o* yon-da-to] omotte-iru-no  
           you-TOP John- NOM what-ACC read-past-comp think-Q  
       c. pro [<sub>CP</sub> John-ga /daremo-ga *nani-o* yon-da-to] omotte-iru-no  
           John-NOM everyone-NOM what-ACC read-past-comp think-Q  
           ‘What do you think that John/everyone read?’

Since topicality is typically associated with subjects, Tomioka also looks at interveners in non-subject position and notes a much weaker effect, demonstrated in (40). This is again as expected since the ATI is not in topic position. However, the scrambled version is still comparatively better. Tomioka ascribes this to the fact that the best position for an ATI is the post wh *tail* portion of the structure:

- (40) a. ?Ken-wa dareka-ni *nani-o* mise-ta-no  
           Ken-TOP someone-DAT what-ACC show-past-Q  
           ‘What did Ken show to someone?’  
       b. Ken-wa *nani-o* dareka-ni t mise-ta-no

The last point addressed by Tomioka is the fact that Japanese NPIs are worse interveners than other quantifiers. Question (38b), which involves an NPI, faces two problems. The first is the occurrence of the ATI *daremo* in the *link* portion of the structure. The other is that it is in a phonologically separate phrase than its licenser, the negation. Tomioka also notes that the improvement in embedded contexts is weaker for NPIs than for other interveners.

Tomioka's proposal is particularly focused on Japanese and Korean but a similar account has been proposed by Grohmann (2006) to handle intervention

in German. Mayr (2014) argues against Grohmann, however, citing examples like (41) where the question makes it clear that *niemand*, “nobody” is the topic of the answer. Similarly, (42) establishes the ability of *wenige N*, “few N” to function as a topic. Since these quantifiers, that are clearly not ATI, are among the set known to create intervention effects, topicality cannot be at issue:

- (41) A: Was hat niemand gemacht?  
       What has nobody done  
       “What did no one do?”  
       B: *Krieg und Frieden* hat zum Beispiel niemand gelesen  
       War and Peace has for instance nobody read  
       “For instance, nobody read *War and Peace*.”
- (42) A: Die Kinder haben alle wenige Bucher gelesen, weil sie faul sind  
       The kids have all few books read because they lazy are  
       B: Moment mal. Wenige Bucher hat nur HANS gelesen  
       Moment just few books has only Hans read  
       “Wait a minute. Only Hans has read few books.”

Mayr’s starting point is the observation that upward entailing quantifiers, unlike downward entailing quantifiers, do not create intervention effects. On this basis, he claims that the crucial property of interveners is non-additivity (Mayr 2014: 517):

- (43) a. *Intervention Effects Generalization (IEG)*: An operator is a problematic intervener if it is non-additive  
       b.  $f$  of type  $\langle \sigma, t \rangle$  is additive if for any  $g, h$  of type  $\sigma$ ,  $f(g \vee h) = f(g) \vee f(h)$   
       c. *No N* (anti-additive):  
            $\neg \exists x(\text{smoke}(x) \vee \text{drink}(x)) \leftrightarrow \neg \exists x(\text{smoke}(x)) \vee \neg \exists x(\text{drink}(x))$   
       d. *Some N* (additive):  
            $\exists x(\text{smoke}(x) \vee \text{drink}(x)) \leftrightarrow \exists x(\text{smoke}(x)) \vee \exists x(\text{drink}(x))$

Intervention effects arise if the proposition resulting from interpreting the *wh* expression as an existential in its surface scope position is not equivalent to the disjunction of the alternatives in its Hamblin–Karttunen denotation.

To see the system at work, consider the following German examples with a non-additive quantifier:

- (44) a. Wen hat **wem** **niemand** t vorgestellt  
       Who has whom no one introduced  
       b. \*Wen hat **niemand** **wem** vorgestellt?  
       Who has no one whom introduced  
       “Who has no one introduced to whom?”

Setting aside the complexities of implementation, the essence of the analysis can be captured with reference to (45). Both (44a) and (44b) would have the same final interpretation (45a), which has the presupposition in (45b). Interpreting the *wh* phrases as existential quantifiers in their surface positions in (44a), we get (45b) again. Interpreting the *wh* phrases as existentials in the surface positions of (44b), however, we get the non-equivalent (45c):<sup>12</sup>

- (45) a. { $\wedge$ nobody introduced a to b,  $\wedge$ nobody introduced c to b,  $\wedge$ nobody introduced a to c, ...}  
 b.  $\exists x \exists y \neg \exists z [\text{introduce}(z, x, y)]$   
*there is some x, there is some y, such that no one introduced x to y*  
 c.  $\exists x \neg \exists z \exists y [\text{introduce}(z, x, y)]$   
*there is someone such that no one introduced him to anyone*

It would be obvious that the position of the *wh* phrases would not make a difference with additive quantifiers like *someone* or referential terms like *John*.

Mayr (2014: 526) also draws attention to a contrast within the class of upward entailing quantifiers that depends on whether the predicate is interpreted collectively or distributively. He adopts the standard view, stemming from Link (1983), that distributivity involves a covert universal operator over VP. The data in (46), due to Bernard Schwarz, provide a particularly interesting illustration:

- (46) a. Wo haben sich mehr als drei Maler wann eine Pizza geteilt?  
 Where have self more than three painters when a pizza shared  
 "Where did more than three painters share a pizza when?"  
 b. \*Wo haben sich mehr als drei Maler wann eine  
 Where have self more than three painters when a  
 Arbeitshose angezogen?  
 dungaree put on  
 "Where did more than three painters put on a dungaree when?"

World knowledge tells us that it is possible for a group to eat a pizza collectively, but a group cannot collectively wear a dungaree. The acceptability of (46a) is predicted under the collective reading since the quantifier *more than n N* is additive. Question (46b) may appear to be a counterexample to the IEG, but Mayr points out that it is actually predicted once distributivity is factored in. It is the hidden universal, not the additive quantifier, that is the culprit here.

Mayr's assumptions about how question meanings are built up differ from our baseline theory but his overall view of the relationship between questions and answers is consonant with it. I refer the reader to the original article to get a sense of how the explanation for the role of (non-)additivity relates to the baseline theory and to the theory of questions that he adopts. I will restrict myself here to

<sup>12</sup> Mayr's implementation follows Chierchia (2006, 2013).

noting that Mayr's position is similar to Tomioka's and Grohmann's insofar as it looks for a principled explanation for distinguishing between quantifiers that do and do not create intervention effects. The differential behavior of quantifiers is a challenge for Beck's approach as it entails distinguishing between quantifiers on the basis of focus sensitivity. However, it should also be mentioned that the cross-linguistic variations that have been noted in the domain of intervention effects do not appear to be easily handled by properties such as anti-topicality or anti-additivity either.

#### 8.2.4 *Section summary*

The starting point of work on intervention is the observation that a quantifier that precedes a *wh* in-situ leads to unacceptability. This unacceptability is overridden if the *wh* can be moved to the left of the quantifier. One influential proposal draws on the focus-based account of question meaning. In spite of its success in addressing the core phenomenon, there are questions that have been raised about its empirical reach. While intervention effects seem to hold across a wide range of languages, there are significant differences between languages as well. Very credible alternative accounts based on refinements or modifications of the intervention data have therefore been proposed. It is unclear at this time whether a single theory of intervention is possible, or even desirable, for capturing the many facets of this phenomenon that have been documented.

### 8.3 Polar, polar-alternative, and alternative questions

We now turn to polar questions and a re-evaluation of their semantics, prompted by a comparison with minimally different alternative questions. The first comparison is with questions that explicitly include the negation of the nucleus proposition as an alternative. The second comparison is with questions that include potentially independent propositions as alternatives. Grammatical and pragmatic differences between these three types of questions suggest that polar questions privilege the nucleus proposition, while alternative questions of both types give equal weight to the listed alternatives. The role of prosody in disambiguating interrogatives with disjunction is discussed, focusing on the semantic and pragmatic functions of different prosodic features. We also discuss constraints on possible answers to alternative questions.

#### 8.3.1 *Polar vs. polar alternative questions*

We have adopted a view of polar questions as denoting a set with the nucleus proposition and its negation. This is explicitly articulated in Hamblin (1973) but Bolinger (1978) notes antecedents going further back. The garden variety polar question (47a), based on its possible answers (47b), is taken to denote (47c). We take the *yes* response to refer to the nucleus proposition, the *no* response to its negation:

- (47) a. Does John want tea?  
 b. Yes (he does) / No (he doesn't)  
 c. { $\wedge$ john wants tea,  $\wedge$ john does not want tea}

This view predicts that a positive polar question like (47a) will be equivalent to its negative version (48a). It also predicts the equivalence of these questions with questions that make the positive and the negative alternatives explicit, as in (49a):

- (48) a. Does John not want tea? / Doesn't John want tea?  
 b. { $\wedge$ john does not want tea,  $\wedge$ john wants tea}
- (49) a. Does John want tea or not? / Does John want tea or does John not want tea?  
 b. { $\wedge$ john wants tea,  $\wedge$ john does not want tea}  $\cup$  { $\wedge$ john does not want tea,  $\wedge$ john wants tea}  $\Rightarrow$  { $\wedge$ john wants tea,  $\wedge$ john does not want tea}

It is obvious, of course, that this picture is too simplistic. Example (48a) is restricted to particular contexts of use, unlike (47a) which is neutral. Someone who is making tea can ask *do you want some tea?* of an addressee without any particular expectation about their preferences. In order to ask *do you not want tea?*, instead, the speaker must have some expectation that they may not be so inclined. This difference relates to the study of bias, which we will take up in Chapter 9. Here we will probe the relationship between positive polar questions and polar alternative questions.<sup>13</sup>

Bolinger (1978) argues against the standard analysis of polar questions because of a number of differences with polar alternative questions. He suggests instead that a polar question denotes a singleton set with only the nucleus proposition. This view has recently been revived by Gawron (2001), van Rooij and Šafářová (2003), Biezma (2009), Farkas and Bruce (2010), Biezma and Rawlins (2012), Roelofsen and Farkas (2015), among others. Though cast in different frameworks, the crucial insight can be represented in terms of denotation sets:

- (50) a.  $\llbracket 47a \rrbracket = \{\wedge \text{john wants tea}\}$   
 b.  $\llbracket 49a \rrbracket = \{\wedge \text{john wants tea}, \wedge \text{john does not want tea}\}$

There is empirical motivation for this distinction. Bolinger notes that only (51a) is a conversation starter:

- (51) a. Do you play golf?  
 b. Do you play golf or not?

<sup>13</sup> Polar alternative questions are not neutral either but their contextual restrictions are different from the contextual restrictions on negative questions.

Biezma and Rawlins (2012) adopt the view that, in fact, (51a) and (51b) have distinct semantic representations at the base, as in (50). In the first case, the negative proposition enters through coercion. Coercion can be ascribed to the requirement that felicitous questions are prompted by ignorance and must present a true choice between alternatives. Their basic premise is that alternative questions do not introduce a new question which participants agree on resolving. Rather, alternative questions are parasitic on the propositional alternatives that are already salient in the discourse, constraining the possible answers to an already established open question. In starting a conversation the dynamics of the situation establishes that there is no prior question to be resolved. A positive polar question gives the addressee maximum freedom of response:

- (52) a. { $\wedge$ you play golf,  $\wedge$ you don't play golf<sub>[IMPLICIT]}</sub>}  
       b. { $\wedge$ you play golf,  $\wedge$ you play tennis<sub>[IMPLICIT]</sub>,  $\wedge$ you play squash<sub>[IMPLICIT]</sub>...}  
       c. { $\wedge$ you play golf,  $\wedge$ you don't play golf}

While the polar question (51a) presents playing golf as a highly likely relevant alternative, other alternatives can be pressed into service. This could be the simple negative proposition (52a), or it could be alternatives naming other relevant activities (52b). Different contexts may make coercion or different alternatives more likely. In the case of the polar alternative question (51b), instead, it is required that the immediate context make exactly two alternatives salient, the ones in (52c). They are therefore not felicitous conversation starters. Rather, they have a somewhat hectoring or badgering overtone which Biezma (2009) dubs the *cornering effect*.<sup>14</sup> They invoke contexts in which the ordinary polar question has already been posed and the addressee has either ignored it or has obfuscated in answering it. In asking (51b), then, the speaker now insists on the question being answered. It is hard to explain this difference if polar questions and polar alternative questions are simple variants of each other. If they differ in their semantics, as shown in (50), there is something to hook a pragmatic explanation to.

The semantic difference reflected in (50) can also explain some hard grammatical differences. The first difference is that a polar alternative question cannot be answered with a *yes* or *no*. If response particles are anaphoric in some sense and require a unique antecedent in the question denotation, a set with a single proposition in it is appropriate but not one which provides more than one such option.<sup>15</sup> By this criterion, only polar questions qualify as appropriate antecedents for response particles.

Another difference has to do with the differential behavior of polar questions and polar alternative questions as antecedents of unconditionals (Gawron 2001;

<sup>14</sup> This effect is less evident or perhaps even missing when polar alternative questions are embedded.

<sup>15</sup> The situation with positive polar questions is descriptively simple: *yes* can only be followed by the nucleus proposition, *no* by its negation. In the case of negative polar questions, the situation is more complex both within and across languages. See Farkas and Bruce (2010), Krifka (2013) and Roelofsen and Farkas (2015).

Biezma and Rawlins 2012) and as complements of dubitative verbs (Karttunen 1977; Huddleston 1994; Biezma and Rawlins 2012). These differences can plausibly be tied to the requirements of the embedding context for a singleton or a plural alternative set (see also Chapter 5):<sup>16</sup>

- (53) a. \*Whether Bill comes we will have a party.  
b. Whether Bill comes or not we will have a party.
- (54) a. I doubt whether Bill will show up.  
b. \*I doubt whether Bill will show up or not.

To this we may also add surprise-predicates, though here the contrast seems restricted to certain modal contexts. Recall from Chapters 3 and 5 that surprise-predicates do not take polar question complements. This also holds for (polar) alternative questions. Example (55) suggests that the restriction is relaxed for polar alternative questions in modal contexts but not for polar questions:<sup>17</sup>

- (55) a. John will be surprised whether \*(or not) Bill shows up/whether Bill shows up \*(or not).  
b. \*John was surprised whether Bill showed up (or not).

Thus the revival of Bolinger's claim that polar questions differ from polar alternative questions in denoting a singleton set seems to be based on solid grounds. With this revision of the semantics of polar questions in place, let us now turn to regular alternative questions.

### 8.3.2 Prosody and disjunction

Alternative questions (Alt-Q) are dependent on the presence of disjunction but the dependence holds only in one direction. It is possible for questions with disjunction to be interpreted as simple polar questions. The crucial role of prosody in determining whether a question with disjunction is a polar question or an Alt-Q is evident in cases where the propositions at issue are logically independent: *you want tea, you want coffee*.

Consider (56a), where the brackets around the phrase *tea or coffee* are intended to indicate that it is treated as a single prosodic unit, and [/] at the end indicates a

<sup>16</sup> Note that (54a) expresses doubts about the overtly expressed positive possibility, not to the negative possibility *Bill will not show up*.

<sup>17</sup> Guerzoni and Sharvit (2014) take *whether or not p* to license NPIs but not *whether p or not*. See also Nicolae (2013). I abstract away from this distinction to focus on the more general point about embedding under surprise-predicates. Apart from the future, I think past counterfactuals may also admit polar alternative questions: *John would have been surprised whether Bill had come or not*. What these sentences seem to imply is that the source of John's surprise is independent of the answer to the question. Incidentally, the constraint seems to be relaxed for regular alternative questions as well: *John will be surprised whether [Bill]<sub>F</sub> or [Sue]<sub>F</sub> shows up*.

rising contour.<sup>18</sup> This leads to a polar question interpretation, as shown by the possibility of the answers in (56b)–(56c):

- (56) a. Do you want [tea or coffee] [/] ?  
       b. Yes, please (tea/coffee/either one/both).  
       c. No, thanks (neither).

Answering *yes* or *no* entails answering whether you want a hot beverage, restricted to the ones mentioned. A positive answer can be followed by naming one or both options or declaring indifference, answering an implicit or explicit follow-up question, *which one*?

The same question results in an Alt-Q reading if accompanied with a prosodic break between the disjuncts and pitch accents on each of them (indicated by the subscript F for focus), and a falling intonational contour [\\]:

- (57) a. Do you want [tea]<sub>F</sub> or [coffee]<sub>F</sub> [\\]?  
       b. #Yes, please/#No, thanks.  
       c. Tea/Coffee/#both/#neither.

As in the case of a polar alternative question, (57a) cannot be answered with *yes* or *no*. Furthermore, the expected answer should pick exactly one of the choices mentioned.

There are, then, three prosodic features that seem to be relevant: the treatment of the disjunction as a single/multiple prosodic unit, pitch accents on individual disjuncts, the final rise/fall. There has been some discussion about the relative importance of these different features in disambiguating between polar questions and Alt-Qs. Bartels (1997) takes both the final fall and the pitch accents on the disjuncts to be determining features. Pruitt (2008) and Pruitt and Roelofsen (2013) present experimental data in favor of a somewhat more nuanced picture. According to Pruitt, the key difference between polar questions and Alt-Qs is in the final intonation. Pruitt and Roelofsen, instead, find that contours with multiple accents and a prosodic phrase break between them are more likely to be interpreted as Alt-Q, compared to contours with a single prominent accent and no prosodic break. Their findings support the claim in Bartels that a final fall is significant but not her claim that a falling contour in the absence of accents on the disjuncts is sufficient for an Alt-Q reading. While final pitch contour is most informative, the accentual characteristics are also important, even though they cannot on their own force Alt-Q readings. Setting aside differences of detail among these authors, we see that all three prosodic features play a role in the disambiguation: prosodic units, pitch accent(s), final rise/fall.

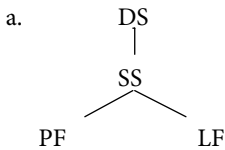
This may be a good point to articulate the formal relationship between prosody and interpretation. The Principles and Parameters model of grammar, which we

<sup>18</sup> There are more precise ways of representing these prosodic features, using for example, the notation of Pierrehumbert (1980) and Beckman and Pierrehumbert (1986).



have adopted in this book, assumes an architecture with no direct connection between the phonological component (PF) and the interpretive component (LF). This is shown in (58a). The same holds for the more current model in (58b), from Truckenbrodt (2012):

(58)



b. semantics ← LF ← s-structure/spellout → prosody/tones → (phonology) phonetics

There is, however, an indirect connection between PF and LF. Relevant features in the syntactic structure are visible for phonological and semantic interpretation, forging a systematic link between the two components. Thus the possibility of semantic and pragmatic explanations based on prosodic features is not surprising on this view of grammar.

Our discussion at various points has relied on this conception of the connection between prosody and interpretation, albeit tacitly. For example, the semantic triggering of alternatives by a focused phrase is due to a syntactic feature on it, the same feature that the PF component uses to give it prosodic prominence. Such features have also been shown to play a role in the given/new distinction and the associated partition of the clause into topic and focus.<sup>19</sup> Put in rough terms, new information is marked by prosodic prominence while old information is not. This was indirectly alluded to in our discussion of anti-topicality and intervention, for example.

The connection between prosody and interpretation is less straightforward with regard to the final fall/rise. A commonly held view is that final fall is associated with assertions and final rise with questions. While this is a good starting point, the situation is more nuanced. An example from Pierrehumbert and Hirschberg (1990: 290), also discussed by Bartels (1997) and Truckenbrodt (2012), illustrates this for the final rise:

(59) My name is Mark Liberman. [/]

It is implausible to think that the speaker is asking about his own name. More likely, (59) would be uttered in a situation in which Mark Liberman approaches a receptionist, somewhat uncertain about the details of his appointment. As the authors put it, what the final rise seems to convey is something along the lines of *am I in the right place for my appointment?* Whatever the nature of this speech act, it is not about the literal content of the utterance, but about a related proposition.

<sup>19</sup> This has also played an important role in discussions directly related to questions. See, for example, Büring (2003, 2007, 2016) and Wagner (2012) where prosody related to topic and focus in declaratives is related to explicit or implicit questions under discussion, in the sense of Roberts (1996).

The proposition achieves discourse salience by virtue of the speaker stating his identity in front of someone in charge of appointments.

The second point about final fall/rise is highlighted by the simple fact that both polar and Alt-Qs are questions requiring some kind of an answer from the addressee. As such, a division in terms of asserting and questioning is not particularly revealing. Final rise has therefore sometimes been taken to signal a continuation of some kind, final fall to signal closure. As Bartels (1997: 100) puts it, “statements bearing a ‘declarative fall’... convey a sense of self-containedness, closedness, or ‘finality’, whereas statements bearing a fall–rise... convey ‘reservations’, openendedness, or ‘continuity’.” In the case of disjunction, for example, final fall signals the exhaustive specification of alternatives without shifting the speech act in the direction of assertion. This view of the discourse contribution of the final rise–fall seems more relevant to the constructions under consideration here.

The relationship between prosody and meaning is an extremely rich area of research and a couple of short paragraphs do not even begin to scratch its surface. The impressionistic pointers given in this subsection were intended to provide just enough background to contextualize the specific analyses of Alt-Qs in Sections 8.3.3 and 8.3.4 and to set the stage for the discussion of non-canonical questions to come in Chapter 9. The interested reader is directed to the following literature where the relationship between prosody and meaning is treated with the thoroughness it deserves: Gussenhoven (1984), Pierrehumbert and Hirschberg (1990), Selkirk (1995), Bartels (1997), Schwarzschild (1999), Kadmon (2001), Büring (2007, 2016), Wagner (2012), Truckenbrodt (2012, 2013), to name just a few.

### 8.3.3 Disjunction in alternative questions

Han and Romero (2004a, 2004b) combine two accounts of disjunction in analyzing Alt-Qs, the movement account of *either/whether* in Larson (1985) and the ellipsis account of *either* in Schwarz (1999):

- (60) a. Did John drink coffee or tea?  
       b.  $Q_i (t_i \text{ or not}) [\text{did John drink } [OP_j \text{ coffee or tea}]]$   
            $\Rightarrow \{^{\wedge}\text{John drank coffee or tea}, ^{\wedge}\text{John didn't drink coffee or tea}\}$   
       c.  $Q_i \text{ Did } t_i [\text{John drink } [\text{coffee}]_F \text{ or } \text{John-drink } [\text{tea}]_F]$   
            $\Rightarrow \{^{\wedge}\text{John drank coffee}, ^{\wedge}\text{John drank tea}\}$

Disjunction is accompanied by an operator in both types of questions. If (60a) is interpreted as a polar question ((60b)), the operator  $OP_j$  has local scope over the disjunction. After  $Q$  is interpreted, we get:  $\{(p \vee q), \neg(p \vee q)\}$ . In the Alt-Q analysis in (60c), the disjunction operator moves to take clausal scope, shown as  $Q_i$ . Within TP, there is ellipsis of the non-focused part in the second disjunct. There is no Y/N operator in the picture and the question denotes the set of alternatives explicitly mentioned in the disjuncts.

Han and Romero also discuss negation in Alt-Q and note a difference based on whether negation is preposed or not. With sentence internal negation ((61a)), polar and alternative question readings are both available, modulo prosody. With

preposed negation, as in (61b), instead, only the polar question reading is available:

- (61) a. Did John not drink [coffee or tea]?/Did John not drink [coffee]<sub>F</sub> or [tea]<sub>F</sub>?  
 b. Didn't John drink [coffee or tea]?/\*Didn't John drink [coffee]<sub>F</sub> or [tea]<sub>F</sub>?

Han and Romero treat the problem with preposed negation as an intervention effect. They show on the basis of a range of facts that preposed negation in English involves *VERUM* focus, an inherently focused epistemic operator cued to the speaker's confidence in the prejacent (see Jacobs 1986b; and Höhle 1992). They explore the following structural possibilities for the Alt-Q version of (61b) and show that each of them is ruled out for different reasons (Han and Romero 2004a: 210):

- (62) a.  $Q_i$  did  $t_i$  [*VERUM*<sub>F</sub> ... *XP*<sub>F</sub>] or [*VERUM*<sub>F</sub> ... *YP*<sub>F</sub>]  
 b.  $Q_i$  did  $t_i$  [*VERUM*<sub>F</sub> ... *XP*<sub>F</sub>] or [*...*<sub>F</sub> *YP*<sub>F</sub>]  
 c.  $Q_i$  did *VERUM*<sub>F</sub> ...  $t_i$  [... *XP*<sub>F</sub>] or [*...*<sub>F</sub> *YP*<sub>F</sub>]

LF (62a) has the focused *VERUM* operator in both disjuncts but it violates the general constraint against deleting focused material at LF (63a). LF (62b), with *verum* missing in the second disjunct, violates the second clause of the Focus Condition in (63b). The final LF in (62c) has the *VERUM* operator take scope over the disjuncts. Moving  $Q_i$  overtly over it, however, creates an intervention configuration of the kind discussed in Section 8.2.2:

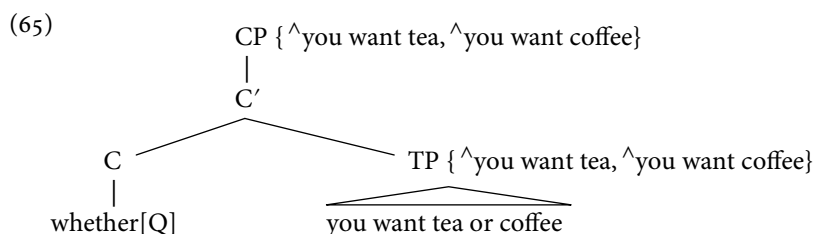
- (63) a. Focus Deletion Constraint: Focus-marked constituents at LF (or their phonological locus) cannot delete at Spell-Out.  
 (Han and Romero 2004a: 199)  
 b. Focus Condition:  $[\alpha \sim C]$  is felicitous if  $C \in \llbracket \alpha \rrbracket^f$  or  $C$  implies a member of  $\llbracket \alpha \rrbracket^f$ .  
 (Han and Romero 2004a: 194)

Beck and Kim (2006) extend the discussion of intervention effects in alternative questions by including focus sensitive operators, as in (64a). Li and Law (2016) show that their modification of the focus-based generalization about intervention also applies here, as shown by (64b):

- (64) a. ?\*Did Peter only give [Mary]<sub>F</sub> [a book]<sub>F</sub> or [a pen]<sub>F</sub>?  
 b. Did Peter only give Mary [a book]<sub>F</sub> or [a pen]<sub>F</sub>?

Let us end with one other account, conducted within the alternative semantic approach of Kratzer and Shimoyama (2006). Biezma and Rawlins (2012) follow Simons (2005) and Alonso-Ovalle (2006) in treating disjunction as the union of the sets denoted by the individual disjuncts:  $\llbracket X \text{ or } Y \rrbracket = \llbracket X \rrbracket \cup \llbracket Y \rrbracket$ . Given the semantics of the *Q* operator in  $C^0$ , we get the set of the alternatives denoted by the disjunction as the meaning of the question. They follow Zimmermann (2000) and

Biezma (2009) in positing a closure operation that captures the presupposition that the stated alternatives include all and only the alternatives that the question denotes:



Alt-Qs have been extensively studied in recent literature but instead of introducing further accounts, we will turn to the final piece of the puzzle about such questions, namely the restrictions on possible answers.

#### 8.3.4 Choosing one, choosing both, choosing none

As previously noted, Alt-Qs do not admit yes/no answers. This is consistent with the view that such responses require singleton propositional sets, that only polar questions denote:

- (66) a. Do you want [tea]<sub>F</sub> or [coffee]<sub>F</sub>?  
 b. Tea, please/Coffee, please.  
 c. #Both/#neither.

It is generally accepted that exactly one of the alternatives should be named in the answer. Thus (66b) is fine but (66c) is not and this requires some discussion.

The problem of appropriately restricting answers to Alt-Qs goes back to Karttunen (1977) who treated them essentially as the union of two polar questions. On the view that a polar question denotes the nucleus proposition and its complement, however, an Alt-Q should denote four propositions, any one of which should be acceptable as an answer. As Karttunen himself noted, this incorrectly lets in the two unacceptable answers *both* and *neither*. The same holds of Groenendijk and Stokhof (1984). More recent accounts of Alt-Qs fall into two broad categories. One allows the question to denote non-alternatives (*both/neither*) and uses pragmatics to rule them out, the other blocks non-alternatives from entering into the question denotation. The first is the line taken by Groenendijk and Roelofsen (2009) and Roelofsen and van Gool (2010), within the framework of inquisitive semantics discussed in Section 8.1.3. The second is the line taken by Biezma and Rawlins (2012).

Groenendijk and Roelofsen (2009) capture the restrictions through pragmatic constraints, which they dub *homogeneity* and *compliance*. *Homogeneity* is supposed to rule out the answer *neither* to (66a). They propose that updates are more

homogeneous if they are more informative and/or less inquisitive. Alt-Qs, compared to polar questions, come out less homogeneous on their account. Since polar questions allow the *neither* answer directly, they are more homogeneous and to be preferred. *Compliance* rules out the possibility of *both* as an answer to (66a). A response is “an optimally compliant response just in case it picks out exactly one of the alternatives proposed by [the question].” By leaving out the option of *both* in the question, the hearer infers that the speaker purposefully excluded the non-compliant response, and therefore the question implicates this proposition to be false. There are many fine points to explore here but the most relevant for present purposes is that the restriction is derived through pragmatic inferences in relation to polar questions.

Roelofsen and van Gool (2010) take a somewhat different line. They maintain the view that Alt-Qs involve the union of polar questions, and therefore include the possibility of non-alternatives as possible answers. However, along with the normal meaning of the question, they admit a second meaning which consists of the propositions a question highlights:  $\llbracket \cdot \rrbracket_H$ . Now, (66a) may denote the four propositions  $\{\wedge \text{you want tea}, \wedge \text{you want coffee}, \wedge \text{you want tea and coffee}, \wedge \text{you want neither tea nor coffee}\}$  but it only highlights  $\{\wedge \text{you want tea}, \wedge \text{you want coffee}\}$ . They assume that focus is critical in effecting this highlighting. In order to block the possibility that both alternatives are included in the answer, they admit a third aspect to the meaning:  $\llbracket \alpha \rrbracket_S$  which is associated with an *Exclusive Strengthening* operator  $\in \chi$  that removes the overlap between the two propositions given explicitly.

Biezma and Rawlins (2012) address the problem by positing the following:

- (67) Constraint on salient propositional alternative sets in a context  $c$
- (i)  $\forall w \in cs_c : \exists p \in \text{SalientAlts}(c) : p(w) = 1$  *Exhaustivity*
  - (ii)  $\forall w \in cs_c : \forall p, q \in \text{SalientAlts}(c) : (p = q \vee \neg(p(w) \wedge q(w)))$  *Mutual Exclusivity*

The first clause ensures that at least one alternative is true and the second clause ensures that at most one is true.

Let us see whether restrictions of the type discussed above are really necessary. It seems to me that if the set of alternatives can be restricted to the ones explicitly given, we already have at our disposal the means to derive the right set of answers. Let us accept that the negative proposition enters the denotation of a polar question to ensure that a question does not denote a singleton set. Alt-Qs are never singletons, so there is no pressure to introduce more propositions into the set. That is, what you hear is what you get. Now, consider the results of applying the answerhood operator of our baseline theory to Alt-Qs. We take underlining, as always, to indicate propositions that are true in the actual world:

- (68) a. Do you want  $[tea]_F$  or  $[coffee]_F$  ?  
 b.  $\{\wedge \text{you want tea}, \wedge \text{you want coffee}\}$   
 c. Ans-D (68b) =  $\wedge \text{you want tea}$

The existence and uniqueness presuppositions of Ans-D rules out *neither* and *both* as direct answers without any further stipulations. Furthermore, it applies to cases where these answers are possible, in an entirely straightforward way:<sup>20</sup>

- (69) a. Do you want [tea]<sub>F</sub>, [coffee]<sub>F</sub> or [both]<sub>F</sub>?  
 b. {^you want tea, ^you want coffee, ^you want tea and coffee}  
 c. Ans-D (69b) = ^you want tea and coffee

In a world where the addressee wants tea as well as coffee, Ans-D correctly delivers *both* as the maximally informative true answer. And, extending even further, it also allows the possibility of *neither*, as long as it is one of the explicitly mentioned options:

- (70) a. Do you want [tea]<sub>F</sub>, [coffee]<sub>F</sub>, [both]<sub>F</sub> or [neither]<sub>F</sub>?  
 b. {^you want tea, ^you want coffee, ^you want tea and coffee,  
^you don't want tea or coffee}  
 c. Ans-D (70b) = ^you don't want tea or coffee

We can capture, without introducing additional principles, the presuppositional account of answers to Alt-Qs that Biezma and Rawlins endorse, following Belnap and Steel (1976), Karttunen and Peters (1976), Higginbotham (1991), and Bartels (1997).

### 8.3.5 Section summary

We looked at properties of disjunctive questions which can either have polar question interpretations or Alt-Q interpretations, depending on prosody. Comparing polar questions to polar alternative questions led to the conclusion that a polar question differs from all other questions in privileging its nucleus proposition. We also took a close look at Alt-Qs, and considered proposals for connecting their prosodic features with their meaning. We considered two accounts of Alt-Q, one based on movement and ellipsis, one based on the union of alternatives generated by disjunction. Finally, we addressed the restriction of possible answers to the stated alternatives. We suggested that this could follow from the existence and uniqueness presuppositions built into Ans-D, also accounting for Alt-Qs that make explicit that neither or both disjuncts may hold in the context of utterance.

## 8.4 Focusing on alternatives

Here we looked at morphological ties between *wh* expressions, indefinites, and focus. These relationships were found to hold cross-linguistically. The connection

<sup>20</sup> Biezma and Rawlins (2012: 388–9) note that examples like (69a) are not tractable, given (67), and leave it as an open problem.

with focus makes the possibility of a common core to a semantics for questions and a semantics for focus appealing. A Hamblin-style semantics of questions seems to mirror this parallelism best. We looked at two proposals in the literature that follow this general approach. The substantive difference from our baseline theory is not in the final output, on which they coincide, but in the way of composing that meaning. Crucially, a *wh* expression on this view does not include an existentially bound variable at any point in the derivation. It is an alternative generator like other focused phrases. The role of the complementizer too is substantively different.  $C^0_{+WH}$  does not necessitate the shift from a declarative/proposition level meaning to an interrogative/set of propositions level meaning. Rather, it is  $C^0_{-WH}$  that is responsible for a shift in types, from an interrogative/set of propositions level meaning to a declarative/proposition level meaning in cases where the sentence includes alternative generators (indefinites and disjunction). We also saw the applicability of this approach to island phenomena, with special reference to Japanese. The proposals discussed here account for the fact that Japanese indeterminate pronouns must be bound by the closest operator. Among the challenges that remain to be addressed is the possibility of long-distance list answers in a *wh* triangle configuration. Such answers crucially require a *wh* to escape the closest operator to get bound by a higher operator at the site of the island or by an operator in the immediately higher clause.

In connecting indefinites and interrogatives we also introduced the framework of inquisitive semantics, which provides a logic for modeling the conversational exchange potentials of declaratives as assertions and interrogatives as questions. Its primary innovation is in allowing declaratives with indefinites or disjunction to cross-classify. Such statements are like assertions, in view of the fact that they eliminate some worlds from the context set. And they are like questions in raising more than one possibility for the addressee to choose from. Thus the canonical tie of the speech act of questioning with a particular syntactic form is relaxed. As far as the meaning of interrogatives is concerned, it does not substantively alter our basic assumptions about their role in discourse.

The second empirical focus of this chapter was the phenomenon of intervention, where the presence of certain operators on the path between a *wh* expression and  $C^0_{+WH}$  results in ungrammaticality. Moving that *wh* expression overtly to the left via scrambling or *wh* movement, avoids the problem. Among the leading explanations of intervention effects is one based on the focus semantics for questions presented in Section 8.1. Under this view the *wh* expression has a focus semantic value but no ordinary semantic value. A  $C^0_{+WH}$  takes the focus semantic value of its sister and turns it into the ordinary semantic value, the meaning of the question. A focus sensitive operator creates a problem because its preadjacent needs to have an ordinary semantic value. Furthermore, the alternatives do not remain available for the  $C^0_{+WH}$  to operate on. Alternative accounts of intervention effects were also discussed, each of which showed the relevance of factors other than the ones identified by the focus-based explanation, highlighting and enriching the empirical terrain that intervention applies to.

The last section of this chapter was devoted to an understanding of polar and alternative questions. We saw arguments against the standard view of polar

questions as denoting a set of two equally weighted propositions. The current understanding is that a polar question privileges the nucleus proposition in some way. This is the property of polar questions, as opposed to Alt-Qs, which makes them amenable to a yes/no response. The negative proposition enters because a proper question cannot denote a singleton set. This repair also allows other alternatives in appropriate contexts. In this respect, polar questions are minimally different from polar alternative questions which make explicit the affirmative and negative versions of a proposition as the only alternatives under discussion. Finally, we looked at regular Alt-Qs and noted their special features, both prosodic and syntactic. Among the various aspects we considered was the fact that they seem to admit answers that choose only one out of the set of alternatives listed in the question. We saw that under a controlled question denotation, the set of possible direct answers can be obtained by applying Ans-D of our baseline theory.

This chapter ends our foray into the core aspects of the semantics and pragmatics of questions, that is, of interrogative syntactic structures used to elicit information. We will conclude our survey by considering in the next chapter some cases that depart from this canonical tie between the syntax of interrogatives and the pragmatics of questioning.



## Non-canonical questions

Our working hypothesis so far has been that questions are interrogatives used to elicit information. In this final chapter we will expand the discussion to include questions that depart from the norm in at least one of the following ways: (i) they elicit information but also convey a bias about what that information is likely to be; (ii) their syntactic form does not conform to the standard interrogative form; (iii) their purpose is to engage in some other kind of speech act instead of, or in addition to, eliciting information. We will look at several types of questions that can be classified as non-canonical using these criteria.

Negative polar questions, as noted in Chapter 8, differ from positive polar questions in conveying a bias. Bias also characterizes declarative questions, questions that have the syntax of assertions but the intonation of questions. Echo questions are included for their disregard of syntactic constraints on movement as well as for their special relation to prior discourse. Rhetorical, ability, and inclination questions are those that would be unremarkable but for the fact that they have the force of assertions or directives, under certain discourse conditions. Finally, tag questions have a dual life, combining declarative and interrogative components in a single locution, calling for an analysis that gives this complexity its due.

I have listed a rather diverse set of constructions as falling under the rubric of non-canonical questions and it is worth keeping in mind that the list is not exhaustive. In spite of their diversity, there are at least two properties these questions share. One, their primary semantic contribution can often be expressed in an arguably simpler alternative form. This suggests that the speaker, in choosing the non-canonical version, intends to convey something beyond the normal semantic contribution by the very choice of that form. Two, they generally resist embedding, at least under responsive predicates. To demonstrate the first point, compare the options for positive polar questions in English and Hindi:

- (1) a. Is John at home?  
b. John is at home?
- (2) anu ghar meN hai?  
Anu home in is  
“Is Anu at home?”

A declarative like (1b) uttered with question intonation conveys a bias that the canonical interrogative form in (1a) does not. Hindi does not distinguish

syntactically between declaratives and interrogatives, using intonation instead to signal the shift from assertions to questions.<sup>1</sup> Example (2) has the same syntactic and prosodic features as (1b) but conveys no bias. Turning to the second point, the unembeddability of most of the constructions listed in the previous paragraph has been noted in the literature but I am not aware if the claim has been made for negative polar questions. They seem to line up with other non-canonical questions in this regard:

- (3) a. John knows whether Mary likes Bill / ?? whether Mary doesn't like Bill.
- b. John discovered whether Mary left / ?? whether Mary didn't leave.

The negative versions improve if the responsive predicate is negated or if the matrix clause has a polar interpretation.<sup>2</sup> This clustering of effects is characteristic of root phenomena and was noted for embedded questions with inversion in Chapter 5. These features provide an underlying unity among the various question types but will not feature prominently in the discussion to follow.

The larger point about non-canonical questions is that they require enhancement of the formal system we have been working with. As laid out in Section 8.1.3, the model of discourse dynamics that is standardly assumed derives from Stalnaker (1972, 1978) and Karttunen (1974). Conversational exchange is treated in terms of incremental additions to the common ground, the information mutually agreed upon by the participants in a discourse.<sup>3</sup> An assertion updates the common ground by eliminating those worlds from the context set (the set of worlds in the intersection of the common ground propositions) where the asserted proposition is false. Of course, this update only goes through if the proposition is accepted by the addressee. This means that the utterance of a declarative sentence does not, in and of itself, update the common ground but is rather a proposal to do so. A question, instead, is a request for an update from the addressee. The speaker declares uncertainty about the status of the nucleus proposition in the actual world. If the speaker's belief that the addressee has the relevant information is validated, the addressee's answer updates the common ground.

There are several different elaborations of this general perspective. Here I will give a very brief sketch of one that connects explicitly with our syntactic and semantic assumptions.<sup>4</sup> Krifka (2014), building on Szabolcsi (1982), posits a speech act operator in ForceP of root clauses (Rizzi 1997):

- (4) a. [<sub>ForceP</sub> ASSERT [<sub>TP</sub> it is raining]]
- b. ASSERT (i)(p)(y)(x)

<sup>1</sup> Hindi has a particle *kyaa* 'what' that can be used in polar questions but it is not obligatory. Some of its properties are discussed in Bhatt and Dayal (2014).

<sup>2</sup> Thanks to Simon Charlow (p.c.) for discussion. See Heycock (2006) for a survey of root phenomena.

<sup>3</sup> See Chierchia and McConnell-Ginet (2000) for an introduction to dynamic semantics.

<sup>4</sup> See also Carlson (1983), Roberts (1996), Asher and Lascarides (2003), Gunlogson (2003), Büring (2007), Farkas and Bruce (2010), and Roelofsen and Farkas (2015), among others.

The proposal in (4a) is reminiscent of the performative hypothesis of Katz and Postal (1964) but the accompanying semantics is quite different. The performative hypothesis would interpret (4a) as *the speaker asserts that it is raining*, a statement that would be true regardless of the weather conditions. Krifka makes a distinction between denotational meaning, located at TP, and illocutionary meaning, located at ForceP. The denotational meaning of TP is the familiar object, a set of worlds. The illocutionary meaning of ForceP, (4b), involves a public declaration by speaker  $x$ , to addressee  $y$ , at index  $i$  that he/she is committed to the truth of  $p$ . The sentence conveys that at the time immediately prior to  $i$ ,  $x$ 's commitment to  $p$  is not public. A question, on this view, has the speech act operator QUEST in ForceP and takes a CP complement with the standard question denotation, a set of propositions. QUEST also effects a change in public commitments, with the speaker putting the addressee under obligation to make assertive commitments regarding the answer to the question. While the denotational meaning is at issue and can be directly addressed, the illocutionary meaning is not at issue and thus cannot be responded to. By separating denotational and illocutionary meanings, Krifka is also able to address restrictions on embedding speech acts, which we have seen is quite relevant to non-canonical questions. This is an admittedly cursory presentation, whose only purpose was to show that an explicit theory of speech acts can be incorporated into the theoretical framework adopted here.

With this background, let us turn now to specific properties of non-canonical questions. My goal here is modest. I would like to show that such questions are amenable to formal analysis and, furthermore, that such analyses build on insights from the study of canonical questions. Unfortunately, space prevents me from presenting full accounts of these question types.

## 9.1 Negation and bias

We pick up on the differences noted in Section 8.3 between positive and negative polar questions and consider the data in greater depth. We look at concrete proposals that have been made for modeling the bias attached to negative polar questions. We also extend the discussion to negative polarity items and explore how their presence leads to bias.

### 9.1.1 Bias in polar questions

The standard semantics of questions assigns the same denotation to affirmative and negative polar questions. In the affirmative case, we have  $\{p, \neg p\}$ . In the negative case, we have  $\{\neg p, \neg\neg p\}$ , which is logically equivalent to  $\{\neg p, p\}$ .<sup>5</sup>

<sup>5</sup> The equivalence holds in classical logic but not in inquisitive semantics (Groenendijk and Roelofsen 2009).

Empirically, this is supported by the fact that they both admit the same possible answers:<sup>6</sup>

- (5) a. Is it raining? / Isn't it raining? / Is it not raining?  
b. Yes, it is (raining) / No, it isn't (raining).

Going beyond possible answers and considering the contexts in which they can be asked, we find that positive and negative polar questions are not interchangeable. Reese and Asher (2010) discuss (6), adapted from Gunlogson (2003):

- (6) a. I have no beliefs on the matter. I just want to know...  
b. Are you / #Aren't you a member of the Communist Party?

Only the positive version in (6b) can be asked in the neutral, out-of-the-blue context that (6a) sets up. One might argue that this is a general feature of negation:

- (7) a. Our first contestant is left-handed.  
b. Our first contestant is not right-handed.

An introduction like (7b) is infelicitous unless there is a presumption of right-handedness. Negation may thus generally impose some sort of givenness on its prejacent.<sup>7</sup> In the case of interrogatives, there has been considerable discussion about bias but it may be worthwhile to relate the bias specific to questions to this feature of negation generally.

Regarding the bias specific to negative questions, consider a context, based on Romero and Han (2004), in which A and B are planning a party. A has the guest list and B the list of people's dietary restrictions. The contrast in (8) is expected if negative questions are necessarily biased:

- (8) A: Let's start with John. Does he eat meat? / #Doesn't he eat?

Now consider the negative question after A and B have gone through the dietary preferences of a few guests. It is acceptable in (9), assuming B has answered in the negative for the last few people. And it is acceptable in (10), if B has answered in the positive for some people. That is, (9) and (10) are set in contexts where the requirement of bias is met:

- (9) A: What about Bill? Doesn't he eat meat *either*?  
B: You guessed correctly—he doesn't.
- (10) A: What about Bill? Doesn't he eat meat *too*?  
B: You're right—he does.

<sup>6</sup> There is considerable cross-linguistic variation in possible responses to negative questions (Krifka 2013; Roelofsen and Farkas 2015).

<sup>7</sup> See Horn (1989) for a historical perspective on the difference between affirmation and negation.

We see that bias is intrinsic to negative questions, but the bias can be to negative answers (9) or to positive answers (10). Particles like *either* and *too*, a diagnostic originating with Ladd (1981), disambiguate between these possibilities. It is worth noting that the speaker may have an expectation about what the answer may be, but these questions remain information seeking questions.<sup>8</sup>

While negative polar questions are the poster child for demonstrating bias, positive polar questions are not completely innocent either. Buring and Gunlogson (2000) present questions in the context of “evidence that has just become mutually available to the participants in the current discourse situation”. *Compelling contextual evidence* excludes the private beliefs of the participants as well as knowledge that is already in the common ground, whether it was introduced linguistically or non-linguistically. If A enters S’s windowless computer room wearing a dripping wet raincoat, it provides compelling contextual evidence for *it is raining*. They note that S’s response is restricted:

- (11) S: What’s the weather like out there? Is it raining?  
       S: #What’s the weather like out there? Is it sunny?

It would appear that positive polar questions are acceptable if there is no evidence with regard to *p*, or if there is evidence for *p*, but not if there is evidence for  $\neg p$ .

### 9.1.2 Inner vs. outer negation

Ladd (1981) provides scenarios to separate out two readings of negative polar questions:

- (12) Context: Bob is visiting Kathleen and Jeff in Chicago while attending the CLS.  
       Bob: I’d like to take you guys out to dinner while I’m here—we’d have time to go somewhere around here before evening session tonight, don’t you think?  
       Kathleen: I guess, but there’s not really any place to go in Hyde Park.  
       Bob: Oh, really, *isn’t there a vegetarian restaurant around here?*  
       Kathleen: No, all we can get is hamburgers and souvlaki.
- (13) Context: Kathleen and Jeff have just come from Chicago on the Greyhound bus to visit Bob in Ithaca.  
       Bob: You guys must be starving. You want to get something to eat?  
       Kathleen: Yeah, *isn’t there a vegetarian restaurant around here?* Moosewood, or something like that?  
       Bob: Gee, you’ve heard about Moosewood all the way out in Chicago, huh?  
       OK, let’s go there.

<sup>8</sup> If A’s expectations are not met, B would have to signal that, maybe with something like *well, actually he does*, for (9).

Ladd gives the following generalization based on (12): Inner negative polar questions are used when "...the speaker has just inferred a proposition  $\neg p$ ... so what is being questioned is the inference  $\neg p$ " (Ladd 1981: 165). He characterizes Kathleen's use of the same negative question in (13) as involving outer negation, "...used to ask for confirmation of something she believes to be true... what is being questioned is the speaker's belief  $p$ " (Ladd 1981: 164–5). The terminology *inner* vs. *outer* refers to the relative position of negation with respect to the question nucleus.

Büring and Gunlogson (2000) probe Ladd's distinction further, providing cross-linguistic morphological evidence in its favor. They pitch the distinction in terms of evidence conditions. *Inner negation* is infelicitous if there is no compelling evidence *for or against*  $p$ , or if there is compelling evidence *for*  $p$ . That is, there has to be compelling evidence *against*  $p$ . The evidence condition for *outer negation*, instead, is that there be *no compelling evidence for*  $p$  (there may be no evidence regarding  $p$ , or there may be evidence *against*  $p$ ). That is, outer negation is claimed to be acceptable in neutral contexts:

- (14) Context: A and S want to go out for dinner. S has been to Moosewood a couple of years back.  
 A: Where do you want to go for dinner? (neutral with respect to  $p$ )  
 S: Isn't there some vegetarian restaurant around here?

Note that what Büring and Gunlogson are calling a neutral context still requires some prior backgrounded belief on the part of S regarding  $p$ . In terms of the distinction made in Section 9.1.1, the negative question in (14) does not have a bias that the answer will be negative, but it is not entirely neutral. In fact, it seems to have a bias towards a positive answer.

Romero and Han (2004) systematically distinguish between preposed and sentential negation with respect to bias:

- (15) a. Isn't Jane coming?  
 b. Is Jane not coming?

They consider (15a) a case of *VERUM* focus (see also Section 8.3). *VERUM* holds of a proposition  $p$ , if the speaker is certain that in all of the worlds in which her conversational goals are met,  $p$  is part of the common ground. Romero and Han gloss this as *FOR-SURE-CGX*  $p$ , with  $x$  indexed to the speaker (Romero and Han 2004: 627). The following demonstrates its workings (Romero and Han 2004: 635–7):

- (16) a.  $[[[_{CP} Q \text{ not } [_{IP} \text{ Jane is coming}]]]] =$   
           {FOR-SURE-CGX Jane is coming,  $\neg$  FOR-SURE-CGX Jane is coming}  
 b.  $[[[_{CP} Q \text{ VERUM } [\text{not } [_{IP} \text{ Jane is coming}]]]] =$   
           {FOR-SURE-CGX  $\neg$  Jane is coming,  $\neg$  FOR-SURE-CGX  $\neg$  Jane is coming}

The outer–inner ambiguity of Ladd's account is due to scope interaction between negation and *VERUM*. Romero and Han derive the backgrounded speaker attitude

associated with negative questions as a conversational implicature. Negative questions are for them meta-conversational moves by definition, since VERUM embeds the modal operator  $\text{CONV}_x$ . Assuming that discourse participants are cooperative, it follows either that the speaker holds some belief  $p$ , which is inconsistent with the addressee's beliefs, or believes  $p$  but lacks sufficient evidence to assert it. The speaker uses this form in order to avoid violating the Maxim of Quality (Grice 1975).

Reese and Asher (2010) argue that an analysis in terms of VERUM has the unwanted consequence that an answer to (16a) amounts to saying: *it is for sure/it is not for sure that Jane is coming*. Whereas the intuition is that answers state the stronger propositions: *Jane is/is not coming*. The results are even more problematic for inner negation. A *yes* answer should mean *Jane is coming*, not *for sure Jane is not coming* and a *no* answer should mean *Jane is not coming*, not *it is not for sure that Jane is not coming* (essentially, *Jane is coming*). But, of course, these objections may be overcome by appropriately distinguishing the conversational contributions of the different components of meaning.

Reese and Asher treat biased questions as combining two speech acts. The biased proposition is a (suitably modalized) assertion which is also a request to the addressee for a response. In analyzing negative questions, then, they take their cue from tag questions:

- (17) a. Jane isn't coming, is she?  
b. Jane is coming, isn't she?

The declarative part is the assertion, while the interrogative part is the question which asks for a response about the asserted proposition. We will discuss this further in Section 9.3.3.

### 9.1.3 Strong NPI and bias

Borkin (1971) noted that questions with strong negative polarity items (18a), unlike those with weak polarity items (18b), are heavily biased towards negative answers:<sup>9</sup>

- (18) a. Did Larry have a drop of whiskey?  
b. Did Larry drink any whiskey?

Further support comes from neutral contexts where only weak polarity items are felicitous:

<sup>9</sup> Note, though, that heavily focusing the weak NPI *any* in (18b) erases the contrast.

- (19) Context: I am trying to buy coffee at a vending machine that takes only coins. I need just one more penny to get my coffee. Bill comes by and I ask him:
- a. #Do you have a red cent?
  - b. Do you have any pennies?

Krifka (1995), following Fauconnier (1975), takes strong NPIs to denote a minimal element on a scale and to introduce a set of alternatives. The negative bias of questions with such minimizers arises as a negative implicature. To illustrate, the speaker is aware that the alternatives to *Larry drank a minimal amount of whiskey*, namely, *Larry drank a shot of whiskey*, *Larry drank several shots of whiskey*, etc., are relevant in (18a). If the question is still unsettled whether Larry drank whiskey or not, then the answers to the alternatives must be negative. The speaker has, then, a pretty strong bias that Larry did not drink any whiskey. They ask the question with a minimizer, giving the nucleus proposition the highest probability of being true, as a way of settling the question.

Van Rooij (2003) considers this issue from the perspective of a general conversational strategy where speakers aim for questions whose answers have roughly equal informativity. He uses the notion of entropy to model it. The idea in brief is that the use of minimizers is a signal about the distribution of the speaker's credence, adding some weight to the positive answer even in contexts that are biased against them, in order to make the question possible.

Guerzoni (2004) follows Heim (1984) in treating questions with minimizers as having a silent *even*. She notes that negative bias correlates with the low endpoint on a scale. When the focused phrase does not indicate which end of the scale it occupies, we get ambiguity:

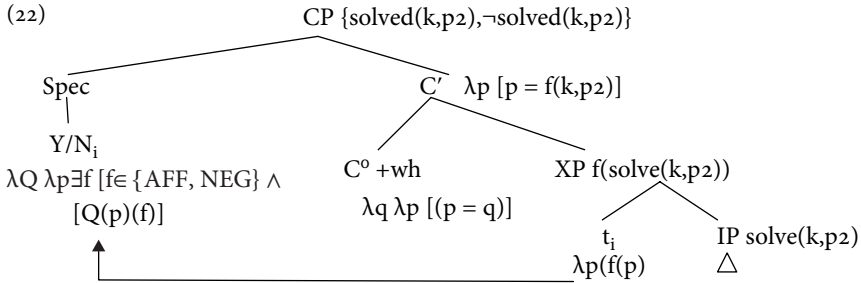
- |      |    |                                                     |                      |
|------|----|-----------------------------------------------------|----------------------|
| (20) | a. | Can Kim <i>even</i> [add 1 + 1]?                    | <i>Negative bias</i> |
|      | b. | Can Kim <i>even</i> [solve this difficult problem]? | <i>Neutral</i>       |
|      | c. | Can Kim <i>even</i> [solve problem 2]?              | <i>Ambiguous</i>     |

Guerzoni's account of this paradigm is based on the view that *even* is a scalar item that can interact scopally with other operators (see Guerzoni (2003: 51–115) for references):

- (21) Even(p) Asserts: p  
 Presupposes: (1)  $\forall q [q \in C \ \& \ \neg p \neq q] \rightarrow q$  more likely than p  
 (2)  $\exists q [q \in C \ \& \ q \text{ is true}]$

*Even* is truth conditionally vacuous but it presupposes that the prejacent must be the least likely alternative in C and that at least one of the members of C is true. Guerzoni's account of negative bias also involves a yes/no operator originating below  $C^0$  but taking scope in Spec CP like regular wh expressions. The yes/no operator quantifies over choice functions that range over the two polarities, *affirmative*, *negative*:





Putting the two together, Guerzoni derives four possible LFs through scope interaction between a silent *even* and the *wh* operator:

- (23) a. LF 1: [even [<sub>t<sub>whether</sub></sub> [Kim solved p<sub>2</sub>]]]  
           a'. Yes answer: even(AFF (Kim solved p<sub>2</sub>))                      *hard presupp*  
           a''. No answer: even(NEG (Kim didn't solve p<sub>2</sub>))            *easy presupp*
- b. LF 2: [<sub>t<sub>whether</sub></sub> [even [Kim solved p<sub>2</sub>]]]  
           b'. Yes answer: AFF (even(Kim solved p<sub>2</sub>))                      *hard presupp*  
           b''. No answer: NEG (even(Kim solved p<sub>2</sub>))                      *hard presupp*

Of the four possible answers, three of them derive from an LF in which *even* has scope over the affirmative proposition. This requires the proposition denoted by the nucleus to be the least likely member of the alternative set, that is, it must be a hard problem. When *even* has scope over the negative proposition, however, the situation is reversed. The presupposition is that the problem must be the easiest. These effects can be explained with reference to the following contexts:

- (24) a. Context 1: p<sub>2</sub> is the hardest problem.  
           The affirmative answer is supported by LF 1 (23a') as well as LF 2 (23b').  
           The negative answer is supported by LF 2 (23b''). In each case, the LF presuppositions align with the context. Hence the perceived neutrality.
- b. Context 2: p<sub>2</sub> is the easiest problem.  
           The negative answer is supported by LF 1 (23a''). The presuppositions of the two LFs that could yield an affirmative answer, (23a') and (23b'), do not align with the context. That is, the question can only be answered felicitously and truthfully with the negative answer. Hence the perceived negative bias.
- c. Context 3: p<sub>2</sub> is neither the hardest nor the easiest.  
           The question is infelicitous. The context does not support the presupposition of any LF since p<sub>2</sub> does not fall at either end of the scale of difficulty.

The negative bias of questions with minimizers, expressions that denote the lowest point on the scale, can be extrapolated from Context 2. The bias obligatorily

associated with strong NPIs is a function of the interaction of different components of meaning, each of which is independently motivated.

#### 9.1.4 Section summary

This section probed the notion of bias, focusing on negation and negative polarity items in polar questions. The kinds of evidence that shape our intuitions were considered and fairly sophisticated accounts to deal with this phenomenon were presented. One key idea is that bias arises as a result of negation or strong NPIs interacting scopally with silent focus operators: *VERUM* or *even*. The work reported here predates the revival of the idea that positive and negative polar questions may not denote the same set of propositions (see Chapter 8). The particulars of these accounts would have to be reworked if positive and negative polar questions are taken to denote sets that privilege the nucleus proposition. Note that we did not discuss *wh* questions with strong NPIs. They will come up in our discussion of rhetorical questions.

## 9.2 Non-canonical interrogation

This section discusses declarative and echo questions. Declarative questions align with negative polar questions in conveying bias. They also align with echo questions with respect to prior discourse. This has sometimes led to a blurring of boundaries between them. We will consider each in turn, and in doing so, we will see how they relate to each other and to the semantics and pragmatics of canonical questions.

### 9.2.1 Declarative questions

Declarative questions, sometimes called rising declaratives, have been studied extensively by Gunlogson (2003), the primary source for the discussion here. The basic paradigm can be illustrated using formal features of syntax and prosody. The interrogative–declarative distinction refers to the presence/absence of inversion while the rise–fall distinction refers to the final intonational contour. Rising intonation is represented with a question mark and falling intonation with a period, following Gunlogson. They correspond to [/?] and [/\] respectively, the convention used in Chapter 8 (see Section 8.3.2 for relevant background):

- |      |                   |                                   |
|------|-------------------|-----------------------------------|
| (25) | a. Is it raining? | <i>Rising polar interrogative</i> |
|      | b. It is raining? | <i>Rising declarative</i>         |
|      | c. It is raining. | <i>Falling declarative</i>        |

Gunlogson establishes that rising declaratives align with falling declaratives in several respects. Both are unacceptable in neutral contexts, for example:

- (26) It's an open question.
- a. Did she lie to the grand jury?
  - b. #She lied to the grand jury?
  - c. #She lied to the grand jury.
- (27) a. Will the incumbent win re-election? It could go either way.
- b. #The incumbent will win re-election? It could go either way.
  - c. #The incumbent will win re-election. It could go either way.

Both are acceptable with bias markers that Huddleston (1994) showed are incompatible with interrogatives:

- (28) a. #Has the manager *of course* been informed?
- b. The manager has *of course* been informed?
  - c. The manager has *of course* been informed.
- (29) a. #Did they *no doubt* misunderstand her intentions?
- b. They *no doubt* misunderstood her intentions?
  - c. They *no doubt* misunderstood her intentions.

However, there are also diagnostics aligning rising declaratives with interrogatives:

- (30) Context: A and B are looking at a co-worker's much-dented car.  
 A: His driving has gotten a lot better.  
 B's response:
- a. Has it? I don't see much evidence of that.
  - b. It has? I don't see much evidence of that.
  - c. It has. #I don't see much evidence of that.
- (31) A: The king of France is bald.  
 B's response:
- a. Is France a monarchy?
  - b. France is a monarchy?
  - c. #France is a monarchy.

Gunlogson explains this distributional pattern by separating the contributions of form and intonation. The model of discourse dynamics she develops distinguishes between the public commitments of individual discourse participants from their shared commitments. Crucially, declaratives are never neutral, in the sense that they require public commitment. Typically, it is the speaker who makes the commitment, leading to a speech act of assertion. The interpretive import of a final rise signals lack of commitment on the part of the speaker. A declarative uttered with rising intonation then meets a potential impasse: the syntax requires commitment, the intonation rules out the speaker as a possible candidate. The impasse is resolved in contexts where the speaker has evidence of the addressee's

public commitment to the nucleus proposition.<sup>10</sup> There is, however, a third component to the explanation, namely context. It must provide evidence of the addressee's commitment to the proposition. To see the relevance of context consider the following non-biased cases of rising declaratives, commonly known as *uptalk*:

- (32) a. I'm Mark Liberman?  
b. I'm Carl? I'll be your waiter tonight?

Unlike the earlier examples, the speaker here has to be fully aware of the answer and must know that the addressee cannot possibly commit to it. These are also contexts in which standard interrogatives are infelicitous. While these cases are independently interesting, their importance here is in establishing the collusion of factors needed for bias to arise.<sup>11</sup>

Gunlogson's account of declarative questions builds on earlier intonational accounts such as Bartels (1997), Steedman (2000), and Pierrehumbert and Hirshberg (1990). Her theoretical assumptions are compatible with the theory of Krifka (2014) mentioned in the introduction to this chapter. Gunlogson is careful to make claims only for English and to allow for cross-linguistic variation. Exploring that very briefly, recall from (2) that a language like Hindi does not syntactically mark the difference between assertion and question. It uses intonation to do precisely what the simple-minded view of intonation suggests. Final rise simply turns a declarative into a question—no reason for bias to enter the picture.<sup>12</sup> If, however, declarative questions cross-linguistically indicate bias only if another option is available, it would suggest that the explanation for bias should include some reference to markedness. Gunlogson ascribes independent meanings to each component and a compositional way of combining them and thus rules out a role for such competition-based effects. I leave this here as a speculative remark.

### 9.2.2 *Echo questions*

Echo questions bear a certain similarity to declarative questions. In fact, many of the examples discussed by Noh (1998) and Iwata (2003) under the rubric of echo questions are, in fact, Gunlogson's declarative questions. And there are accounts of the two that look remarkably alike. Truckenbrodt (2009), for example, extends Gunlogson's account of declarative questions by including overt reference to the

<sup>10</sup> This predicts differences with respect to biased negative polar questions, mentioned in Gunlogson but not analyzed in detail.

<sup>11</sup> See Section 8.3.2 for discussion of (32a).

<sup>12</sup> One might say, perhaps provocatively, that Hindi mirrors the English situation in (32), suggesting a correlation between the availability of an alternative locution and the emergence of bias. To get a biased version of Hindi (2), additional intonational focus is needed on the PP. Playing devil's advocate, this may be because the prosodically enhanced version can now be compared to the prosodically simpler form.

discourse context. On this view, (33a) has the logical form in (33b) and is, in effect, the question in (33c):

- (33) a. France is a monarchy?  
 b.  $Q [A \text{ says } [\text{that France is a monarchy}]]$   
 c. Are you saying that France is a monarchy?

One could characterize this as a quasi-echo treatment of rising declaratives. Truckenbrodt's approach, however, is not to be taken literally as an echo-question approach. Gunlogson noted that (33a) could be a response to the statement *the king of France is bald*, addressing the presupposition of the previous utterance rather than the statement itself. Both Gunlogson and Truckenbrodt also emphasize that declarative questions can respond to non-linguistic but salient propositions: seeing someone get up and start picking up their things, one might felicitously ask *you're leaving?* Echo questions, on the other hand, require linguistic triggers.

Setting that difference aside for the moment, the account of echo questions in Ginzburg and Sag (2000: 259–60) is remarkably similar. Small caps in (34a) indicate the pitch accent associated with echo wh phrases:

- (34) a. You like WHO?  
 b. Who did you say (just now) that you like?

Ginzburg and Sag treat echo, or reprise questions as they call them, as regular questions. They provide supporting evidence from paraphrases with 'normal' fronted interrogatives:

- (35) a. Tracy: You like WHO?  
 b. Tracy was amazed at what I told her, so she asked me who it was that I had said I liked.

The view that echo questions belong with regular questions also characterizes, to some extent, the analysis given earlier in this book. Comorovski (1996) notes correlations between echo questions and D-linking. Dayal (1996) posits a question formation operation that can apply to prosodically prominent/D-linked wh expressions (see Chapters 4 and 7). The relevant operator is located in a projection immediately above CP:

- (36) a.  $[_{CP^*} OP_{ECHO} [_{CP} \text{ }_{+/-wh} ]]$   
 b.  $[[OP_{ECHO}]] = \lambda Z \lambda Q [\exists x_1 \dots \exists x_n [Q = Z(x_1) \dots (x_n)]]$

Recall that the Echo operator is given scope over CP to allow for both first order and second order questions. The two cases are illustrated below, with subscripting indicating inaudible strings:<sup>13</sup>

<sup>13</sup> Such questions can also express incredulity at the previous utterance. One could treat incredulity questions in terms of indirect speech acts that do not require any response from the addressee, or

- (37) a. John saw <sub>Mirabelle</sub>.  
 b. John saw WHO?  
 c. {<sup>^</sup>John saw Mirabelle, <sup>^</sup>John saw Anabelle}
- (38) a. Who saw <sub>Mirabelle</sub>?  
 b. Who saw WHO?  
 c. {who saw Mirabelle? who saw Anabelle?}

If the lower CP is interpreted as a proposition, binding of the embedded wh by  $OP_{ECHO}$  yields a set of propositions—an ordinary question. If the lower CP is interpreted as a set of propositions, binding of the embedded wh by  $OP_{ECHO}$  yields a set of sets of propositions, that is, a set of questions. Where echo questions depart from regular questions is in their answerhood conditions. Ans-C/D, from Dayal (1996), builds on ideas in Comorovski about the relationship of echo questions to discourse. It replaces reference to truth in Ans-D with reference to previously uttered propositions:

- (39)  $Ans-C/D_{ECHO}(Q_{ECHO}) = \{p[p \in Q \wedge \text{previously-uttered}'(p)]\}$

Ans-C/D applies to both first order and second order questions to yield the right results.<sup>14</sup>

The most interesting and also the most challenging aspect of echo questions, however, is their morpho-syntactic profile, seen in the following examples from Cooper (1983). Echo wh phrases may or may not move ((40a)), but when they stay in-situ, they are impervious to standard constraints on scope taking ((40b)–(40c)). The latter are illustrative of what is considered metalinguistic uses of echo questions (see also Janda 1985):

- (40) a. A: I saw Mirabelle today.  
 B: WHO did you see? / You saw WHO? /
- b. A: This animal is totally quercivorous?  
 B: This animal is totally WHAT?  
 B': \*WHAT is this animal totally?  
 B'': \*Totally WHAT is this animal?
- c. A: He voiced a querimony.  
 B: He voiced a WHAT imony?  
 B': \*WHAT did he voice a imony?

perhaps, in terms of responses reasserting the prior proposition and noting that this proposition is surprising.

<sup>14</sup> Echo questions are known not to embed. In Chapter 7, I argued for the possibility of embedding an operator similar to  $OP_{ECHO}$  to derive long-distance list answers. To capture the unembeddability of echo questions, then, we would have to locate the problem with the embeddability of Ans-C/D. I leave it as an open problem for now. Echo questions may also allow for paraphrases of the trigger, another issue that we will set aside here.

Artstein (2002) proposes a focus-based account of echo questions and takes on the challenge posed by the metalinguistic cases of echo questions. He captures (41b), for example, by defining meanings for parts of words:<sup>15</sup>

- (41) a. Bill is an orthodontist.  
 b. Bill is a <sup>WHAT</sup>-dontist?  
 c.  $\llbracket \text{what} \rrbracket^f = \{\text{ortho}, \text{perio}\}$   
 d.  $\llbracket \text{dontist} \rrbracket^o = \llbracket \beta \text{dontist} \rrbracket^o$  if  $\beta \text{dontist}$  is a word and  $\llbracket \beta \text{dontist} \rrbracket^o \in D_{\langle e, t \rangle}$   
     undefined otherwise.  
 e.  $\llbracket \text{what -dontist} \rrbracket^f = \{\alpha(\beta) \mid \alpha \in \llbracket \text{dontist} \rrbracket^f \rightarrow \beta \in \llbracket \text{what} \rrbracket^f\}$   
      $\Rightarrow \{\llbracket \text{orthodontist} \rrbracket^o, \llbracket \text{periodontist} \rrbracket^o \dots\}$

Note that the alternatives here are based on phonological form. There is also an interpretive component to the calculation of alternatives since *-dontist* will impose semantic constraints on the morphemes that can combine with it. Once there is a meaning like (41e), the rest of the interpretation follows straightforwardly. Artstein's account, thus, ingeniously incorporates metalinguistic cases of echo questions into core grammar.

### 9.2.3 Section summary

Here we saw that declarative questions and echo questions are prosodically marked in ways that signal their non-canonical status. Declarative questions are biased requests for information, much like negative polar questions, considered earlier. Echo questions are requests for clarification of prior utterances. Both are questions in the sense that they require an answer, but the truth requirement for answers is relevant only to declarative questions. Echo questions instead are cued to prior discourse. Their most important property is their disregard for syntactic, or even morphological constraints on scope. We saw, however, that it is possible to provide a compositional account even for these cases.

## 9.3 Indirect speech acts

So far in this book we have been concerned with interrogatives used for the speech act of questioning, requiring the addressee to respond in a way that settles an uncertainty about the nucleus proposition. We will now consider rhetorical questions, ability/inclination questions, and tag questions. In each case, the interrogative is used in a context where the information that the interrogative would normally elicit is already available to the speaker and, quite possibly, the addressee. The result is an indirect speech act.

<sup>15</sup> Ginzburg and Sag do not consider such cases part of the core grammar of English.

### 9.3.1 Rhetorical questions

A rhetorical question presents the mirror image of a declarative question. Syntactically an interrogative, it does not seem to be a request for any kind of information. Rather, it seems to be an assertion. The question in (42a), spoken by a member on the semantics search committee, is likely to be interpreted as a rhetorical question equivalent to *Onavi knows nothing about semantics*. The same question in (42b), spoken by a member on the syntax search committee, asks about Onavi's knowledge of the related field of semantics:

- (42) a. I don't think we should have Onavi on our short list. What does he know about semantics?  
 b. Onavi looks like an interesting syntactician. What does he know about semantics?

Caponigro and Sprouse (2007) present this paradigm to argue that there is no formal distinction between a rhetorical and an ordinary question. The issue it raises is whether rhetorical questions need to be recognized as a formal entity distinct from ordinary questions or whether the same question yields a rhetorical interpretation in one context and an ordinary information seeking interpretation in another. In other words, is the distinction semantic in nature or is the flexibility in the use of the interrogative?<sup>16</sup> I will review three broad approaches to the phenomenon, using the terms *rhetorical question* (RQ) and *ordinary question* (OQ) as convenient labels, without taking a stand on their formal status.

Sadock (1971) pointed out some differences between RQs and OQs. The use of *after all* and *yet* is only compatible with RQ interpretations:

- (43) a. After all, who helped Luca when he was in trouble?  
 b. Who helped Luca when he was in trouble? Yet he managed to become what he is now.

Phrases like *I'm really curious* or *I really don't know*, instead, are only compatible with OQ interpretations:

- (44) a. I'm really curious: who helped Luca when he was in trouble?  
 b. I really don't know: Who helped Luca when he was in trouble?

A question with a strong NPI, which we saw in Section 9.1 to introduce negative bias, can only be interpreted as an RQ:

<sup>16</sup> This leads to an interesting question. Can there be grammaticalized forms that can only be used for rhetorical questions? Xu (2012) argues that Chinese *nandao* questions have such specialized uses but in later ongoing work re-analyzes them as inherently *biased*. The connection between *bias* and rhetorical uses is a promising line to pursue.



- (45) a. After all, who lifted a finger to help Luca?  
 b. Who gave a damn when Paolo was in trouble? Yet, he made it.
- (46) a. #I'm really curious: Who lifted a finger to help Luca?  
 b. #I really don't know: Who gave a damn when Paolo was in trouble?

Sadock's diagnostics establish that RQs have the characteristics of assertions rather than questions. In these examples, the negation of the nucleus proposition is asserted: *no one helped Luca*, *no one lifted a finger to help Luca*, etc. This fits in with Bartels' (1997) observation that wh-RQs have a fall at the end, unlike wh-OQs which have a final rise. These findings, however, are noted not to be definitive. It is worth emphasizing that Sadock's diagnostics do not provide an answer to whether RQs and OQs have distinct denotations. They simply show that the discourse contribution of RQs aligns with that of assertions.

Among those who claim a semantic distinction between RQs and OQs are Ladusaw (1979) and Gutierrez Rexach (1997) who, differences aside, take RQs to denote the empty set. Han (2002) also argues for a semantic distinction. She posits an operation that takes a question nucleus and turns it into a negative proposition, in essence treating the wh expression as a negative quantifier:

- (47) a. (After all) Who helped Luca?  
 b.
- $$\begin{array}{c}
 \text{CP } \neg \exists x[\text{person}(x) \wedge \text{helped}(x, l)] \\
 \swarrow \quad \searrow \\
 \text{DP } \lambda X [\neg \exists x[\text{person}(x) \wedge X(x)] \quad \text{C}' \lambda x_i[\text{helped}(x, l)] \\
 | \qquad \qquad \qquad \triangle \\
 \text{Who}_i \rightarrow \text{nobody}_i \quad \text{t}_i \text{ helped Luca}
 \end{array}$$

A problem with Han's approach, pointed out by Rohde (2006) and Caponigro and Sprouse (2007), is that it can only handle questions with a clear negative bias. But rhetorical questions are known to be compatible with positive assertions as well:

- (48) a. Is the Pope Catholic?  $\Rightarrow$  The Pope is Catholic.  
 b. Does the sun rise in the east?  $\Rightarrow$  The sun rises in the east.

Rohde shifts the discussion from the formal properties of RQs to the contexts in which they can be used. An interrogative is interpreted as an RQ rather than an OQ if the speaker and addressee share prior commitments, in the sense of Gunlogson (2003), to similar and obvious answers. In such contexts they can be used to assert the affirmative or the negative version of the nucleus proposition. Furthermore, they can be used to make a higher order assertion, one which taps into the very obviousness of the answer, to respond to an implicit or explicit question under discussion. If it is known that Bill is a leading capitalist and a naive

speaker asks question (49a), the addressee can respond with a regular statement or with an interrogative she can count on to be understood rhetorically:

- (49) a. A: Does Bill believe in the free market?  
       B: Obviously, he does.  
       B': Is the Pope Catholic?

Rhetorical questions, then, show that an interrogative asked in a context where the answer is obvious need not represent a failed conversational move. Rather, it can be put to use to make an effective assertion.

### 9.3.2 *Ability/inclination questions*

An emphatic assertion is not the only possible conversational contribution of an interrogative whose answer is obvious. The well-known example of ability questions shows the possibility of using interrogatives as directives, a request or an order:

- (50) a. Can you pass the salt?  
       b. Could you take out the garbage?

Searle (1975) introduces ability questions in his discussion of indirect speech acts. They have been discussed more recently by Asher and Lascarides (2001) and Reese (2007).<sup>17</sup>

Strictly speaking, these are polar questions which could very well be interpreted canonically. This is made clear by changing to a third person subject ((51a)), or by changing the tense ((51b)). In both cases, the directive is lost:

- (51) a. Can Bill pass the salt?  
       b. Could you have passed the salt?

This is further corroborated by the (im)possibility of adding request-particles:

- (52) a. Can you, please, pass the salt?  
       b. \*Can Bill, please, pass the salt?

Note that the contexts in which interrogatives are interpreted as directives are those in which the answer to the question is obvious to both speaker and addressee, and moreover the answer is a positive one. Question (53), where the obvious answer is negative, cannot be interpreted as a directive:

- (53) Can you lift this truck?

<sup>17</sup> See also Morgan (1978), Horn and Bayer (1984), and Asher (2007).

The reason (50a) can be interpreted as a request is because the obvious answer meets the felicity conditions for the speech act of request. The reason (53) cannot be so interpreted is because the obvious answer does not meet those requirements. If the conditions are right, (53) could be interpreted either as an information question (maybe the addressee has a machine for lifting heavy objects) or as a rhetorical question (maybe the question under discussion is such that it requires a clearly negative response). To be interpreted as a directive, the speaker must believe it to be possible for the addressee to lift the truck and also be the beneficiary of the proposed act in some way.

Searle notes that it is not necessary for the speaker of such interrogatives to know the answer. Example (54) can be interpreted as an indirect request, contingent on a positive response:

- (54) Could you please lend me a dollar?

A final point to note about interrogatives as directives is their idiomatic character. (55a)–(55b) are semantically equivalent but only the former qualifies as a directive:

- (55) a. Can you (please) hand me that hammer?  
b. Are you able to (\*please) hand me that hammer?

As Searle notes, while conversational principles can explain the emergence of indirect speech acts, the conventional aspect of the phenomenon needs to be recognized. Asher and Lascarides (2001) model such questions formally as a complex speech act, combining the properties of a question and a directive. We will look at their account of complex speech acts in relation to tag questions in the next section.

Let me end this discussion by noting that questions that purportedly ask for the addressee's inclinations also belong in this category:

- (56) a. Would you mind moving over?  
b. Would you mind turning up the heat?

Here the actual desires of the addressee are not at issue but the interrogative sets up a situation in which the speaker conveys her own desires as a way of suggesting a course of action to the addressee. I am not sure if this kind of question has been formally analyzed.

### 9.3.3 *Tag questions*

Returning to tag questions, Ladd (1981) identifies three types of tag questions, two of which he characterizes as nuclear and one as post-nuclear. The distinction is based on whether there is a prosodic break between the declarative anchor and the interrogative tag.

The two nuclear tag questions are shown in (57) and (58), uttered in a context where A and B are attempting a task at which neither is proficient, but at which Julie is known to excel:

- (57) a. A: Julie wouldn't do it that way.  
b. B: Well, Julie isn't here, / is she.

The tag follows after an intonation break, indicated by / and is uttered with a final fall. The speaker is certain about the proposition in the anchor. The function of the tag is to get A to acknowledge this fact, thereby conveying the irrelevance of A's statement about Julie's abilities.

In (58), the tag is uttered with a final rise and its function is to elicit confirmation of the fact expressed by the anchor. It can only be used in contexts where B is not completely certain about Julie's absence but has a weak belief that she isn't present:

- (58) a. A: Can Julie do it for us?  
b. B: Well, Julie isn't here, / is she?

Finally, a tag can be uttered without a break or with just a weak break after the anchor, indicated by =. Post-nuclear tag questions can be neutral. Note that a neutral interpretation is only possible with a negative anchor:

- (59) a. A: We need to find someone who has done this before.  
b. B: Julie isn't here = is she?

Reese (2007) and Reese and Asher (2010) build on Ladd (1981) and Huddleston and Pullum (2002) in their analysis of tag questions. They consider the nuclear cases to involve two illocutionary acts, an *assertion* and a *question* and the post-nuclear case to involve a complex act which they represent as *assertion.question*. Setting aside the details of implementation, the key aspect of the explanation rests on defining appropriate discourse relations between the components of such combined speech acts.

They first establish that nuclear tag questions are compatible with Sadock's test for assertion:

- (60) a. The conference should be exceptional this year. After all, Julie is coming / isn't she.  
b. The conference might be sub-par this year. After all, Julie isn't coming / is she.

They also show that nuclear tag questions are not neutral since they do not pass the *by any chance* test. They are questions, however, as shown by the *tell me* test:

- (61) a. #Jane {is/isn't} coming, by any chance / {isn't/is} she.  
b. Tell me, Jane {is/isn't} coming / {isn't/is} she.

Reese and Asher also establish that post-nuclear tag questions have different properties, based on whether the anchor is positive or negative.<sup>18</sup> Those with positive anchors behave more or less like nuclear tag questions. The diagnostics of *after all* and *tell me* establish that they are assertions and questions. The inclusion of a positive polarity item in a post-nuclear tag question with a negative anchor coerces a neutral reading, as shown in (62):

(62) Jane isn't coming {too/\*either}, by any chance = is she?

They claim that in such cases, negation scopes over the speech act and functions metalinguistically. As such, it blocks the inference that the speaker believes the proposition, which then allows the tag to elicit a response like an ordinary interrogative.

Tag questions, as analyzed by Reese and Asher, instantiate an interesting type of construction with both declarative and interrogative forms and which can come with a range of prosodic patterns. They therefore present an ideal case study for the investigation of possible combinations of speech acts. It goes without saying that tag questions also have interesting properties that are strictly semantic, which we set aside in keeping with the focus of this chapter.

#### 9.3.4 Section summary

In this section we looked at interrogatives used for speech acts other than, or in addition to, eliciting information. Rhetorical questions can be fruitfully analyzed either as biased questions (negative or positive) whose answer is known to speaker and addressee or as ordinary questions whose answer is also patently obvious. In either case, under appropriate conditions, this fact can be exploited to make a forceful assertion. We also looked at contexts in which ability questions can be used as directives. This happens if the answer that is manifest in the context provides (one of) the necessary conditions for the directive. Finally, we looked at tag questions which combine an assertion along with a request for acknowledgement/confirmation. In all such cases, we found that the canonical meaning of the question is an essential ingredient in the explanation.

### 9.4 Beyond information seeking

In this chapter we looked at negative polar questions, polar questions with strong negative polarity items, declarative questions, echo questions, rhetorical questions, directive questions, ability questions, and tag questions. The list is long and varied but there are several strands that tie them together, strands that emerge when we look at them against the backdrop of canonical questions. There are two dimensions, in particular, along which we can understand non-canonical

<sup>18</sup> Tag questions can be positive on the anchor and the tag, *Julie is here, is she?* See Reese (2007).

questions. One is the background of beliefs against which a question is asked, the other is its discourse contribution.

The standard question presupposes ignorance on the part of the speaker and a presumption of knowledge on the part of the addressee. We know, of course, that a speaker who already has the relevant information can felicitously pose the question to an addressee whose knowledge is in doubt. This is the case of exam/quiz questions, which we did not specifically discuss here. We looked instead at intermediate cases where the speaker has enough information to form expectations about a possible answer but not enough to feel confident about it. Several of the constructions discussed here fell into the class of such biased questions. While there may be specific pieces of the structure that conspire to create bias, the overall goal is still to gain information. In that sense, biased questions conform to the speech act of questioning, while possibly including an assertive component with regard to the proposition it is biased towards. Going further on the information spectrum, if a question is asked in a context where the answer is obvious to both speaker and addressee, it can no longer function as a request for information. It can then take on rhetorical force and function as an emphatic assertion. Or, assuming that the obvious answer has the appropriate property, it can be interpreted as a directive. We also looked at questions that are requests for repetition, the typical echo question which requires a crucial shift in the definition of an answer. Or the echo question may simply be an expression of surprise at the prior utterance. In all such cases, we saw that there are cues from syntax, prosody, and context that alert us to the fact that the speaker is departing from the norm.

That the bulk of this book was devoted to describing the seemingly vanilla cases of questioning is proof of their rich and intricate nature and of their centrality in the grammar of natural language. Our exploration of issues related to non-canonical questions was comparatively limited but hopefully it was sufficient to establish that departures from the norm are fully amenable to formal explanation. What they require is an articulated theory of the interplay between syntax, semantics, pragmatics, and prosody, highlighting a point we made at the very beginning, that questions are a quintessential interface phenomenon.



# Afterword

We have reached the end of our investigation into the semantics and pragmatics of questions, and it is time for me to step back and take stock of what I have and have not accomplished. The first is easy to do: I can simply repeat the concluding section of each chapter, but I will not bore the reader with that. The second is harder, by definition. There are many issues and contributions that I am well aware I have not touched upon or not engaged with at the level they deserve. In particular, I am keenly aware that insights from work conducted in alternative research paradigms have not been covered adequately and in some cases have not been covered at all. And I have most certainly simply missed many interesting contributions within the paradigms I am familiar with, for even with the resources afforded by the internet, there is always something out there that one has not yet discovered.<sup>1</sup> There are, however, some conscious choices I have made in putting together this survey and I'll try to share the reasoning behind a few of them here.

One might reasonably have expected a survey of questions to be conducted within a larger study of *wh* constructions. We briefly touched upon the connection between free relatives and interrogatives in Chapter 5, but I stopped at identifying and distinguishing between them. There are many studies that actually explore the semantic overlap between questions, relative clauses, correlatives, unconditionals, and pseudoclefts—to name just a few. No doubt taking these studies into consideration would have added an interesting dimension to our understanding of questions. Research on the semantics and pragmatics of questions is quite extensive, as we have seen, so that extending in this direction would surely have come at the cost of leaving out core aspects of the study of questions.

Within the domain of questions, I did not mention *sluicing*, an extremely interesting phenomenon that lies at the interface between syntax and semantics. Our understanding of this topic relies on a fully articulated theory of questions but I would venture to say that work on *sluicing* has not prompted significant shifts in our view of the semantics of questions. Its primary contribution has been in deepening our knowledge of how *ellipsis* works in natural language and in probing the difference in constraints on *elided* and *non-elided* structures. We might perhaps compare *sluicing* with *scope marking*, another topic at the intersection of syntax and semantics, but one that I have referred to more than once.

<sup>1</sup> For example, Uegaki (2015), of particular relevance to Chapters 3 and 5, came to my attention too late to incorporate into the discussion. Similarly, I became aware of Heim (2012) and Theiler (2014) only at the copyediting stage.



This was prompted by the impact that insights from the analysis of scope marking have had on some key aspects of the semantics of questions.

Perhaps the area of inquiry that I most regret not exploring properly is the relationship between prosody and interpretation. We touched upon it briefly in the last two chapters, in relation to the study of polar vs. alternative questions and in relation to the study of bias, but prosody has broader relevance in the domain of questions. It is sensitive to the types of discourse within which questions are asked and answered. Thus, the prosody–meaning–use triptych would also have benefited from the inclusion of information structural perspectives on questions and answers.

Even within the semantics of questions, there are topics I barely touched upon, the role of modals and discourse particles, for example. In this as elsewhere, the one overarching constant was the constraint of time and space. As it is, the book has ended up longer than it was supposed to be. A final caveat is in order. Of the topics I have discussed at some length in the foregoing chapters, many issues are very much under debate. Probably most of what I have classified as open problems will be background material by the time this book comes out in print. The reader is requested to keep this in mind.

On a personal note, this book should have been completed several years ago. While this is not something I am particularly proud of, I have found the proverbial silver lining in the delay. In reading up for this survey I have learnt a great deal from recent work on this rich, multi-faceted, and yet basic natural language activity of asking and answering. When I look at how much has been achieved in 40 some years of research, it seems to me that no one could ask, rhetorically or otherwise: *what can truth conditional semantics tell us about non-declaratives?*

Well, perhaps there is one intonational contour that would make it possible to ask this question.<sup>2</sup> I leave it to the reader to figure out how that fits into the theory of questions!

<sup>2</sup> Hint: the speaker is quite likely to follow up with: *You've got to be kidding!*

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